

SeqStudio™ Genetic Analyzer Instrument and Software

USER GUIDE

for use with:

SeqStudio™ Data Collection Software v1.2.5

SeqStudio™ Plate Manager

SeqStudio™ Remote Monitoring App

SAE Administrator Console v2.1 or v2.2

SeqStudio™ Genetic Analyzer Cartridge (Cat. No. A33671)

SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. A41331)

Publication Number MAN0018646

Revision D



For Research Use Only. Not for use in diagnostic procedures.

ThermoFisher
S C I E N T I F I C



Life Technologies Holdings Pte Ltd |
Block 33 |
Marsiling Industrial Estate Road 3 |
#07-06, Singapore 739256

Products manufactured at this site:

- SeqStudio™ Genetic Analyzer
- SeqStudio™ Data Collection Software
- SeqStudio™ Plate Manager (desktop)
- SeqStudio™ Genetic Analyzer Cartridge
- SeqStudio™ Genetic Analyzer Cartridge v2
- Security, Auditing, and E-signature (SAE) module



Life Technologies Corporation |
200 Oyster Point Blvd |
South San Francisco, California 94080 USA

Products manufactured at this site:

- SeqStudio™ Remote Monitoring App



Life Technologies Corporation |
1455 Concord Street |
Framingham, Massachusetts 01701 USA

Products manufactured at this site:

- SeqStudio™ Genetic Analyzer Cathode Buffer Container

Revision history: MAN0018646 D (English)

Revision	Date	Description
D	22 April 2026	Updated for use with the BigDye™ Terminator Plus Cycle Sequencing Kit. Added compatibility with InnoviGene™ Suite, SAE v2.2, and Windows™ 11. Corrected the instrument front panel indicator information and the median PUP description. Added instructions to avoid performing a software upgrade or powering off the instrument when a cartridge is loaded. Added troubleshooting information for a loss of power to the instrument.
C	15 May 2024	Updates for SeqStudio™ Data Collection Software v1.2.5: • Information about AnyDye-8 was added. • A procedure to accept a failed HID install run was added.
B.0	23 June 2022	Updates for SeqStudio™ Data Collection Software v1.2.4: • Connect cloud-based platform was updated to Thermo Fisher™ Connect Platform. • Procedures for the Export Status screen were added. • The SAE server connection procedure was updated to include HTTPS as the default selection. • Changes to SeqStudio™ Plate Manager v2.0 were added. See Chapter 5, "Create or modify a plate setup from the SeqStudio™ Plate Manager". • The publication number for the <i>SAE Administrator Console v2.1 User Guide for Capillary Electrophoresis Products</i> (Pub. No. MAN0025849) was updated. • USB Wifi Dongle installation instructions were added. • The software compatibility for SAE v2.0 was updated to include v1.1 to v1.2.3. • SeqScreener Gene Edit Confirmation and Microsatellite Analysis Software were added to the list of secondary analysis apps on the Thermo Fisher™ Connect Platform. • The United Kingdom Conformity Assessment mark was added.
A.0	4 November 2019	New user guide for SeqStudio™ Data Collection Software v1.2. The v1.2 software includes options for HID kits. This user guide replaces Pub. No. MAN0016138.

The information in this guide is subject to change without notice.

DISCLAIMER: TO THE EXTENT ALLOWED BY LAW, THERMO FISHER SCIENTIFIC INC. AND/OR ITS AFFILIATE(S) WILL NOT BE LIABLE FOR SPECIAL, INCIDENTAL, INDIRECT, PUNITIVE, MULTIPLE, OR CONSEQUENTIAL DAMAGES IN CONNECTION WITH OR ARISING FROM THIS DOCUMENT, INCLUDING YOUR USE OF THE DOCUMENT OR THE PRODUCT.

Important Licensing Information: These products may be covered by one or more Limited Use Label Licenses. By use of these products, you accept the terms and conditions of all applicable Limited Use Label Licenses.

TRADEMARKS: All trademarks are the property of Thermo Fisher Scientific and its subsidiaries unless otherwise specified. Microsoft, Windows, and Edge are trademarks of the Microsoft group of companies. Dell and OptiPlex are trademarks of Dell Inc. or its subsidiaries. Intel and Core are trademarks of Intel Corporation. Apple and Safari are trademarks of Apple Inc. Google, Google Play, and Chrome are trademarks of Google LLC. Mozilla and Firefox are trademarks of the Mozilla Foundation.

©2019-2026 Thermo Fisher Scientific Inc. All rights reserved.

Contents

■ CHAPTER 1	Product information	12
	Instrument overview	13
	Parts of the instrument	14
	Instrument front panel indicator	15
	SeqStudio™ Genetic Analyzer consumables	15
	Cartridge	16
	SeqStudio™ Genetic Analyzer Cathode Buffer Container	20
	Radio frequency identification	21
	Important notice regarding use of consumables that exceed supported limits	22
	Software features	22
	SeqStudio™ Plate Manager overview	24
	Computer and operating system options	25
	Recommended browsers	25
	SeqStudio™ Remote Monitoring App overview	26
	Network connection options	27
	Network and password security requirements	27
	Network configuration and security	27
	Password security	28
	Experiment types	28
	Data output	29
	Secondary analysis software	30
	Workflow: Thermo Fisher™ Connect Platform or desktop	32
	Workflow: instrument	33
■ CHAPTER 2	Prepare the samples and the instrument	34
	Precautions for use	35
	Power on the instrument	35
	Sign in	36
	Sign in with the Guest instrument profile	36
	Sign out	37
	Parts of the home screen	37

Prepare the samples	38
Sample preparation guidelines	38
Plate layout and loading guidelines	39
Prepare the plate	40
Prepare the tubes	40
Prepare the instrument	41
Install the USB Wifi Dongle	41
Check the consumables status	42
Load the CBC, the sample plate, and the cartridge	43
 ■ CHAPTER 3 Use the instrument with the Thermo Fisher™	
Connect Platform	45
Understanding instrument and Thermo Fisher™ Connect Platform interaction	45
"Connect" versus "link"	45
First user who links is assigned administrator role	45
Local versus Thermo Fisher™ Connect Platform instrument profiles	46
If you link when you <i>are</i> signed in to the instrument	46
If you link when you <i>are not</i> signed in to the instrument	48
Thermo Fisher™ Connect Platform instrument profile roles and functions	49
For more information on using the Thermo Fisher™ Connect Platform	50
Thermo Fisher™ Connect Platform administrators for an instrument	50
Thermo Fisher™ Connect Platform administrator functions	50
Assign instrument administrator role to other users	50
Manage the users and administrators of your instrument	51
Disconnect individual users from an instrument	52
Register and obtain a Thermo Fisher™ Connect Platform account	53
Link the instrument to your Thermo Fisher™ Connect Platform account	54
Change your own Thermo Fisher™ Connect Platform instrument profile PIN	55
Set up email notifications from the instrument	56
 ■ CHAPTER 4 Use the instrument with the Security, Auditing, and	
E-signature (SAE) Administrator Console	58
Overview of the Security, Auditing, and E-signature (SAE) Administrator	
Console components	59
Compatibility: SAE and SeqStudio™ instrument software versions	60
Configuring the SAE module	60
Instrument functionality when SAE is enabled or disabled	60
Enable SAE on the instrument (administrator only)	60
Disable SAE (administrator only)	61
Disable SAE on the SeqStudio™ Genetic Analyzer	62
Sign in with SAE enabled	63
Specify audit reasons	64

E-signature requirements when you start a run	64
Sign a plate setup	66
View E-signature records for a plate setup	66
Sign a dye calibration or an install run	67
View E-signature records for a dye calibration or an install run	67
Signing in after automatic screen locking	67
Use the instrument when the SAE server is offline	68
Auditing of imported plate setups	68
SAE error messages and actions	69
 ■ CHAPTER 5 Create or modify a plate setup from the SeqStudio™	
Plate Manager	70
Options for creating plate files in the Plate Manager software	70
Overview of plate setup settings	71
Set up a plate using default settings (Plate Manager)	72
Access the Plate Manager	72
Create or open a plate setup PSM file	74
Enter plate properties	75
Assign wells: Sample and run information	76
Keyboard shortcuts for the sample table	77
Save a plate setup in the Plate Manager	79
Additional plate settings (Plate Manager)	80
Specify replicate injections	80
Edit analysis settings	80
Modify the default file name convention	81
Hide or share a plate (Plate setup security)	81
Define custom fields	82
 ■ CHAPTER 6 Create or modify a plate setup from the instrument	83
PSM and CSV plate setup files for import into the instrument	84
Shared (public), hidden (my plates), and guest plate setup files	85
Overview of plate setup settings	85
(Optional) Set up for auto export of sample data files (AB1 and FSA)	86
Set up a plate using default settings (instrument)	87
Create or import a plate setup	87
Enter plate properties	88
Assign wells: run module, size standard, dye set, and kit	90
Assign wells: sample name, sample type, and custom fields	92
Set optional plate settings (instrument)	93
Specify replicate injections	93
Modify analysis settings	94
Modify the default file name convention	94

Hide or share a plate (Plate setup security)	95
Define custom fields	96
(<i>Optional</i>) View the injection list, change injection settings or order, or specify replicates and re-injections	97
CHAPTER 7 Start and monitor a run	98
Load the plate or the tube assembly	98
Select a plate setup and start a run	99
Automatic file cleanup	100
Lock the touchscreen	101
Monitor a run from the Thermo Fisher™ Connect Platform	101
Open the Remote Monitoring App from Instrument Connect App	101
Open the Remote Monitoring App from the Plate Manager	103
View notifications from the instrument on your Thermo Fisher™ Connect Platform account	105
View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform	105
Monitor a run from a mobile device	109
Monitor a run from the instrument	112
(<i>Optional</i>) View the injection list, change injection settings or order, or specify replicates and re-injections	112
View the run status	112
View real-time results	113
Pause a plate or cancel or stop injections	114
View results when the run is completed	115
Unload the plate or the tube assembly	116
CHAPTER 8 View and analyze results	118
View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform	118
View results on the instrument	119
View results when the run is completed	119
Fragment/HID analysis results	121
Sequence analysis results	123
EPT plot	126
View results for a previously run plate (run history)	127
Data quality alerts	128
Siz calling and basecalling quality alerts	128
Edit injection parameters and re-inject samples	129
Export a report (QC report)	129
Export results from the instrument (sample data files and QC reports)	130
Analyze data	130
(<i>If needed</i>) View the export status for sample data files	131
View and export sample data files at the plate-level	131
View and export individual sample data files	132

■	CHAPTER 9	Manage the software (Plate Manager)	136
		Add a custom dye calibration to the Plate Manager	136
		Download a plate setup template as a CSV file	136
		Save a plate setup as a PDF	137
		Save a plate setup as a CSV file	137
		Create a plate setup template	137
		Delete plate setups	138
		Manage email notifications in the Thermo Fisher™ Connect Platform	139
		Manage run modules	139
		Manage size standards	140
		Manage analysis settings	140
		Fragment/HID analysis settings (size calling)	141
		Sequencing settings (base calling)	143
■	CHAPTER 10	Manage the software (instrument)	144
		Connect the instrument to a network drive	144
		Determine IP address for a computer on a network	144
		Create folders and enable network folder sharing	145
		Connect to a network drive	146
		Set up a default Thermo Fisher™ Connect Platform location for opening plate setups	146
		Set up a default Thermo Fisher™ Connect Platform location for saving results (auto export)	147
		Set up a default network location for opening plate setups	148
		Set up a default network location for saving results (auto export)	149
		Link the instrument to your Thermo Fisher™ Connect Platform account	151
		Lock the touchscreen	152
		Manage plate setups	153
		Export or delete a plate setup (PSM file)	153
		Import a plate setup from a CSV or PSM file	154
		Manage run settings (instrument)	154
		Manage run modules	154
		Manage size standard definitions	157
		Manage analysis settings	158
		Modify the default file name convention	162
		Manage instrument settings	163
		Display instrument hardware and software information	163
		Change the instrument name	163
		Enable Demo mode	164
		Manage date and time settings	164
		Manage the network configuration	165
		Check for software updates (administrator only)	166

Manage instrument profiles on the instrument	167
Local instrument profile roles and functions	167
Create a local instrument profile for another user (administrator only)	167
Change the role of a local instrument profile (administrator only)	168
Delete an instrument profile from an instrument (administrator only)	168
Delete the PIN for a local instrument profile (administrator only)	169
Create your own local instrument profile	169
Change your own local instrument profile PIN	169
Manage storage space	170
Automatic file cleanup	170
Export or delete a plate setup (PSM file)	170
Export results from the instrument (sample data files and QC reports)	171
Delete a run history	171
Back up user data (plates and results) (administrator only)	171
 ■ CHAPTER 11 Maintain the instrument	 173
Regular maintenance tasks	173
Check the cathode buffer fill level	173
Clean the instrument exterior and touchscreen	174
Clean the autosampler	174
Manage the cartridge	175
Cartridge storage	175
Set cartridge storage mode (administrator only)	177
Check or export the Instrument-cartridge history	177
Check or export the cartridge history	177
Fill the capillary array and refresh the polymer delivery system	178
Remove the cartridge	178
Insert the cartridge	179
Store the cartridge	180
SeqStudio™ Integrated Capillary Protector	181
Long-term on-instrument cartridge storage	182
Install cathode buffer	183
SeqStudio™ Genetic Analyzer Cathode Buffer Container storage	183
Remove the SeqStudio™ Genetic Analyzer Cathode Buffer Container	183
Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)	183
Insert the SeqStudio™ Cathode Buffer Container	184
 ■ APPENDIX A Troubleshooting	 186
Troubleshooting resources	186
Troubleshooting workflow	187
Export log files for plates, install runs, injections, and instrument	187
View the raw data and the EPT plot	188
Data quality alerts	188

Sizecalling and basecalling quality alerts	189
Check size standard quality	189
Instrument troubleshooting	190
Cartridge troubleshooting	192
Sample and data troubleshooting	193
EPT data	199
Software troubleshooting	199
Service tools	202
 ■ APPENDIX B Link the instrument to your Thermo Fisher™ Connect Platform account—detailed instructions	 203
Workflow: Set up the instrument for Thermo Fisher™ Connect Platform access	203
Network requirements	203
Link the instrument from a mobile device	204
Create a Thermo Fisher™ Connect Platform PIN and generate the QR code on the instrument	204
Register the instrument with the Instrument Connect App	205
Link the instrument using a link code	205
Set up email notifications from the instrument	209
 ■ APPENDIX C Dye calibration and install standard checks	 211
Calibrate dyes	211
Overview of system dye set and custom dye set calibration	211
System versus custom dyes	212
Factory, auto, and manual calibration	212
Determine if manual calibration is required	212
Perform a system dye calibration	214
Perform a custom dye calibration	216
View the dye calibration history	221
Import or export a custom dye set	222
Perform an install run	223
Overview of install checks	223
Perform an install run	224
View install run results	225
Accept a failed HID install run	228
Export the install run report	229
View the install run history	232
 ■ APPENDIX D Create and run an HID plate	 233
HID analysis requirements	233
Pull-up reduction feature	234
Resolution algorithm	235

Workflow: HID analysis	236
HID install run	237
HID run module, dye sets, size standard definitions, and kits	237
Manage kits (HID applications only)	238
■ APPENDIX E Run modules	239
Run modules, read lengths, size ranges, and run times	239
■ APPENDIX F Parts and materials	241
Required materials not supplied	241
■ APPENDIX G Instrument specifications and layout	243
Instrument dimensions	243
Instrument clearances	243
Environmental requirements	244
Electrical requirements	245
Electrical protective devices	245
Network requirements	246
Safety requirements	246
Safety practices	246
Required safety equipment	247
■ APPENDIX H Safety	248
Symbols on this instrument	248
Conformity symbols	249
Safety alerts on this instrument	250
Location of safety labels	250
Safety information for instruments not manufactured by Thermo Fisher Scientific	251
Instrument safety	251
General	251
Physical injury	251
Electrical safety	252
Cleaning and decontamination	252
Instrument component and accessory disposal	252
Laser	253
Safety and electromagnetic compatibility (EMC) standards	254
Safety compliance	254
EMC	254
Environmental design	255
Radio compliance	255

Chemical safety	256
Biological hazard safety	256
■ APPENDIX I Documentation and support	257
Related documentation	257
Customer and technical support	257
Limited product warranty	258
Index	259



Product information

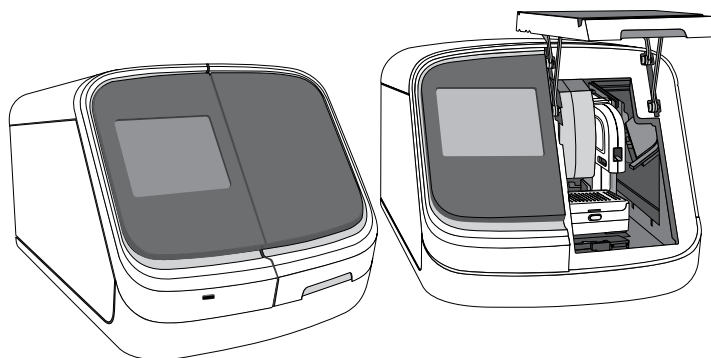
■ Instrument overview	13
■ Parts of the instrument	14
■ SeqStudio™ Genetic Analyzer consumables	15
■ Software features	22
■ SeqStudio™ Plate Manager overview	24
■ SeqStudio™ Remote Monitoring App overview	26
■ Network connection options	27
■ Network and password security requirements	27
■ Experiment types	28
■ Data output	29
■ Secondary analysis software	30
■ Workflow: Thermo Fisher™ Connect Platform or desktop	32
■ Workflow: instrument	33

Instrument overview

The Applied Biosystems™ SeqStudio™ Genetic Analyzer with SeqStudio™ Data Collection Software is a fluorescent dye-labeled genetic analysis system using capillary electrophoresis technology. It enables both sequencing and fragment analysis applications without the need to switch polymer type or capillary array length.

The instrument uses a self-contained, replaceable cartridge with:

- A 4-capillary array
- A universal polymer capable of performing sequencing and fragment analysis
- A polymer delivery system (PDS)
- Anode buffer



The SeqStudio™ Genetic Analyzer automatically:

- Performs an optical alignment each time a cartridge is inserted.
- Performs an automatic spectral calibration adjustment (auto calibration) for each sample to correct for spectral overlap.

The instrument is compatible with 96-well Standard plates and 8-strip Standard tubes.

The SeqStudio™ Genetic Analyzer is a stand-alone instrument. It is run directly from the touchscreen with SeqStudio™ Data Collection Software and does not require a computer. Plate setup can be done directly on the touchscreen, on a computer with SeqStudio™ Plate Manager, or on the Thermo Fisher™ Connect Platform. A run can be monitored directly on the instrument touchscreen or remotely on the Thermo Fisher™ Connect Platform.

Note: The Thermo Fisher™ Connect Platform is not supported for HID analysis.

Parts of the instrument

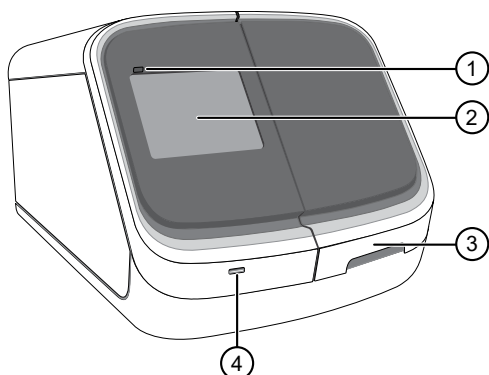


Figure 1 Front of the instrument

- ① Front panel indicator—Shows the status of the instrument
- ② Touchscreen—User interface
- ③ Door—Provides access to the cartridge, the cathode buffer, and sample plate or tubes
- ④ USB port

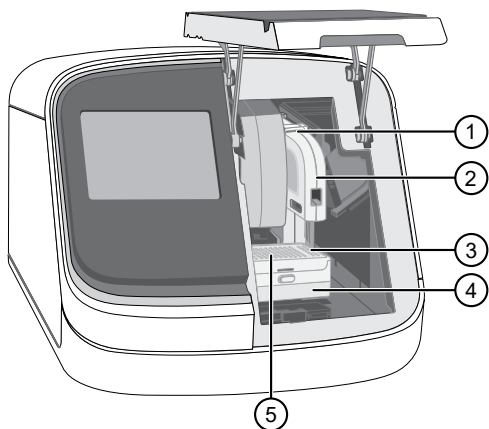


Figure 2 Interior of the instrument

- ① Cartridge rails
- ② Cartridge
- ③ Cathode buffer (located inside the autosampler)
- ④ Autosampler (contains the plate or tube holder and the cathode buffer)
- ⑤ Plate or tube holder

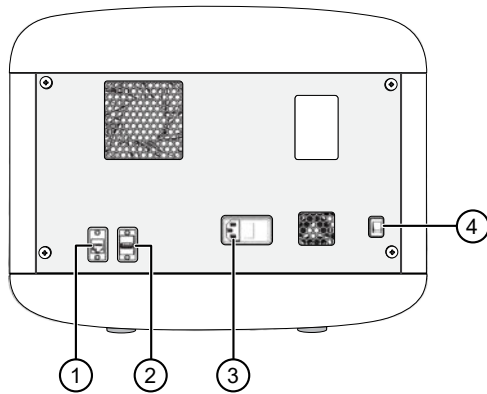


Figure 3 Rear of the instrument

- ① RJ45 ethernet port
- ② USB port for use with Wifi Dongle (dongle not shown)
- ③ Power receptacle
- ④ On/Off switch

Instrument front panel indicator

Indicator	Instrument status
All lights off	Powered off
Blue light (blinking)	Starting up (initializing)
Blue light (solid)	The instrument is initialized and operating normally
Amber light (blinking)	Instrument is in an error state

SeqStudio™ Genetic Analyzer consumables

Description	Cat. No.	Amount	Storage
SeqStudio™ Genetic Analyzer Cartridge	A33671	1 cartridge with an optical cover and a SeqStudio™ Integrated Capillary Protector attached for shipment and storage.	See Table 1 on page 18
SeqStudio™ Genetic Analyzer Cartridge v2	A41331	1 cartridge with an optical cover and a SeqStudio™ Integrated Capillary Protector attached for shipment and storage.	See Table 2 on page 19
SeqStudio™ Genetic Analyzer Cathode Buffer Container	A33401	1 package of 4	See page 183
SeqStudio™ Integrated Capillary Protector	A31923	1 (single-use) for future storage	See page 182

Cartridge

The SeqStudio™ Genetic Analyzer uses a 4-capillary, self-contained, replaceable cartridge. The following cartridges are available:

Cartridge	No. of injections	No. of samples	Storage
SeqStudio™ Genetic Analyzer Cartridge (Cat. No. A33671)	125 injections	500 samples	See Table 1 on page 18
SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. A41331)	250 injections	1,000 samples	See Table 2 on page 19

Each cartridge:

- Contains a capillary array, a polymer delivery system, polymer, and anode buffer. See Figure 4 and Figure 5.
- Is shipped with an optical cover and an Integrated Capillary Protector. See Figure 6.
- Has a radio frequency identification (RFID) tag, which is used by the instrument to track remaining usage and expiration.

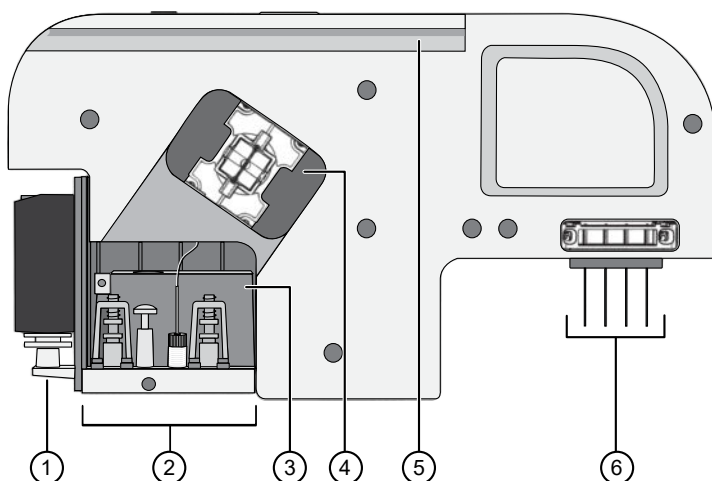


Figure 4 Parts of the cartridge

- ① Universal polymer—Supplies polymer to the polymer delivery pump.
- ② Polymer delivery system—Pumps polymer into the capillary array.
- ③ Anode buffer reservoir—Genetic Analysis running buffer to support electrophoresis.
- ④ Optical detection window
- ⑤ Cartridge track—To insert the cartridge into the instrument.
- ⑥ Capillary array (cathode end)—Tips of the four capillaries that enable electrophoretic separation of fluorescent-labeled DNA fragments.

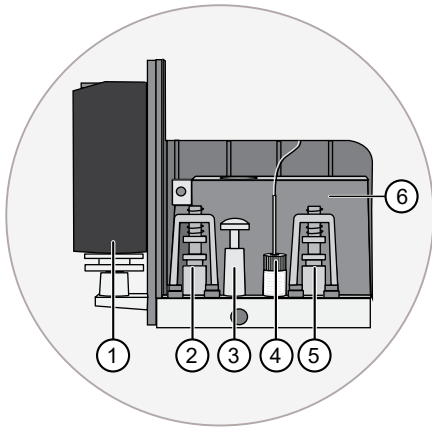


Figure 5 Parts of the polymer delivery system

- ① Universal polymer
- ② Polymer valve
- ③ Syringe
- ④ Capillary fitting (anode end)
- ⑤ Buffer valve
- ⑥ Anode buffer reservoir

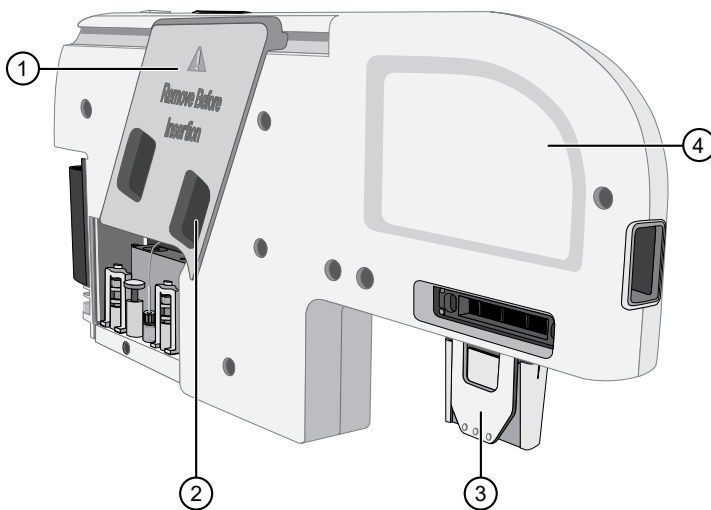


Figure 6 Cartridge with optical cover and Integrated Capillary Protector

- ① Optical cover
- ② Optical cover hand hold (for removing the optical cover)
- ③ Integrated Capillary Protector
- ④ Cartridge hand hold

IMPORTANT! Remove the Integrated Capillary Protector before installing the cartridge into the instrument. Installing the cartridge with the ICP in place can damage the capillary array.

Cartridge storage

Table 1 Storage information for the SeqStudio™ Genetic Analyzer Cartridge (Cat. No. [A33671](#))

Condition	Description
Shipping	Shipped at 2–8°C. Store upright at 2–8°C upon receipt. Save the white storage box and optical cover for off-instrument cartridge storage.
On-instrument storage	For routine use, can be used and stored on the instrument for up to 4 months. If you store the cartridge on-instrument: • The instrument must be powered on. • A SeqStudio™ Cathode Buffer Container must also be installed. The instrument keeps the components under the following conditions when it is powered on and in Cartridge storage mode: • Optical detection window—Covered • Capillary array electrodes—Submerged in cathode buffer (buffer must be above the fill line in the SeqStudio™ Cathode Buffer Container) • Polymer—Chilled • Anode buffer—Ambient temperature IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.
Off-instrument storage	For intermittent use, can be stored off-instrument until the expiry date on the label or up to 4 months after first use. Store upright at 2–8°C, with an integrated capillary protector (ICP) and optical cover installed (see “Store the cartridge” on page 180). Note: After you remove the cartridge from the instrument, install an ICP within a few minutes. Avoid cartridge exposure to ambient temperature.
Reuse	Can be removed from an instrument then inserted again on the same instrument or a different instrument, if it was stored properly at 2–8°C and has not expired or exceeded 125 injections. Information about the cartridge installation and usage is retained in the cartridge history (Settings ▶ Cartridge ▶ Instrument–cartridge history).

Table 2 Storage information for the SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. [A41331](#))

Condition	Description
Shipping	Shipped at 2–8°C. Store upright at 2–8°C upon receipt. Save the white storage box and optical cover for off-instrument cartridge storage.
On-instrument storage	For routine use, can be used and stored on the instrument for up to 6 months. If you store the cartridge on-instrument: • The instrument must be powered on. • A SeqStudio™ Cathode Buffer Container must also be installed. The instrument keeps the components under the following conditions when it is powered on and in Cartridge storage mode: • Optical detection window—Covered • Capillary array electrodes—Submerged in cathode buffer (buffer must be above the fill line in the SeqStudio™ Cathode Buffer Container) • Polymer—Chilled • Anode buffer—Ambient temperature IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.
Off-instrument storage	For intermittent use, can be stored off-instrument until the expiry date on the label or up to 6 months after first use. Store upright at 2–8°C, with an integrated capillary protector (ICP) and optical cover installed (see “Store the cartridge” on page 180). Note: After you remove the cartridge from the instrument, install an ICP within a few minutes. Avoid cartridge exposure to ambient temperature.
Reuse	Can be removed from an instrument then inserted again on the same instrument or a different instrument, if it was stored properly at 2–8°C and has not expired or exceeded 250 injections. Information about the cartridge installation and usage is retained in the cartridge history (Settings ▶ Cartridge ▶ Instrument–cartridge history).

SeqStudio™ Genetic Analyzer Cathode Buffer Container

The SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC) contains running buffer for capillary electrophoresis. The container has two compartments. The rear compartment provides the cathode buffer for electrophoresis. The front compartment is for capillary wash and waste.

The CBC requires a Reservoir Septa.

See “Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)” on page 183.

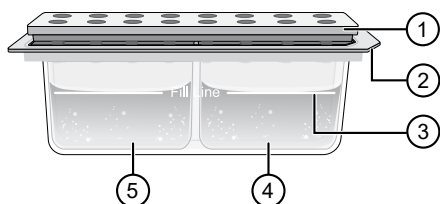


Figure 7 SeqStudio™ Genetic Analyzer Cathode Buffer Container with Reservoir Septa

- ① Reservoir Septa
- ② Notch (inserted towards the rear right in the autosampler)
- ③ Fill line (replace the CBC when the buffer is at the fill line)
- ④ Cathode buffer compartment
- ⑤ Waste and wash compartment

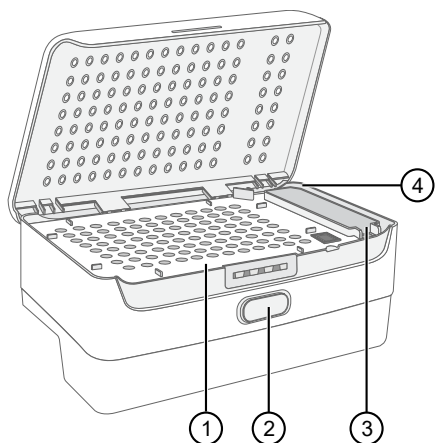


Figure 8 Autosampler

- ① Location of plate or tubes
- ② Release button
- ③ Location of SeqStudio™ Cathode Buffer Container
- ④ Location to position the SeqStudio™ Cathode Buffer Container notch

Radio frequency identification

The cartridge and cathode buffer have radio–frequency identification (RFID) tags. The instrument reads and tracks:

- Expiry date (shelf life)
- Remaining injections (usage)
- Serial number (cartridge only)
- Lot number (cathode buffer only)

Table 3 RFID tag read/write events and consumables status updates

Component	RFID read	RFID write	Consumables status update
SeqStudio™ Genetic Analyzer Cartridge	When a cartridge inserted. The first time a new cartridge is loaded into an instrument, the manufacturer expiry date is listed for Expiration time in the Consumable status screen.	• The Expiration time is reset from the manufacturer expiry date to 4 months from the current date of installation. • After the cartridge has been loaded and before it is ejected, its cartridge history record is updated to include its usage on this instrument. • During each injection, injection count and remaining polymer volume are updated.	Every 8 hours and/or before each run ^[1] : • When a cartridge is within 2 weeks of expiry date. • When the number of injections is approaching the limit of 125 injections.
SeqStudio™ Genetic Analyzer Cartridge v2		• The Expiration time is reset from the manufacturer expiry date to 6 months from the current date of installation. • After the cartridge has been loaded and before it is ejected, its cartridge history record is updated to include its usage on this instrument. • During each injection, injection count and remaining polymer volume are updated.	Every 8 hours and/or before each run ^[1] : • When a cartridge is within 2 weeks of expiry date. • When the number of injections is approaching the limit of 250 injections.
Cathode buffer	• When the autosampler initializes. • When a plate is retracted.	• The first time a new CBC is loaded on an instrument, the installation date is recorded on the CBC. • After each injection, the injection count is updated.	Every 8 hours and/or before each run ^[1] : • When the CBC is within 2 days of expiry date. • When the number of injections is approaching the limit of 125 injections.

^[1] If either limit is met, an email notification is also sent to any Thermo Fisher™ Connect Platform users who are linked to this instrument

Important notice regarding use of consumables that exceed supported limits

BEFORE DISMISSING THE WARNING THAT THE CONSUMABLES HAVE REACHED SUPPORTED LIMITS AND CONTINUING WITH OPERATION OF THE INSTRUMENT, PLEASE READ AND UNDERSTAND THE FOLLOWING IMPORTANT NOTICE AND INFORMATION:

Life Technologies does not recommend the use of consumables that exceed supported limits. The recommended limits are designed to promote the production of high quality data and minimize instrument downtime. Reagent and consumable lifetime minimum performance are based on testing and studies that use reagents and consumables that have not exceeded supported limits.

The use of consumables beyond the supported limits may impact data quality or cause damage to the instrument or capillary array. The cost of repairing such damage is NOT covered by any Life Technologies product warranty or service plan. Customer use of expired consumables is at customer's own risk and without recourse to Life Technologies. For example, product warranties do not apply to defects resulting from or repairs required due to misuse, neglect, or accident including, without limitation, operation outside of the environmental or use specifications or not in conformance with Life Technologies instructions for the instrument system, software, or accessories.

Please see your specific service contract or limited product warranty for exact language regarding coverage and ask your Life Technologies representative if you have further questions.

Software features

The instrument can be operated directly from the touchscreen using the SeqStudio™ Data Collection Software. The touchscreen allows scrolling and zooming by pinch.

Plates can be set up and saved on a computer using the SeqStudio™ Plate Manager running on a desktop or on the Thermo Fisher™ Connect Platform. These plate setups can be saved to and accessed from the instrument from:

- A network drive
- A USB drive
- The Thermo Fisher™ Connect Platform

You can monitor runs directly from the instrument or from the SeqStudio™ Remote Monitoring App.

Table 4 Features of the software

Feature	SeqStudio™ Data Collection Software	SeqStudio™ Plate Manager (desktop)	SeqStudio™ Plate Manager (Thermo Fisher™ Connect Platform)	Remote Monitoring App
Plate setup				
Create a new plate setup	✓	✓	✓	—
Enter plate properties	✓	✓	✓	—
Set up plate wells	✓	✓	✓	—

Table 4 Features of the software (continued)

Feature	SeqStudio™ Data Collection Software	SeqStudio™ Plate Manager (desktop)	SeqStudio™ Plate Manager (Thermo Fisher™ Connect Platform)	Remote Monitoring App
Advanced options for plate setup				
Edit sample properties	✓	✓	✓	—
Edit plate setup	✓	✓	✓	—
Manage size standards	✓	✓	✓	—
Manage run modules (including edit a run module)	✓	✓	✓	—
Advanced options for plate properties				
Adjust fragment/HID analysis parameters	✓	✓	✓	—
Adjust sequence parameters	✓	✓	✓	—
Adjust the file naming format	✓	✓	✓	—
Select injection options	✓	✓	✓	—
Create custom dye set	✓	— (to import, open a plate setup containing a custom dye set)	— (to import, open a plate setup containing a custom dye set)	—
Edit during a run or after a run is complete				
Monitor run	✓	—	—	✓
Pause or cancel a run	✓	—	—	✓
Edit injection parameters and re-inject samples	✓	—	—	✓
View results				
View run result details	✓	—	—	✓
View PUP score	✓	—	—	✓
View trace score (sequence analysis only)	✓	—	—	✓
View size quality (fragment/HID analysis only)	✓	—	—	✓

Table 4 Features of the software (continued)

Feature	SeqStudio™ Data Collection Software	SeqStudio™ Plate Manager (desktop)	SeqStudio™ Plate Manager (Thermo Fisher™ Connect Platform)	Remote Monitoring App
View Contiguous Read Length (sequence analysis only)	✓	—	—	✓
View and export results				
View real-time results	✓	—	—	✓
Adjust the graphical view	✓	—	—	✓
Auto export or manually export data files to the Thermo Fisher™ Connect Platform, a network drive, or a USB	✓	—	—	—
Export a results report	✓	—	—	✓

SeqStudio™ Plate Manager overview

The SeqStudio™ Plate Manager is a stand-alone software. It allows you to set up and save plates that you can open and run on the instrument. SeqStudio™ Plate Manager can also be used with other instruments; see “Access the Plate Manager” on page 72.

The SeqStudio™ Plate Manager is available:

- On the Thermo Fisher™ Connect Platform as an app, with access to the SeqStudio™ Remote Monitoring App
- At [thermofisher.com](https://www.thermofisher.com), for download and installation on a computer
- On a USB, for installation on a computer

Install the Plate Manager by following the instructions in the install wizard.

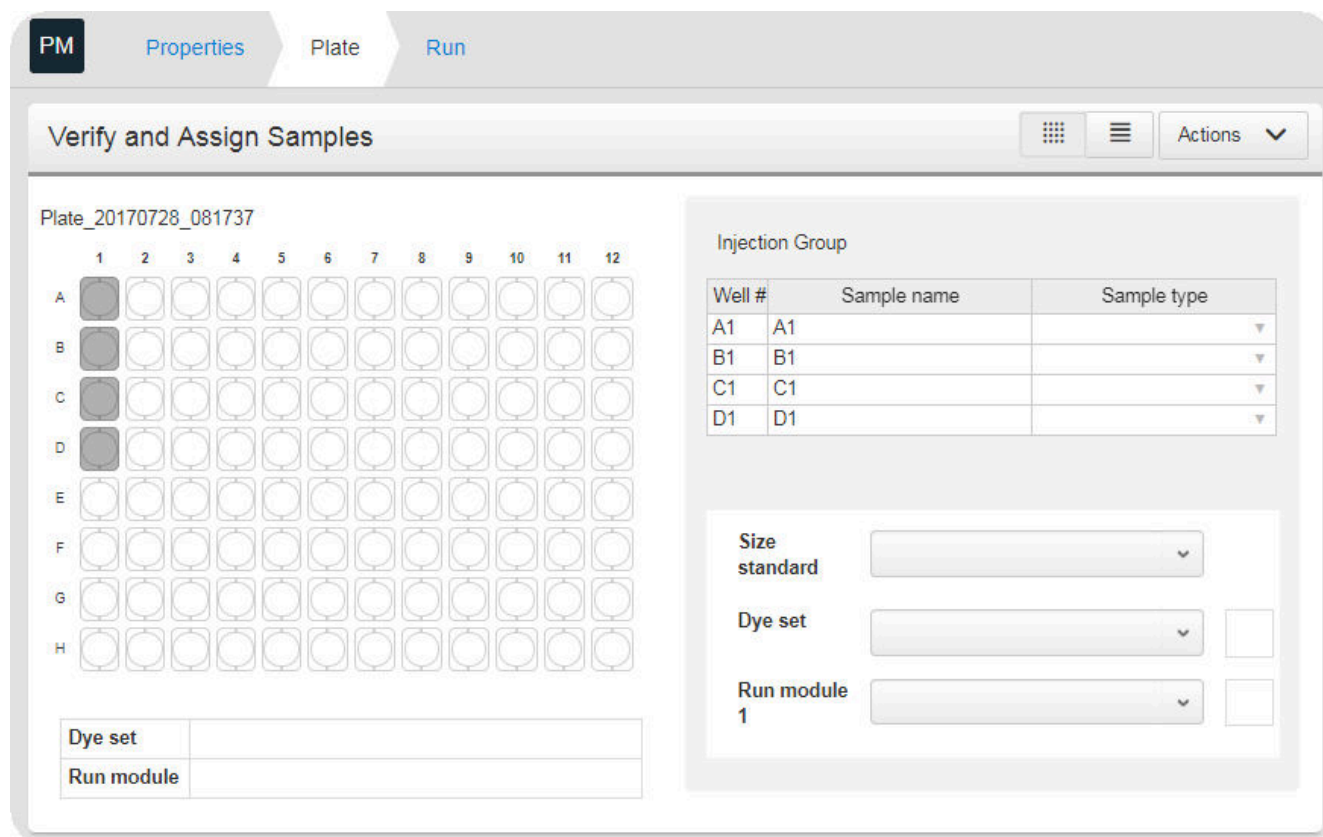


Figure 9 Plate Manager on the Thermo Fisher™ Connect Platform or desktop

Computer and operating system options

- PC with Windows™ 11
- PC with Windows™ 10

Recommended browsers

The following browsers are recommended to use the Plate Manager app on the Thermo Fisher™ Connect Platform or a desktop computer:

- Mozilla™ Firefox™ Version 32.0.3+
- Google Chrome™ Version 38.02+
- Apple™ Safari™ Version 7+
- Microsoft™ Edge 10+ (Windows™ 10)

The use of other browsers or other versions can result in reduced functionality and improper display of information.

SeqStudio™ Remote Monitoring App overview

The Remote Monitoring App allows you to monitor the status of instrument runs from a remote location.

The Remote Monitoring App is available:

- As an app on the Thermo Fisher™ Connect Platform, with direct access from the Plate Manager or the InstrumentConnect.
- At thermofisher.com, for download and installation on a mobile device.

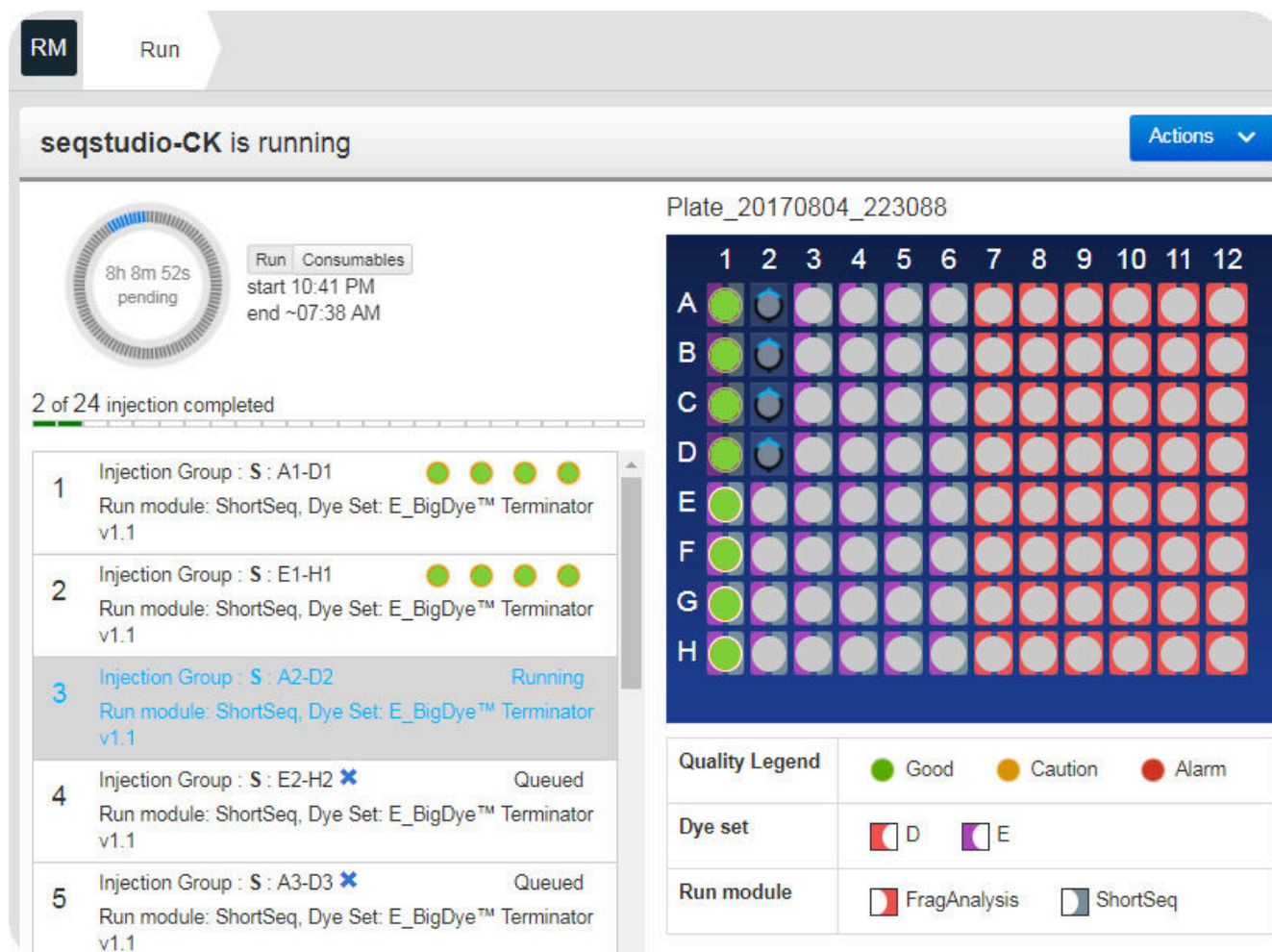


Figure 10 Remote Monitoring App on the Thermo Fisher™ Connect Platform

Network connection options

The SeqStudio™ Genetic Analyzer can be connected to a network or computer in the following configurations:

Thermo Fisher™ Connect Platform connection	Local area network (LAN) connection	Direct connection
Wired or wireless 	Wired or wireless 	Wired 

Network connection	Requirement
Wireless	A wireless adapter (also referred to as a dongle) is provided with the instrument. The wireless connection conforms to 802.11 b/g/n wireless standards.
Wired	The instrument is factory-configured for IPv4 TCP/IP communication and uses an Ethernet adapter (10/100 Mbps) with an RJ45-type connector for local area network (LAN) connection. - An active, tested network jack must be in place before the scheduled installation date. - The assigned IT or network specialist from your organization must be available during the installation to help connect the instrument to your network.
Direct	No network is required.

Network and password security requirements

Network configuration and security

The network configuration and security settings of your laboratory or facility (such as firewalls, anti-virus software, network passwords) are the sole responsibility of your facility administrator, IT, and security personnel. This product does not provide any network or security configuration files, utilities, or instructions.

If external or network drives are connected to the software, it is the responsibility of your IT personnel to ensure that such drives are configured and secured correctly to prevent data corruption or loss. It is the responsibility of your facility administrator, IT, and security personnel to prevent the use of any unsecured ports (such as USB, Ethernet) and ensure that the system security is maintained.

Password security

Thermo Fisher Scientific strongly recommends that you maintain unique passwords for all accounts in use on this product. All passwords should be reset upon first sign in to the product. Change passwords according to your organization's password policy.

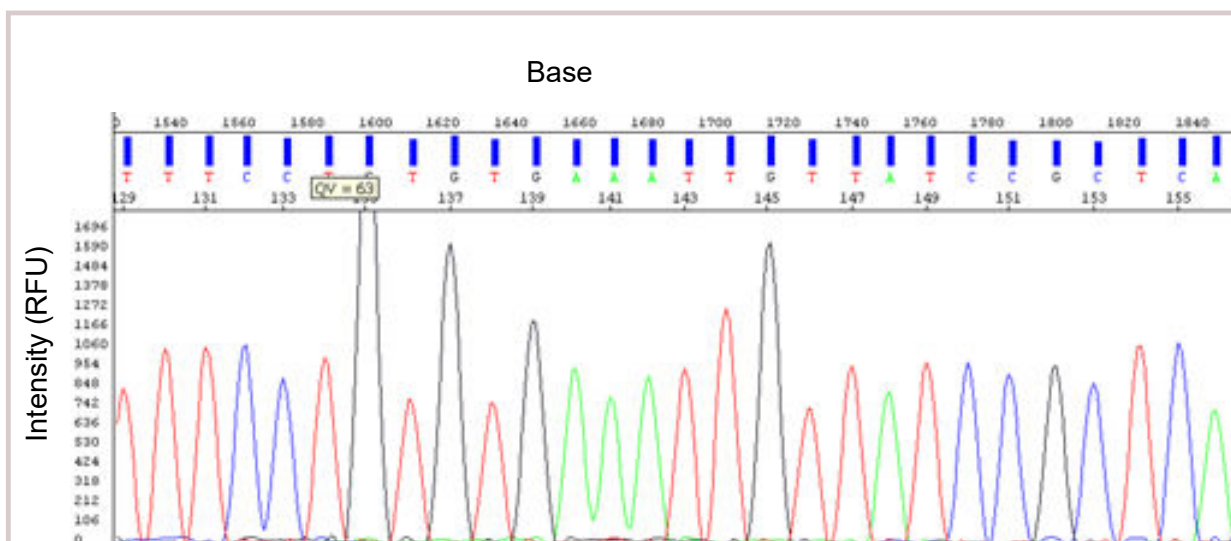
It is the sole responsibility of your IT personnel to develop and enforce secure use of passwords.

Experiment types

Sequencing

Sequencing is the determination of the base-pair sequence of a DNA fragment by the formation of extension products of various lengths amplified through PCR.

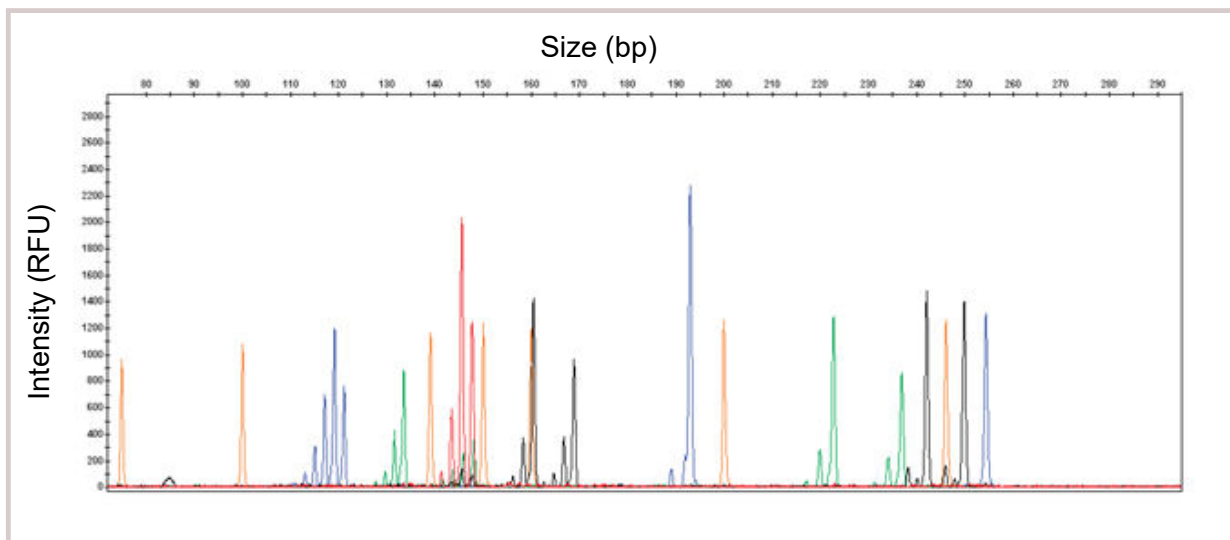
For more information, see *DNA Sequencing by Capillary Electrophoresis Chemistry Guide Second Edition* (Pub. No. 4305080).



Fragment/HID analysis

Fragment/HID analysis is the determination of the size of fragments. It uses the size standard in each sample to create a standard curve for each sample. It then determines the relative size of each dye-labeled fragment in the sample by comparing fragments with the standard curve for that specific sample.

For more information, see *DNA Fragment Analysis by Capillary Electrophoresis User Guide* (Pub. No. 4474504).



Data output

SeqStudio™ Data Collection Software generates an electropherogram (intensity plot) for each dye that is based on the migration of DNA fragments through the capillaries during a run.

The AB1 and FSA file formats can be analyzed by secondary analysis software.

Secondary analysis software

Secondary analysis software is available for desktop computers and on your Thermo Fisher™ Connect Platform account.

Visit thermofisher.com/connect for the latest available secondary analysis applications.

Note: Data from the SeqStudio™ Genetic Analyzer may be labeled as "3200" in secondary analysis software.

Secondary analysis apps on the Thermo Fisher™ Connect Platform

Analysis	App	Description
Sequencing	Quality Check (QC) module 	- Automatically checks sequence trace quality. - Provides a results summary that is based on quality parameter settings. - Auto-flags lower-quality traces for further inspection.
	Variant Analysis (VA) module 	- Finds variants in samples that are sequenced on Applied Biosystems™ genetic analyzers. - Reports variants at genomic coordinates. - Allows export of variant calls in standard Variant Call Format.
	Next-generation Confirmation (NGC) module 	- Confirms next-generation sequencing (NGS) variants using CE technology. - Allows visualization of the variants that are detected by both NGS and CE platforms. - Allows export of confirmed variants in standard Variant Call Format.
Fragment analysis	Sizing Analysis Module Peak Scanner™ Software 	Performs peak sizing.
	SeqScreener Gene Edit Confirmation 	Analyzes Sanger sequencing data from CRISPR-Cas9 experiments.
	Microsatellite Analysis Software 	Analyzes a mixture of DNA fragments, separated by size, on supported capillary electrophoresis systems.

Desktop secondary analysis software

IMPORTANT! Older versions of the desktop secondary analysis software cannot analyze data files generated by the SeqStudio™ Genetic Analyzer. Contact Support for information on obtaining the latest versions of software.

Analysis	Software	Minimum version required
Sequencing and fragment analysis	InnoviGene™ Suite	2.0 ^[1]
Sequencing	Sequencing Analysis Software	6.2
	SeqScape™ Software	3.2
	Variant Reporter™ Software	2.2
	Minor Variant Finder Software	1.2
Fragment analysis	GeneMapper™ Software	5.1
HID analysis	GeneMapper™ ID-X Software	1.6

^[1] Version 1.0 for sequencing only.

Workflow: Thermo Fisher™ Connect Platform or desktop

Get started

Prepare the instrument
(page 41)

Prepare the samples
(page 38)

Create a plate setup on the Thermo Fisher™ Connect Platform or desktop

Access the Plate Manager
(page 72)

Create or open a plate setup PSM file
(page 74)

Enter plate properties
(page 75)

Assign wells: Sample and run information
(page 76)

Start and monitor a run

On the instrument: Load the plate or the tube assembly
(page 98)

On the instrument: Select a plate setup and start a run
(page 99)

Monitor a run from the Thermo Fisher™ Connect Platform
(page 101)

Monitor a run from a mobile device
(page 109)

Monitor a run from the instrument
(page 112)

View and analyze results

View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform
(page 105)

View results on the instrument
(page 119)

Export results from the instrument (sample data files and QC reports)
(page 130)

Analyze data
(page 130)

(If needed) View the export status for sample data files
(page 131)

Workflow: instrument

Get started

Prepare the instrument
(page 41)

Prepare the samples
(page 38)

Create a plate setup on the instrument

Create or import a plate setup
(page 87)

Enter plate properties
(page 88)

Assign wells: run module, size standard, dye set, and kit
(page 90)

Assign wells: sample name, sample type, and custom fields
(page 92)

Start and monitor a run

On the instrument: Load the plate or the tube assembly
(page 98)

Select a plate setup and start a run
(page 99)

Monitor a run from the instrument
(page 112)

View and analyze results

View results on the instrument
(page 119)

Export results from the instrument (sample data files and QC reports)
(page 130)

Analyze data
(page 130)

(If needed) View the export status for sample data files
(page 131)

2

Prepare the samples and the instrument

■ Precautions for use	35
■ Power on the instrument	35
■ Sign in	36
■ Sign in with the Guest instrument profile	36
■ Sign out	37
■ Parts of the home screen	37
■ Prepare the samples	38
■ Prepare the instrument	41

Precautions for use



CAUTION! PHYSICAL INJURY HAZARD. Do not remove the instrument cover. There are no components inside the instrument that you can safely service yourself. If you suspect a problem, contact technical support.



CAUTION! Moving parts.



CAUTION! FIRE HAZARD. For continued protection against the risk of fire, replace fuses only with listed and certified fuses of the same type and rating as those currently in the instrument.



CAUTION! Hot surface.



CAUTION! Piercing hazard.



CAUTION! Potential biohazard.



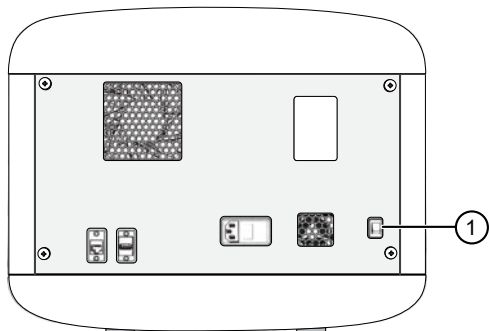
CAUTION! Risk of electrical shock.

Power on the instrument

IMPORTANT!

- Do not power on the instrument until it has been installed and set up by a Thermo Fisher Scientific representative.
 - Do not power off the instrument when a cartridge is loaded.
-

Press the On/Off switch on the rear panel.



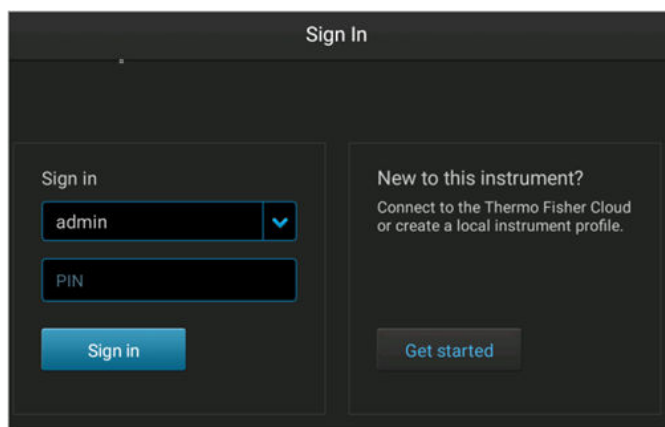
IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.

① On/Off switch

Sign in

1. If another user is signed in, touch  in the home screen, then touch **Sign out**.
2. In the **Sign in** screen, touch the down arrow.
3. Touch your instrument profile, then enter your PIN.

Note: If the instrument is left unattended for 120 minutes, the instrument profile is signed out.



Sign in with the Guest instrument profile

1. In the **Sign In** screen, touch the down arrow.
2. In the bottom left of the screen, touch the **Guest** button. No PIN is required.

Sign out

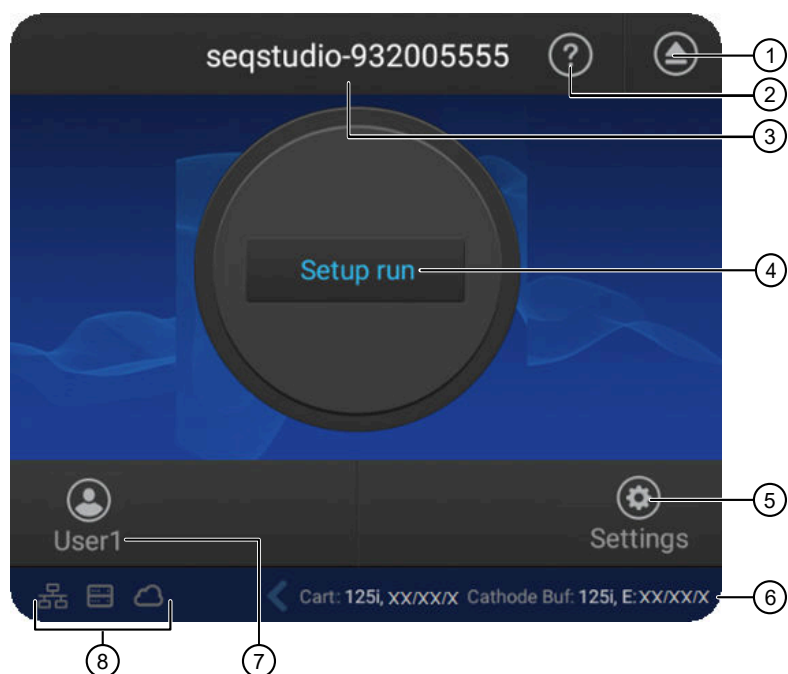
In the home screen:

1. Touch .
2. Touch **Sign out**.

Note: If a run is in progress, **Lock the instrument** is displayed instead of **Sign out**.

3. Touch **Yes** to confirm.

Parts of the home screen

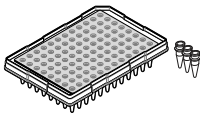
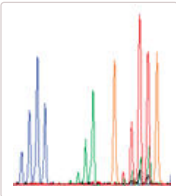


- ① Eject icon
- ② Help
- ③ Instrument name
- ④ Status dial—Touch to create a plate setup.
- ⑤ **Settings**—Touch to view previous results (**Run History**) or configure the instrument.
- ⑥ Status of consumables—See “Check the consumables status” on page 42.
- ⑦ Current user
- ⑧ Connectivity icons

Connectivity icon	Status
 or 	The instrument is connected to a wired or wireless network.
	The instrument is connected to network drive.
	The current instrument profile is linked to the Thermo Fisher™ Connect Platform.

Prepare the samples

Sample preparation guidelines

Item	Guidelines
Plates and tubes 	- Use MicroAmp™ Optical 96-Well Reaction Plate or MicroAmp™ Reaction Tubes with a tray and retainer set. IMPORTANT! Fast plates are not compatible with the SeqStudio™ Genetic Analyzer. Fast plates will damage the cartridge. - Use the appropriate septa for plates and tubes. - See “Required materials not supplied” on page 241 for more information.
Fragment and HID analysis sample preparation 	- Prepare the samples as recommended by the kit for fragment analysis. - Use a 10–20 µL sample volume. - Ensure that Hi-Di™ Formamide is fresh. - Hi-Di™ Formamide should not undergo more than two freeze-thaw cycles (one to aliquot and one for use). - Use the same day after thawing. - For more information, see <i>DNA Fragment Analysis by Capillary Electrophoresis User Guide</i> (Pub. No. 4474504).

(continued)

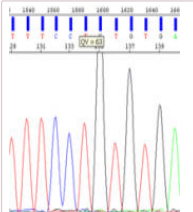
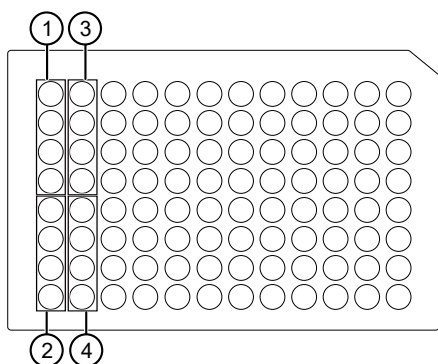
Item	Guidelines
Sequence analysis sample preparation 	• Prepare sequencing reactions according to kit instructions, and purify the extension products with ethanol precipitation, spin columns, or the BigDye XTerminator™ Purification Kit. – If ethanol precipitation or spin columns are used, dry the samples in a vacuum centrifuge without heat or at low heat for 10–15 minutes or until dry. Note: Do not over dry the samples. – Resuspend in 10–20 µL of Hi-Di™ Formamide. • Use a 65 µL or 130 µL sample volume for samples that are prepared with the BigDye XTerminator™ Purification Kit. See <i>BigDye XTerminator™ Purification Kit User Guide</i> (Pub. No. 4374408). IMPORTANT! Use the appropriate run modules for samples prepared with BigDye XTerminator™ Purification Kit. See “Run modules, read lengths, size ranges, and run times” on page 156. • Use a 10–20 µL sample volume for samples that are prepared with Hi-Di™ Formamide. • Ensure that Hi-Di™ Formamide is fresh. – Hi-Di™ Formamide should not undergo more than two freeze-thaw cycles (one to aliquot and one for use). – Use the same day after thawing. • Do not resuspend samples in water, which can decrease sample stability. • For more information, see <i>DNA Sequencing by Capillary Electrophoresis Chemistry Guide Second Edition</i> (Pub. No. 4305080).

Plate layout and loading guidelines

- Samples are stable for 16–24 hours on the instrument.
- Load a maximum of 48 samples per plate if you use a long run module (Long Seq, Long Seq BDX, and Long Frag Analysis). The long fragment analysis run modules can take >24 hours to run an entire 96-well plate.
- Add samples to plates in columns. The default injection order is: A1–D1, E1–H1, A2–D2, E2–H2....A12–D12, E12–H12.

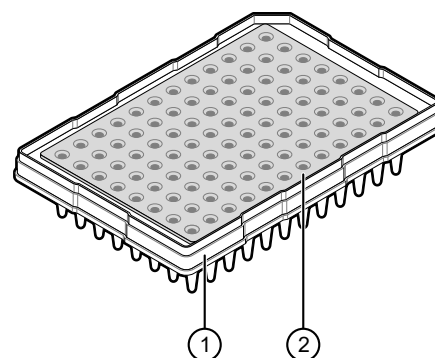


- ① Injection group 1, wells A1–D1
- ② Injection group 2, wells E1–H1
- ③ Injection group 3, wells A2–D2
- ④ Injection group 4, wells E2–H2

Prepare the plate

On a clean and level surface:

1. Prepare the sample according to your application protocol, then pipet the sample into the plate.
2. Place a septum onto the plate.
 - a. Align the holes of the septa with the wells.
 - b. Press gently until the septum is inserted into position in each well.
3. Centrifuge the plate assembly briefly to collect the contents at the bottom of each well.
Centrifuge the plate assembly again if the contents are not at the bottom of the wells.
4. Heat-denature as required.
5. Place the plate on ice for 3 minutes.



- ① Plate
- ② Septum

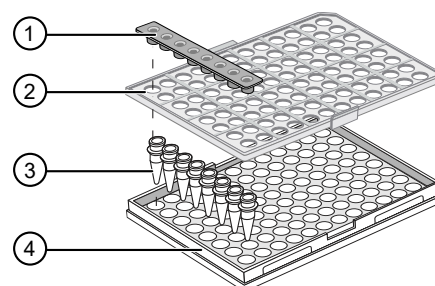
Load the plate onto the instrument immediately or keep the plate on ice and protected from light until it is loaded onto the instrument.

Prepare the tubes

On a clean and level surface:

1. Place the tubes in the MicroAmp™ 96-well tray, then place the tray retainer over the tubes.
2. Prepare the sample according to your application protocol, then pipet the sample into the tubes.

3. Place a septum on the tubes.
 - a. Align the holes of the septa with the tubes.
 - b. Press gently until the septum are inserted into position in each tube.
4. Centrifuge the tube assembly briefly to collect the contents at the bottom of each tube.
Centrifuge the tube assembly again if the contents are not at the bottom of the tubes.
5. Heat-denature as required.
6. Place the tube assembly on ice for 3 minutes.



- ① Septum
- ② 96-well retainer
- ③ 8-tube strip
- ④ MicroAmp™ 96-well tray

Load the tube assembly onto the instrument immediately or keep the tubes on ice and protected from light until they are loaded onto the instrument.

Prepare the instrument

Install the USB Wifi Dongle

1. Plug the USB Wifi Dongle into the back panel of the instrument.

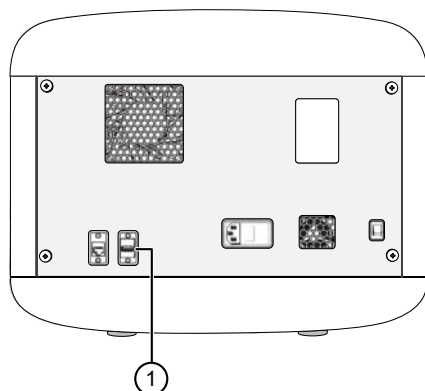


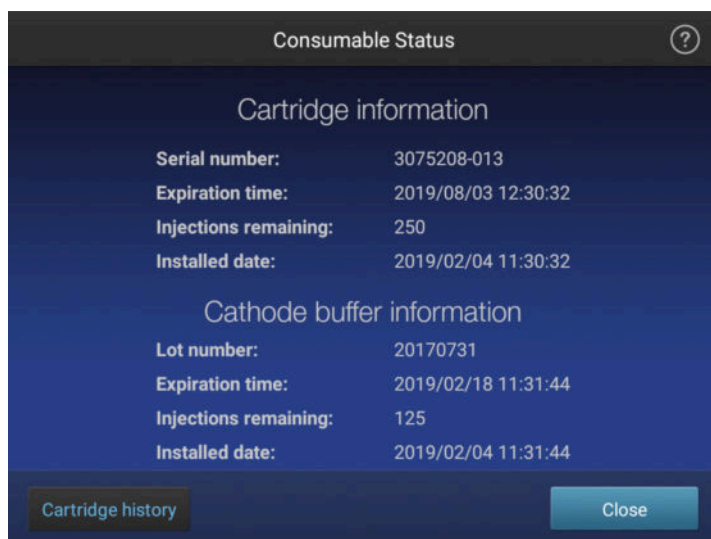
Figure 11 Rear of the instrument

- ① USB port for use with Wifi Dongle (dongle not shown)

2. (Recommended) Restart the instrument.

Check the consumables status

1. Tap  **Settings** ► **Consumable status**.
2. Review the **Consumable Status** screen.
 - a. Ensure that:
 - Sufficient consumables are installed for the run.
 - Installed consumables have not exceeded their expiry date.



Display	Cartridge	Cathode buffer
NA	Not installed on the instrument.	Not installed on the instrument.
White	OK for use.	OK for use.
Yellow	OK for use, but: • 1–25 injections remain for the capillary array, <i>or</i> • It will exceed the manufacturer's expiry date within 2 weeks, <i>or</i> • It will exceed the maximum number of days allowed on the instrument within 2 weeks^[1]	OK for use, but: • 1–25 injections remain for the cathode buffer, <i>or</i> • It will exceed the manufacturer's expiry date within 2 days <i>or</i> • It will exceed the maximum number of days allowed on the instrument (14 days) within 2 days
Red	The cartridge has expired: • No injections remain for the capillary array, <i>or</i> • It has exceeded the manufacturer's expiry date, <i>or</i> • It has exceeded the maximum number of days allowed on the instrument^[1]	The buffer has expired: • No injections remain for the cathode buffer, <i>or</i> • It has exceeded the manufacturer's expiry date <i>or</i> • It has been installed on the instrument for >14 days

^[1] The maximum number of days allowed on the instrument is 120 days for the SeqStudio™ Genetic Analyzer Cartridge (Cat. No. [A33671](#)) and 180 days for the SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. [A41331](#)). For more information, see Table 1.

- b. If a new SeqStudio™ Cathode Buffer Container is required, see “Insert the SeqStudio™ Cathode Buffer Container” on page 184.
 - c. If a new cartridge is required, see “Insert the cartridge” on page 179
3. Tap **Close**, then tap .

Load the CBC, the sample plate, and the cartridge

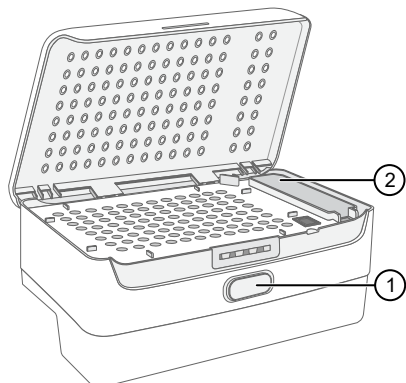
The **Eject plate** command is disabled for a few minutes after you insert a cartridge. If you are loading the CBC, sample plate, and cartridge at the same time, you can save time by loading the CBC and sample plate before you insert the cartridge.

For information on loading the individual components, see:

- “Install cathode buffer” on page 183
- “Load the plate or the tube assembly” on page 98
- “Insert the cartridge” on page 179

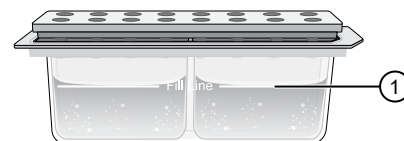
In the home screen:

1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid, then remove the CBC.



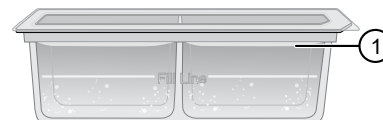
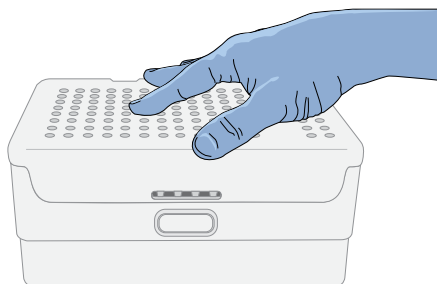
- ① Release button
- ② Location of CBC

3. Check the buffer fill level:
 - a. Remove the CBC.
 - b. Ensure that the level of buffer is above the fill line.
If the buffer is at or below the fill line, see “Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)” on page 183 and “Insert the SeqStudio™ Cathode Buffer Container” on page 184.
If the buffer is above the fill line, reinsert the CBC.



- ① Replace if buffer is at or below the fill line

4. Place the plate or tube assembly firmly in the autosampler.
5. Close the autosampler lid: Press down on the center of the lid or press down on both sides of the lid with equal pressure until the lid clicks shut.



① New CBC buffer level

6. Touch **Retract plate**, then close the instrument door when prompted.
7. Touch  **Eject cartridge**, then open the instrument door when prompted.
8. Hold the cartridge at the hand hold above the capillaries, then pull to remove it from the instrument.
9. Insert a new cartridge (see “Insert the cartridge” on page 179).
10. Close the instrument door.



Use the instrument with the Thermo Fisher™ Connect Platform

■ Understanding instrument and Thermo Fisher™ Connect Platform interaction	45
■ Thermo Fisher™ Connect Platform administrators for an instrument	50
■ Register and obtain a Thermo Fisher™ Connect Platform account	53
■ Link the instrument to your Thermo Fisher™ Connect Platform account	54
■ Change your own Thermo Fisher™ Connect Platform instrument profile PIN	55
■ Set up email notifications from the instrument	56

Note: The Thermo Fisher™ Connect Platform is not supported for HID analysis.

Understanding instrument and Thermo Fisher™ Connect Platform interaction

"Connect" versus "link"

The words "connect" and "link" are used interchangeably in the software.

In one location you touch a **Connect** button, in another location you touch a **Link** button.

Both actions do the same thing:

- Connect the instrument to the InstrumentConnect on the Thermo Fisher™ Connect Platform *and*
- Link your local instrument profile to your Thermo Fisher™ Connect Platform account.

First user who links is assigned administrator role

The first user who links the instrument to the Thermo Fisher™ Connect Platform is automatically assigned the Thermo Fisher™ Connect Platform administrator role for the instrument (even if the user has a standard local profile).

Additional instrument administrators can be assigned, and user roles can be changed after linking.

For more information, see “Thermo Fisher™ Connect Platform instrument profile roles and functions” on page 49 and “Thermo Fisher™ Connect Platform administrators for an instrument” on page 50.

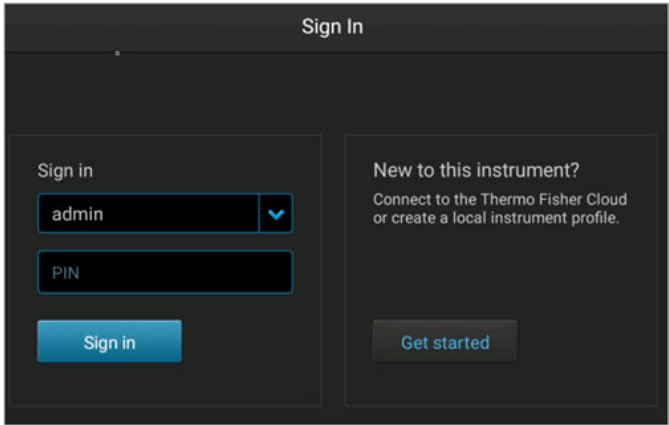
Local versus Thermo Fisher™ Connect Platform instrument profiles

- Local instrument profile—All plates and results are stored on the instrument under a local instrument profile.
- Thermo Fisher™ Connect Platform instrument profile—When you link your local instrument profile to your Thermo Fisher™ Connect Platform account, a Thermo Fisher™ Connect Platform instrument profile is created. With a Thermo Fisher™ Connect Platform instrument profile:
 - You can save results and data files directly to your Thermo Fisher™ Connect Platform account, access plate setups that you create in the SeqStudio™ Plate Manager on the Thermo Fisher™ Connect Platform or desktop, and monitor instrument runs from the Thermo Fisher™ Connect Platform.
 - All plate setups, data files, and results are automatically copied to your Thermo Fisher™ Connect Platform account if the plate setup **Save location** is set to **Cloud**.

If you link when you are signed in to the instrument

In this scenario, your local instrument profile name is created manually on the instrument before you link. Your local instrument profile name differs from your Thermo Fisher™ Connect Platform instrument profile name.

If you link to the Thermo Fisher™ Connect Platform when you are signed in to the instrument:

Phase	Steps that occur
Before you link:	• You enter your local instrument profile name in the Sign In screen.  • Your local instrument profile (<i>UserABC</i>) is displayed in the home screen of the instrument. • All plates and results that you create are accessible only when you are signed in with your local instrument profile.

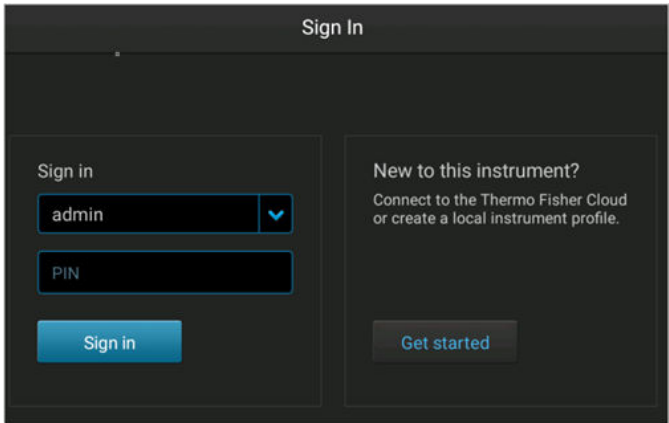
(continued)

Phase	Steps that occur
When you link:	- You use an option described in “Link the instrument to your Thermo Fisher™ Connect Platform account” on page 151 to link. - If this is the first time you link, a Thermo Fisher™ Connect Platform instrument profile is created using the <i>FirstNameLastInitial</i> of the user name from your thermofisher.com account. Example: <i>User1@thermofisher.com</i> First name is <i>User</i>, Last name is <i>Gray</i>. Thermo Fisher™ Connect Platform account username is <i>User G</i>. - Your local instrument profile (<i>UserABC</i>) is linked to your Thermo Fisher™ Connect Platform account (<i>User1@thermofisher.com</i>). - Your Thermo Fisher™ Connect Platform instrument profile (<i>User G</i>.) replaces your local instrument profile.
After you link:	- Your Thermo Fisher™ Connect Platform instrument profile (<i>User G</i>.) and  is displayed in the home screen of the instrument. - Plates and results from your local instrument profile can be copied to the Thermo Fisher™ Connect Platform (see “Export results from the instrument (sample data files and QC reports)” on page 130). - New plates and results are saved under your Thermo Fisher™ Connect Platform instrument profile. - Your Thermo Fisher™ Connect Platform instrument profile name (<i>User G</i>. ) is available for selection in the Sign In screen.
If your Thermo Fisher™ Connect Platform account is unlinked:	- Your local instrument profile (<i>UserABC</i>) is displayed in the home screen of the instrument. - Plates and results that were saved under your Thermo Fisher™ Connect Platform instrument profile are accessible under your local instrument profile. - New plates and results are saved under local instrument profile and can be copied to the Thermo Fisher™ Connect Platform (see “Export results from the instrument (sample data files and QC reports)” on page 130). - Your local instrument profile name (<i>UserABC</i>) is available for selection in the Sign In screen.

If you link when you *are not* signed in to the instrument

In this scenario, your local instrument profile name is created automatically on the instrument before you link. The same user name is used for your local instrument profile and your Thermo Fisher™ Connect Platform instrument profile. Plates and results are accessible when you sign in with either profile.

If you link to the Thermo Fisher™ Connect Platform when you *are not* signed in to the instrument:

Phase	Steps that occur
Before you link:	In the Sign In screen, you touch Get Started ▶ Connect. 
When you link:	- You use an option described in “Link the instrument to your Thermo Fisher™ Connect Platform account” on page 151 to link. - If this is the first time you link, a Thermo Fisher™ Connect Platform instrument and a local instrument profile (with standard role) are created with the same name using the <i>FirstNameLastInitial</i> of the user name from your thermofisher.com account. Example: <i>User1@thermofisher.com</i> First name is <i>User</i>, Last name is <i>Gray</i>. Thermo Fisher™ Connect Platform account username is <i>User G</i>. - Your local instrument profile (<i>User G.</i>) is linked to your Thermo Fisher™ Connect Platform account (<i>User1@thermofisher.com</i>). - Your Thermo Fisher™ Connect Platform instrument profile (<i>User G.</i>) replaces your local instrument profile.
After you link:	- Your Thermo Fisher™ Connect Platform instrument profile (<i>User G.</i>) and  is displayed in the home screen of the instrument. - Plates and results from your local instrument profile can be copied to the Thermo Fisher™ Connect Platform (see “Export results from the instrument (sample data files and QC reports)” on page 130). - New plates and results are saved under your Thermo Fisher™ Connect Platform instrument profile. - Your Thermo Fisher™ Connect Platform instrument profile name (<i>User G.</i> ) is available for selection in the Sign In screen.

(continued)

Phase	Steps that occur
If your Thermo Fisher™ Connect Platform account is unlinked:	- Your local instrument profile (<i>User G.</i>) is displayed in the home screen of the instrument. - Plates and results that were saved under your Thermo Fisher™ Connect Platform instrument profile are accessible under your local instrument profile. - New plates and results are saved under local instrument profile and can be copied to the Thermo Fisher™ Connect Platform (see “Export results from the instrument (sample data files and QC reports)” on page 130). - Your local instrument profile name (<i>User G.</i>) is available for selection in the Sign In screen.

IMPORTANT! If you sign in with a local profile, without linking to the Thermo Fisher™ Connect Platform, sign out, then link using **Get Started ▶ Connect**, you can potentially have two local instrument profiles with different names. Plates and results created when you are signed in with one local instrument profile are not accessible when you are signed in with the other local instrument profile.

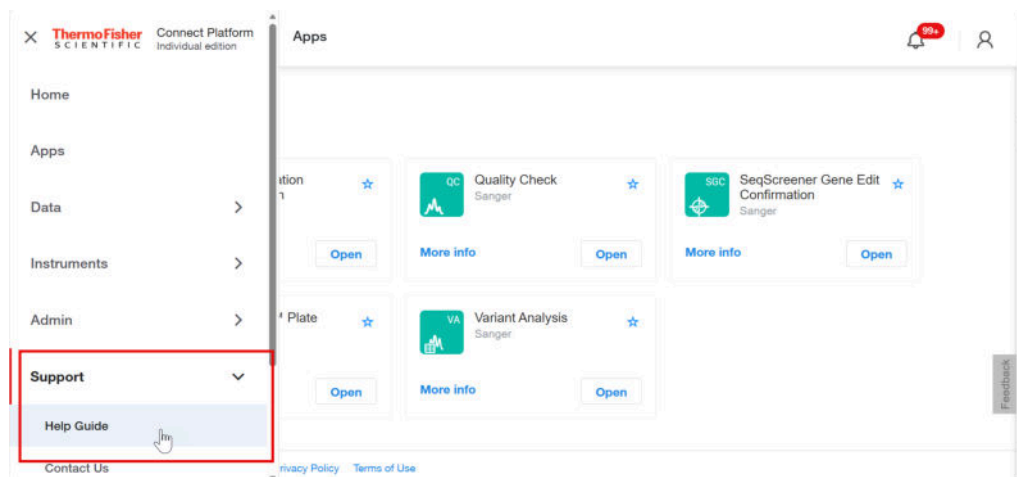
Thermo Fisher™ Connect Platform instrument profile roles and functions

Instrument profile	Location	Functions allowed
Standard	Thermo Fisher™ Connect Platform ^[1]	- Create, save, open, import, and run plate setups - Create and modify run settings - View and export results
Administrator	Thermo Fisher™ Connect Platform ^[1]	All the permissions of a local administrator profile, plus the following functions performed in InstrumentConnect: - Access the Manage users function on the Thermo Fisher™ Connect Platform to see a list of all instrument profiles that are linked to the instrument. - Assign the Thermo Fisher™ Connect Platform administrator role to one or more users. - Remove a user from an instrument. - Disconnect the instrument from InstrumentConnect. - Change the instrument name.

^[1] The first user who links their local instrument profile to their Thermo Fisher™ Connect Platform account is assigned a Thermo Fisher™ Connect Platform profile with administrator role.

For more information on using the Thermo Fisher™ Connect Platform

In the top left of the screen you are viewing in the Thermo Fisher™ Connect Platform, click ☰, then select **Help guide**.



Thermo Fisher™ Connect Platform administrators for an instrument

Thermo Fisher™ Connect Platform administrator functions

The first user who links the instrument to the Thermo Fisher™ Connect Platform is automatically assigned the Thermo Fisher™ Connect Platform administrator role for the instrument (even if the user has a standard local profile).

At least one Thermo Fisher™ Connect Platform administrator is required for each instrument.

A Thermo Fisher™ Connect Platform administrator can perform the following tasks from the InstrumentConnect:

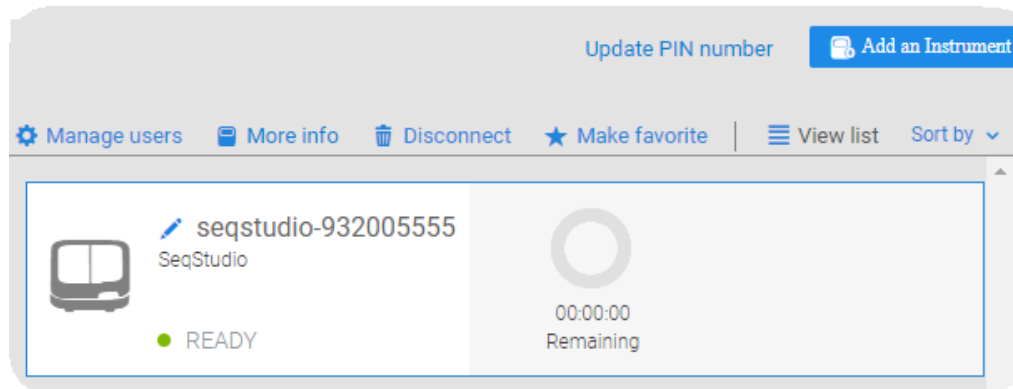
- Access the **Manage users** function on the Thermo Fisher™ Connect Platform to see a list of all instrument profiles that are linked to the instrument.
- Assign the Thermo Fisher™ Connect Platform administrator role to one or more users.
- Remove a user from an instrument.
- Disconnect the instrument from InstrumentConnect.
- Change the instrument name.

Assign instrument administrator role to other users

A Thermo Fisher™ Connect Platform administrator for an instrument can assign instrument administrator role to other users.

1. Sign in to thermofisher.com/connect.
2. Click  to access InstrumentConnect.

3. Select the instrument.



Note: The **Manage users** and other administrator functions are not displayed until you select an instrument.

4. To assign the Admin role to a user, click **Manage users**, then select the Administrator check box for the user.

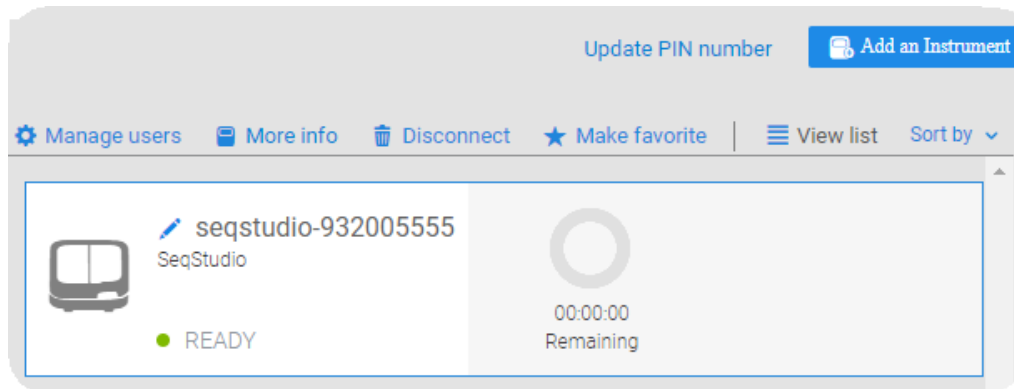
Manage the users and administrators of your instrument

Any user with a Thermo Fisher™ Connect Platform administrator profile can manage users for an instrument or disconnect an instrument from InstrumentConnect.

If a Thermo Fisher™ Connect Platform administrator...	The software...
Assigns Admin role to a user	Allows the user to perform all Thermo Fisher™ Connect Platform administrator functions (see “Thermo Fisher™ Connect Platform administrator functions” on page 50).
Removes a user	Unlinks the instrument from their Thermo Fisher™ Connect Platform account.
Disconnects the instrument	- Unlinks the instrument from all user Thermo Fisher™ Connect Platform accounts. - Removes the instrument from InstrumentConnect.

1. Sign in to apps.thermofisher.com.
2. Click  to access InstrumentConnect.

3. Select the instrument.



Note: The **Manage users** and other administrator functions are not displayed until you select an instrument.

4. To assign the Admin role to a user or to remove a user, click **Manage users**, then:

To...	Do this...
Assign the Admin role to an additional user	Select the Admin checkbox, then click Close .
Remove a user	Click  , then click Confirm .

Disconnect individual users from an instrument

You cannot disconnect individual users from an instrument.

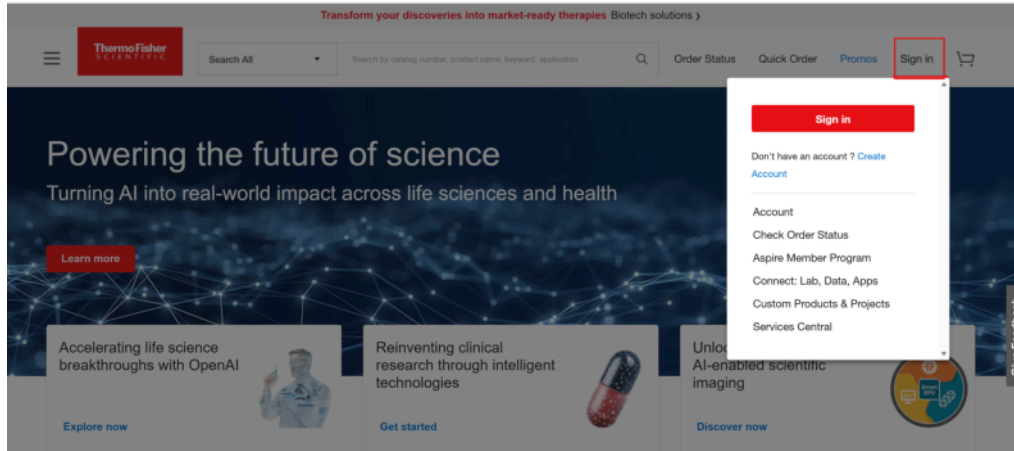
To *disconnect* a user, or to unlink the instrument from a Thermo Fisher™ Connect Platform account, you must disconnect the instrument from InstrumentConnect. Doing so unlinks *all* instrument profiles and removes the instrument from InstrumentConnect.

You can *remove* a user from the instrument, however, doing so deletes the user data from the instrument.

For more information, see “Manage the users and administrators of your instrument” on page 51

Register and obtain a Thermo Fisher™ Connect Platform account

1. Go to thermofisher.com.
2. On the home page, select **Sign In ► Create Account**.

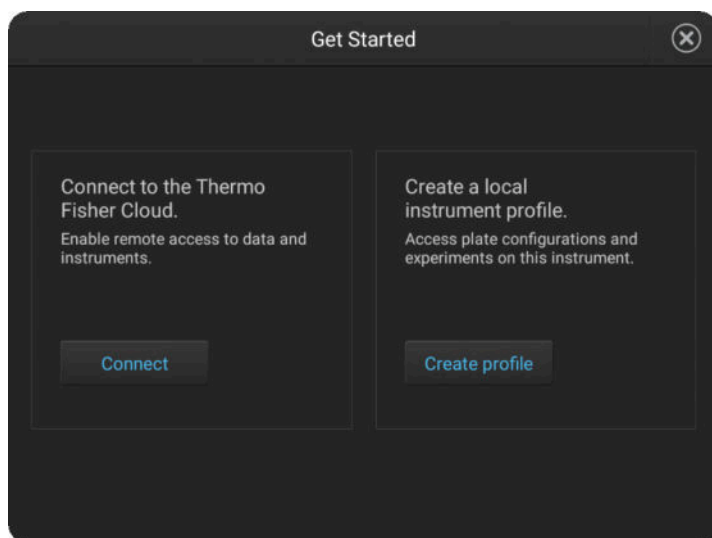


3. Fill in the information, then click **Create account**.

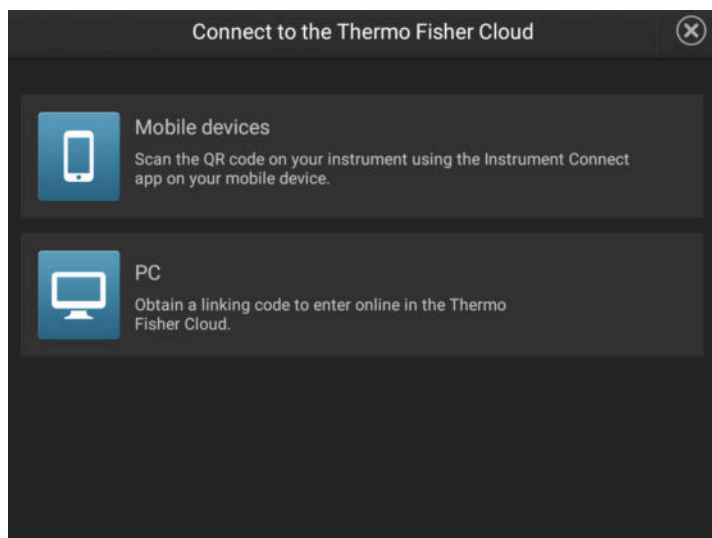
Link the instrument to your Thermo Fisher™ Connect Platform account

Note: For detailed information on linking the instrument to your Thermo Fisher™ Connect Platform account, see Appendix B, “Link the instrument to your Thermo Fisher™ Connect Platform account—detailed instructions”.

1. If a user is signed in, touch , then touch **Sign out**.
2. In the **Sign In** screen, touch **Get started ▶ Connect**.



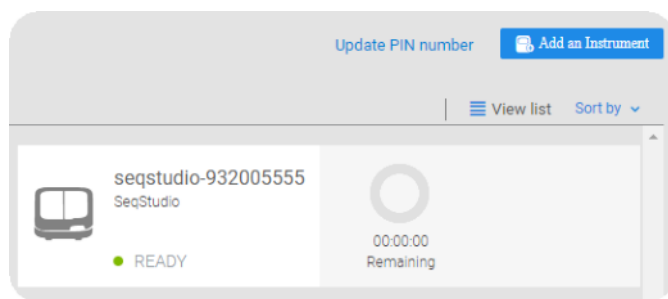
3. In the **Connect to the Cloud** screen, touch a connection option.



Option	Action
 Mobile devices	Note: Before selecting this option, install and sign in to the InstrumentConnect app on your mobile device. In the Connect to the Cloud screen: - Touch Mobile devices. - Hold the camera on your mobile device over the QR code that is displayed on the touchscreen. - Click Close.
 PC	In the Connect to the Cloud screen, a link code is displayed. On a computer: - Access the Thermo Fisher™ Connect Platform. - Access InstrumentConnect. - Click Add instrument. - Select SeqStudio. - Enter the link code.

Change your own Thermo Fisher™ Connect Platform instrument profile PIN

1. Sign in to thermofisher.com/connect.
2. Click  to access InstrumentConnect.
3. Click **Update PIN number**.

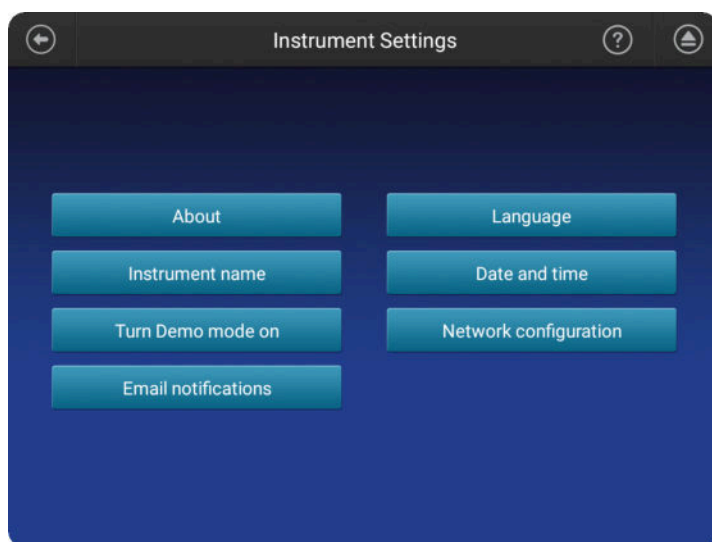


Set up email notifications from the instrument

When an instrument is linked to your Thermo Fisher™ Connect Platform account, email notifications are automatically sent to your Thermo Fisher™ Connect Platform account email address.

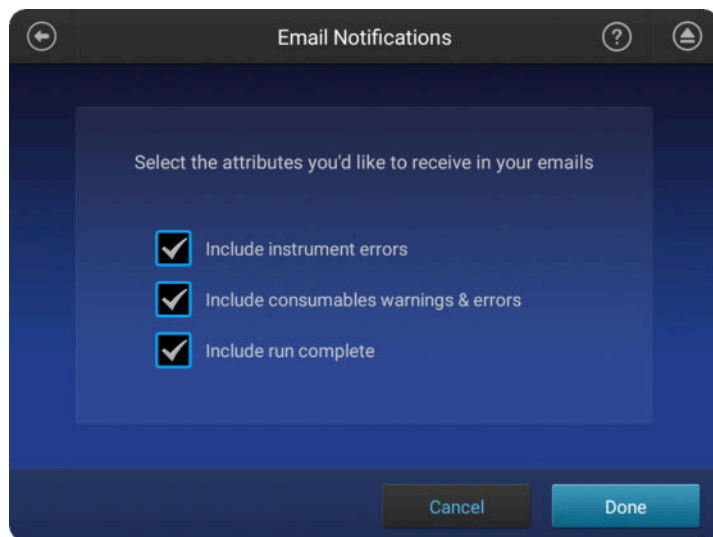
Perform this procedure to disable any of the default notifications.

1. Sign in to the instrument with your Thermo Fisher™ Connect Platform instrument profile and PIN.
2. In the home screen of the instrument, touch ⓘ **Settings** ▶ **Instrument settings** ▶ **Email notifications**.



Note: If you are signed in with a local instrument profile instead of a Thermo Fisher™ Connect Platform instrument profile, the **Email notifications** button is not displayed on the **Instrument Settings** screen.

3. In the **Email notifications** screen, select or deselect the options for which you want to receive email notifications, then touch **Done**.



Use the instrument with the Security, Auditing, and E-signature (SAE) Administrator Console

■ Overview of the Security, Auditing, and E-signature (SAE) Administrator Console components .	59
■ Compatibility: SAE and SeqStudio™ instrument software versions	60
■ Configuring the SAE module	60
■ Instrument functionality when SAE is enabled or disabled	60
■ Enable SAE on the instrument (administrator only)	60
■ Disable SAE (administrator only)	61
■ Disable SAE on the SeqStudio™ Genetic Analyzer	62
■ Sign in with SAE enabled	63
■ Specify audit reasons	64
■ E-signature requirements when you start a run	64
■ Sign a plate setup	66
■ View E-signature records for a plate setup	66
■ Sign a dye calibration or an install run	67
■ View E-signature records for a dye calibration or an install run	67
■ Signing in after automatic screen locking	67
■ Use the instrument when the SAE server is offline	68
■ Auditing of imported plate setups	68
■ SAE error messages and actions	69

The instrument software is compatible with SAE Administrator Console v2.2 and SAE Administrator Console v2.1.

Overview of the Security, Auditing, and E-signature (SAE) Administrator Console components

The Security, Auditing, and E-signature (SAE) Administrator Console includes three components:

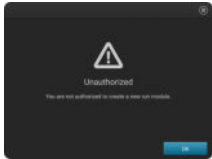
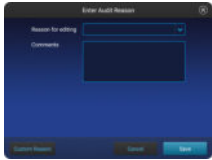
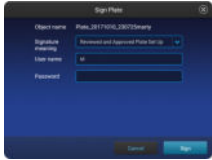
- **SAE Administrator Console**—Tool that is used by an SAE administrator to configure the SAE module.
- **SAE server** (server)—Service that runs in the background and stores SAE settings, user accounts, audit records, and e-signature records. By default, the SAE server is installed on the same computer as the SAE Administrator Console.
- **SAE screens** (client)—Screens that are displayed in an application (sign in, audit, and e-signature), for example, the SeqStudio™ Data Collection Software, and that require user input.

The Security, Auditing, and E-signature (SAE) Administrator Console provides the following SAE functionality on the instrument:

- **System security**—Controls user sign in and access to functions
- **Auditing**—Tracks changes and actions performed by users.
- **E-signature** —Allows users to provide an electronic signature (user name and password) when performing certain functions.

Depending on the way that your SAE administrator configures these features:

- Some of the features and functions described in later chapters of this guide may not be accessible to you.
- You may see dialog boxes and prompts when you use the software (examples shown below).

Dialog box	Description
	You do not have permission to perform a function.
	An action is set up for auditing and requires you to specify a reason for the action.
	An action is set up for electronic signature and allows you to enter your password to allow the action.

Compatibility: SAE and SeqStudio™ instrument software versions

SAE version	Instrument software version
2.2	v.1.2.5 and later
2.1	v.1.2.4 and later
2.0	v.1.1 to v.1.2.3

Configuring the SAE module

For information on configuration, see your SAE user guide.

Instrument functionality when SAE is enabled or disabled

To	Sign in to the instrument with profile type	Effect on instrument
Enable SAE	- Local instrument administrator to access the SAE screen - SAE administrator to enable SAE	- Sign in is limited to users with SAE accounts. Local instrument profiles cannot be used to run the instrument. - Auditing and E-signature functions are active (if they are enabled in the SAE Admin Console). - Plate setup and run functions for a user are determined by the SAE account settings. - Thermo Fisher™ Connect Platform and remote monitoring functions are disabled.
Disable SAE	- SAE administrator - Local instrument administrator	- Sign in is limited to users with local instrument profiles. - Auditing and E-signature functions are not performed. - Thermo Fisher™ Connect Platform and remote monitoring functions are enabled.

Enable SAE on the instrument (administrator only)

Before you enable SAE, install the SAE Administrator Console on a computer with a static IP address.

This procedure requires a local instrument administrator profile and an SAE account with administrator role.

1. Sign in to the instrument with a local instrument administrator profile.
2. Tap  **Settings** ▶ **SAE settings** ▶ **Connection settings**.

3. Enter the server IP address of the computer on which the SAE Administrator Console is installed.

IMPORTANT! Specify a computer with a static IP address. For more information, see “Determine IP address for a computer on a network” on page 144.

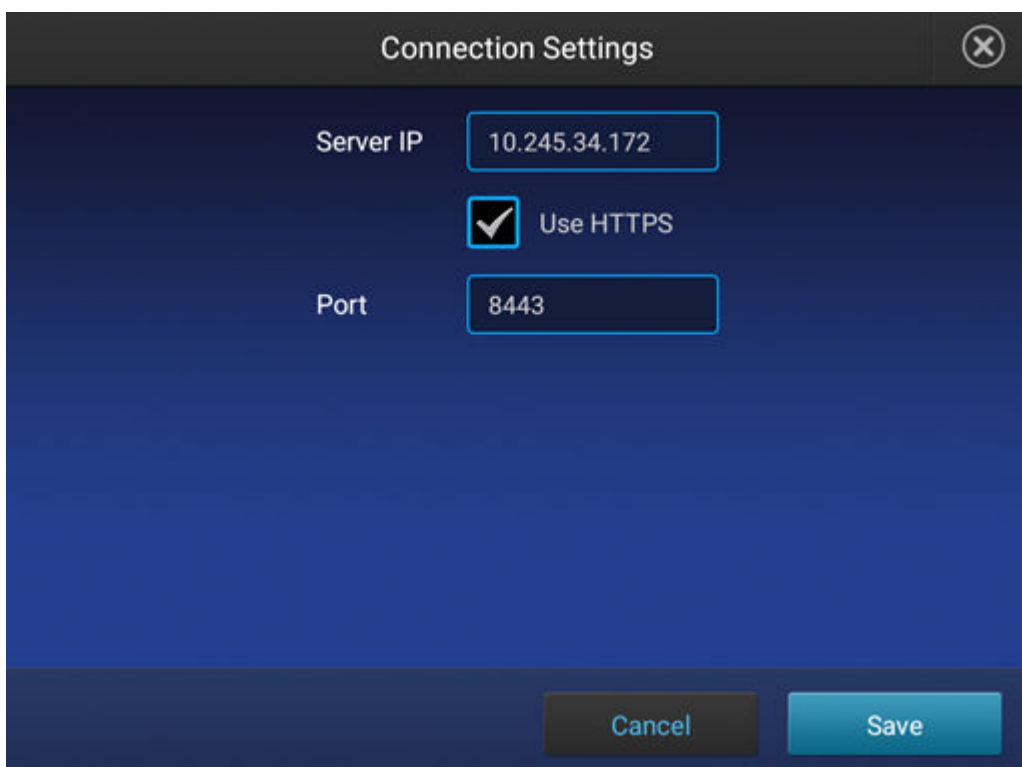
4. Select the **Use HTTPS** checkbox. Use the default port number, or a port number that you obtain from your network administrator.

IMPORTANT! Do not deselect the **Use HTTPS** checkbox, unless the SAE server has been configured to use HTTP. (If you deselect the checkbox, the software automatically uses HTTP.)

5. Tap **Save**.

6. Tap **Enable SAE**. Enter your SAE administrator account user name and password, then tap **Enable SAE**.

The instrument automatically restarts.



Connection Settings

Server IP 10.245.34.172

☒ Use HTTPS

Port 8443

Cancel Save

Disable SAE (administrator only)

This procedure requires a local administrator profile or an SAE administrator account.

- If you are using an SAE administrator account:
 - a. In the home screen, touch **Settings** ▶ **SAE** ▶ **Disable SAE**.
 - b. Enter your SAE administrator account username and password.

- If you are using a local administrator profile:
 - a. If another user is signed in, touch  in the home screen, then touch **Sign out**.
 - b. In the **Sign In** screen under **Local sign in**, touch **Sign in**.
 - c. Enter your local administrator profile name and PIN.
 - d. In the **Settings** screen, touch **SAE**, then touch **Disable SAE**.

Disable SAE on the SeqStudio™ Genetic Analyzer

This procedure requires an SAE account with administrator role.

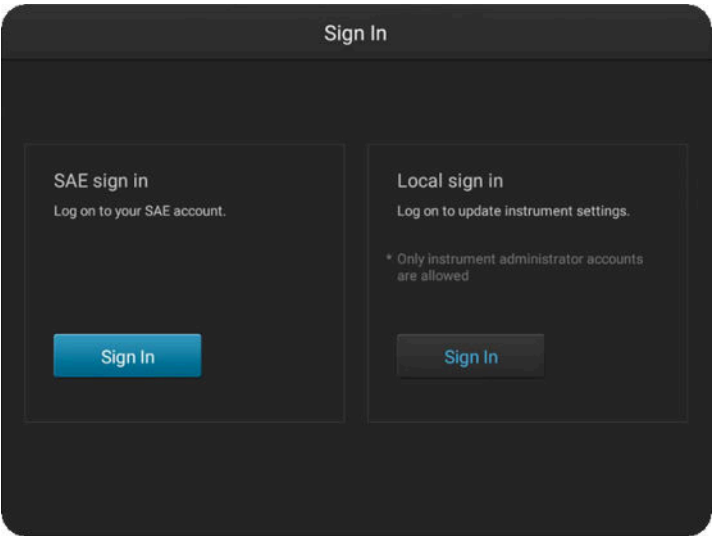
1. Sign in to the instrument with an SAE account with administrator role.
2. Touch  **Settings** ▶ **SAE**.
3. Touch **Disable SAE**.
4. Enter your SAE administrator account user name and password, then touch **Disable SAE**.

Sign in with SAE enabled

If the instrument is left unattended for 120 minutes, the instrument profile is signed out.

1. If another user is signed in, touch  in the home screen, then touch **Sign out**.
2. In the **Sign in** screen, touch **Sign In**, then enter your SAE username and password.

To access	Touch Sign in under
All instrument functions allowed by your SAE account	SAE
Administrator access to functions on the Settings screen	Local (requires an administrator instrument profile)



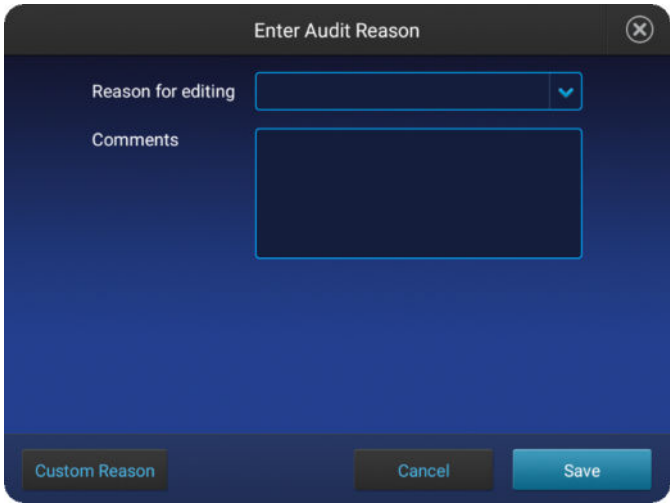
The home screen is displayed.

Specify audit reasons

Depending on the way that your SAE administrator configures audit settings, the **Enter Audit Reason** screen may be displayed when you make changes to a plate setup or a run module.

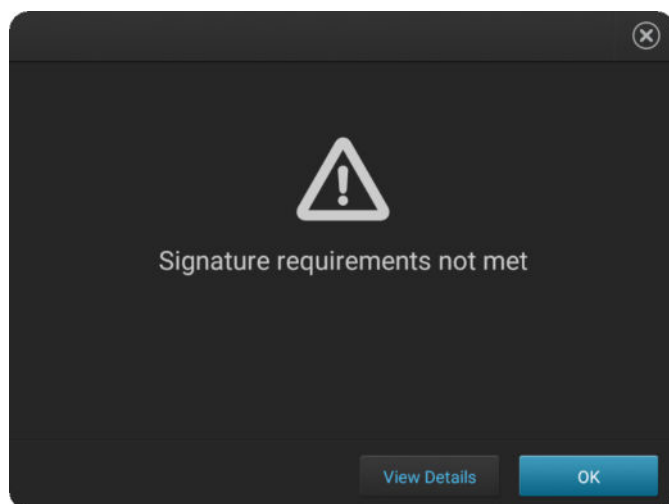
Select a reason or add a custom reason.

Note: The **Custom Reason** button is not displayed if audit settings are configured to require users to select a reason.

A dialog box titled "Enter Audit Reason" with a close button (X) in the top right corner. It contains two main sections: "Reason for editing" with a dropdown menu, and "Comments" with a large text area. At the bottom, there are three buttons: "Custom Reason" (disabled), "Cancel", and "Save".

E-signature requirements when you start a run

Depending on the way your SAE administrator has configured your instrument, E-signatures may be required when you start a run. If starting a run is configured to require signatures, a message is displayed when you start a run.



Touch **View Details** to show the pending signatures.

The **Object Name** column lists the items that require signatures:

If the Object Name lists	The object is	For information, see
A dye set name	The most recent dye calibration for the dye set specified in the plate setup.	"Sign a dye calibration or an install run" on page 67
An install standard name	The most recent install run for the application specified in the plate setup.	"Sign a dye calibration or an install run" on page 67
A plate name	Plate setup	"Sign a plate setup" on page 66

The example below indicates that E-signature is configured to require 4 signatures when you start a run, 1 signature is complete and 3 signatures are pending:

- 2 administrator signatures for dye calibration (dye set name is listed two times):
 - 1 signature that specifies the "Reviewed and approved" E-signature meaning (complete, because 1 signature is required [Signatures Required=1] and 1 signature has been logged [Current=1]).
 - 1 signature that specifies the "Checked consumables" meaning (incomplete).
- 1 administrator signature for the plate setup that specifies the "Reviewed and approved" E-signature meaning (incomplete).
- 1 administrator signature for install run that specifies the "Reviewed and approved" E-signature meaning (incomplete).

Meaning	Role	Object Name	Signatures Required	Current
Reviewed and Approved Dye Calibration	Administra...	E_BigDye™ Term...	1	1
Checked consumables	Administra...	E_BigDye™ Term...	1	0
Reviewed and Approved Plate Setup	Administra...	Plate_20171012...	1	0
Reviewed and Approved Install Run	Administra...	BDT v1.1	1	0

Close

Sign a plate setup

1. Create or open a plate setup, make changes as needed, then save the plate setup.
2. In the Plate Properties tab, touch **More Options**, then touch the **Sign Plate** button at the bottom left of the screen.
3. Select a **Signature meaning**.
4. Enter your password, then touch **Sign**.

Sign Plate

Object name Plate_20171010_230725marty

Signature meaning Reviewed and Approved Plate Set Up

User name M

Password

Cancel Sign

View E-signature records for a plate setup

1. Open a plate setup.
2. In the **Plate Properties** tab, touch **More Options**, then touch the **Signing records**.

Status	Meaning
Current	The plate has not been modified after the plate was signed.
Obsolete	The plate has been modified after the plate was signed.

Sign a dye calibration or an install run

1. Tap  **Actions** ▶ **Maintenance**.
2. Tap **Calibration** ▶ **Calibration history** or **Install run** ▶ **Install run history**.
3. Tap a dye set or install run to view the data.
4. Tap **Select to sign**, then select an item.
5. Tap **Sign**, enter your password, then touch **Sign**.
 A signed item is labeled with . Touch the  to display all signature records for the item.

View E-signature records for a dye calibration or an install run

1. Tap  **Actions** ▶ **Maintenance**.
2. Tap **Calibration** ▶ **Spectral calibration history** or **Install run** ▶ **Install run history**.
3. Tap a dye set or install run to view the data.
4. Tap **Actions** ▶ **Display Signatures**.
5. Tap **Select to sign**, then select an item.
6. Tap the  to display all signature records for the item.

Signing in after automatic screen locking

Depending on the way your SAE administrator has configured your instrument, the instrument touchscreen may automatically lock after a specified duration.

The users who can sign in when the screen is locked are determined by run status:

If a run is	These users can sign in
In process	• User who started the run • Administrator • Users with Log into Timed Out User Sessions permission
Complete	Any user

Use the instrument when the SAE server is offline

If your SAE administrator has configured your instrument to allow use when the SAE server is offline (**Client offline login** System setting in the SAE module), you can use the instrument for the period of time specified by the SAE administrator for **Client offline login**.

Note: If you have not previously signed in to the instrument with your SAE account, you cannot sign in when the SAE server is offline.

All SAE records are retained if the instrument is disconnected from an SAE server. When the instrument is reconnected to the SAE server, SAE records are uploaded to the server.

The following functions are not available when the SAE server is offline:

- Account lockout, password reminder, mandatory password change
- Disable SAE
- Change Password

Auditing of imported plate setups

If you import a plate setup, the following auditing occurs.

Note: You can import only one plate setup at a time.

If the file was created on...	And your instrument has SAE enabled and is configured to...	Then the following occurs
A different instrument with SAE enabled	— —	• The plate is imported. • Auditing is restarted.
A different instrument with SAE disabled	Open files from non-SAE systems=Forbidden	• An "Incomplete audit trail" message is displayed. • The plate is not imported.
A different instrument, and the plate setup was: • Created with SAE or plate auditing enabled • Modified when SAE or plate auditing was disabled	Open files from non-SAE systems=Allowed	• An "Incomplete audit trail" message is displayed. • You can touch Yes to import the plate.

SAE error messages and actions

Message	Possible cause	Action
Not authorized to...	Your SAE account does not specify permission to perform the function.	Contact your SAE administrator.
Unable to connect to SAE server. Check current connections.	The SAE server connection settings are incorrect.	1. Check the SAE server IP address (see “Determine IP address for a computer on a network” on page 144). 2. In the instrument Sign In screen, sign in with a local administrator account. 3. Set the correct IP address (Settings ▶ SAE settings ▶ Connection settings).
	There is a problem with the computer on which the SAE Administrator Console is installed or a problem with the network.	Troubleshoot computer or network problems.
	The computer on which the SAE Administrator Console has a dynamic IP address that is disconnecting the server when the computer is restarted.	Set a static IP address on the computer.
Incomplete audit trail (when you import a plate setup)	There is a gap in the audit trail. For example, the plate setup was created with SAE enabled. The plate setup was then modified with SAE disabled.	If your instrument is configured with Open files from non-SAE systems=Allowed , you can import the plate setup.



Create or modify a plate setup from the SeqStudio™ Plate Manager

■ Options for creating plate files in the Plate Manager software	70
■ Overview of plate setup settings	71
■ Set up a plate using default settings (Plate Manager)	72
■ Additional plate settings (Plate Manager)	80

Options for creating plate files in the Plate Manager software

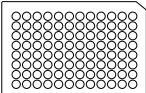
The Plate Manager software (desktop) is an application for creating plate files for importing to an instrument.

The Plate Manager software (cloud) has the same features as the desktop option, plus the additional feature of accessing SeqStudio™ Remote Monitoring App. This application requires connection to the Thermo Fisher™ Connect Platform.

In this chapter, the Plate Manager software is referred to in the following ways:

- If a section is specific to a platform, the software is identified as Plate Manager software (desktop) or Plate Manager software (cloud).
- If a section applies to both platforms, the software is identified as Plate Manager software.

Overview of plate setup settings

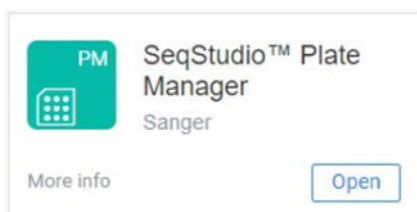
Category	Setting	Notes
Plate 	- Plate properties - Plate set up security (Shared or Hidden) - File name convention - Analysis settings	The analysis settings are saved with the plate setup as follows: - Hidden plates—The analysis settings are saved with the plate setup, and are used the next time that you open or create a hidden plate. The analysis settings will not be available to other system users. - Shared plates—The analysis settings are saved with the plate setup, and are used the next time that you open or create a shared plate. The analysis settings will be available to other system users. User-created analysis settings: - Users can name and save analysis settings. - If you create analysis settings in the Plate Manager (desktop), the settings will be available to other users. - If you create analysis settings in the Plate Manager (Thermo Fisher™ Connect Platform), the settings will not be available to other users.
Injection group (a set of 4 wells) 	<i>(Fragment/HID analysis only)</i> Size standard - Run module - Application type (mixed plate only) - Dye set <i>(HID analysis only)</i> Kit (for marker-to-marker pull-up reduction)	User-created size standards and dye sets are accessible to all users. User-created run modules: - Users can name and save run modules. - If you create run modules in the Plate Manager (desktop), the modules will be available to other users. - If you create run modules in the Plate Manager (Thermo Fisher™ Connect Platform), the modules will not be available to other users.
Well 	- Sample name <i>(Fragment/HID analysis only)</i> Sample type <i>(Sequencing only)</i> Specimen and amplicon - Custom fields 1–5	Specimen and Amplicon fields are useful in secondary analysis software applications that organize sample data files based on amplicon and specimen information. - Custom fields are text fields in which you can include additional sample attributes or identifiers. Custom fields can be used in file name conventions or by some secondary analysis applications.

Set up a plate using default settings (Plate Manager)

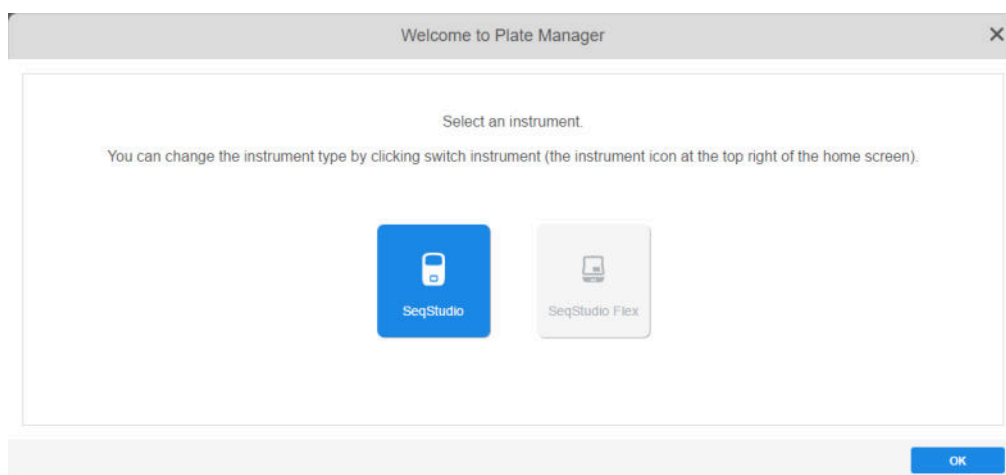
Access the Plate Manager

Access the Plate Manager on the Thermo Fisher™ Connect Platform

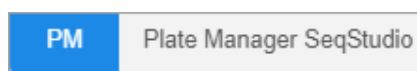
1. Sign in to apps.thermofisher.com.
2. In the **My apps** list, select **Plate Manager**.
If **Plate Manager** is not listed under **My apps**, scroll down in the **All Apps** list.



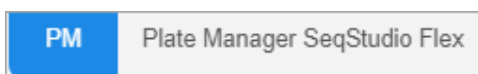
3. (First use only) In the **Select an instrument** screen, select **SeqStudio**, then click **OK**.
The instrument type determines the plate type that is created. You can change the instrument type at any time in the home screen by clicking  (**Switch instrument**) at the top right of the screen.



The selected instrument type is displayed at the top left of the Plate Manager software home screen.



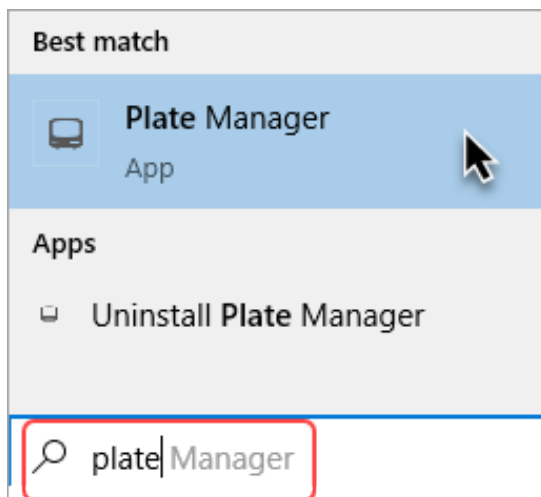
Note: You can also use the Plate Manager software with a SeqStudio™ Flex instrument. To change the default instrument type, click  (**Switch instrument**) at the top right of the screen, then select **SeqStudio Flex**.



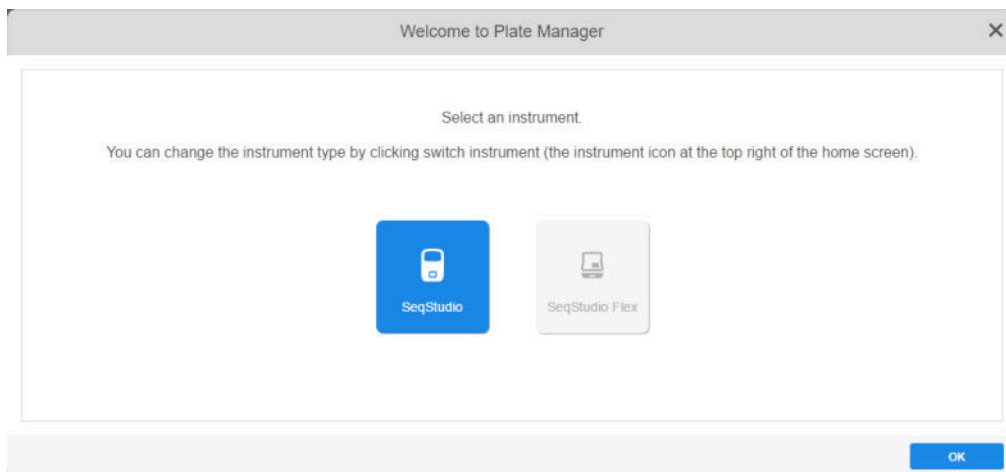
For information, see *SeqStudio™ Flex Series Genetic Analyzer with Instrument Software v1.2 User Guide* (Pub. No. 100104689).

Access the Plate Manager software (desktop)

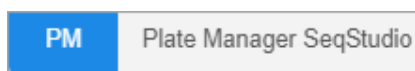
1. In the Windows™ desktop, click , type **Plate Manager**, then select **Plate Manager** from the start menu.



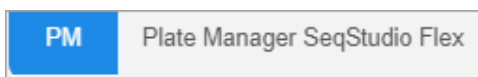
2. (First use only) In the **Select an instrument** screen, select **SeqStudio**, then click **OK**.
The instrument type determines the plate type that is created. You can change the instrument type at any time in the home screen by clicking  (**Switch instrument**) at the top right of the screen.



The selected instrument type is displayed at the top left of the Plate Manager software home screen.



Note: You can also use the Plate Manager software with a SeqStudio™ Flex instrument. To change the default instrument type, click  (**Switch instrument**) at the top right of the screen, then select **SeqStudio Flex**.



For information, see *SeqStudio™ Flex Series Genetic Analyzer with Instrument Software v1.2 User Guide* (Pub. No. [100104689](#)).

Create or open a plate setup PSM file

1. Click  to display the home screen.
2. In the **Plate setup** screen, create or open a plate setup:
If you are running the Plate Manager on the Thermo Fisher™ Connect Platform.

Click...	To...
Create a plate file	• Create a new plate setup. • Create a plate setup from a template.
Open from cloud	Open a plate setup that you created in Plate Manager on your Thermo Fisher™ Connect Platform account.
Open from local drive	• Open a plate setup that you created in Plate Manager on your computer (PSM file). • Open a plate setup that you created in Plate Manager in another application (CSV file).

If you are running the Plate Manager on the desktop:

Click...	To...
New	Create a new plate setup or to create a plate setup from a template.
Open	Open a plate setup that you created in Plate Manager (PSM file) or in another application (CSV file).

Note: In the home screen, **Recent plate files** is displayed. For quick access, navigate to a recent plate setup file of interest.

Enter plate properties

In the **Properties** tab:

1. (Optional) Edit the **Plate name**, **Barcode**, or **Owner**.

2. Select an option in the **Plate setup security** field.
 - **Hidden**—Prevents other users from using or accessing the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a hidden plate. However, the analysis settings will not be available to other system users.
 - **Shared**—Allows other users to access and edit the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a shared plate. Your analysis settings will be available to other system users.
3. Select the **Application type: Sequencing, Fragment analysis, HID, or Mixed plate (sequencing & fragment)**.
A mixed plate allows you to specify fragment analysis and sequence analysis settings on the same plate.
4. Select the analysis settings. If needed, edit the settings (see “Edit analysis settings” on page 80).
5. (Optional for Sequencing or Mixed plate) Select the **I am analyzing my data with Sanger variant analysis software** checkbox.
The amplicon and specimen fields are added to the **Plate** view, and the attributes are automatically added to the default file name conventions (see “Modify the default file name convention” on page 94).

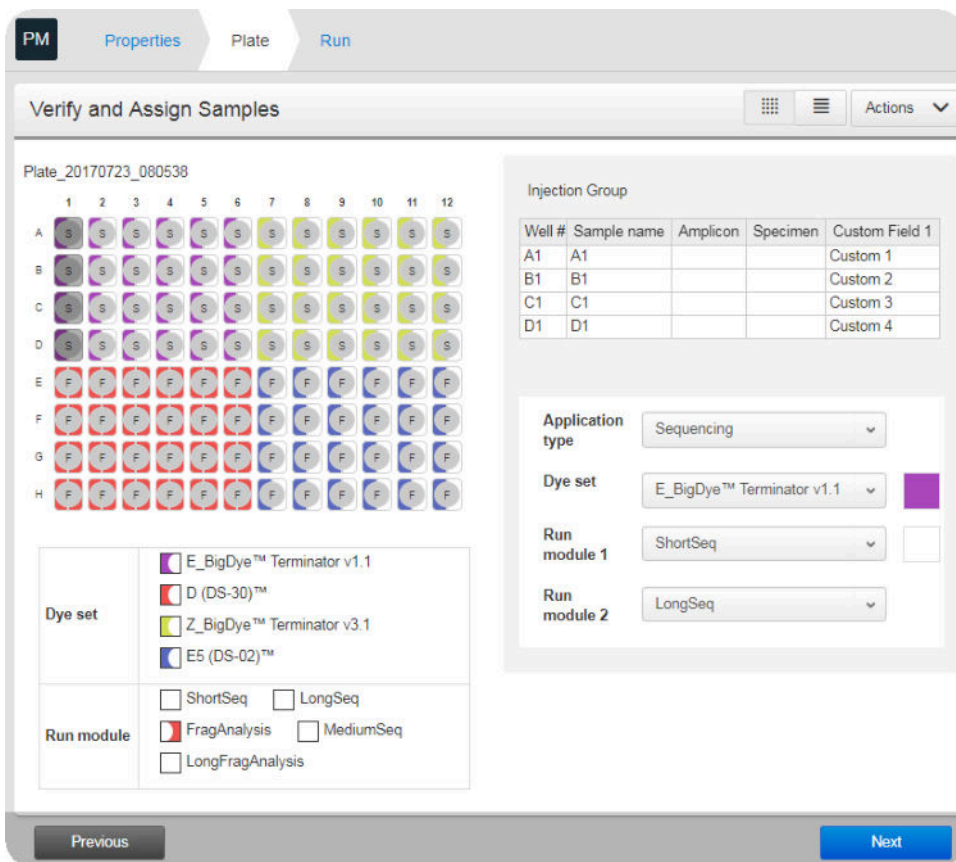
This feature is useful in secondary analysis software applications that organize files based on amplicon and specimen information (Thermo Fisher™ Connect Platform applications: Variant Analysis (VA) module, Next-generation Confirmation (NGC) module; desktop applications: SeqScape™ Software, Variant Reporter™ Software, Minor Variant Finder Software).

6. Click **Next**.

Assign wells: Sample and run information

This procedure describes using the **Plate** view. Alternatively, you can click  and use keyboard short-cuts to enter information. See “Keyboard shortcuts for the sample table” on page 77.

1. In the **Plate** screen, select one or more injection groups: Click a well to select one injection group; shift+click or click+drag to select multiple injection groups.
Each set of 4 wells on the plate is referred to as an injection group.
The default injection order is: A1-D1, E1-H1, A2-D2, E2-H2....A12-D12, E12-H12.



Verify and Assign Samples

Plate_20170723_080538

Well #	Sample name	Amplicon	Specimen	Custom Field 1
A1	A1			Custom 1
B1	B1			Custom 2
C1	C1			Custom 3
D1	D1			Custom 4

Application type: Sequencing

Dye set: E_BigDye™ Terminator v1.1

Run module 1: ShortSeq

Run module 2: LongSeq

Dye set:
☒ E_BigDye™ Terminator v1.1
☐ D (DS-30)™
☐ Z_BigDye™ Terminator v3.1
☐ E5 (DS-02)™

Run module:
☐ ShortSeq ☐ LongSeq
☐ FragAnalysis ☐ MediumSeq
☐ LongFragAnalysis

Previous Next

2. If you are creating a mixed plate, select the **Application type** for the selected injection groups.

3. (Optional) Edit the default **Sample name** (well) for each well.

Note: If multiple injection groups are selected when you edit the **Sample name** or **Sample type**, *both wells* with the same capillary number are edited. For example, if injection groups A1–D1 and E1–H1 are selected, and you change the **Sample name** to "Test" for well E1, the **Sample name** for well A1 is also changed to "Test".

4. (Fragment/HID analysis only) Select the **Sample type** for each well: **Allelic ladder**, **Negative control**, **Positive control**, or **Sample**.
5. (Fragment/HID analysis only) Select a size standard for the selected injection groups.
6. (HID analysis only) Select a kit for the selected injection groups (enables marker-to-marker pull-up reduction).
7. Select a **Dye set** for the selected injection groups.
8. Select a **Run module** for the selected injection groups.

Note: **Run module 1** is used for the first injection you specify. If you specify replicate injections, additional run module fields are added. For information, see "Specify replicate injections" on page 93.

For more information, see "Run modules, read lengths, size ranges, and run times" on page 156.

9. Click **Next** and proceed to "Save a plate setup in the Plate Manager" on page 79.

Keyboard shortcuts for the sample table

Keys	Action
Navigation	
Arrow keys	Move to the cell above, below, to the right, or to the left of the current cell.
Tab	Move to the cell to the right of the current cell.
Shift+Tab	Move to the cell to the left of the current cell.
Home	Move to the first cell in the row.
End	Move to the last cell in the row.
Ctrl+Home	Move to the first cell in the column.
Ctrl+ End	Move to the last cell in the column.
Selection	
Ctrl+A	Select all.
Shift+Arrow keys	Extend the selection of the cell above, below, to the right, or to the left of the current cell.
Shift+Home	Select all cells in the row to the right including the current cell.

(continued)

Keys	Action
Shift+End	Select all cells in the row to the left including the current cell.
Ctrl+Shift+Home	Select all cells in the column to the top cell including the current cell.
Ctrl+ Shift+End	Select all cells in the column to the bottom cell including the current cell.
Editor	
Enter	Open and close the cell editor.
F2	Clear the cell contents and open the cell editor.
Esc	Cancel editing and close the cell editor.
Backspace	Delete.
Delete	Clear the cell contents.
Ctrl+C	Copy.
Ctrl+X	Cut.
Ctrl+V	Paste.
Ctrl+Enter	Fill all selected cells with edited cell's value: Select a range of cells, then press F2 to edit the first cell in the selection. Type a value, then press Ctrl+Enter to fill all selected cells with the value.
Ctrl+Z	Undo.
Ctrl+Y	Redo.

Save a plate setup in the Plate Manager

After you assign wells to a plate setup:

1. In the **Save the plate setup** dialog box, modify any settings as needed.

The screenshot shows the 'Save the plate setup' dialog box. It has a title bar with the text 'Save the plate setup'. Inside, there are several sections: 'Plate name' with a text input field containing 'Plate_20170723_080538'; 'File name convention' with two lines of text: 'Sequencing file : Well position_Sample name_Amplicon_Specimen_20170723_081724.ab1' and 'Fragment analysis file : Well position_Sample name_Sample type_20170723_081724.fsa', followed by a note '* Note: All files names are suffixed with a date and time stamp.'; 'Sequencing analysis settings' with a dropdown menu showing 'Default analysis settings'; and 'Fragment analysis settings' with a dropdown menu also showing 'Default analysis settings'. At the bottom, there are buttons for 'Edit file name convention', 'Edit analysis settings', 'Close', and 'Save'. A 'Home' button is at the bottom left, and a 'Monitor my run' button is at the bottom right.

2. Click **Save**.

The screenshot shows the 'Plate' tab in the SeqStudio Plate Manager. The top navigation bar has tabs for 'PM', 'Properties', 'Plate', and 'Run'. The 'Plate' tab is selected. Below the navigation bar, the text 'Plate : Plate_20170724_080601' is displayed. The main area shows 'Step 1' with a large checkmark icon. Below the checkmark, the text 'Your plate is now available to run.' is followed by a list of four steps: 1. Go to SeqStudio™ Genetic Analyzer. 2. Import the plate setup file. 3. Insert the plate in the instrument and start the run. 4. Go to InstrumentConnect or a mobile device to monitor the run. A 'Feedback' button is on the right side. At the bottom, there are 'Home' and 'Monitor my run' buttons.

Note: The **Monitor my run button** is available only in the Thermo Fisher™ Connect Platform app.

If you are running the Plate Manager on the...	The plate setup is saved as a...
Thermo Fisher™ Connect Platform	Plate setup that you can open from the Thermo Fisher™ Connect Platform and run on the instrument.
Desktop	PSM file that you can open from a network drive or a USB and run on the instrument.

Proceed to Chapter 7, “Start and monitor a run”.

Additional plate settings (Plate Manager)

Specify replicate injections

In the **Plate** screen:

1. Select **Actions ▶ Add injection**.
2. Select one or more injection groups, then select a run module for the replicate injections.
You can add up to 5 replicate injections (for a total of 6 injections). **Run module 1** is used for the first injection you specify. Additional run module fields are added for replicate injections.

To remove a replicate injections, select **Actions ▶ Remove injection**.

Edit analysis settings

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

1. In the **Properties** tab, click **Setting Details**, or select **Actions ▶ Manage analysis settings**.
2. To create new analysis settings:
 - a. Select the default analysis settings or user-created analysis settings, then click **Copy**.
 - b. Enter a name and edit settings as needed (see “Fragment/HID analysis settings (size calling)” on page 141 or “Sequencing settings (base calling)” on page 143).
 - c. Click **Save**.
3. As needed, select an analysis setting of interest, then click **Edit** or **Delete** (user-created settings only).
4. Click **Close**.

Modify the default file name convention

The file name convention determines how the data files (AB1 or FSA) associated with a plate are named.

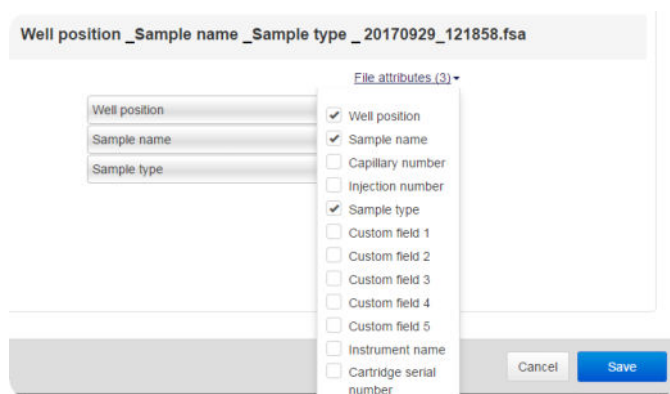
The default file name convention is:

Application	Default settings
Fragment/HID analysis	<well>_<sample name>_<sample type>_<date and timestamp>. .fsa
Sequence analysis	<well>_<sample name>_<date and timestamp>.ab1
Sequence analysis with the Sanger variant analysis option selected	<well>_<sample name>_<amplicon>_<specimen>_<date and timestamp>.ab1

To change the default settings, in the **Properties** tab:

1. Select **Actions ▶ Edit the file name convention**.
2. Click **File attributes**, then select the attributes to include in the data file name.

Note: The timestamp attribute cannot be deselected; it is always included in the data file name.



For information on creating custom fields to include in file name conventions, see “Define custom fields” on page 96.

3. (Optional) Click-drag ≡ to move an attribute to another position.
4. Click **Save**.

Hide or share a plate (Plate setup security)

In the **Properties** screen:

Select:

- **Hidden**—Prevents other users from using or accessing the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a hidden plate. However, the analysis settings will not be available to other system users.
- **Shared**—Allows other users to access and edit the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a shared plate. Your analysis settings will be available to other system users.

Define custom fields

Custom fields are text fields in which you can include additional sample attributes or identifiers. Custom fields can be used in file name conventions or by some secondary analysis applications.

In the **Plate** tab:

1. Select **Action ▶ Add custom field**.
You can add up to 5 custom fields.
2. Enter information in the custom field in the table at the right of the plate.

To remove a custom field, select **Actions ▶ Remove custom field**.



Create or modify a plate setup from the instrument

■ PSM and CSV plate setup files for import into the instrument	84
■ Shared (public), hidden (my plates), and guest plate setup files	85
■ Overview of plate setup settings	85
■ (Optional) Set up for auto export of sample data files (AB1 and FSA)	86
■ Set up a plate using default settings (instrument)	87
■ Set optional plate settings (instrument)	93

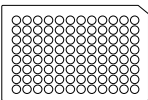
PSM and CSV plate setup files for import into the instrument

Format	Contains...	When to use...
PSM		
Create by: Saving a plate setup with the SeqStudio™ Plate Manager desktop or Thermo Fisher™ Connect Platform	- Size standard <i>names and definitions</i> - Run module <i>names and settings</i> - Dye set <i>names and settings</i> - Analysis settings <i>names and settings</i> - All remaining plate setup field settings: Well ID, Sample name, Application type and so on.	To create a plate setup that contains all the properties needed to save and run the plate setup. Note the following: - If the size standard, analysis settings, or run module in the PSM file do not exist on the instrument, they are automatically created when the PSM file is imported. - If the Plate setup security is set to Hidden: - The plate setup is available for selection on the instrument only to the user who created the PSM file.
CSV		
Create by: - Saving as CSV in the SeqStudio™ Plate Manager desktop or Thermo Fisher™ Connect Platform - Downloading a template from the SeqStudio™ Plate Manager desktop or Thermo Fisher™ Connect Platform	- Size standard <i>names only</i> - Dye set <i>name only</i> - Run module <i>names only</i> - All remaining plate setup field settings: Well ID, Sample name, Application type and so on.	To update properties of the plate setup (such as sample name) <i>after</i> the size standard, dye set, and run module for the plate setup have been created or imported on the instrument. When you import a CSV file: - If the size standard, dye set, or run module in the CSV file <i>do not</i> exist on the instrument: - An error message is displayed. - The CSV file is not imported on the instrument. - If the size standard or run module in the CSV file <i>do</i> exist on the instrument: - The CSV file is imported on the instrument. - The settings from the size standard, dye set, and run module that exist on the instrument are used (because the CSV file contains the names of these elements, but not the settings). - The Plate setup security is set to Hidden.

Shared (public), hidden (my plates), and guest plate setup files

Create by setting Plate setup security to...	Accessible to users...	Stored in folder on instrument...	Analysis settings used...	File name convention used...
Shared	All users, including guest	Public	Analysis settings saved in the plate setup	Last settings specified by the signed-in user (not saved with plate setup)
Note: If a guest user edits a shared plate setup, the plate must be saved under a new name, and the Plate setup security is set to Hidden and cannot be changed.				
Hidden	Only the user who created the plate setup	My plates	Last settings specified by the signed-in user (not saved with plate setup)	Last settings specified by the signed-in user (not saved with plate setup)
Guest plate setup security is automatically set to Hidden and cannot be changed	Guest user only	My plates	Last settings specified by the signed-in user (not saved with plate setup)	Last settings specified by the signed-in user (not saved with plate setup)

Overview of plate setup settings

Category	Setting	Notes
Plate 	- Plate properties - Plate set up security (Shared or Hidden) - File name convention - Analysis settings	The analysis settings are saved with the plate setup as follows: - Hidden plates—The analysis settings are saved with the plate setup, and are used the next time that you open or create a hidden plate. The analysis settings will not be available to other system users. - Shared plates—The analysis settings are saved with the plate setup, and are used the next time that you open or create a shared plate. The analysis settings will be available to other system users. User-created analysis settings: - Users can name and save analysis settings. - If you create analysis settings in the instrument software, the settings will not be available to other users.

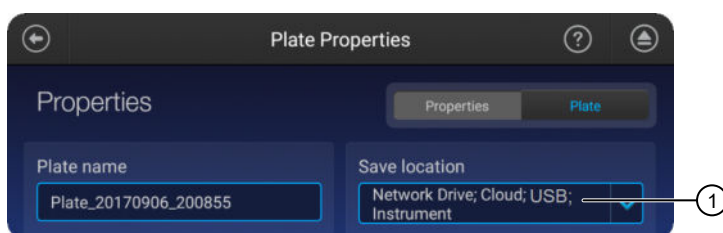
(continued)

Category	Setting	Notes
Injection group (a set of 4 wells) ○ ○ ○ ○	(Fragment/HID analysis only) Size standard - Run module - Application type (mixed plate only) - Dye set (HID analysis only) Kit (for marker-to-marker pull-up reduction)	User-created size standards and dye sets are accessible to all users. User-created run modules: - Users can name and save run modules. - If you create run modules in the instrument software, the modules will not be available to other users.
Well ○	- Sample name (Fragment/HID analysis only) Sample type (Sequencing only) Specimen and amplicon - Custom fields 1–5	Specimen and Amplicon fields are useful in secondary analysis software applications that organize sample data files based on amplicon and specimen information. - Custom fields are text fields in which you can include additional sample attributes or identifiers. Custom fields can be used in file name conventions or by some secondary analysis applications.

(Optional) Set up for auto export of sample data files (AB1 and FSA)

By default, sample data files (AB1 and FSA) are saved to the instrument.

When you create a plate setup, you can also set the **Save location** to **Cloud**, **Network Drive**, and/or **USB**.



① Save location

When the plate is run, the instrument automatically exports the sample data files to the save locations.

Before you can select these save locations, set up the instrument:

- “Link the instrument to your Thermo Fisher™ Connect Platform account” on page 151
- “Connect to a network drive” on page 146
- Insert a USB into the USB port on the front of the instrument (Figure 1 on page 14)

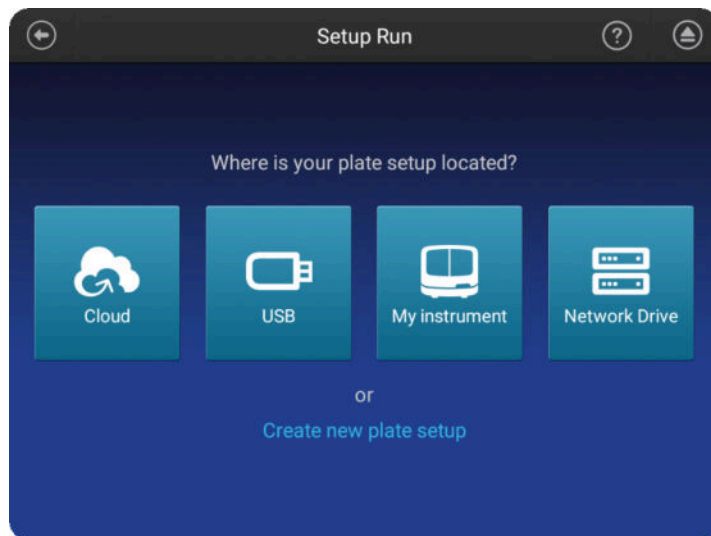
Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

Set up a plate using default settings (instrument)

Create or import a plate setup

In the home screen:

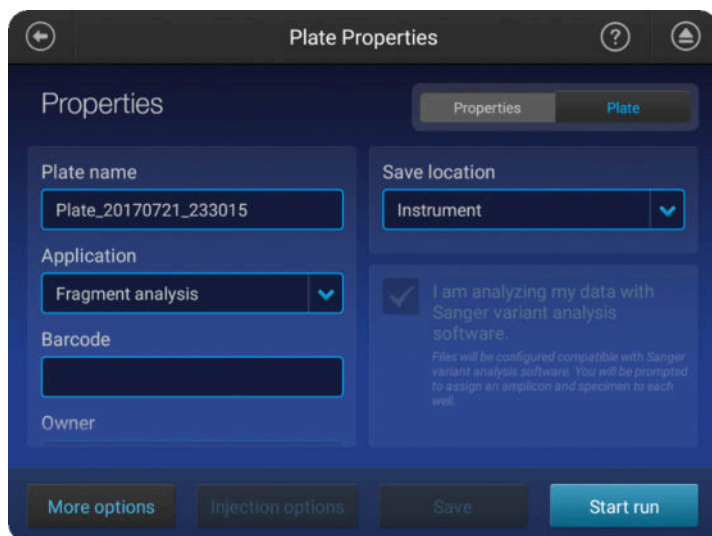
1. Touch **Set up run**.
2. Create, open, or import a plate setup:



To	Procedure
Create a new plate setup	- Touch Create new plate setup. - See “Enter plate properties” on page 88.
Open an existing plate setup on the instrument	- Touch  My Instrument. - Touch: My plates folder to select hidden plates that you have created. Public folder to select (1) shared plates that were created by any user or (2) any plates that were created by a Guest user. For more information, see “Shared (public), hidden (my plates), and guest plate setup files” on page 85.
Import a plate setup	- Touch  Cloud,  USB, or  Network Drive. - Select:  PSM file  CSV file For more information, see “PSM and CSV plate setup files for import into the instrument” on page 84.

Enter plate properties

- At the top-right of the **Plate properties** screen, touch the **Properties** tab.
- Touch the **Plate name** field, then enter the plate name.



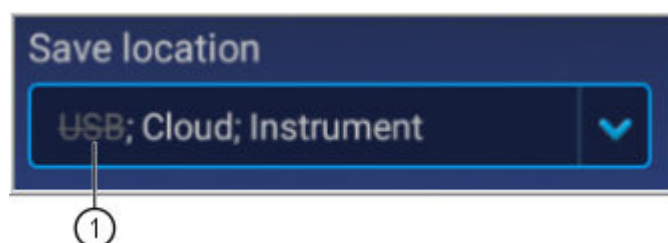
- Touch **Applications**, then select **Sequencing**, **Fragment analysis**, **HID**, or **Mixed plate** (allows you to specify fragment analysis and sequence analysis settings on the same plate).
- (Optional) Touch the **Barcode** field, then use a scanner to scan the barcode.

5. (Optional) Touch the **Owner** field, then enter the plate owner name.
6. (Optional) Touch **More options** to check other settings (for example, **Plate setup security**, **Analysis settings**, and **File name convention**).
7. (Optional) Touch **Save location**, then select a location for the plate setup.

The plate setup is always saved to the instrument. In addition, you can save the plate to the Thermo Fisher™ Connect Platform, a network, or a USB, which will auto export the sample data files.

IMPORTANT! To view analyzed data in the Remote Monitoring App on the Thermo Fisher™ Connect Platform, you must save the plate setup to the Thermo Fisher™ Connect Platform.

Note: (Sequencing and Fragment analysis) If you save a plate setup to the Thermo Fisher™ Connect Platform, a network, or a USB, then access the plate setup at a later time when the instrument is not linked to the Thermo Fisher™ Connect Platform, a network, or a USB, the save location is displayed with strikethrough text.

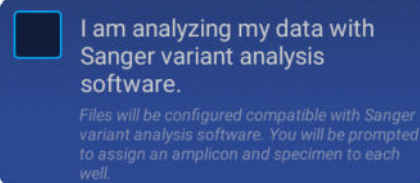


① Original location to which the plate was saved, but is no longer accessible by the instrument.

8. (Optional for Sequencing or Mixed plate) Touch the **I am analyzing my data with Sanger variant analysis software** checkbox.

The amplicon and specimen fields are added to the **Plate** view, and the attributes are automatically added to the default file name conventions (see “Modify the default file name convention” on page 94).

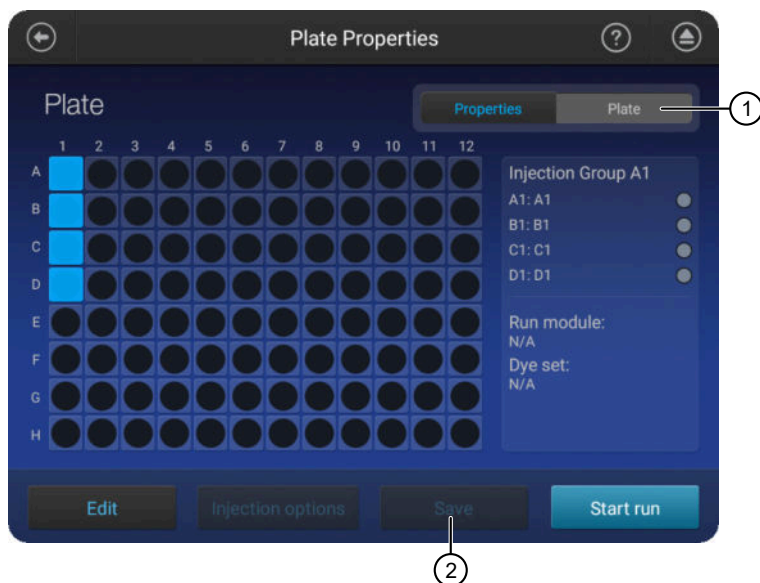
This feature is useful in secondary analysis software applications that organize files based on amplicon and specimen information (Thermo Fisher™ Connect Platform applications: Variant Analysis (VA) module, Next-generation Confirmation (NGC) module; desktop applications: SeqScape™ Software, Variant Reporter™ Software, Minor Variant Finder Software).



Assign wells: run module, size standard, dye set, and kit

Note: You can assign these settings to one or more injection groups at the same time.

1. In the **Plate properties** screen, touch **Plate** at the top-right of the screen.



- ① Plate tab
- ② Save button is not enabled until you create an injection group

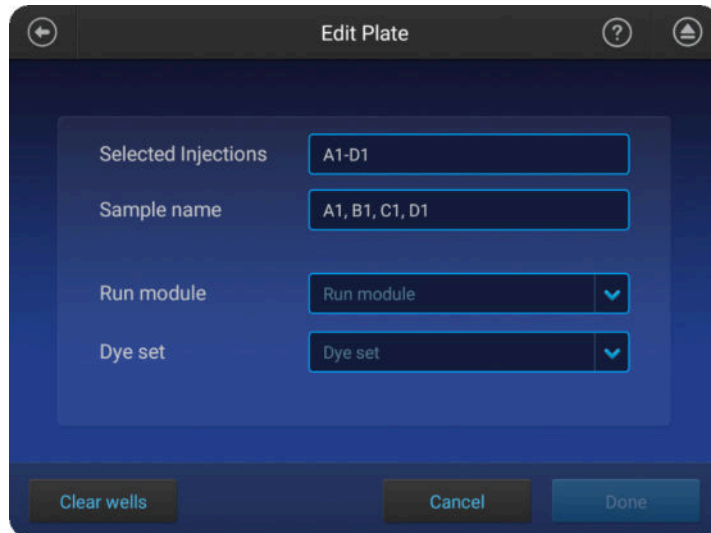
Each set of 4 wells on the plate is referred to as an injection group.

An injection group is identified by the first well in the set of 4 (for example, Injection Group A1 contains wells A1–D1).

The default injection order is: A1-D1, E1-H1, A2-D2, E2-H2....A12-D12, E12-H12.

2. Select injection groups, then touch **Edit**.
 - Touch a well to select a single injection group.
 - Touch and drag to select multiple injection groups or the entire plate.

Note: All settings in the remaining steps will be assigned to all selected injection groups. If different injection groups require different settings, repeat these steps for each injection group.



3. If you are creating a mixed plate, select the **Application type** for the wells.
4. Touch **Run modules**, then select a run module.
For more information, see “Run modules, read lengths, size ranges, and run times” on page 156.
5. (Fragment/HID analysis only) Touch **Size standard**, then select a size standard for the injection group.
6. Swipe up to display the rest of the screen.
7. Touch **Dye set**, then select a dye set for the selected injection groups.
8. (HID analysis only) Touch **Kit**, then select a kit for the selected injection groups.
9. Touch **Done**.

Assign wells: sample name, sample type, and custom fields

Note: You can assign these settings to only one well at a time.

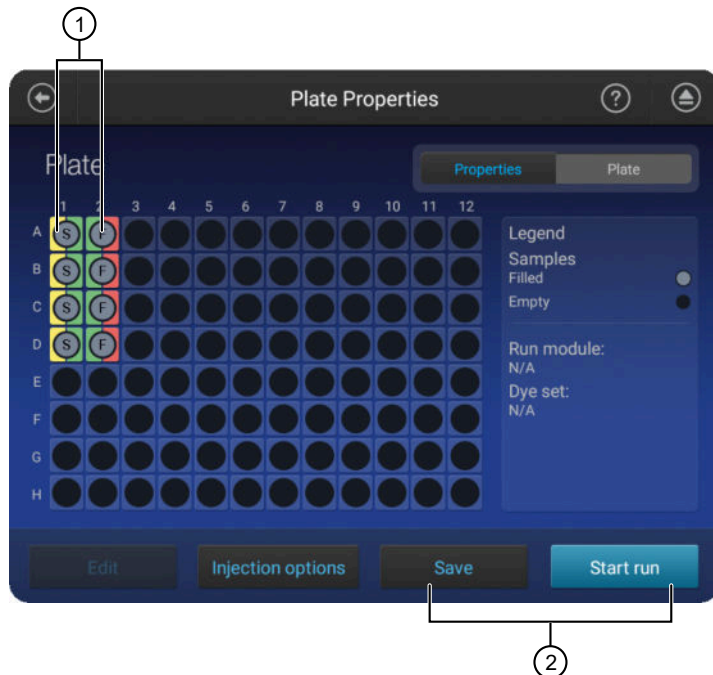
1. In the **Edit plate** screen, touch **Sample name** to display the well attributes fields.

	Well ID	Sample
Sample	A1	A1
Run module	B1	B1
Dye set	C1	C1
Custom field 1	D1	D1
Custom field 2		
Custom field 3		
Custom field 4		
Custom field 5		

Cancel Done

2. Touch a setting, then enter the definition for the selected wells:
 - *(Fragment/HID analysis only)* **Sample type**—**Sample**, **Positive Control**, **Negative Control**, or **Allelic Ladder**.
 - **Custom fields**—Text fields to include additional sample attributes or identifiers that can be used by secondary analysis applications.
 - *(Sequencing only)* **Amplicon and Specimen**—Amplicon and Specimen names for Sanger Sequence analysis, if you selected the option in the **Plate Properties** screen.

3. Touch **Done** to close the screen then **Done** to close the **Edit Plate** screen.



- ① Application type—Sequencing, Fragment or HID; designated by S, F, or H
- ② Save the plate or start the run

4. Touch **Save** to save the plate to run at a later time, or touch **Start Run**.

Set optional plate settings (instrument)

Specify replicate injections

In the **Plate properties** screen:

1. Touch **Injection options**.
2. Touch an injection group, then touch **Edit and re-inject** to add replicates to the injection list.
3. (Optional) Modify **Run module**, **Injection time**, **Injection voltage**, **Run time**, or **Run voltage** for the injections.

IMPORTANT! The changes to the run conditions apply only to the replicate injection. The changes are not saved to the run module.

4. Touch **Done**.

Modify analysis settings

In the **Properties** tab of the **Plate properties** screen:

1. Touch **More options** ▶ **Analysis settings**.

2. Select a setting.

For more information, see “Manage analysis settings” on page 158.

Note: The last setting selected is used as the default for new plates.

3. Touch **Done**.

Modify the default file name convention

The default file name convention determines how the data files (AB1 or FSA) associated with a plate are named.

The default file name convention is:

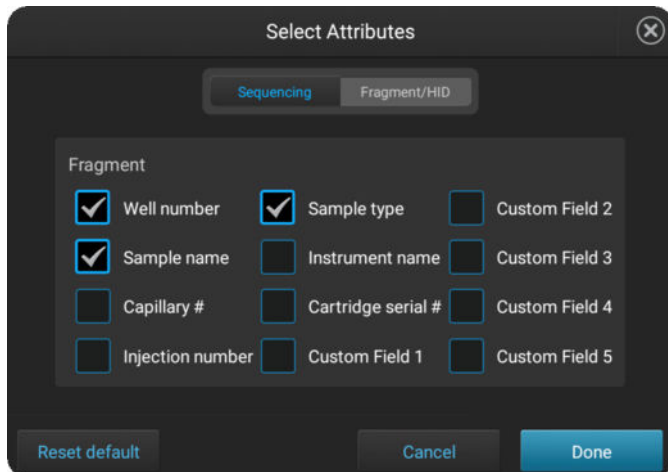
Application	Default settings
Fragment/HID analysis	<well>_<sample name>_<sample type>_<date and timestamp>. .fsa
Sequence analysis	<well>_<sample name>_<date and timestamp>.ab1
Sequence analysis with the Sanger variant analysis option selected	<well>_<sample name>_<amplicon>_<specimen>_<date and timestamp>.ab1

1. Access the **File name convention** screen:

From	Action
Plate properties screen	Select the Properties tab, then touch More options ▶ File name convention .
Home screen	Touch  Settings ▶ Run settings ▶ File name convention .

2. Touch **Attributes**.

3. Select the attributes to include in the data file name.



For information on creating custom fields to include in file name conventions, see “Define custom fields” on page 96.

Note: The timestamp attribute cannot be deselected; it is always included in the data file name.

4. Touch **Done**.
5. Touch and drag attributes up or down in the list.
6. Touch **Done**.

Hide or share a plate (Plate setup security)

For more information, see “Shared (public), hidden (my plates), and guest plate setup files” on page 85.

1. Touch **More Options** ▶ **Plate Setup security**.
2. Touch an option:
 - **Hidden**—Prevents other users from using or accessing the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a hidden plate. However, the analysis settings will not be available to other system users.
 - **Shared**—Allows other users to access and edit the plate on the instrument. Your analysis settings are saved with the plate setup, and are used the next time that you open or create a shared plate. Your analysis settings will be available to other system users.
3. Touch **Done**, then touch **Save**.

Define custom fields

Custom fields are text fields in which you can include additional sample attributes or identifiers. Custom fields can be used in file name conventions or by some secondary analysis applications.

In the **Plate** tab of the **Plate properties** screen:

1. Select injection groups, then touch **Edit**.
 - Touch a well to select a single injection group.
 - Touch and drag to select multiple injection groups or the entire plate.
2. Touch **Sample name** to display the injection group and well attributes fields.

The screenshot shows the 'Edit Plate' screen with three main input fields: 'Selected Injections' (containing 'A3-D3'), 'Sample name' (containing 'A3, B3, C3, D3'), and 'Application type' (a dropdown menu). A blue border highlights the 'Sample name' field, and a circled '1' points to it.

- 1 Touch the sample name field

The screenshot shows the 'Plate_20170722_225336' screen. It features a table with two columns: 'Well ID' and 'Sample'. The 'Sample' column contains input fields for 'A1', 'B1', 'C1', and 'D1'. The 'Sample' column is highlighted with a blue border. At the bottom of the screen are 'Cancel' and 'Done' buttons.

Well ID	Sample
A1	A1
B1	B1
C1	C1
D1	D1

3. Touch a custom field, then enter the definition for the selected wells.
4. Click **Done**.

(Optional) View the injection list, change injection settings or order, or specify replicates and re-injections

In the **Plate properties** screen:

1. Touch **Injection options**.
2. Touch an injection group, then configure the injection list:
 - Touch and drag an injection group to a new location in the injection list.
 - Touch **Inject first**—Moves the selected injection group to the top of the injection list.
 - Touch **Edit and re-inject**—Adds replicates or re-injections to the injection list. You can also modify **Run module**, **Injection time**, **Injection voltage**, **Run time**, or **Run voltage** for these injections.

Note: *Replicate* indicates a duplicate—the replicate has the same parameters as the original injection. *Re-injection* indicates a copy of the injection, but with user-modified parameters.

Note: The changes to the run conditions apply only to the replicate or re-injection. The changes are not saved to the run module.

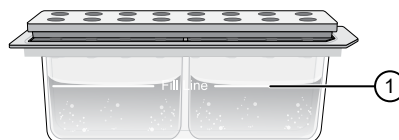
3. Touch **Done**.

■ Load the plate or the tube assembly	98
■ Select a plate setup and start a run	99
■ Lock the touchscreen	101
■ Monitor a run from the Thermo Fisher™ Connect Platform	101
■ Monitor a run from a mobile device	109
■ Monitor a run from the instrument	112
■ Unload the plate or the tube assembly	116

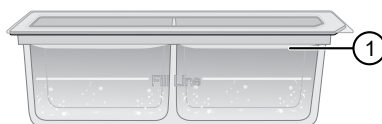
Load the plate or the tube assembly

In the home screen:

1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid.
3. Check the buffer fill level:
 - a. Remove the CBC.
 - b. Ensure that the level of buffer is above the fill line.
If the buffer is at or below the fill line, see “Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)” on page 183 and “Insert the SeqStudio™ Cathode Buffer Container” on page 184.
If the buffer is above the fill line, reinsert the CBC.
4. Place the plate or tube assembly firmly in the autosampler.

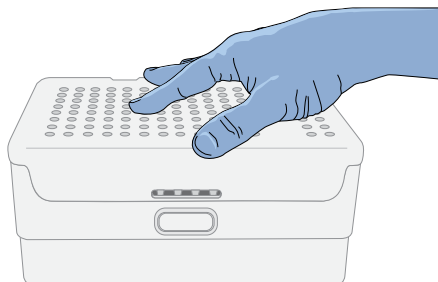


① Replace if buffer is at or below the fill line



① New CBC buffer level

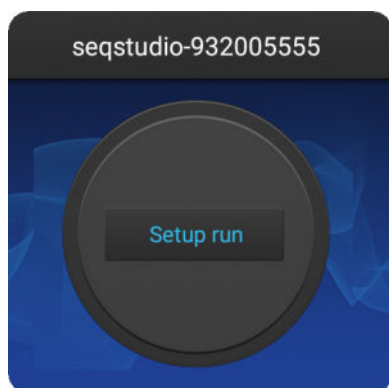
5. Close the autosampler lid: Press down on the center of the lid or press down on both sides of the lid with equal pressure until the lid clicks shut.



6. Touch **Retract plate**, then close the instrument door when prompted.

Select a plate setup and start a run

After you load the plate in the instrument (see “Load the plate or the tube assembly” on page 98):



1. In the instrument home screen, touch **Setup run**.
2. Select the location of your plate setup, then select the plate setup.
For information on plate setup files, see “PSM and CSV plate setup files for import into the instrument” on page 84 and “Shared (public), hidden (my plates), and guest plate setup files” on page 85.
3. Verify that settings are as needed.
The **Save location** must specify **Cloud** if you want to view analyzed data when you monitor the run from the Thermo Fisher™ Connect Platform.

4. Touch **Start run**.

- If a new cartridge was inserted before the run, the instrument performs an optical alignment before starting the run.
- During the run, the instrument performs an automatic spectral calibration adjustment (auto calibration) for each sample to correct for spectral overlap.
- During a run, the user who started the run or an administrator can lock the touchscreen to prevent other users from using the instrument. If the touchscreen is locked, only the user who started the run or an administrator can sign in to the instrument.

Proceed to:

- “Monitor a run from the Thermo Fisher™ Connect Platform” on page 101
- “Monitor a run from a mobile device” on page 109
- “Monitor a run from the instrument” on page 112

Automatic file cleanup

Before starting a run, the instrument calculates the total amount of storage space required to save the run to the instrument. If the required storage space is not available, the instrument deletes files associated with the oldest exported plates until sufficient space is available.

Note: Only complete plates that have been auto exported (saved to Thermo Fisher™ Connect Platform, network, or USB) or manually exported (using ⚙ **Settings** ▶ **Run History** ▶ **plate name** ▶ **Export**) are deleted.

If the required storage space is not available and no plates have been exported, the instrument displays a notification indicating that there is not enough storage space.

You can export plates and delete plates, then start the run again.

Lock the touchscreen

During a run, you can lock the touchscreen to prevent other users from using the instrument. This feature is not available to Guest users.

Only the user who locked the touchscreen or an administrator can sign in to the instrument if the touchscreen is locked.

1. Touch .
2. Touch **Profile**.
3. Touch **Lock instrument**.

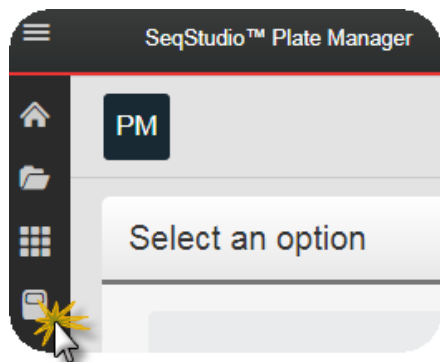
Note: If a run is not in progress, **Sign out** is displayed instead of **Lock instrument**.

Monitor a run from the Thermo Fisher™ Connect Platform

Open the Remote Monitoring App from Instrument Connect App

A run is accessible from InstrumentConnect for 24 hours after the run is complete, or until another run is started.

1. Sign in to thermofisher.com/connect.
2. Click  to access InstrumentConnect.



3. Click the run status dial to display the Remote Monitoring App.

The screenshot displays the Remote Monitoring App interface. At the top, the header shows 'InstrumentConnect' and 'Powered by Thermo Fisher Cloud'. Below the header, there's a navigation bar with 'Instruments' and 'PCs & smartphones'. The main area shows a card for 'seqstudio-CK' with a 'SeqStudio' icon and a 'RUNNING' status. A circular progress indicator shows '00:04:50 Remaining'. Below this, the 'Run' tab is active, showing 'seqstudio-CK is running'. A circular progress indicator shows '8h 8m 52s pending'. To the right, a table shows the run progress for 'Plate_20170804_223088' with columns 1-12 and rows A-H. The table uses color-coded circles to indicate the status of each well: Green for Good, Yellow for Caution, and Red for Alarm. A 'Quality Legend' section at the bottom right explains the color coding and the run module used.

Run Details:

- Run: seqstudio-CK
- Status: RUNNING
- Remaining: 00:04:50

Run Progress:

- 8h 8m 52s pending
- 2 of 24 injection completed

Injection Groups:

- Injection Group : S : A1-D1
Run module: ShortSeq, Dye Set: E_BigDye™ Terminator v1.1
- Injection Group : S : E1-H1
Run module: ShortSeq, Dye Set: E_BigDye™ Terminator v1.1
- Injection Group : S : A2-D2 **Running**
Run module: ShortSeq, Dye Set: E_BigDye™ Terminator v1.1
- Injection Group : S : E2-H2 **Queued**
Run module: ShortSeq, Dye Set: E_BigDye™ Terminator v1.1
- Injection Group : S : A3-D3 **Queued**
Run module: ShortSeq, Dye Set: E_BigDye™ Terminator v1.1

Plate_20170804_223088

	1	2	3	4	5	6	7	8	9	10	11	12
A	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
B	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
C	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
D	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
E	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
F	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
G	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good
H	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good	Good

Quality Legend:

- Good (Green)
- Caution (Yellow)
- Alarm (Red)

Dye set:

- D (Red)
- E (Purple)

Run module:

- FragAnalysis (Red)
- ShortSeq (Blue)

Figure 12 Remote Monitoring App

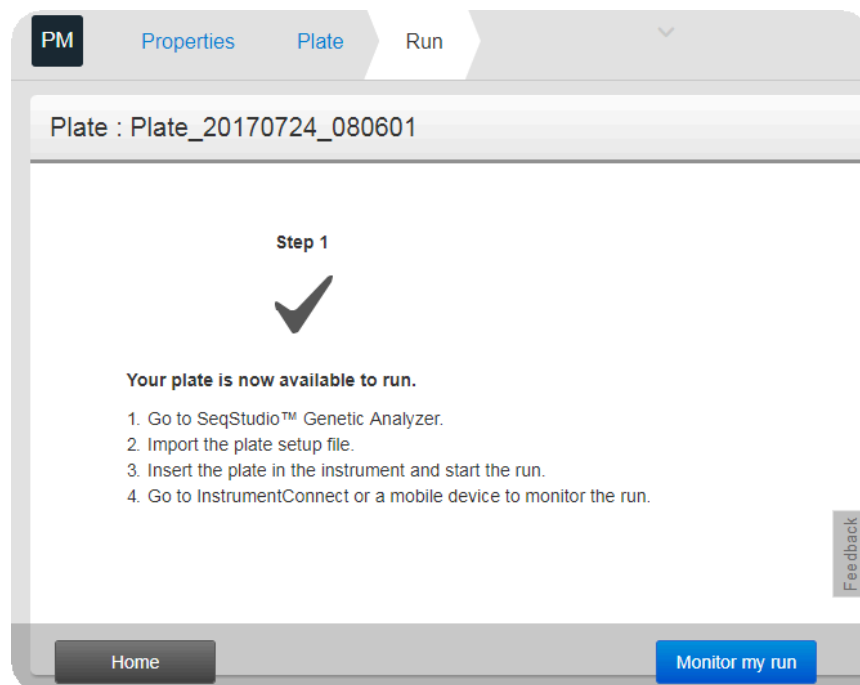
Open the Remote Monitoring App from the Plate Manager

A run is accessible from the Plate Manager for 24 hours after the run is complete, or until another run is started.

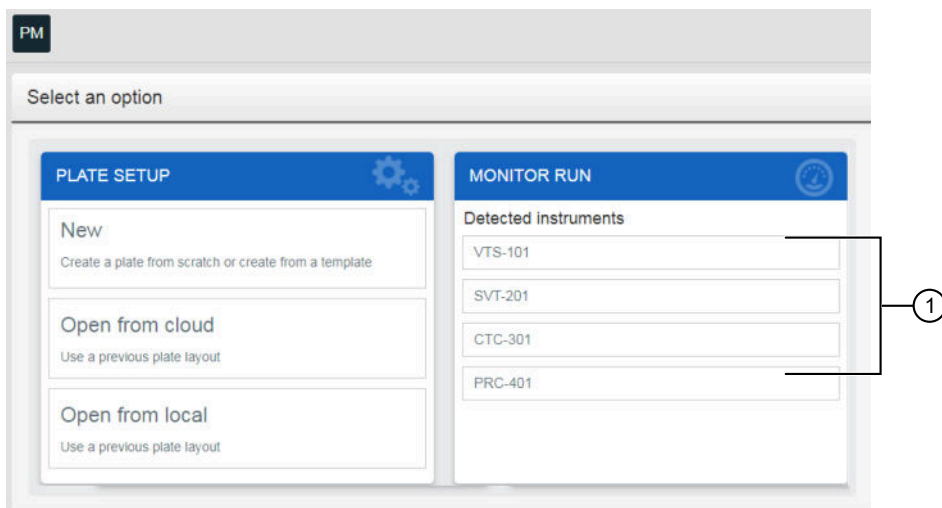
You can open the Remote Monitoring App immediately after you save a plate setup or at a later time.

In the Plate Manager on the Thermo Fisher™ Connect Platform:

- To open the Remote Monitoring App immediately after you save a plate setup and start the run, click **Monitor my run**.



- To open the Remote Monitoring App at a later time, click **PM**, then select an instrument or click to access InstrumentConnect.



- ① Select an instrument or click to access InstrumentConnect.

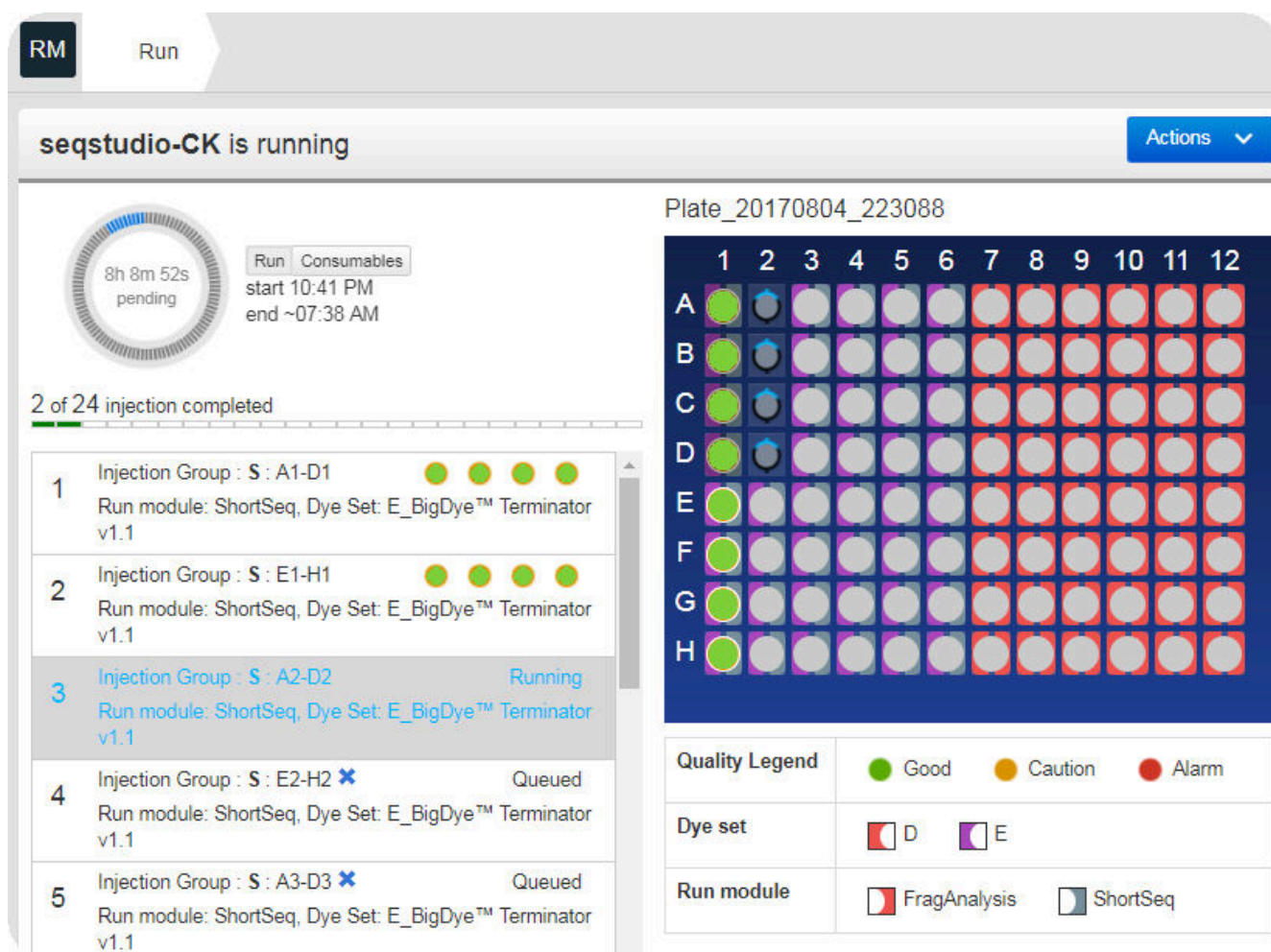


Figure 13 Remote Monitoring App

View notifications from the instrument on your Thermo Fisher™ Connect Platform account

1. In any screen in the Thermo Fisher™ Connect Platform, click .



2. Click a notification, then click **Dismiss** or **Dismiss all** to dismiss the notification.

View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform

1. Open the Remote Monitoring App (see “Open the Remote Monitoring App from Instrument Connect App” on page 101).
2. Click an injection group in the injection list or the plate view.

The status dials are color-coded for quality alerts:

-  —All QC tests passed.
-  —At least 1 warning quality alert was triggered.
-  —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

Injection Group (4) E2-H2

ActionsBack

Application

Dye Set

Run Module

Size standard

FRAGMENT_ANALYSIS

GS



FragAnalysis

GS600LIZ

Well Details

Raw

EPT

Analyzed

Status	Well #	Sample Name	Sample Type	Quality Alerts
 Good	E2	E2	Sample	
 Good	F2	F2	Sample	
 Good	G2	G2	Sample	
 Good	H2	H2	Sample	

Note: The **Analyzed** tab is disabled if the **Save location** for the plate setup is not set to **Cloud** or if the injection group has not finished running.

3. In the **Quality alerts** screen, click the **Raw**, **EPT**, or **Analyzed** tab to view data.
4. As needed, select **Actions** ▶ **Re-inject group**, select the **Run module** and settings, then click **Inject**.

Pause or cancel an injection in the Remote Monitoring App

Select:

- **Actions ▶ Pause plate** to pause the run after the current injection is complete.
- **Actions ▶ Stop current injection** to immediately stop the injection.

Edit injection group run settings

In the **Results** screen:

1. Select **Actions ▶ Edit injection group**.
2. Edit settings as needed, then click **OK**.

Re-inject or delete an injection group

In the **Results** screen:

Select **Actions ▶ Re-inject group** or **Actions ▶ Delete injection group**.

Export a QC report

The **QC Report** command is not available for an injection until the injection is complete. The **QC Report** command is not available for any injections if the plate setup **Save location** does not specify **Cloud**.

In the **Remote Monitor** screen:

Select **Actions ▶ QC Report**.

A PDF of the plate QC report is generated.

Remote Monitoring App raw trace

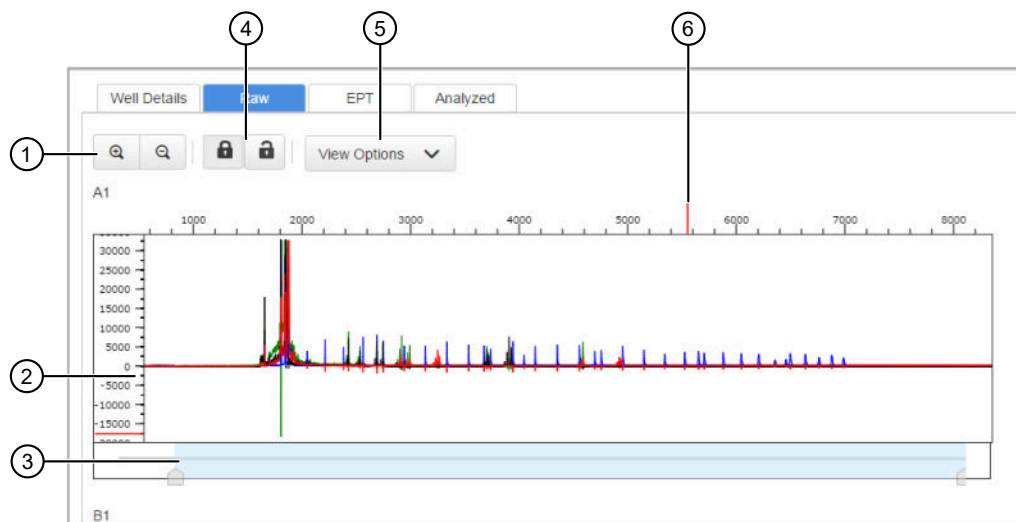


Figure 14 Fragment analysis raw trace

- ① Zoom in/out.
- ② Raw trace.
- ③ Thumbnail trace—Click-drag to view another region of the trace.
- ④ Lock/unlock trace zooming for all traces in the injection group.
- ⑤ View Options—Select the dye color to display; set vertical scaling.
- ⑥ Cursor position indicator (red line).

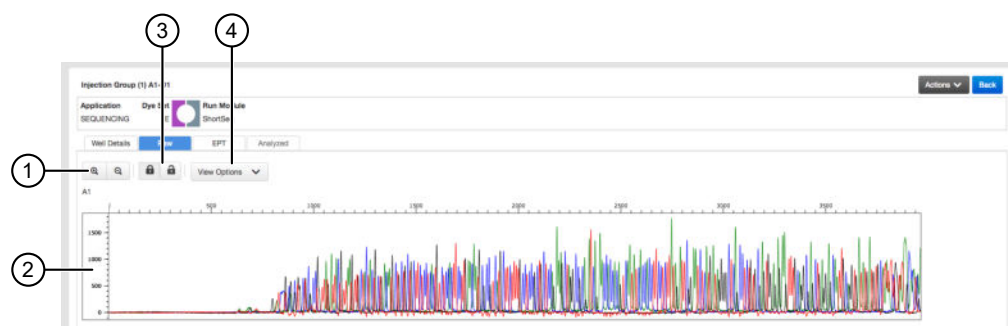
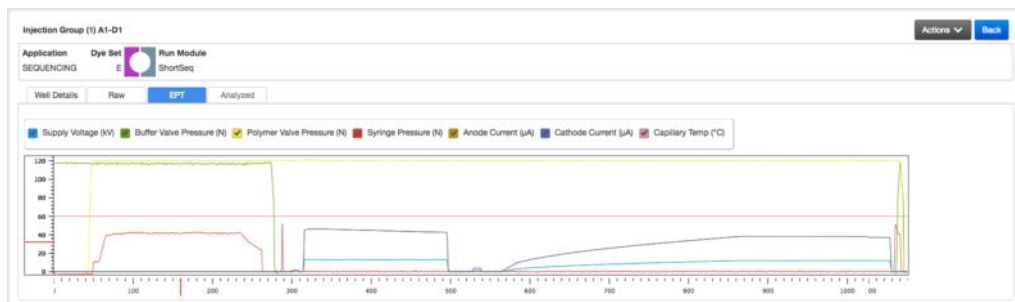


Figure 15 Sequence analysis raw trace

- ① Zoom in/out.
- ② Raw trace.
- ③ Lock/unlock trace zooming for all traces in the injection group.
- ④ View Options—Select the basecalls to display; set vertical scaling.

Remote Monitoring App EPT trace

The EPT view (ElectroPhoresis Telemetry) shows instrument data conditions (currents, temperatures, electrophoresis voltage) as a function of time.



Remote Monitoring App analyzed trace

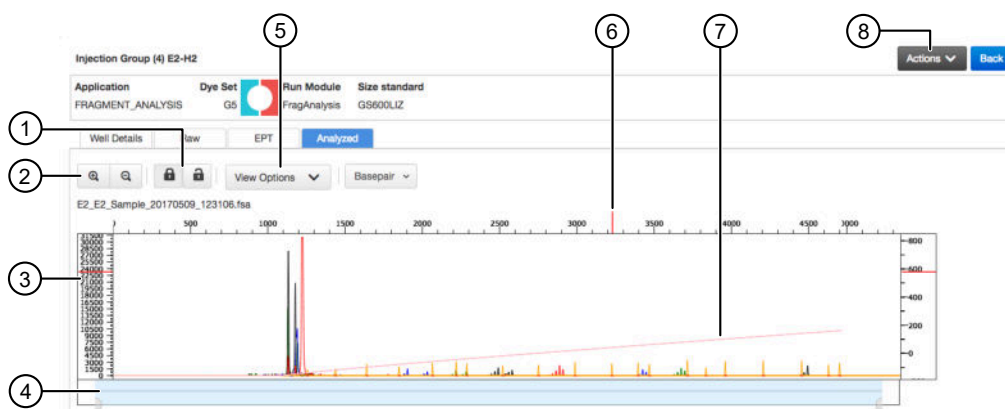


Figure 16 Fragment analysis analyzed trace

- ① Lock/unlock trace zooming for all traces in the injection group.
- ② Zoom in/out.
- ③ Analyzed trace.
- ④ Thumbnail trace—Click-drag to view another region of the trace.
- ⑤ View Options—Select the dye colors to display; set vertical scaling.
- ⑥ Cursor position indicator (red vertical and horizontal tick marks outside trace).
- ⑦ Size standard curve (red line).
- ⑧ Actions—Select commands to pause and cancel injections.

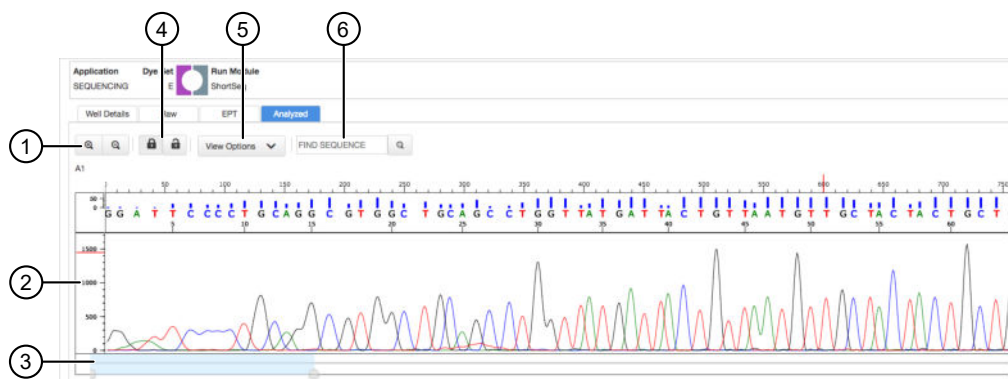


Figure 17 Sequence analysis analyzed trace

- ① Zoom in/out.
- ② Analyzed trace.
- ③ Thumbnail trace—Click-drag to view another region of the trace.
- ④ Lock/unlock trace zooming for all traces in the injection group.
- ⑤ View Options—Select the basecalls to display; show/hide quality bars and values; set vertical scaling.
- ⑥ Search for a sequence.

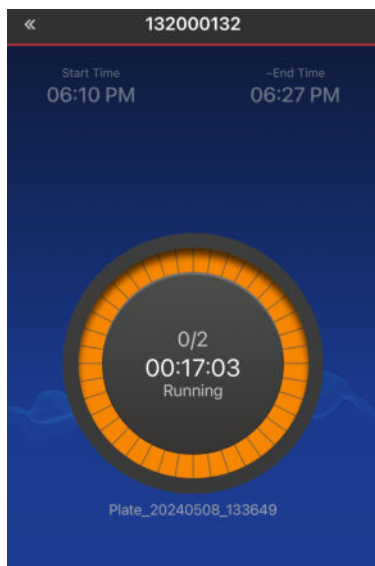
Monitor a run from a mobile device

Before you begin, see “Link the instrument from a mobile device” on page 204.

1. On your mobile device, launch InstrumentConnect.
2. Touch the instrument to monitor.
3. Swipe left to view consumable status.



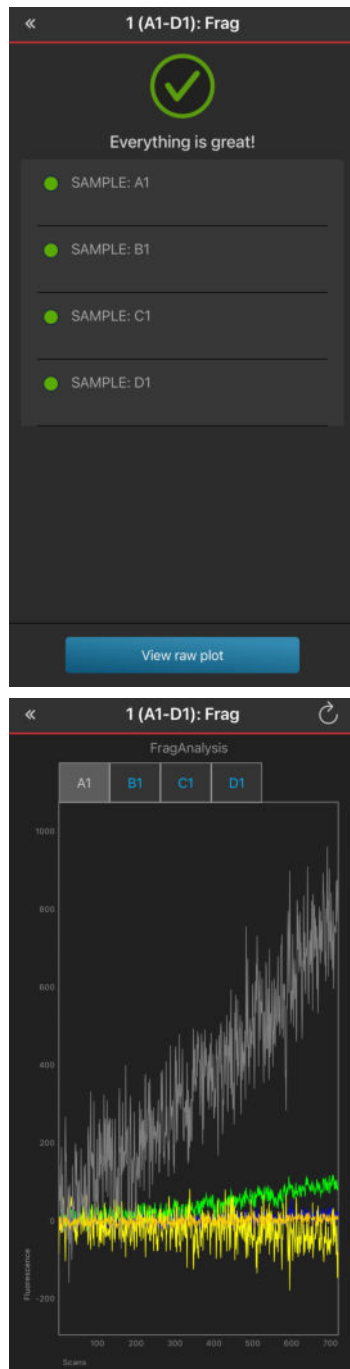
4. Touch the status dial to view the injection list.



The screenshot shows the SeqStudio mobile app interface with the injection list. At the top, there is a header bar with a back arrow and the sample ID '132000132'. Below the header, the 'Time Remaining' is '00:16:51'. Below the time, there is a legend: 'Seq = Sequencing' and 'Frag = Fragment Analysis'. The main display shows a table with two columns: 'Injection group' and 'Quality'.

Injection group	Quality
1 (A1-D1): Frag	00:08:10
2 (A2-D2): Frag	QUEUED

5. Touch an injection group to display quality alerts, then touch **View raw plot** to view the data.



- Swipe left to view the entire trace.
- Pinch-zoom to expand the trace.

Monitor a run from the instrument

(Optional) View the injection list, change injection settings or order, or specify replicates and re-injections

In the **Plate properties** screen:

1. Touch **Injection options**.
2. Touch an injection group, then configure the injection list:
 - Touch and drag an injection group to a new location in the injection list.
 - Touch **Inject first**—Moves the selected injection group to the top of the injection list.
 - Touch **Edit and re-inject**—Adds replicates or re-injections to the injection list. You can also modify **Run module**, **Injection time**, **Injection voltage**, **Run time**, or **Run voltage** for these injections.

Note: *Replicate* indicates a duplicate—the replicate has the same parameters as the original injection. *Re-injection* indicates a copy of the injection, but with user-modified parameters.

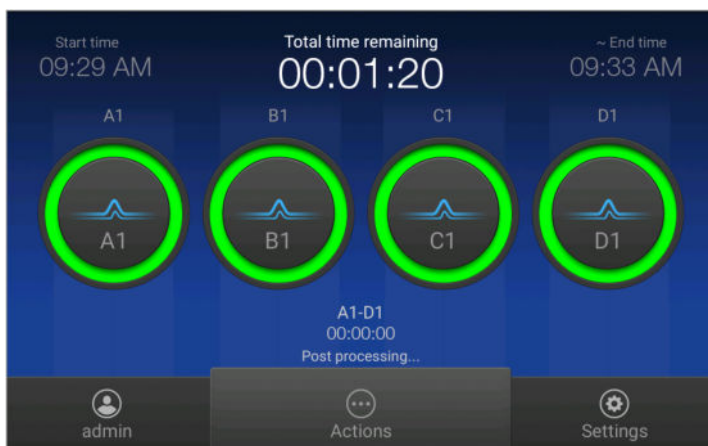
Note: The changes to the run conditions apply only to the replicate or re-injection. The changes are not saved to the run module.

3. Touch **Done**.

View the run status

During a run, the run status screen is displayed.

View the run time information and the status dial for each capillary.



The status dials are color-coded for quality alerts:

- ● —All QC tests passed.
- ● —At least 1 warning quality alert was triggered.
- ● —At least 1 failing quality alert was triggered.

If an injection group is set to re-inject, the number of the current injection is displayed on the status dials.

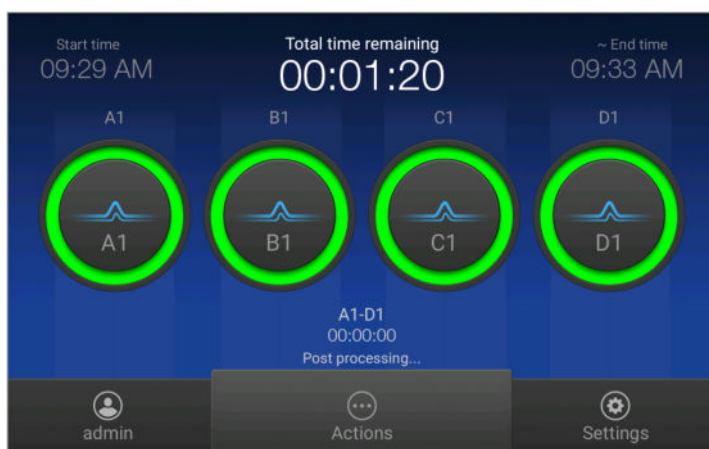
For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

View real-time results

During a run, touch one of the injection dials to display the trace for the selected capillary.

See “Fragment/HID analysis trace” on page 121 or “Sequence analysis trace” on page 123 for information.



The status dials are color-coded for quality alerts:

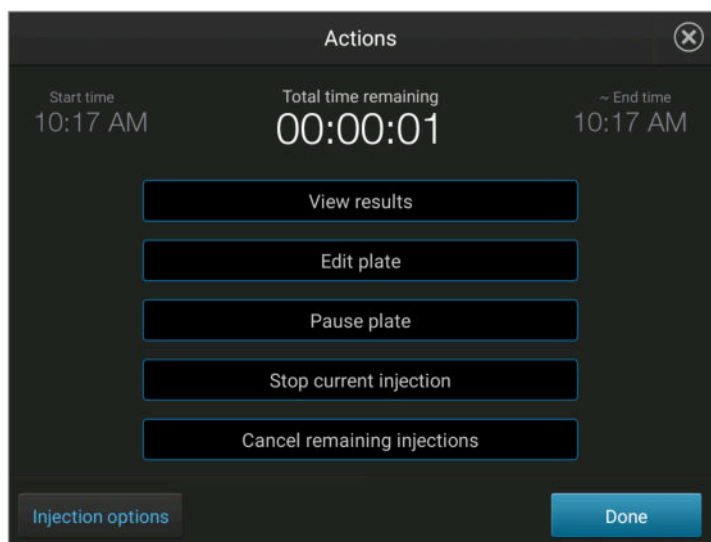
- ● —All QC tests passed.
- ● —At least 1 warning quality alert was triggered.
- ● —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

Pause a plate or cancel or stop injections

1. In the run status screen, touch  **Actions**.



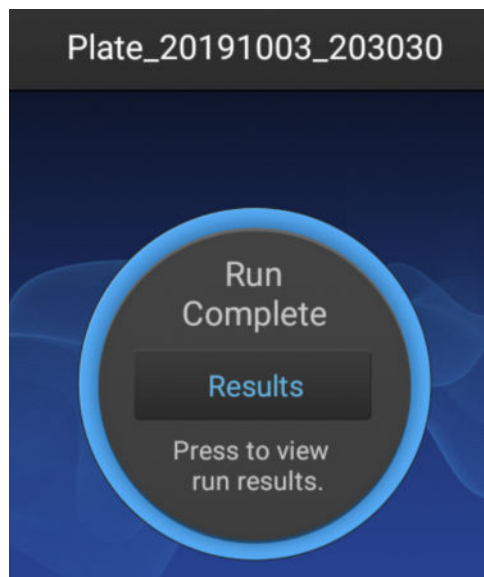
2. Manage the plate or injections:

Touch	To
View results	View the list of injections and status.
Edit plate	For injections that have not yet been run, edit Sample Name , Run Module , Dye Set , Size Standard , Sample Type , and custom fields.
Pause plate	Stop the run after the current injection is complete. Touch Resume to continue the run.
Stop current injection	Stop the current injection immediately. Touch Resume to continue the run.
Cancel remaining injections	Specify whether to cancel the run immediately or after the current injection is complete.
Injection options	Move an injection to the top of the injection list, edit run module information, and/or reinject samples.

View results when the run is completed

View results for the plate when the run is completed (all injections are finished).

1. Touch **Results** to view the run results.



2. Touch **List view**.

Each injection group displays a QC color for each capillary:

- ● —All QC tests passed.
- ● —At least 1 warning quality alert was triggered.
- ● —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

3. Touch an injection group.

4. View the results in the **Run Result Details** screen, or touch  for well details.

See “Fragment/HID analysis results” on page 121 or “Sequence analysis results” on page 123 for detailed information.

5. Touch a sample file name.

If the data triggered any quality alerts, a QC alerts screen is displayed. See “Data quality alerts” on page 128 and “Sizecalling and basecalling quality alerts” on page 128 for detailed information.

Touch **View data** to display the analyzed trace for the sample. See “Fragment/HID analysis trace” on page 121 or “Sequence analysis trace” on page 123 for detailed information.

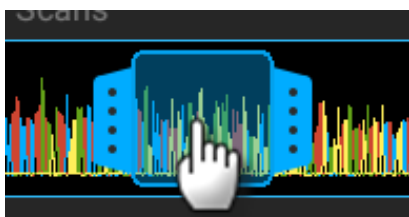
6. Touch and drag the thumbnail view of the analyzed trace (below the trace) to scroll left or right.

7. (Optional) Adjust the graphical view (see “Adjust the trace display” on page 116).
8. Touch > or < to scroll to the raw data or **EPT Plot** (ElectroPhoresis Telemetry).
See “EPT plot” on page 126.

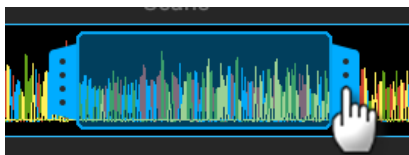
Adjust the trace display

See “View results when the run is completed” on page 115 to access results.

- Drag one finger to pan to the left or right.
- Zoom in and out by pinching and expanding with two fingers.
- Touch  on the left border of the trace, then touch a dye to deselect.
- Touch  on the right border of the trace, then touch  **Zoom In**,  **Zoom Out**, or  **Fit to screen** to adjust the display.
- Drag the center of the pane in thumbnail view to scroll left or right.



- Drag the right or left handle of the pane to zoom horizontally.



Unload the plate or the tube assembly

When the run is complete:

1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid.
3. Remove the plate or tube assembly.

4. Close the autosampler lid: Press down on the center of the lid or press down on both sides of the lid with equal pressure until the lid clicks shut.



5. Touch **Retract plate**, then close the instrument door when prompted.



View and analyze results

- View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform 118
- View results on the instrument 119
- Export results from the instrument (sample data files and QC reports) 130
- Analyze data 130
- (If needed) View the export status for sample data files 131

View results in the Remote Monitoring App on the Thermo Fisher™ Connect Platform

1. Open the Remote Monitoring App (see “Open the Remote Monitoring App from Instrument Connect App” on page 101).

2. Click an injection group in the injection list or the plate view.

The status dials are color-coded for quality alerts:

- ● —All QC tests passed.
- ● —At least 1 warning quality alert was triggered.
- ● —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

Injection Group (4) E2-H2				
Application	Dye Set	Run Module	Size standard	
FRAGMENT_ANALYSIS	G5	FragAnalysis	GS600LZ	
Well Details Raw EPT Analyzed				
Status	Well #	Sample Name	Sample Type	Quality Alerts
● Good	E2	E2	Sample	
● Good	F2	F2	Sample	
● Good	G2	G2	Sample	
● Good	H2	H2	Sample	

Note: The **Analyzed** tab is disabled if the **Save location** for the plate setup is not set to **Cloud** or if the injection group has not finished running.

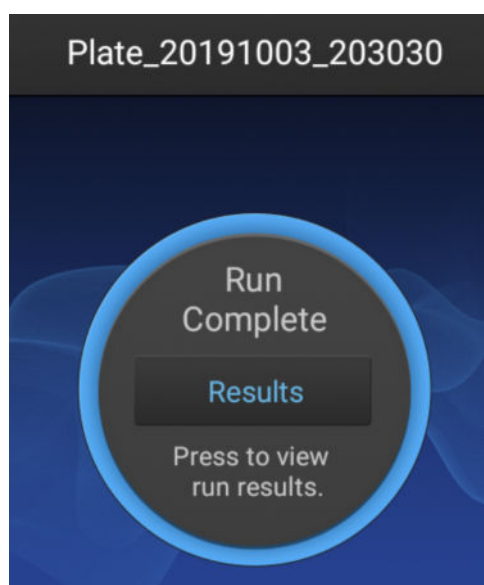
3. In the **Quality alerts** screen, click the **Raw**, **EPT**, or **Analyzed** tab to view data.
4. As needed, select **Actions** ▶ **Re-inject group**, select the **Run module** and settings, then click **Inject**.

View results on the instrument

View results when the run is completed

View results for the plate when the run is completed (all injections are finished).

1. Touch **Results** to view the run results.



2. Touch **List view**.

Each injection group displays a QC color for each capillary:

- ● —All QC tests passed.
- ● —At least 1 warning quality alert was triggered.
- ● —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128

3. Touch an injection group.
4. View the results in the **Run Result Details** screen, or touch  for well details.
See “Fragment/HID analysis results” on page 121 or “Sequence analysis results” on page 123 for detailed information.

5. Touch a sample file name.

If the data triggered any quality alerts, a QC alerts screen is displayed. See “Data quality alerts” on page 128 and “Sizecalling and basecalling quality alerts” on page 128 for detailed information.

Touch **View data** to display the analyzed trace for the sample. See “Fragment/HID analysis trace” on page 121 or “Sequence analysis trace” on page 123 for detailed information.

6. Touch and drag the thumbnail view of the analyzed trace (below the trace) to scroll left or right.

7. (Optional) Adjust the graphical view (see “Adjust the trace display” on page 116).

8. Touch > or < to scroll to the raw data or **EPT Plot** (ElectroPhoresis Telemetry).

See “EPT plot” on page 126.

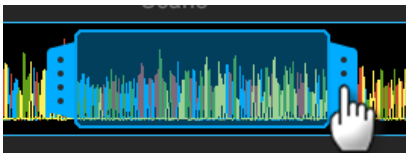
Adjust the trace display

See “View results when the run is completed” on page 115 to access results.

- Drag one finger to pan to the left or right.
- Zoom in and out by pinching and expanding with two fingers.
- Touch  on the left border of the trace, then touch a dye to deselect.
- Touch  on the right border of the trace, then touch  **Zoom In**,  **Zoom Out**, or  **Fit to screen** to adjust the display.
- Drag the center of the pane in thumbnail view to scroll left or right.

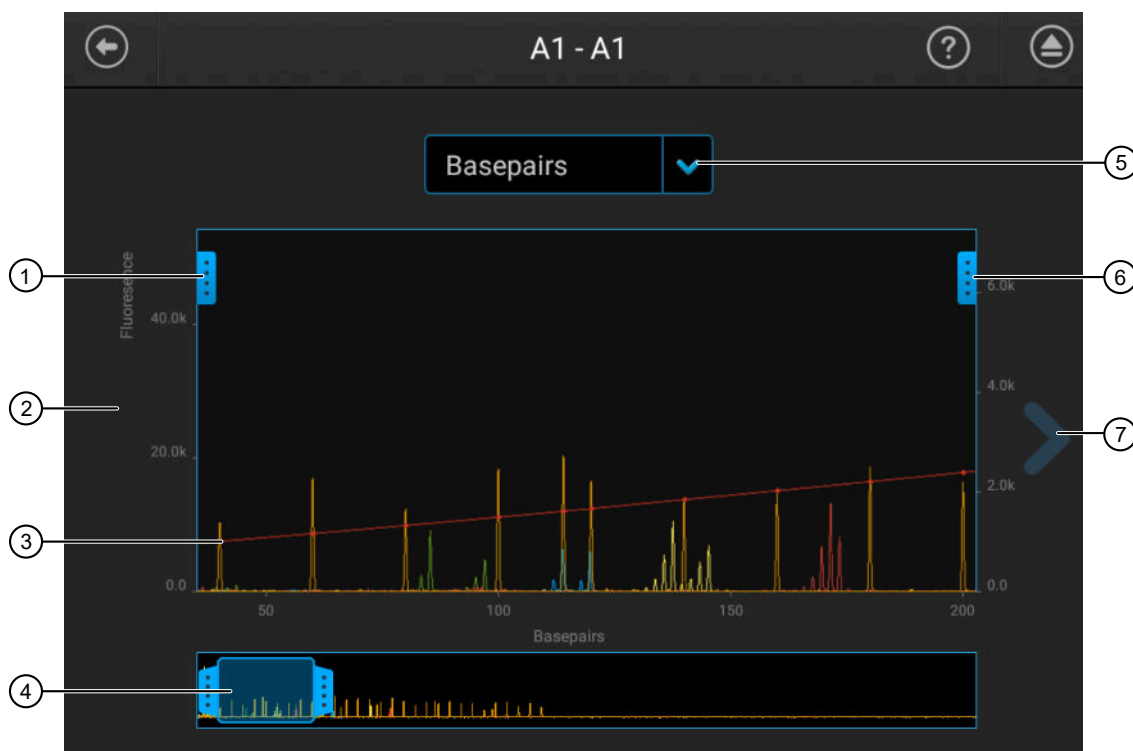


- Drag the right or left handle of the pane to zoom horizontally.



Fragment/HID analysis results

Fragment/HID analysis trace



- ① Trace color hide/show—Touch to open, then touch a color to hide or show.
- ② Analyzed trace
- ③ Size standard curve (red line)
- ④ Thumbnail trace—Drag the center of the pane in the thumbnail trace to display another trace area in the top pane. Drag the right or left handle of the pane to zoom horizontally.
- ⑤ Basepair or scan display selection.
- ⑥ Zoom tools—Touch to open.
- ⑦ Next trace tool—Touch to view the raw trace or EPT for the well.

Fragment/HID analysis results

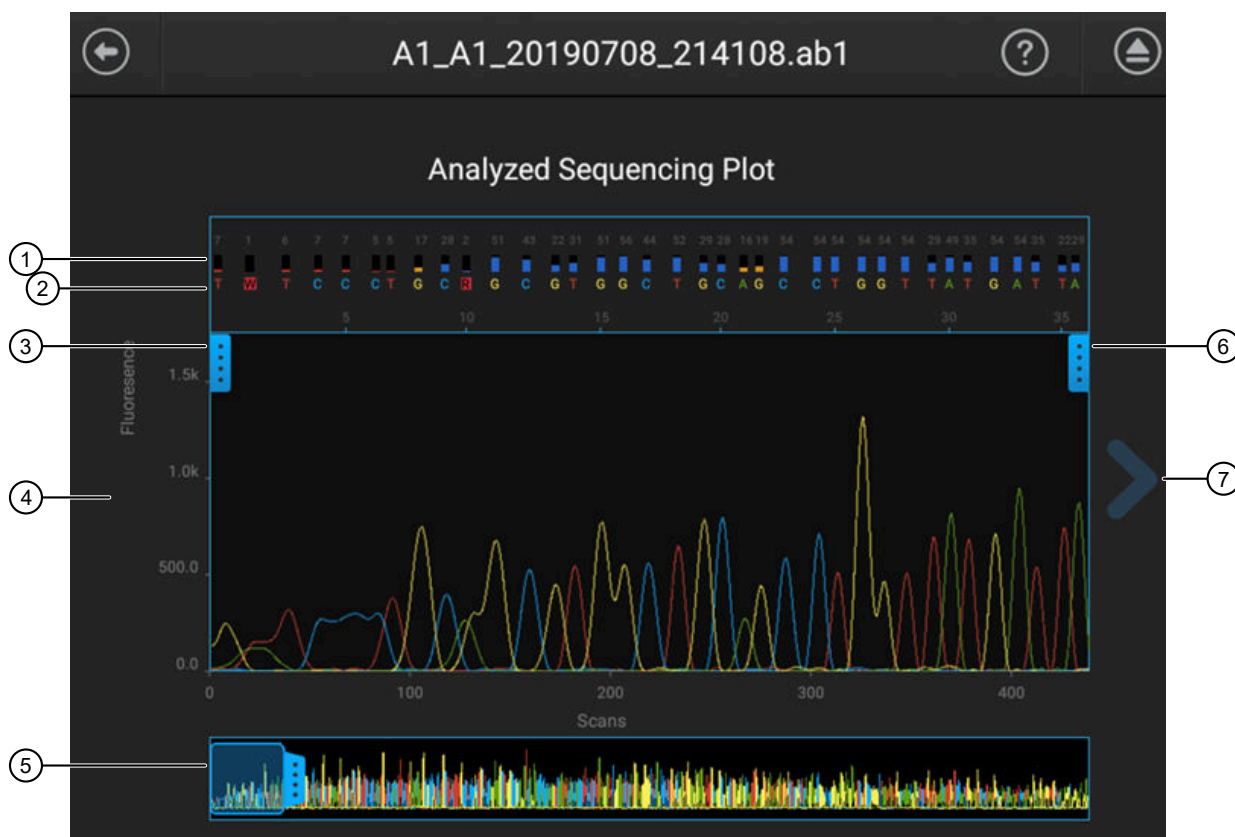
Setting	Description
Quality	QC flags ● —All QC tests passed. ● —At least 1 warning quality alert was triggered. ● —At least 1 failing quality alert was triggered.

(continued)

Setting	Description
SQ	Size Quality - If SQ is ≤ 0.75, it passes the QC test and does not trigger a quality alert (●). - If SQ is 0.25–0.74, it triggers a warning quality alert (●). - If SQ is < 0.25, it triggers a failing quality alert (●). How Size Quality is determined The Size Quality algorithm evaluates the similarity between the fragment pattern for the size standard dye specified in the size standard definition and the actual distribution of size standard peaks in the sample, calculates an interim SQ (a value between 0 and 1).
Exported	The sample has been auto exported (saved to the Thermo Fisher™ Connect Platform, network, or USB) or manually exported (using ⓘ Settings ▶ Run History ▶ <i>plate name</i> ▶ Export).
	Well details (see below).
Well details	
Sample file name, QC flag, sample name, sample type, settings used to acquire the data (size standard, run module and dye set), Sizing quality value, and export status.	

Sequence analysis results

Sequence analysis trace



- ① Quality Value bars and values:
 - Pure basecall with QV ≥ 20
 - Pure basecall with QV 15–19
 - Pure basecall with QV < 15
 - Mixed basecall
- ② Bases—Mixed bases are highlighted in red. You can set the secondary peak threshold, as a percentage of the primary peak, for consideration as a mixed base by the basecalling algorithm. Reaching this threshold is a necessary but not sufficient condition for arriving at a mixed base determination. See “Sequencing settings (base calling)” on page 143.
- ③ Trace color hide/show—Touch to open, then touch a color to hide or show.
- ④ Analyzed trace
- ⑤ Thumbnail trace—Drag the center of the pane in the thumbnail trace to display another trace area in the top pane. Drag the right or left handle of the pane to zoom horizontally.
- ⑥ Zoom tools—Touch to open.
- ⑦ Next trace tool—Touch to view the raw trace or EPT for the well.

Sequencing results

Result	Description
Quality	QC flags ● — All QC tests passed. ● — At least 1 warning quality alert was triggered. ● — At least 1 failing quality alert was triggered.
CRL (Contiguous Read Length)	The longest uninterrupted segment of basecalls with an average Quality Value (QV) ≥ 20. In addition to evaluating the QV of a basecall, the software considers the QV of adjacent basecalls within a 21-bp moving window to determine a contiguous read length based on quality values: the software starts from the 5' end and calculates the average QV across a moving window size of 21, sliding 1 bp at a time, to the 3' end. The resulting longest contiguous segment is determined as the CRL. Note: The contiguous read length passing criteria for install checks is an uninterrupted segment of bases with an average Quality Value (QV) of 30.
Exported	The sample has been auto exported (saved to the Thermo Fisher™ Connect Platform, network, or USB) or manually exported (using  Settings ▶ Run History ▶ plate name ▶ Export).
	Well details (see below).
Well details	
Sample file name, QC flag, sample name, CRL, signal strength, Median PUP, Trace score, run module and dye set used to acquire the data, and export status.	
Signal strength	The average relative fluorescence unit (RFU) for all dyes combined across the electropherogram in the raw data.
Trace score	The average basecall Quality Value (QV) of basecalls in the clear range sequence of a trace. The <i>clear range</i> is the region of the sequence that remains after excluding the low-quality or error-prone sequence at the 5' and 3' ends. The clear range is calculated by the KB Basecaller using QVs.
Median PUP (pull-up peak)	A measure of noise or pull-up that is determined by taking the median of the ratios of signal strength calculated for each basecalled peak: primary peak/secondary peak under the primary peak. A higher value indicates less baseline or secondary noise. A lower value indicates an elevated baseline or secondary noise. Example 1: Main called base signal strength is 1,000 RFU and the largest secondary peak beneath it is 10 RFU; PUP=100 Example 2: Main called base signal strength is 1,000 RFU and the largest secondary peak beneath it is 100 RFU; PUP=10

Understanding Quality Values (QVs)

Quality value ranges

The color of a QV bar indicates the QV of a base.

-  Pure basecall with QV ≥ 20
-  Pure basecall with QV 15–19
-  Pure basecall with QV < 15
-  Mixed basecall

Pure base versus mixed base QVs

Pure bases and mixed bases have the same probability of error for the associated basecall ($10^{-q/10}$). Note the following:

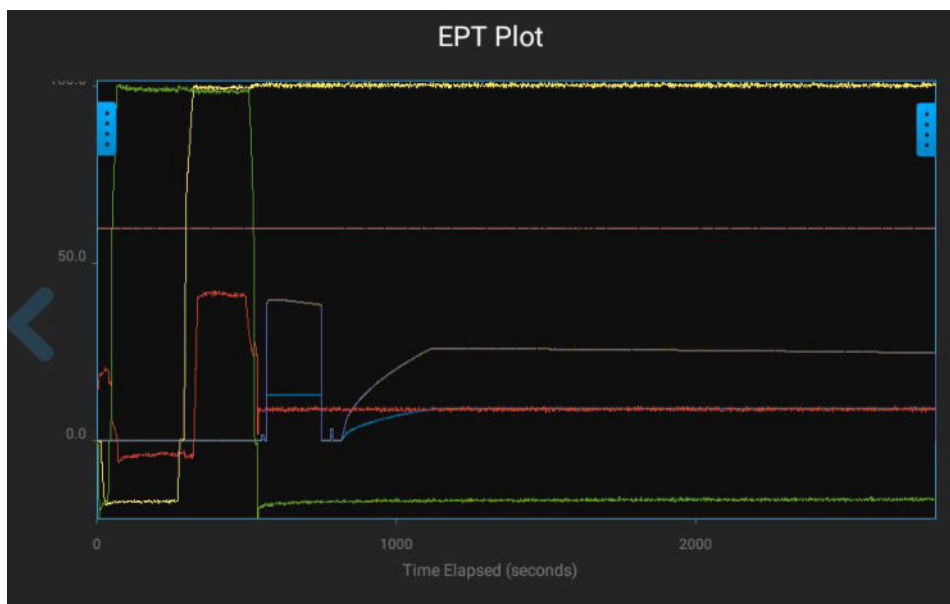
- High-quality pure bases typically have QVs of 20 or higher.
- The distribution of quality values for mixed bases differs dramatically from that of pure bases.
- Mixed bases have a maximum QV of 20.
- Review all mixed base calls.

Quality values (QV) and probability of error (Pe)

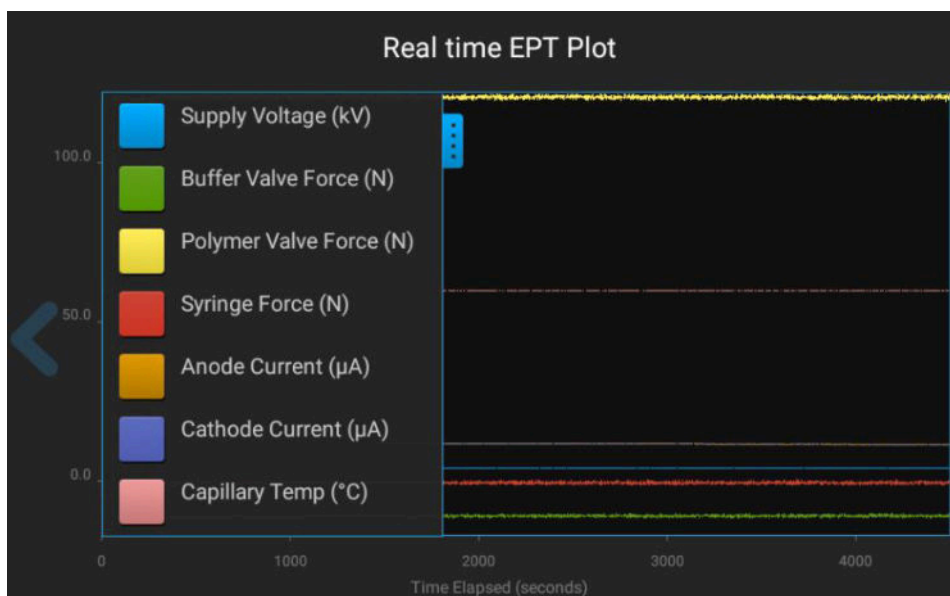
QV	Pe	QV	Pe
1	79.0%	30	0.10%
5	32.0%	35	0.032%
10	10.0%	40	0.010%
15	3.2%	45	0.0032%
20	1.0%	50	0.0010%
25	0.32%	60	0.00010%

EPT plot

The EPT view (ElectroPhoresis Telemetry) shows instrument data conditions (currents, temperatures, electrophoresis voltage) as a function of time.



Touch  on the left border of the plot to display the legend.



View results for a previously run plate (run history)

In the home screen:

1. Touch  **Settings** ▶ **Run history**.
2. Touch a plate name, then touch **View**.
If you select more than one plate name, the **View** button is dimmed.
The **Run History** screen is displayed.

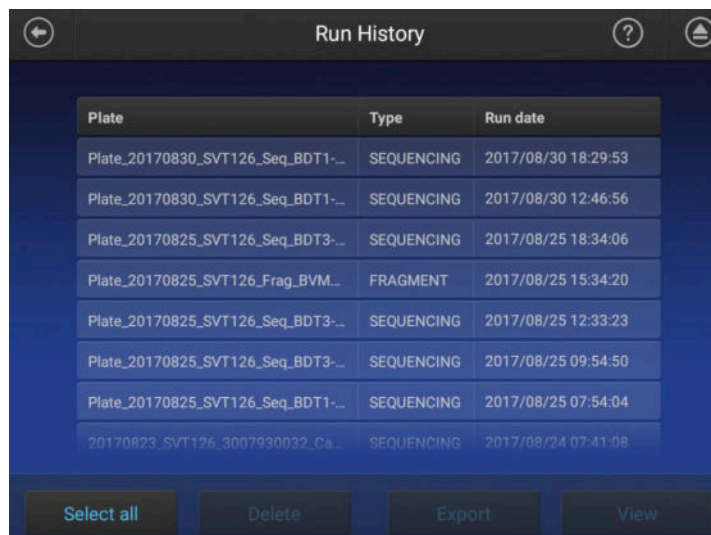


Plate	Type	Run date
Plate_20170830_SVT126_Seq_BDT1-...	SEQUENCING	2017/08/30 18:29:53
Plate_20170830_SVT126_Seq_BDT1-...	SEQUENCING	2017/08/30 12:46:56
Plate_20170825_SVT126_Seq_BDT3-...	SEQUENCING	2017/08/25 18:34:06
Plate_20170825_SVT126_Frag_BVM...	FRAGMENT	2017/08/25 15:34:20
Plate_20170825_SVT126_Seq_BDT3-...	SEQUENCING	2017/08/25 12:33:23
Plate_20170825_SVT126_Seq_BDT3-...	SEQUENCING	2017/08/25 09:54:50
Plate_20170825_SVT126_Seq_BDT1-...	SEQUENCING	2017/08/25 07:54:04
20170823_SVT126_3007930032_Ca...	SEQUENCING	2017/08/24 07:41:08

Select all Delete Export View

3. Touch a sample file name.
4. View the results in the **Run history** screen, or touch  to view well details.
See “Fragment/HID analysis results” on page 121 or “Sequence analysis results” on page 123 for information.
5. Touch a sample file name, then touch **View**.
If you select more than one sample file name, the **View** button is dimmed.
If the data triggered any quality alerts, a QC alerts screen is displayed.
For information on quality alerts, see:
Click **View data** to display the trace for the sample. See “Fragment/HID analysis trace” on page 121 or “Sequence analysis trace” on page 123.
6. Touch and drag the thumbnail view of the analyzed trace (below the trace) to scroll left or right.
7. Touch  or  to scroll to the raw data or **EPT Plot**.

Data quality alerts

Quality alert	Description	Action
Offscale peaks. Adjust the injection parameters and/or the sample concentration.	At least 10 scans have saturated the CCD camera.	- Reduce the injection voltage or time. - Dilute the sample.
No sample was detected.	Poor signal-to-noise ratio with low signal detected.	- Verify that the sample volume follows recommendations in the user manual. - Troubleshoot upstream PCR and sequencing steps.

Sizecalling and basecalling quality alerts

Table 5 Sizecalling quality alerts

Quality alert	Description	Action
Sizing quality value is low due to poor size standard peak quality. Peak height uniformity is low or the fitting quality in sizing is poor.	Low resolution or poor quality data is present.	- Re-inject the sample. - If the problem persists, check the sample quality.
Sizecaller found broad peak(s) in the size standard peak(s).		
Sizing quality value is in the intermediate range; check size standard data quality.		
The number of size standard peaks detected is less than what is defined in the size standard.	Size standard definition includes peaks that are not present in the sample. Example: Sample peaks are detected up to 500 bp, but the size standard definition includes peak sizes that are >500 bp.	Use or create a size standard definition with the appropriate number of peaks and peak sizes.
The analysis range is too small. Correct the analysis range in analysis settings and re-analyze in secondary analysis software or re-inject sample.	Various causes.	- Analyze the data in a secondary analysis software with a corrected analysis range. - Re-inject the sample.

Table 6 Basecalling quality alerts

Quality alert	Description	Action
Basecalling failed due to poor quality data.	Poor quality data is present.	• Re-inject the sample. • If the problem persists, prepare fresh sample. • Troubleshoot upstream PCR and sequencing steps.

Edit injection parameters and re-inject samples

You can edit injection parameters and re-inject samples during a run or after a run is complete.

1. Access **Injection options**.

- During a run—Touch  **Actions** ▶ **Edit plate**.
- After a run—Touch **Results**.

2. Touch an injection group, then configure the injection list:

- Touch and drag an injection group to a new location in the injection list.
- Touch **Inject first**—Moves the selected injection group to the top of the injection list.
- Touch **Edit and re-inject**—Adds replicates or re-injections to the injection list. You can also modify **Run module**, **Injection time**, **Injection voltage**, **Run time**, or **Run voltage** for these injections.

3. Touch **Done**.

Note: The changes are not applied until you touch **Done**.

Export a report (QC report)

This function allows you to export a QC report for the current plate. To export a QC report for a previously run plate, export a run history (see “Export results from the instrument (sample data files and QC reports)” on page 130).

When a run is complete, in the home screen:

1. Touch **Results**.
2. Touch **Export report**.
3. Select a save location.
4. Navigate to, then select a location, then touch **Export**.

Export results from the instrument (sample data files and QC reports)

In the home screen:

1. Touch  **Settings** ▶ **Run history**.
2. Select one or more plates from the **Run History** table.
3. Touch **Export**.
4. Select a save location.

The following data is exported for the plate:

- Fragment/HID analysis—FSA file for each sample.
- Sequencing—AB1 file for each sample.
- Plate QC report in CSV and PDF format.

Note: You can also export sample data if you select a plate, then select **View** ▶ **Export**. If you select a plate, only an FSA or AB1 file for each analyzed sample is exported.

Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

Analyze data

1. Export results (see “Export results from the instrument (sample data files and QC reports)” on [page 130](#)) or use auto exported data.

Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

2. Use an appropriate fragment analysis or sequencing application to analyze the data.

Note: Data from the SeqStudio™ Genetic Analyzer may be labeled as "3200" in secondary analysis software.

For more information, see “Secondary analysis software” on [page 30](#)

(If needed) View the export status for sample data files

If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen.

View and export sample data files at the plate-level

1. Touch  **Settings** ▶ **Export status**.

The software displays a list of plates that contain one or more sample data files (AB1 and FSA) that failed to export to the expected save location (Cloud, Network Drive, and/or USB).

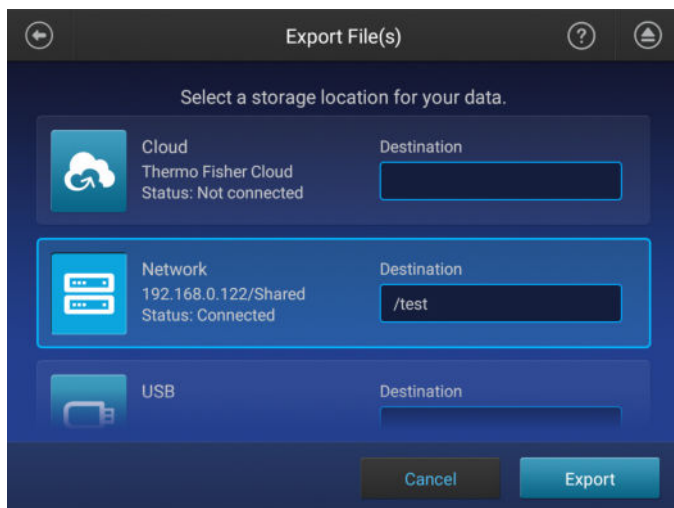
2. Touch one or more plates of interest, or touch **Select all** to select all the plates listed, then touch **Export**.



Plate	Type	Exported	Run date
Plate_20210811_194351	FRAGMENT	Failed	2021/09/27 17:04:36

Select all Export View

3. In the **Export File(s)** screen, touch a save location for all sample data files in the selected plates, then touch **Export**.



4. Close the **Files exported successfully** message.

The exported plates are removed from the **Export Status** screen.

Note: If you do not manually export a plate, or the export fails, the plate will automatically be removed from the **Export Status** screen after 30 days.

View and export individual sample data files

1. Touch **⚙ Settings** ▶ **Export status**.

The software displays a list of plates that contain one or more sample data files (AB1 and FSA) that failed to export to the expected save location (Cloud, Network Drive, and/or USB).

2. Touch a plate of interest, then touch **View**.

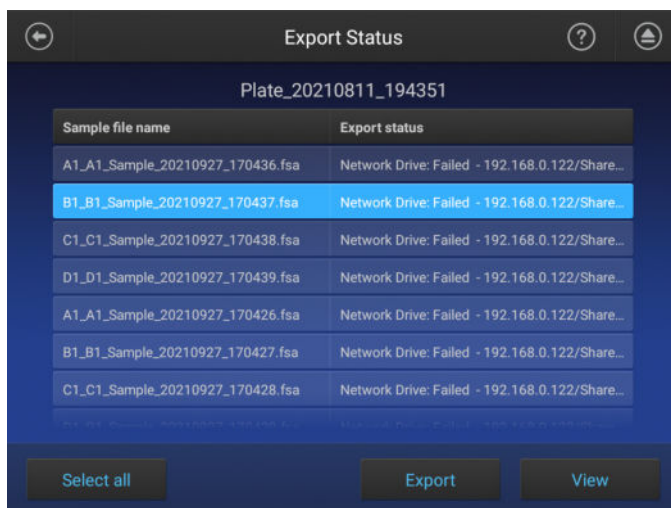
Note: If you touch more than one plate, or touch **Select all**, the **View** button is disabled.



Plate	Type	Exported	Run date
Plate_20210811_194351	FRAGMENT	Failed	2021/09/27 17:04:36

Select all Export View

The **Export Status** screen displays all sample data files in the selected plate that failed to export.



Plate_20210811_194351

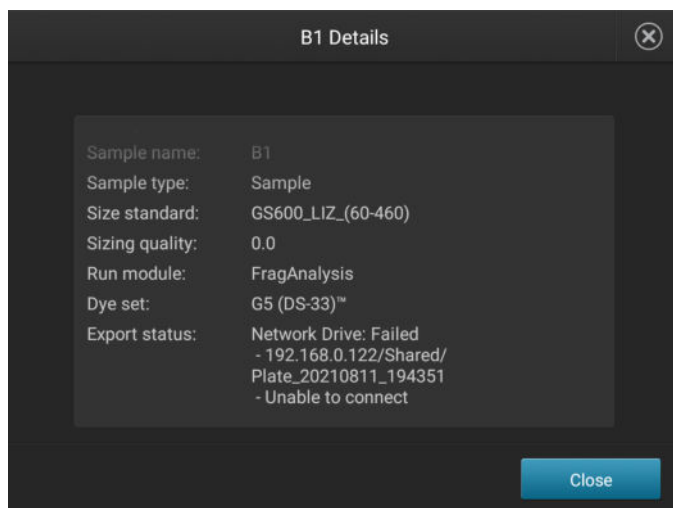
Sample file name	Export status
A1_A1_Sample_20210927_170436.fsa	Network Drive: Failed - 192.168.0.122/Share...
B1_B1_Sample_20210927_170437.fsa	Network Drive: Failed - 192.168.0.122/Share...
C1_C1_Sample_20210927_170438.fsa	Network Drive: Failed - 192.168.0.122/Share...
D1_D1_Sample_20210927_170439.fsa	Network Drive: Failed - 192.168.0.122/Share...
A1_A1_Sample_20210927_170426.fsa	Network Drive: Failed - 192.168.0.122/Share...
B1_B1_Sample_20210927_170427.fsa	Network Drive: Failed - 192.168.0.122/Share...
C1_C1_Sample_20210927_170428.fsa	Network Drive: Failed - 192.168.0.122/Share...

Select all Export View

3. (Optional) View sample details.

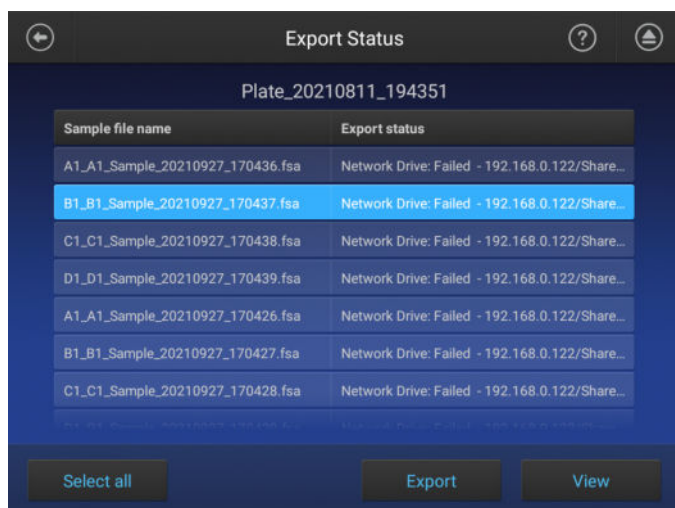
- a. Touch a sample data file of interest, then touch **View**.

If you touch more than one file, or touch **Select all**, the **View** button is disabled.

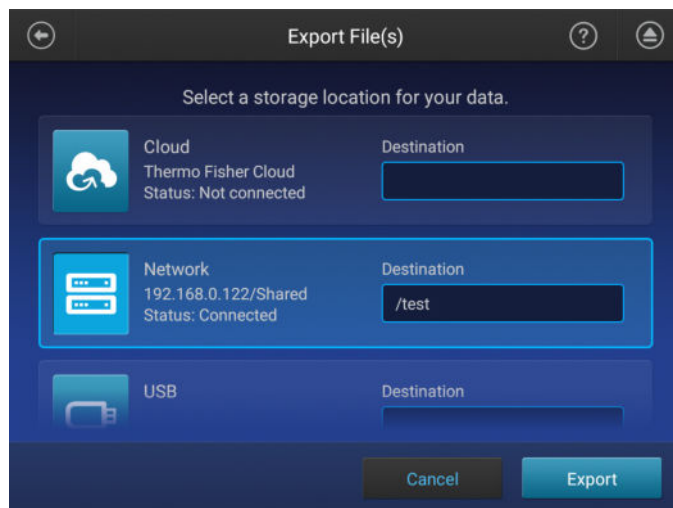


- b. Touch **Close** to return to the **Export Status** screen.

4. Touch one or more sample data files of interest, or touch **Select all** to select all the files listed, then touch **Export**.



5. In the **Export File(s)** screen, touch a save location for the selected sample data files, then touch **Export**.



6. Close the **Files exported successfully** message.

The exported sample data files are removed from the **Export Status** screen.

Note: If you do not manually export a sample data file, the file will automatically be removed from the **Export Status** screen after 30 days.



Manage the software (Plate Manager)

■ Add a custom dye calibration to the Plate Manager	136
■ Download a plate setup template as a CSV file	136
■ Save a plate setup as a PDF	137
■ Save a plate setup as a CSV file	137
■ Create a plate setup template	137
■ Delete plate setups	138
■ Manage email notifications in the Thermo Fisher™ Connect Platform	139
■ Manage run modules	139
■ Manage size standards	140
■ Manage analysis settings	140

Add a custom dye calibration to the Plate Manager

On the instrument:

1. Open a plate setup that specifies the custom dye set of interest.
2. Export a plate setup that specifies the custom dye.
3. Open the exported plate setup in the Plate Manager.

The custom dye is imported and is available for selection when you create new plates.

Download a plate setup template as a CSV file

1. Click **PM** to display the home screen.
2. Click **Create a plate file**, select **Create from template**, then click **Download** for the template of interest.

Note: You cannot add plate setups to the list of templates. However, you can create a plate setup with "Template" suffix, and use it as a starting point for creating new plate setups.

Note: In the home screen, **Recent plate files** is displayed. For quick access, navigate to a recent plate setup file of interest.

Save a plate setup as a PDF

In the **Plate** tab:

Select **Actions** ▶ **Print plate view**.

The plate view and table view are exported.

Save a plate setup as a CSV file

In the **Plate** tab:

1. Select **Actions** ▶ **Save as CSV**.
2. Navigate to and select a storage destination.

The plate setup is saved as a CSV file. The names of the dye set, run module, and size standard are included in the CSV file.

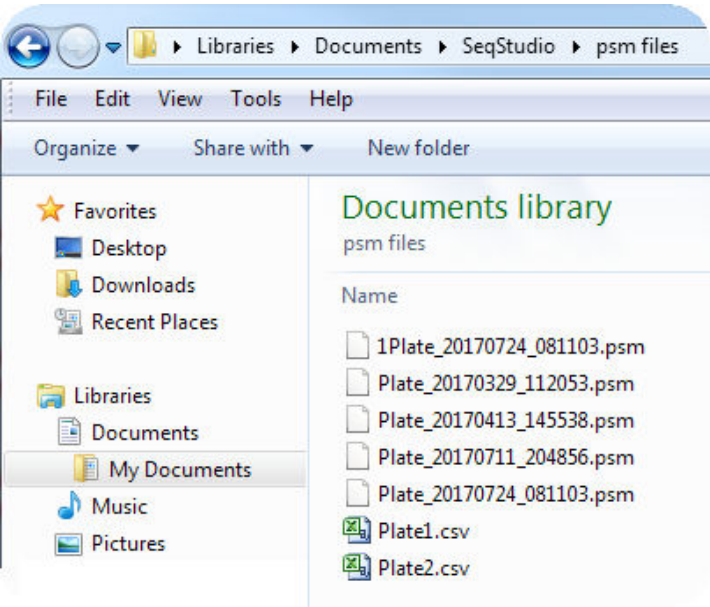
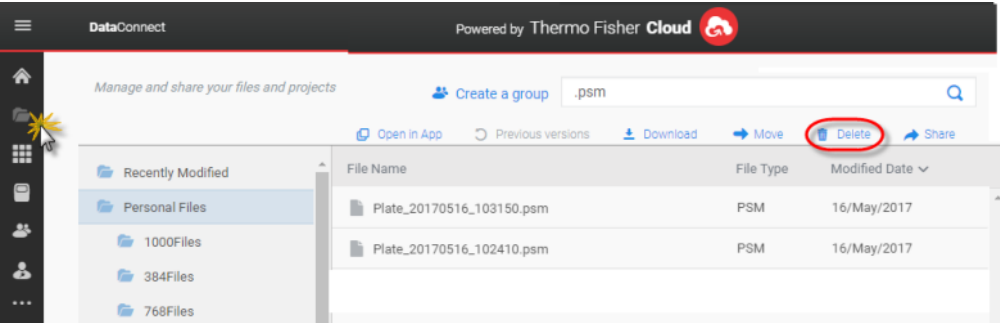
Create a plate setup template

Note: You cannot add templates to the list of templates you select when you create a plate setup. However, you can create a plate setup with "Template" suffix, then use it as a starting point for creating new plate setups.

1. Click **New**, select **Create from template**, then click **Download** for the template of interest.
2. Modify the CSV file as needed.
3. Save the plate setup with a "Template" suffix (example: Plate_FragAnalysis_Template), or other identifier.

Delete plate setups

To delete plate setups from the Plate Manager:

On the...	Use...
Desktop	Windows™ Explorer to delete PSM or CSV files. Note: The location shown in the figure is an example. Users can save PSM files in any location. 
Thermo Fisher™ Connect Platform	Click  in the Thermo Fisher™ Connect Platform, then use DataConnect to delete PSM files. 

Manage email notifications in the Thermo Fisher™ Connect Platform

1. In any screen in the Thermo Fisher™ Connect Platform, click .



2. Click **Settings**.
3. Select or deselect your email address.

Manage run modules

This function manages the list of run modules that you can select from when you create a plate setup.

To assign a run module to a plate, see “Assign wells: Sample and run information” on page 76.

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

For more information, see “Run modules, read lengths, size ranges, and run times” on page 156.

1. In the **Plate** tab, select **Actions ▶ Manage run modules**.
2. To create a new run module:
 - a. Select a default run module or a user-created run module, then click **Copy**.
 - b. Enter a run module name.
 - c. Enter values, click **Advanced** to enter additional settings, then click **Save**.

See “Run module settings” on page 155 for detailed information.

3. As needed, select a run module of interest, then click **Edit** or **Delete** (user-created run modules only).

Manage size standards

This function manages the list of size standard definitions that you can select from during plate set up.

To assign a size standard definition to a plate, see “Assign wells: Sample and run information” on page 76.

A size standard defines the sizes in basepairs of known fragments. It is used to generate a standard curve. The standard curve is used to determine the sizing of fragments in unknown samples.

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

Size standard definitions are accessible to all users.

1. In the **Plate** tab, select **Actions ▶ Manage size standards**.
2. To create a new size standard:
 - a. Select a default size standard or a user-created size standard, then click **Copy**.
 - b. Enter a name and select a dye color, then edit the fragment sizes (basepairs).
 - c. Click **Save**.
3. As needed, select a size standard of interest, then click **Edit** or **Delete** (user-created size standards only).

Manage analysis settings

In the **Properties** tab:

1. Select **Actions ▶ Analysis settings**.
2. To create new analysis settings:
 - a. Select the default analysis settings or user-created analysis settings, then click **Copy**.
 - b. Enter a name and edit settings as needed (see “Fragment/HID analysis settings (size calling)” on page 141 or “Sequencing settings (base calling)” on page 143).
 - c. Click **Save**.
3. As needed, select an analysis setting of interest, then click **Edit** or **Delete** (user-created settings only).

Fragment/HID analysis settings (size calling)

Setting	Description
Size calling method	• Local Southern—(<i>default</i>) Determines the fragment sizes using the reciprocal relationship between fragment length and electrophoretic mobility. • Global Southern—Compensates for standard fragments with anomalous electrophoretic mobility (similar to least squares methods). • 2nd LSQ (2nd Order Least Squares)—Uses regression analysis to build a best-fit size calling curve. • 3rd LSQ (3rd Order Least Squares)—Uses regression analysis to build a best-fit size calling curve. • Cubic Spline Interpolation—Forces the sizing curve through all the known points of the selected size standard.
Analysis range	• Full Range—(<i>default</i>) To analyze the entire scan region as collected by the genetic analysis instrument, including the primer peak. • Partial Range—To analyze only data points within a specified range. Enter Start Point in data points after the primer peak and before the first required size standard peak. Enter a Stop Point after the last required size standard fragment. Start and Stop points may vary from instrument to instrument and platform to platform. View raw data to determine the appropriate analysis range. Data points outside the specified analysis range are ignored. Note: Ensure the Analysis Range contains all size standard fragments included in the Sizing Range.
Sizing range	The size range (in base pairs) appropriate for the kit you are using: • Full Range for the software to analyze fragments of all sizes in the Analysis Range. • Partial Range for the software to analyze only fragments within a specified range. Enter a Start Size and a Stop Size appropriate for the size standard used.
Peak amplitudes	The peak height threshold (RFU) for peak detection for each dye color. Peaks below the threshold are not detected. For example, if you use the default values of 175 RFU, peaks with heights equal to or greater than 175 RFU are detected. Peaks with heights below 175 RFU are still displayed in the electropherogram plots but are not detected or labeled. Note: Use the same peak amplitude thresholds in secondary analysis software.
Primer peak	If the primer peaks in your application obscure peaks of interest, select Present. This instructs the algorithm to ignore primer peaks. Primer peaks are still displayed in the trace. If this setting does not allow detection of the 20- and 40-mer peaks for samples that use the GS600 LIZ™ size standard, running samples with the GS600_LIZ_(60-600) or other size standards that include lower bp starting points may allow detection of the peaks.

(continued)

Setting	Description
Common settings	
Smoothing	Select an option to smooth the outline of peaks and reduce the number of false peaks detected: • None (default) to apply no smoothing. Best if the data display sharp, narrow peaks of interest. • Light to provide the best results for typical data. Light smoothing slightly reduces peak height. • Heavy for data with very sharp, narrow peaks of interest. Heavy smoothing can significantly reduce peak height.
Baseline Window	Specify a window to adjust the baseline signals of all detected dye colors to the same level for an improved comparison of relative signal intensity. Note the following: • A small baseline window relative to the width of a cluster, or grouping of peaks spatially close to each other, can result in shorter peak heights. • Larger baseline windows relative to the peaks being detected can create an elevated baseline, resulting in peaks that are elevated or not resolved to the baseline.
Minimum Peak Half Width	Specify the minimum full peak width at half maximum Peak Height required for peak detection. The range is 2 to 99 data points.
Peak Window Size	Enter a window width in data points for peak detection sensitivity. If more than one peak apex is within the window, all are labeled as a single peak. Note the following: • The maximum value is the number of data points between peaks. • The Peak Window Size setting is limited to odd numbers. To increase peak detection sensitivity: Increase polynomial degree, decrease peak window size. To decrease peak detection sensitivity: Decrease polynomial degree, increase peak window size.
Polynomial Degree	Polynomial Degree cannot be greater than Peak Window Size. Adjust to affect the sensitivity of peak detection. You can adjust this parameter to detect a single base pair difference while minimizing the detection of shoulder effects and/or noise. The peak detector calculates the first derivative of a polynomial curve fitted to the data within a window that is centered on each data point in the analysis range. Using curves with larger polynomial degree values allows the curve to more closely approximate the signal and, therefore, captures more of the peak structure in the electropherogram.

(continued)

Setting	Description
Slope Thresholds Peak Start and End	• Peak Start—The peak starts when the first derivative (slope of the tangent) in the beginning of the peak signal before the inflection point becomes equal to or exceeds the Peak Start value. This threshold is set to 0 by default, which means that the peak will normally start at the leftmost point where the slope of the tangent is closest to 0° (horizontal line). A value other than 0 moves the peak start point toward its center. The value entered must be non-negative. • Peak End—The peak ends when the first derivative (slope of the tangent) in the end of the peak signal after the inflection point becomes equal to or exceeds the Peak End value. This value is set to 0 by default, which means that the peak will normally end at the rightmost point where the slope of the tangent is closest to 0° (horizontal line). A value other than 0 moves the peak end point toward its center. The value entered in this field must be non-positive. Using curves with larger polynomial degree values allows the curve to more closely approximate the signal and, therefore, captures more of the peak structure in the electropherogram.

Sequencing settings (base calling)

The default settings are optimized for sequencing of PCR amplicons from diploid genes, which are expected to have Mixed bases and which should end At PCR Stop.

For sequencing from plasmid templates, which are of pure sequence (no Mixed bases) and have longer sequence read times than PCR products, create new analysis settings and *disable* (uncheck) **At PCR Stop** and **Mixed base threshold** checkboxes.

Setting	Description
Quality Threshold	• Basecall assignment (ambiguous bases): – Do not assign Ns to basecalls – Assign Ns to basecalls with QV<5— Bases with a QV less than the threshold display N instead of the base letter • End base—Last base on which to perform basecalling: – At PCR Stop – After X number of bases – After X number of Ns in X number of bases – After X number of Ns Note: If you have PCR products with sequences that end while data is still being collected, select the At PCR Stop checkbox.
Mixed bases threshold	When enabled, determines the secondary peak height ratio where the secondary peak is considered a potential mixed base. Reaching the threshold is a necessary but not sufficient condition for the basecalling algorithm to call a mixed base.
Clear range methods	• Use quality values—Sets a window with a specified number of allowed low-quality bases by removing bases until there are <X number of bases per Z number of bases with QV <Y. • Use base positions—Specifies the first and last base in the range to consider, or trims the specified number of bases from the 3' end. • Mask base positions before—Specifies the base position before which to disregard bases.

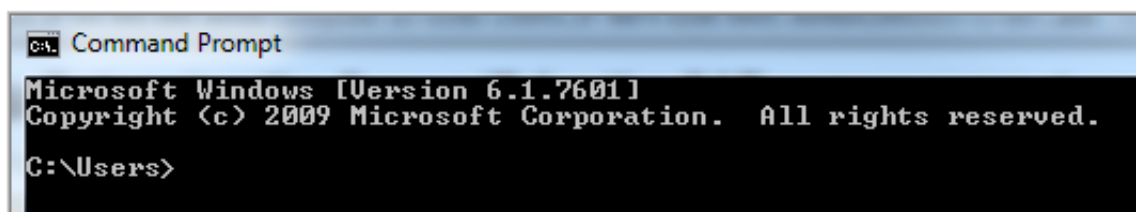
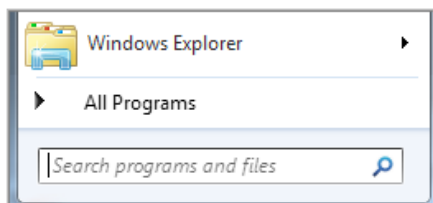
■ Connect the instrument to a network drive	144
■ Link the instrument to your Thermo Fisher™ Connect Platform account	151
■ Lock the touchscreen	152
■ Manage plate setups	153
■ Manage run settings (instrument)	154
■ Manage instrument settings	163
■ Manage instrument profiles on the instrument	167
■ Manage storage space	170

Connect the instrument to a network drive

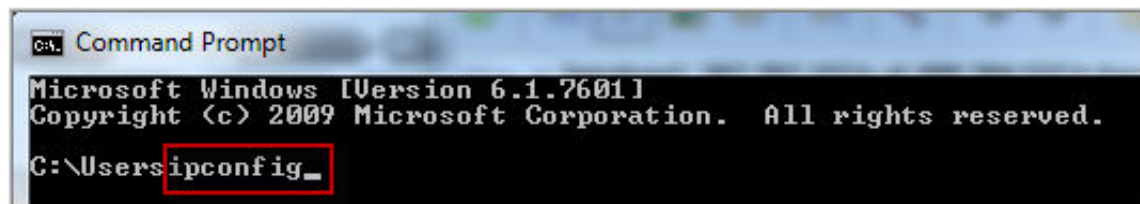
Determine IP address for a computer on a network

On the destination computer:

1. In the Windows™ desktop, click .
2. In the search field at the bottom of the pane, type **command prompt**, then press **Enter**.

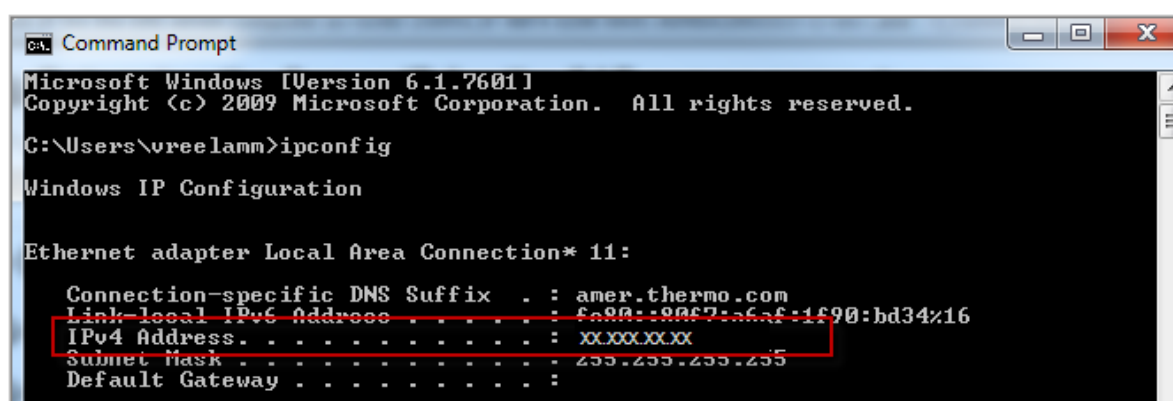


3. At the command prompt, type **ipconfig**, then press **Enter**.



4. Note the IP address listed.

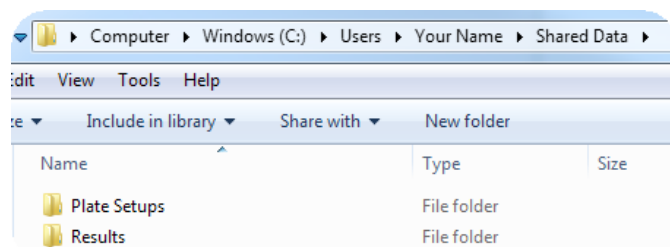
Note: The location and number of digits in your IP address may differ from the **IPv4 Address** example shown below.



Create folders and enable network folder sharing

In the Windows™ desktop:

1. On a Windows™ computer, server, or network drive, create a folder to store your plates and results. Example: **C:/Users/Your Name/SharedData**.
2. Create subfolders in the **SharedData** folder. Example: **PlateSetups** and **Results**.



3. Right click on a folder, then select **Sharing > Advanced** or **Share with > Advanced sharing**.
4. Select the **Sharing** tab.
5. Click **Advanced sharing**.
6. Select **Share this folder**.

7. Click **Permissions**, then select **Full Control** or **Read, Write**, and **Delete** options.

IMPORTANT! Without these permissions, the instrument cannot auto export results when a run is complete.

8. Click **OK**, click **OK**, then click **Close**.

Connect to a network drive

See your laboratory administrator for the information you need to connect to a network drive.

From any screen that displays  **Network drive** or **Save location** as an option, you can connect to the drive for the first time.

IMPORTANT! Before saving to the network drive, ensure that the folder is shared (see “Create folders and enable network folder sharing” on page 145).

1. Touch  **Network drive** or **Save location** field.
2. Touch the **Destination** field, then touch the appropriate field to enter the IP address and shared folder name, then touch **Done**.

Note: Do not include the drive name in the drive location. For example, if you created folder *C:/Users/Your Name/SharedData*, type *IPaddress/Users/Your Name/SharedData*, (do not include C:).

For more information, see “Determine IP address for a computer on a network” on page 144.
For MicroSEQ™ ID Software For SeqStudio™ Genetic Analyzer, the shared folder should be *IP address/SeqstudioData*.

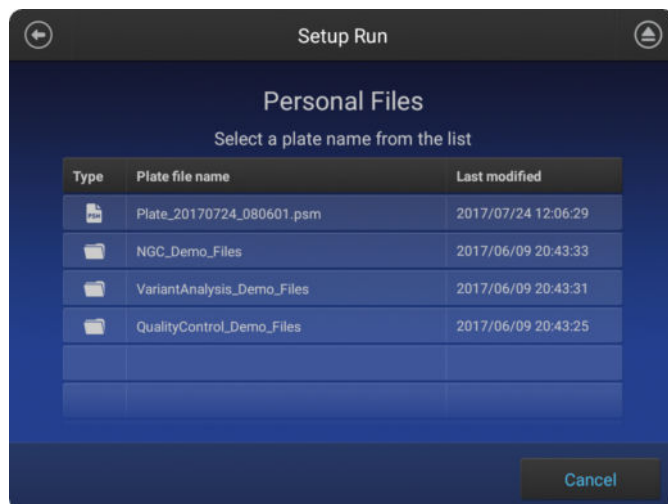
3. If necessary, touch the appropriate fields to enter a domain name, username, and password.
4. Touch **Connect**.

Set up a default Thermo Fisher™ Connect Platform location for opening plate setups

Before you begin, create a plate setup (PSM file or CSV file) in the Thermo Fisher™ Connect Platform location that you want to set as the default location.

To set the default location, you must select a plate setup.

1. In the home screen of the instrument, touch **Setup run**, then touch  **Cloud**.
The **Setup Run** screen is displayed.



2. Select a folder (if needed), then select a plate file.
The plate file is imported.
3. Touch to return to the **Setup Run** screen.

Set up a default Thermo Fisher™ Connect Platform location for saving results (auto export)

1. In the home screen of the instrument, touch **Setup run**, then touch **Create new plate setup**.
2. Touch **Save location**.
3. Touch the **Cloud Destination** field.
The **Select Directory** screen is displayed.
4. Select or create a folder.
5. Touch **Done**.

Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

Set up a default network location for opening plate setups

Before you begin, create a plate setup (PSM file or CSV file) in the shared directory on the computer that you want to set as the default location.

To set the default location, you must select a plate setup.

1. Determine the IP address of the computer on which you created shared folders (see “Determine IP address for a computer on a network” on page 144 and “Create folders and enable network folder sharing” on page 145).
2. In the home screen of the instrument, touch **Setup run**, then touch  **Network Drive**.

If your instrument profile	This screen is displayed	How to proceed
Is not connected to a network drive	Connect	1. In the Network Destination field, enter the IP address of the computer followed by the folder names you created. Example: 10.43.32.82/SharedData/Results. 2. If required by your network, enter Domain, User Name, and Password to access the shared location. 3. Touch Connect. The Setup Run screen is displayed.
Is connected to a network drive	Select Directory	Proceed to step 3.

3. In the **Setup Run** screen:

Touch	To
Set up import	Change the Network Destination you specified in the previous step.
A folder name	Navigate to the location you want to set as the default.
A plate name	Import the plate.

The **Setup Run** screen is displayed.

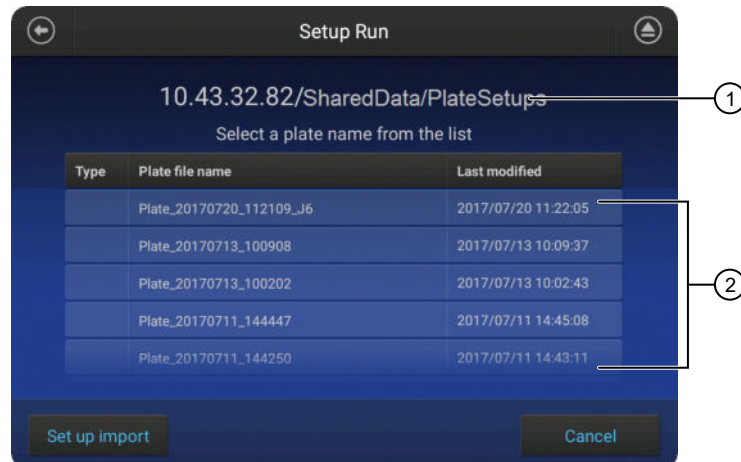


Figure 18 Setup Run screen example

- ① Example IP address and folder locations
- ② List of plates (will be blank if you have not saved plates to this location)

Set up a default network location for saving results (auto export)

1. Determine the IP address of the computer on which you created shared folders (see “Determine IP address for a computer on a network” on page 144 and “Create folders and enable network folder sharing” on page 145).
2. In the home screen of the instrument, touch **Setup run**, then touch **Create new plate setup**.
3. Touch **Save location**.
4. Touch the network **Destination** field.

If your instrument profile	This screen is displayed	How to proceed
Is not connected to a network drive	Connect	1. In the Network Destination field, enter the IP address of the computer followed by the folder names you created. Example: 10.43.32.82/SharedData/Results. 2. If required by your network, enter Domain, User Name, and Password to access the shared location. 3. Touch Connect. The Select Directory screen is displayed.
Is connected to a network drive	Select Directory	Proceed to step 5.

5. In the **Select Directory** screen:

Touch	To
New folder	Create a subfolder.
Set up export	Change the Network Destination you specified in the previous step.
Select this folder	Select the directory shown at the top of the screen.

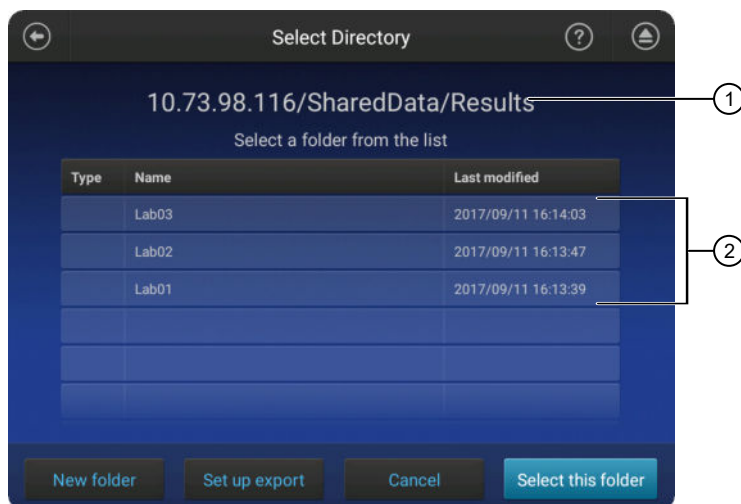


Figure 19 Select Directory screen example

- ① Example IP address and folder locations
- ② List of folders

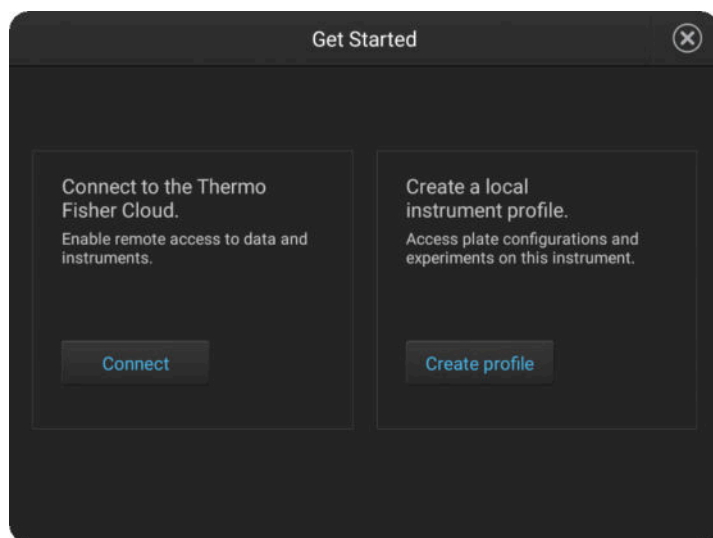
6. In the **Save Destination** screen, touch **Done**.

Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

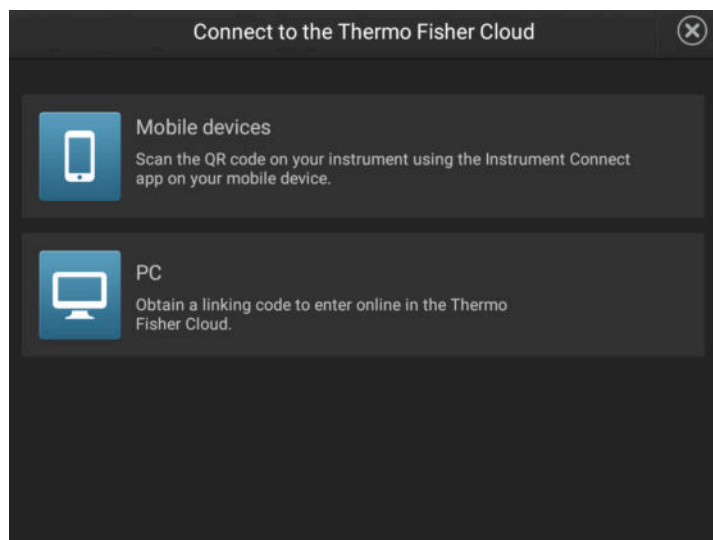
Link the instrument to your Thermo Fisher™ Connect Platform account

Note: For detailed information on linking the instrument to your Thermo Fisher™ Connect Platform account, see Appendix B, “Link the instrument to your Thermo Fisher™ Connect Platform account—detailed instructions”.

1. If a user is signed in, touch , then touch **Sign out**.
2. In the **Sign In** screen, touch **Get started ▶ Connect**.



3. In the **Connect to the Cloud** screen, touch a connection option.



Option	Action
 Mobile devices	Note: Before selecting this option, install and sign in to the InstrumentConnect app on your mobile device. In the Connect to the Cloud screen: - Touch Mobile devices. - Hold the camera on your mobile device over the QR code that is displayed on the touchscreen. - Click Close.
 PC	In the Connect to the Cloud screen, a link code is displayed. On a computer: - Access the Thermo Fisher™ Connect Platform. - Access InstrumentConnect. - Click Add instrument. - Select SeqStudio. - Enter the link code.
 Instrument	In the Connect to the Cloud screen, enter your account information, then click Link account.

Lock the touchscreen

During a run, you can lock the touchscreen to prevent other users from using the instrument. This feature is not available to Guest users.

Only the user who locked the touchscreen or an administrator can sign in to the instrument if the touchscreen is locked.

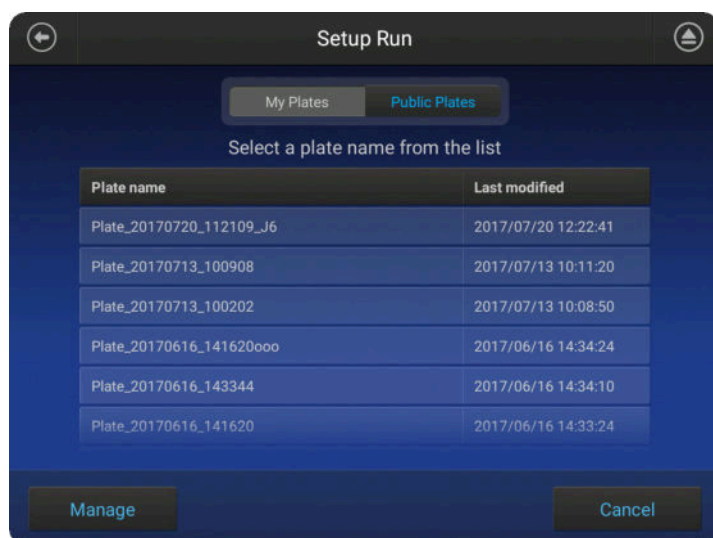
1. Touch .
2. Touch **Profile**.
3. Touch **Lock instrument**.

Note: If a run is not in progress, **Sign out** is displayed instead of **Lock instrument**.

Manage plate setups

Export or delete a plate setup (PSM file)

1. Touch **Set up run**, then touch  **My instrument**.
2. Touch **Manage** at the bottom left of the screen.



3. Touch a plate, then touch **Export** or **Delete**.
4. If you touched:
 - **Export**, select a save location, then touch **Export**.
 - **Delete**, then touch **Yes** to delete the plate setup.

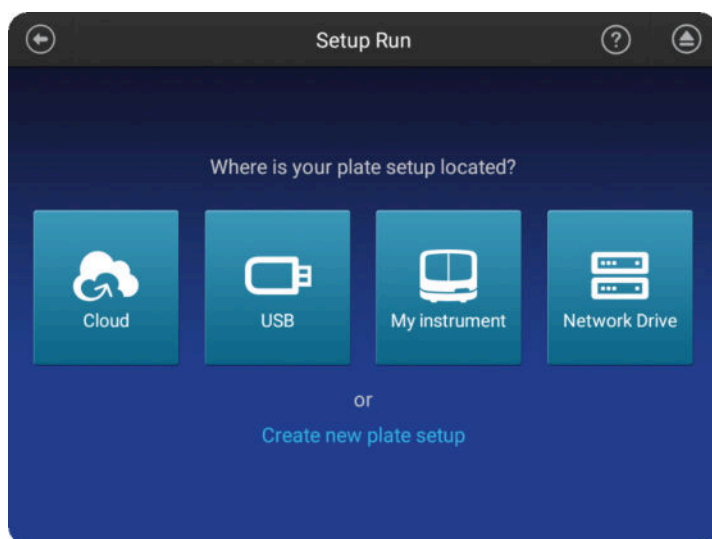
Import a plate setup from a CSV or PSM file

Note: You can create a plate setup in CSV or PSM file in the Plate Manager (desktop or Thermo Fisher™ Connect Platform).

Before importing a CSV file, see “PSM and CSV plate setup files for import into the instrument” on page 84.

In the home screen:

1. Touch **Set up run**, then touch  **Cloud**,  **Network**, or  **USB**.



2. Navigate to, then select a CSV or PSM file.

Manage run settings (instrument)

Manage run modules

This function manages the list of run modules that you can select from when you create a plate setup.

To assign a run module to a plate, see “Assign wells: run module, size standard, dye set, and kit” on page 90.

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

Run modules are accessible only to the user who creates them.

For more information, see “Run modules, read lengths, size ranges, and run times” on page 156.

1. Access the **Manage run modules** screen:

From	Action
Plate properties tab	Touch More options ► Manage run modules .
Home screen	Touch ⓘ Settings ► Run settings ► Run modules .

2. To create a new run module:
 - a. Touch a default run module or a user-created run module to use as a starting point, then touch **Copy**.
 - b. Enter values, then touch **Next**.
 - c. Enter a name, touch **Advanced** to enter additional settings, then **Done**.

See “Run module settings” on page 155 for detailed information.

3. As needed, touch a run module of interest, then touch **Edit** or **View**.

Note: The **Edit** button is dimmed if a run is in progress.

Run module settings

Setting	Description
Run module name	Name of the module
Capillary temperature (°C)	Temperature setting for the capillary array throughout run
Prerun voltage (Volts)	Voltage setting for pre-run before sample injection
Prerun time (seconds)	Prerun time
Injection voltage (Volts)	Voltage for sample injection
Injection time (seconds)	Sample injection time
Run voltage (Volts)	Final sample electrophoresis separation run voltage
Run ramp duration (seconds)	Time required to reach the Run voltage Note: Data collection does not start until this time elapses.
Run time (seconds)	Length of time that data is collected after the Run ramp duration elapses

Run modules, read lengths, size ranges, and run times

Table 7 Sequencing run modules for standard sequencing

Run module	Contiguous read length (CRL) ^[1]	QV threshold	Approximate run time
ShortSeq	≥350	QV30	30 minutes
ShortSeq_BDX			
MediumSeq	≥500	QV30	45 minutes
MediumSeq_BDX			
LongSeq	≥800	QV20	~ 2 hours
LongSeq_BDX			

^[1] CRL was determined using the Long Read Sequencing standard. A minimum of 90% of analyzed sequences with an average QV ≥QV threshold were observed.

IMPORTANT! Use BDX run modules only if you prepare samples with BigDye XTerminator™ Purification Kit. Use non-BDX run modules for samples purified with other methods.

Table 8 Fragment analysis run modules

Run module	Resolution range	Approximate run time	Sizing precision	Compatible size standards
SNaPshot	40–120 bp	25 minutes	40–120: <0.5	GeneScan™ 120 LIZ™ Size Standard
FragAnalysis	60–460 bp ^[1]	45 minutes	60–460: <0.15	All except GeneScan™ 1200 LIZ™ Size Standard
LongFragAnalysis ^[2]	60–600 bp ^[1]	<2 hours	60–460: <0.15 461–600: <0.3 601–800: >0.45	- GeneScan™ 600 LIZ™ Size Standard v2.0 - GeneScan™ 1200 LIZ™ Size Standard

^[1] Resolution Range: The range of bases over which the resolution (peak spacing interval divided by the peak width at half-max in a GS600 or GS1200 LIZ size standard sample sized with a third order fit) is ≥1. The table shows the resolution range in ≥90% of samples.

^[2] Load a maximum of 48 samples per plate if you use a long run module.

Table 9 HID run modules

Run module	Resolution range	Approximate run time	Sizing precision	Compatible size standards
HIDAnalysis	60–470 bp ^[1]	39 minutes	60–470: <0.15	- GeneScan™ 500 LIZ™ Size Standard^[2] - GeneScan™ 600 LIZ™ Size Standard v2.0

^[1] Resolution Range: The range of bases over which the resolution (peak spacing interval divided by the peak width at half-max in a GS600 size standard sample sized with a third order fit) is ≥1.

^[2] The GeneScan™ 500 LIZ™ Size Standard was not included in the HID validation of the instrument.

Note: The following size standards have not been validated for use with the instrument. A default size standard definition is not provided in the software.

- GeneScan™ 350 ROX™ Size Standard
- GeneScan™ 400HD ROX™ Dye Size Standard

Manage size standard definitions

This function manages the list of size standard definitions that you can select from during plate set up.

To assign a size standard definition to a plate, see “Assign wells: run module, size standard, dye set, and kit” on page 90.

A size standard defines the sizes in basepairs of known fragments. It is used to generate a standard curve. The standard curve is used to determine the sizing of fragments in unknown samples.

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

Size standard definitions are accessible to all users.

1. Access the **Manage size standards** screen:

From	Actions
Plate properties tab	Touch More options ▸ Manage size standards .
Home screen	Touch ⚙ Settings ▸ Run settings ▸ Size standards .

2. To create a new size standard:
 - a. Touch a default size standard or a user-created size standard, then touch **Copy**.
 - b. Enter a name and select a dye color.
 - c. As needed, touch any of the following, then touch **Done**:

Note: To change an existing value, add a new value, then delete the original value.

- One or more fragment size values, then touch **Delete**.
- **Add** to add a value.

3. As needed, touch a size standard of interest, then touch **Edit** or **View**.

Manage analysis settings

Edit fragment/HID analysis settings

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

Analysis settings are accessible only to the user who creates them.

1. Access the **Analysis settings** screen:

From	Action
Plate properties tab	Touch More options ▶ Analysis settings , then select the settings to assign to the plate setup.
Home screen	Touch ⓘ Settings ▶ Run settings ▶ Analysis settings , then manage the list of settings that you can select from during plate set up.

2. To create new analysis settings:
 - a. Touch the default analysis settings or user-created analysis settings, then touch **Copy**.
 - b. Enter a name and edit settings as needed (see “Fragment/HID analysis settings (size calling)” on page 141).
 - c. As needed, touch **View** or **Delete** (user-created settings only).
3. Touch **Done**.

Edit sequencing settings

Factory-installed items cannot be edited or deleted. To create a new item from a factory-installed item, copy, edit, then save the new item.

Analysis settings are accessible only to the user who creates them.

1. Access the **Analysis settings** screen:

From	Action
Plate properties tab	Touch More options ▶ Analysis settings , then select the settings to assign to the plate setup.
Home screen	Touch ⓘ Settings ▶ Run settings ▶ Analysis settings , then manage the list of settings that you can select from during plate set up.

2. To create new analysis settings:
 - a. Touch the default analysis settings or user-created analysis settings, then touch **Copy**.
 - b. Enter a name and edit settings as needed (see “Sequencing settings (base calling)” on page 143).
 - c. As needed, touch **View** or **Delete** (user-created settings only).
3. Touch **Done**.

Fragment/HID analysis settings (size calling)

Setting	Description
Size calling method	• Local Southern—(<i>default</i>) Determines the fragment sizes using the reciprocal relationship between fragment length and electrophoretic mobility. • Global Southern—Compensates for standard fragments with anomalous electrophoretic mobility (similar to least squares methods). • 2nd LSQ (2nd Order Least Squares)—Uses regression analysis to build a best-fit size calling curve. • 3rd LSQ (3rd Order Least Squares)—Uses regression analysis to build a best-fit size calling curve. • Cubic Spline Interpolation—Forces the sizing curve through all the known points of the selected size standard.
Analysis range	• Full Range—(<i>default</i>) To analyze the entire scan region as collected by the genetic analysis instrument, including the primer peak. • Partial Range—To analyze only data points within a specified range. Enter Start Point in data points after the primer peak and before the first required size standard peak. Enter a Stop Point after the last required size standard fragment. Start and Stop points may vary from instrument to instrument and platform to platform. View raw data to determine the appropriate analysis range. Data points outside the specified analysis range are ignored. Note: Ensure the Analysis Range contains all size standard fragments included in the Sizing Range.
Sizing range	The size range (in base pairs) appropriate for the kit you are using: • Full Range for the software to analyze fragments of all sizes in the Analysis Range. • Partial Range for the software to analyze only fragments within a specified range. Enter a Start Size and a Stop Size appropriate for the size standard used.
Peak amplitudes	The peak height threshold (RFU) for peak detection for each dye color. Peaks below the threshold are not detected. For example, if you use the default values of 175 RFU, peaks with heights equal to or greater than 175 RFU are detected. Peaks with heights below 175 RFU are still displayed in the electropherogram plots but are not detected or labeled. Note: Use the same peak amplitude thresholds in secondary analysis software.
Primer peak	If the primer peaks in your application obscure peaks of interest, select Present. This instructs the algorithm to ignore primer peaks. Primer peaks are still displayed in the trace. If this setting does not allow detection of the 20- and 40-mer peaks for samples that use the GS600 LIZ™ size standard, running samples with the GS600_LIZ_(60-600) or other size standards that include lower bp starting points may allow detection of the peaks.

(continued)

Setting	Description
Common settings	
Smoothing	Select an option to smooth the outline of peaks and reduce the number of false peaks detected: • None (default) to apply no smoothing. Best if the data display sharp, narrow peaks of interest. • Light to provide the best results for typical data. Light smoothing slightly reduces peak height. • Heavy for data with very sharp, narrow peaks of interest. Heavy smoothing can significantly reduce peak height.
Baseline Window	Specify a window to adjust the baseline signals of all detected dye colors to the same level for an improved comparison of relative signal intensity. Note the following: • A small baseline window relative to the width of a cluster, or grouping of peaks spatially close to each other, can result in shorter peak heights. • Larger baseline windows relative to the peaks being detected can create an elevated baseline, resulting in peaks that are elevated or not resolved to the baseline.
Minimum Peak Half Width	Specify the minimum full peak width at half maximum Peak Height required for peak detection. The range is 2 to 99 data points.
Peak Window Size	Enter a window width in data points for peak detection sensitivity. If more than one peak apex is within the window, all are labeled as a single peak. Note the following: • The maximum value is the number of data points between peaks. • The Peak Window Size setting is limited to odd numbers. To increase peak detection sensitivity: Increase polynomial degree, decrease peak window size. To decrease peak detection sensitivity: Decrease polynomial degree, increase peak window size.
Polynomial Degree	Polynomial Degree cannot be greater than Peak Window Size. Adjust to affect the sensitivity of peak detection. You can adjust this parameter to detect a single base pair difference while minimizing the detection of shoulder effects and/or noise. The peak detector calculates the first derivative of a polynomial curve fitted to the data within a window that is centered on each data point in the analysis range. Using curves with larger polynomial degree values allows the curve to more closely approximate the signal and, therefore, captures more of the peak structure in the electropherogram.

(continued)

Setting	Description
Slope Thresholds Peak Start and End	• Peak Start—The peak starts when the first derivative (slope of the tangent) in the beginning of the peak signal before the inflection point becomes equal to or exceeds the Peak Start value. This threshold is set to 0 by default, which means that the peak will normally start at the leftmost point where the slope of the tangent is closest to 0° (horizontal line). A value other than 0 moves the peak start point toward its center. The value entered must be non-negative. • Peak End—The peak ends when the first derivative (slope of the tangent) in the end of the peak signal after the inflection point becomes equal to or exceeds the Peak End value. This value is set to 0 by default, which means that the peak will normally end at the rightmost point where the slope of the tangent is closest to 0° (horizontal line). A value other than 0 moves the peak end point toward its center. The value entered in this field must be non-positive. Using curves with larger polynomial degree values allows the curve to more closely approximate the signal and, therefore, captures more of the peak structure in the electropherogram.

Sequencing settings (base calling)

The default settings are optimized for sequencing of PCR amplicons from diploid genes, which are expected to have Mixed bases and which should end At PCR Stop.

For sequencing from plasmid templates, which are of pure sequence (no Mixed bases) and have longer sequence read times than PCR products, create new analysis settings and *disable* (uncheck) **At PCR Stop** and **Mixed base threshold** checkboxes.

Setting	Description
Quality Threshold	• Basecall assignment (ambiguous bases): – Do not assign Ns to basecalls – Assign Ns to basecalls with QV<5— Bases with a QV less than the threshold display N instead of the base letter • End base—Last base on which to perform basecalling: – At PCR Stop – After X number of bases – After X number of Ns in X number of bases – After X number of Ns Note: If you have PCR products with sequences that end while data is still being collected, select the At PCR Stop checkbox.
Mixed bases threshold	When enabled, determines the secondary peak height ratio where the secondary peak is considered a potential mixed base. Reaching the threshold is a necessary but not sufficient condition for the basecalling algorithm to call a mixed base.
Clear range methods	• Use quality values—Sets a window with a specified number of allowed low-quality bases by removing bases until there are <X number of bases per Z number of bases with QV <Y. • Use base positions—Specifies the first and last base in the range to consider, or trims the specified number of bases from the 3' end. • Mask base positions before—Specifies the base position before which to disregard bases.

Modify the default file name convention

The default file name convention determines how the data files (AB1 or FSA) associated with a plate are named.

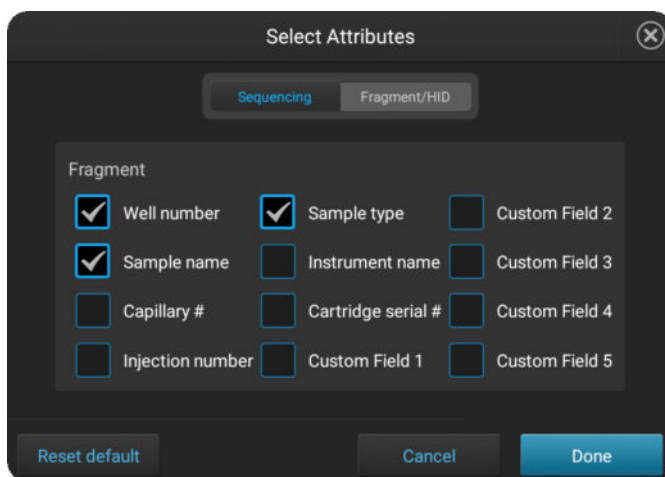
The default file name convention is:

Application	Default settings
Fragment/HID analysis	<well>_<sample name>_<sample type>_<date and timestamp> .fsa
Sequence analysis	<well>_<sample name>_<date and timestamp>.ab1
Sequence analysis with the Sanger variant analysis option selected	<well>_<sample name>_<amplicon>_<specimen>_<date and timestamp>.ab1

1. Access the **File name convention** screen:

From	Action
Plate properties screen	Select the Properties tab, then touch More options ▸ File name convention .
Home screen	Touch  Settings ▸ Run settings ▸ File name convention .

2. Touch **Attributes**.
3. Select the attributes to include in the data file name.



For information on creating custom fields to include in file name conventions, see “Define custom fields” on page 96.

Note: The timestamp attribute cannot be deselected; it is always included in the data file name.

4. Touch **Done**.

5. Touch and drag attributes up or down in the list.
6. Touch **Done**.

Manage instrument settings

Display instrument hardware and software information

In the home screen:

1. Touch ⓘ **Settings** ▶ **Instrument settings** ▶ **About** to access the instrument information:
 - **Model name**
 - **Ethernet IP address**
 - **Ethernet MAC address**
 - **Wireless IP address**
 - **Wireless MAC address**
 - **Instrument software release**
 - **Instrument serial number**
2. (Optional) Touch **EULA** to display the end-user license agreement or touch **Details** to display additional instrument information.

Change the instrument name

If the instrument is linked to the Thermo Fisher™ Connect Platform, only a Thermo Fisher™ Connect Platform administrator for the instrument can change the instrument name. The instrument name can be changed by a Thermo Fisher™ Connect Platform administrator on the instrument or in InstrumentConnect.

In the home screen:

1. Touch ⓘ **Settings** ▶ **Instrument Settings** ▶ **Instrument name**.
2. Touch the **Instrument Name** field, enter an instrument name, then touch **Done**.
3. Touch **OK**.

When you change the instrument name, the software unlinks all instrument profiles.

Enable Demo mode

Demo mode allows you to use all features of the system and provides simulated real-time data and results. You cannot save plates or settings in Demo mode.

In the home screen:

1. Touch ⓘ **Settings** ▶ **Instrument settings** ▶ **Turn Demo mode on**.
2. Touch **OK** to allow the instrument to restart.
The instrument automatically restarts in Demo mode.

Manage date and time settings

In the home screen:

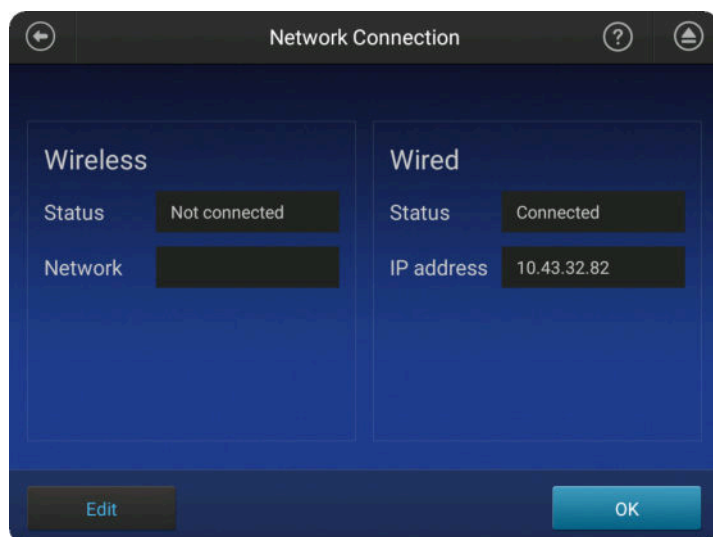
1. Touch ⓘ **Settings** ▶ **Instrument settings** ▶ **Date and Time**.
2. Slide the control to select a time-setting option:
 - The instrument automatically detects the time via the network.
 - Enter the time manually.
3. Touch **Time zone**, then select a time zone.
4. Touch **Date/Format**, then select the display order for the month, day, and year.
5. Touch **Time/Format**, then select a 12-Hour or 24-Hour time display format.
6. Click **OK**.

Manage the network configuration

Note: For a direct connection between the instrument and a computer, set up a wired connection.

In the home screen:

1. Touch  **Settings** ▶ **Instrument settings** ▶ **Network Configuration**.



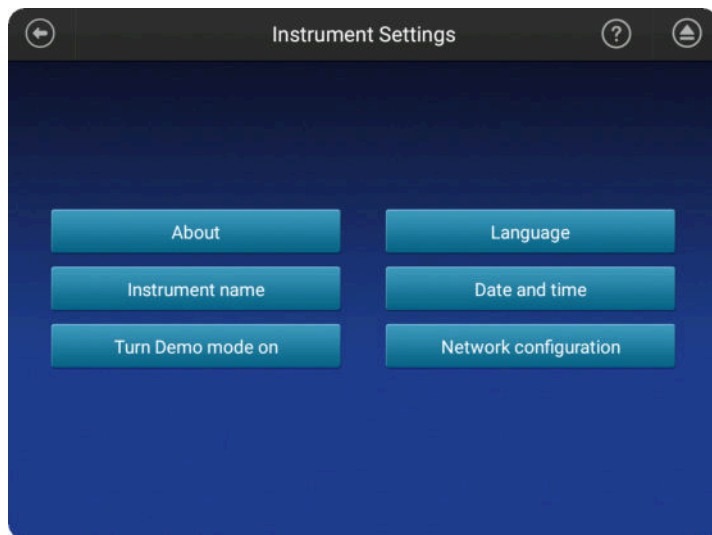
2. Touch **Edit**, or touch one of the network settings fields.
3. (Optional) Edit the **Wireless** network settings.
 - a. Select a **Network** that was automatically detected by instrument.
 - b. Enter a password, if prompted.
 - c. Click **Join**.
 - d. Click **OK** when the authentication is completed.
4. (Optional) Edit the **Wired** network settings.
 - a. Select **DHCP** or **Static IP**.
An IP address is automatically assigned if **DHCP** is selected.
 - b. (Static IP only) Enter an **IP address**, **Subnet mask**, **Default gateway**, **Primary DNS server**, and **Secondary DNS server**.
For more information, see “Determine IP address for a computer on a network” on page 144.
5. Touch **OK**.

Check for software updates (administrator only)

IMPORTANT! Do not perform a software upgrade on the instrument when a cartridge is loaded.

In the home screen:

1. Touch  **Settings** ▶ **Instrument settings** ▶ **About**.



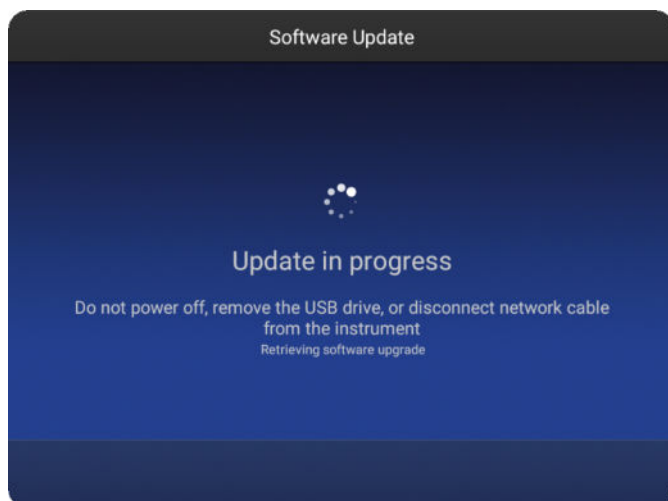
2. Touch **Check for updates**.

If the software update is located on a USB, it may take 10–15 seconds for the instrument to recognize the USB.

3. Touch:

- **Update** if an update is available.
- **Cancel** if there is no update available.

A message is displayed during the software updated, then the instrument automatically restarts.



Manage instrument profiles on the instrument

Local instrument profile roles and functions

Instrument profile	Location	Functions allowed
Standard	Local ^[1]	- Create, save, open, import, and run plate setups - Create and modify run settings - View and export results
Administrator	Local ^[1]	All standard user functions, plus: - Create or delete an instrument profile - Change an instrument profile from Standard to Administrator access - Reset an instrument profile PIN for another user - Unlock instrument touchscreen - Backup user data (plates and results) - Delete the Thermo Fisher™ Connect Platform instrument profile from the instrument, which removes the instrument from the InstrumentConnect. The user can link the instrument to the Thermo Fisher™ Connect Platform again using a local instrument profile. After a user relinks to the Thermo Fisher™ Connect Platform, the Thermo Fisher™ Connect Platform instrument profile is displayed on the home screen and the instrument is listed in the InstrumentConnect.
Guest	Local	All standard user functions, except: - Cannot link to the Thermo Fisher™ Connect Platform - Cannot modify a public plate setup Note: Standard instrument profiles cannot access Guest instrument profile plate setups unless the Plate Setup Security is set to Shared.

^[1] The first user who signs in to the instrument is assigned a local profile with administrator role.

For more information, see Chapter 3, “Use the instrument with the Thermo Fisher™ Connect Platform”.

Create a local instrument profile for another user (administrator only)

An instrument profile can be assigned standard or administrator roles. For more information, see “Change the role of a local instrument profile (administrator only)” on page 168.

In the home screen:

1. Touch .
2. Touch **All accounts**.
3. Touch **Add Profile**.
4. Touch **User name**, enter an instrument profile name, then touch **Done**.

5. Touch **PIN (4 digits required)**, enter a four-digit numerical PIN, then touch **Enter**.
6. Touch **Confirm PIN**, reenter the PIN, then touch **Enter**.
7. Touch **Create profile**.
8. Touch **Done**.

Change the role of a local instrument profile (administrator only)

In the home screen:

1. Touch .
2. Touch **All accounts**.
3. Touch the account of interest.
4. Slide the control from **Standard** to **Administrator** or from **Administrator** to **Standard**.
5. Touch **Done**.

Delete an instrument profile from an instrument (administrator only)

Local administrators and Thermo Fisher™ Connect Platform administrators can delete local and Thermo Fisher™ Connect Platform profiles.

IMPORTANT! Before proceeding, back up user data to retain plates and results (“Back up user data (plates and results) (administrator only)” on page 171).

In the home screen:

1. Touch .
2. Touch **All accounts**.
3. Touch the instrument profile to delete.
4. Touch **Delete account**.
5. Touch **Yes** to confirm.
6. Touch **Done**.

Delete the PIN for a local instrument profile (administrator only)

Local administrators and Thermo Fisher™ Connect Platform administrators can delete the PIN for a local instrument profile.

In the home screen:

1. Touch .
2. Touch **All accounts**.
3. Touch the instrument profile of interest.
4. Touch **Delete PIN**.
The user will be prompted for a new pin upon the next sign in.
5. Touch **Yes** to confirm.
6. Touch **Done**.

Create your own local instrument profile

If you are not signed in to the instrument when you link the instrument to your Thermo Fisher™ Connect Platform account, the software creates a local instrument profile with the **Standard** role using the *FirstName LastName* of your Thermo Fisher™ Connect Platform account.

For more information, see “Link the instrument to your Thermo Fisher™ Connect Platform account” on page 151.

Change your own local instrument profile PIN

Sign in to access these features (see “Sign in” on page 36).

1. Touch .
2. Touch **Edit**.
3. Touch **Old PIN**, enter your current PIN, then touch **Enter**.
Touch the **Show PIN** checkbox to switch the PIN display on or off.
4. Touch **PIN (4 digits required)**, enter a new four-digit numerical PIN, then touch **Enter**.
5. Touch **Confirm PIN**, reenter your new PIN, then touch **Enter**.
6. Touch **Done**.

Manage storage space

Automatic file cleanup

Before starting a run, the instrument calculates the total amount of storage space required to save the run to the instrument. If the required storage space is not available, the instrument deletes files associated with the oldest exported plates until sufficient space is available.

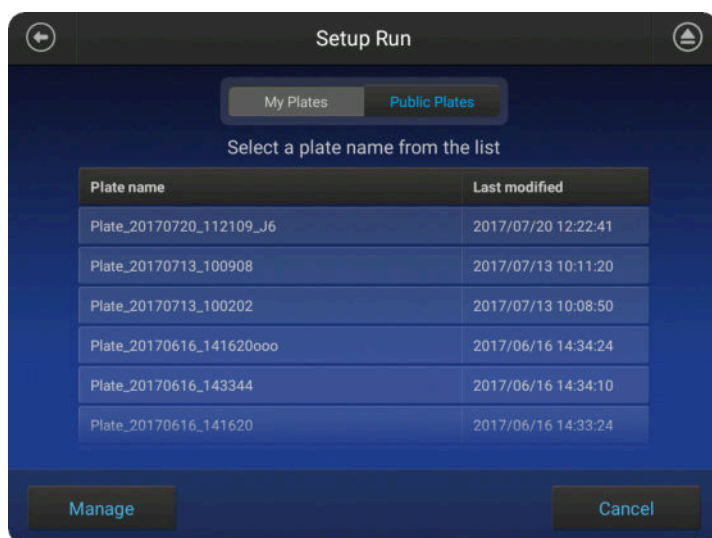
Note: Only complete plates that have been auto exported (saved to Thermo Fisher™ Connect Platform, network, or USB) or manually exported (using **Settings** ▶ **Run History** ▶ **plate name** ▶ **Export**) are deleted.

If the required storage space is not available and no plates have been exported, the instrument displays a notification indicating that there is not enough storage space.

You can export plates and delete plates, then start the run again.

Export or delete a plate setup (PSM file)

1. Touch **Set up run**, then touch  **My instrument**.
2. Touch **Manage** at the bottom left of the screen.



3. Touch a plate, then touch **Export** or **Delete**.
4. If you touched:
 - **Export**, select a save location, then touch **Export**.
 - **Delete**, then touch **Yes** to delete the plate setup.

Export results from the instrument (sample data files and QC reports)

In the home screen:

1. Touch  **Settings** ▶ **Run history**.
2. Select one or more plates from the **Run History** table.
3. Touch **Export**.
4. Select a save location.

The following data is exported for the plate:

- Fragment/HID analysis—FSA file for each sample.
- Sequencing—AB1 file for each sample.
- Plate QC report in CSV and PDF format.

Note: You can also export sample data if you select a plate, then select **View** ▶ **Export**. If you select a plate, only an FSA or AB1 file for each analyzed sample is exported.

Note: If sample data files (AB1 and FSA) are not exported to the expected save location (Cloud, Network Drive, and/or USB), you can open the **Export Status** screen to view failed exports at the plate- or sample-level. You can also re-export the files from the **Export Status** screen. See [page 131](#).

Delete a run history

In the home screen:

1. Touch  **Settings** ▶ **Run history**.
2. Select one or more plates from the Run History table.
3. Touch **Delete**, then touch **OK** to confirm.

Note: Run histories for the oldest exported plates are automatically deleted if sufficient storage space is not available when you start a run.

Back up user data (plates and results) (administrator only)

1. Select **Settings** ▶ **Maintenance and service** ▶ **Back up user data**.
2. Enter a name for the folder that will contain the backup data.
3. Touch the instrument profiles for which you want to back up plate and result information.
4. Touch **Backup**.
5. Specify a storage location, then touch **Export**.
A folder containing plate setup and sample data file is backed up to the specified location.

(Optional) After backing up, you can:

- Delete the plates (see “Delete a run history” on page 171).
- Open the CSV files in the folder on another instrument or in the Plate Manager.

■ Regular maintenance tasks	173
■ Manage the cartridge	175
■ Install cathode buffer	183

Regular maintenance tasks

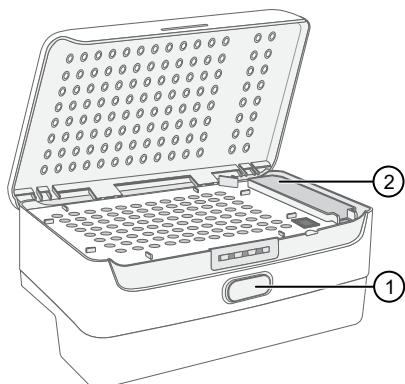
Check the cathode buffer fill level

When a cartridge is installed in the instrument, the volume of buffer in the SeqStudio™ Cathode Buffer Container (CBC) must be above the fill line.

Check the cathode buffer fill level before each run. It is recommended to check one time each week if the instrument is not in use.

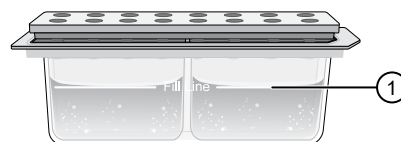
In the home screen:

1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid, then remove the CBC.



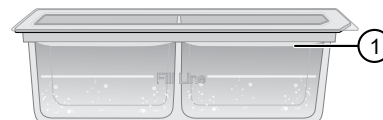
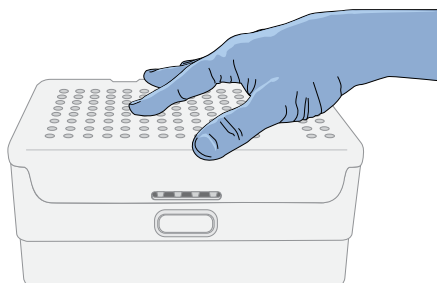
- ① Release button
② CBC

3. Ensure that the level of buffer is above the fill line.
If the buffer is at or below the fill line, see “Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)” on page 183 and “Insert the SeqStudio™ Cathode Buffer Container” on page 184.



- ① Replace if buffer is at or below the fill line

4. Reinstall the CBC.
5. Close the autosampler lid: Press down on the center of the lid or press down on both sides of the lid with equal pressure until the lid clicks shut.



① New CBC buffer level

6. Touch **Retract plate**, close the instrument door, then touch **Done** when the **Consumables Status** screen is displayed.

Clean the instrument exterior and touchscreen

Power off the instrument before cleaning.

IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.

- Wipe the exterior of the instrument with a lint-free cloth and deionized water.

IMPORTANT! Do not allow any moisture to reach the interior of the instrument through the door.

- Wipe the touchscreen with a lint-free cloth and glass cleaner.

Clean the autosampler

The autosampler is attached to the instrument.

Clean spills to prevent a build-up of crystallized polymer or dried salt from the buffers.

Power off the instrument before cleaning the autosampler.

1. Wipe the exterior of the autosampler with a lint-free cloth and deionized water.

IMPORTANT! Do not pour liquid directly on to the autosampler. This will damage the heater at the bottom of the cathode buffer reservoir.

Note: Do not use detergents or solvents.

2. Use a lint-free cloth to absorb any liquid spilled in the individual wells of the autosampler.

Manage the cartridge

Cartridge storage

Table 10 Storage information for the SeqStudio™ Genetic Analyzer Cartridge (Cat. No. [A33671](#))

Condition	Description
Shipping	Shipped at 2–8°C. Store upright at 2–8°C upon receipt. Save the white storage box and optical cover for off-instrument cartridge storage.
On-instrument storage	For routine use, can be used and stored on the instrument for up to 4 months. If you store the cartridge on-instrument: • The instrument must be powered on. • A SeqStudio™ Cathode Buffer Container must also be installed. The instrument keeps the components under the following conditions when it is powered on and in Cartridge storage mode: • Optical detection window—Covered • Capillary array electrodes—Submerged in cathode buffer (buffer must be above the fill line in the SeqStudio™ Cathode Buffer Container) • Polymer—Chilled • Anode buffer—Ambient temperature IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.
Off-instrument storage	For intermittent use, can be stored off-instrument until the expiry date on the label or up to 4 months after first use. Store upright at 2–8°C, with an integrated capillary protector (ICP) and optical cover installed (see “Store the cartridge” on page 180). Note: After you remove the cartridge from the instrument, install an ICP within a few minutes. Avoid cartridge exposure to ambient temperature.
Reuse	Can be removed from an instrument then inserted again on the same instrument or a different instrument, if it was stored properly at 2–8°C and has not expired or exceeded 125 injections. Information about the cartridge installation and usage is retained in the cartridge history (Settings ▶ Cartridge ▶ Instrument–cartridge history).

Table 11 Storage information for the SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. [A41331](#))

Condition	Description
Shipping	Shipped at 2–8°C. Store upright at 2–8°C upon receipt. Save the white storage box and optical cover for off-instrument cartridge storage.
On-instrument storage	For routine use, can be used and stored on the instrument for up to 6 months. If you store the cartridge on-instrument: • The instrument must be powered on. • A SeqStudio™ Cathode Buffer Container must also be installed. The instrument keeps the components under the following conditions when it is powered on and in Cartridge storage mode: • Optical detection window—Covered • Capillary array electrodes—Submerged in cathode buffer (buffer must be above the fill line in the SeqStudio™ Cathode Buffer Container) • Polymer—Chilled • Anode buffer—Ambient temperature IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.
Off-instrument storage	For intermittent use, can be stored off-instrument until the expiry date on the label or up to 6 months after first use. Store upright at 2–8°C, with an integrated capillary protector (ICP) and optical cover installed (see “Store the cartridge” on page 180). Note: After you remove the cartridge from the instrument, install an ICP within a few minutes. Avoid cartridge exposure to ambient temperature.
Reuse	Can be removed from an instrument then inserted again on the same instrument or a different instrument, if it was stored properly at 2–8°C and has not expired or exceeded 250 injections. Information about the cartridge installation and usage is retained in the cartridge history (Settings ▶ Cartridge ▶ Instrument–cartridge history).

Set cartridge storage mode (administrator only)

If a cartridge is installed on the instrument, the instrument will enter **Cartridge storage mode** after the time you specify.

The instrument keeps the components under the following conditions when it is powered on and in **Cartridge storage mode**:

- **Optical detection window**—Covered
- **Capillary array electrodes**—Submerged in cathode buffer (buffer must be above the fill line in the SeqStudio™ Cathode Buffer Container)
- **Polymer**—Chilled
- **Anode buffer**—Ambient temperature

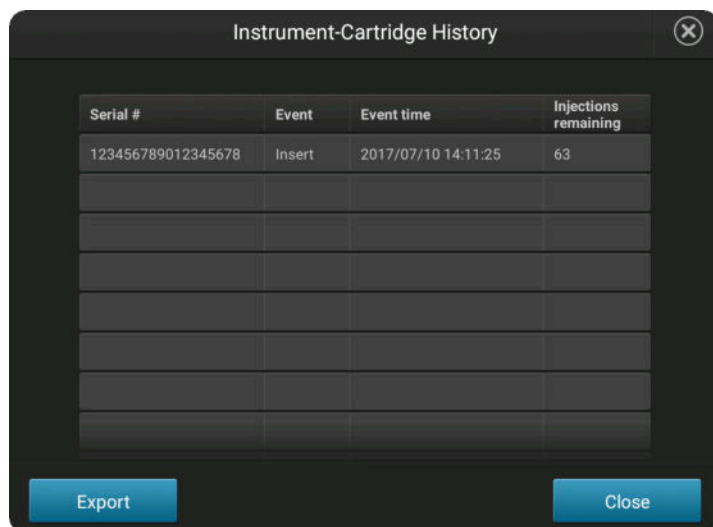
In the home screen:

1. Touch ⓘ **Settings** ▶ **Cartridge** ▶ **Cartridge Storage Mode**.
2. Select the duration of instrument inactivity before the instrument enters Cartridge Storage Mode.
3. Touch **OK**.

Check or export the Instrument-cartridge history

This function lists the cartridges that have been installed on this instrument.

1. Touch ⓘ **Settings** ▶ **Cartridge** ▶ **Instrument-cartridge history**.



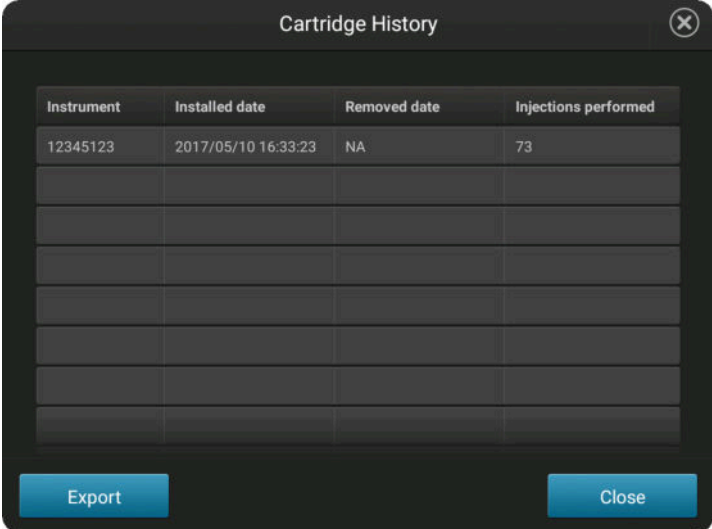
Serial #	Event	Event time	Injections remaining
123456789012345678	Insert	2017/07/10 14:11:25	63

2. (Optional) Touch **Export**.

Check or export the cartridge history

This function lists the instruments on which the cartridge has been installed.

Touch  **Settings** ▶ **Consumables status**, then touch **Cartridge history**.



Instrument	Installed date	Removed date	Injections performed
12345123	2017/05/10 16:33:23	NA	73

Fill the capillary array and refresh the polymer delivery system

The functions accessed from **Cartridge maintenance** are performed automatically during a run. Do not use these commands to manually perform these functions unless instructed to do so in troubleshooting or by Support.

In the home screen:

1. Touch  **Settings** ▶ **Cartridge** ▶ **Cartridge maintenance**.
2. Select **Fill array** or **Refresh PDS**.
3. Touch  to close the screen when the function ends.

Remove the cartridge

Perform these steps if you are replacing the cartridge or storing it off-instrument. Removal of the cartridge after each run is not required.

For appropriate off-instrument storage conditions, see the following tables:

- For the SeqStudio™ Genetic Analyzer Cartridge, see Table 1 on page 18
- For the SeqStudio™ Genetic Analyzer Cartridge v2, see Table 2 on page 19

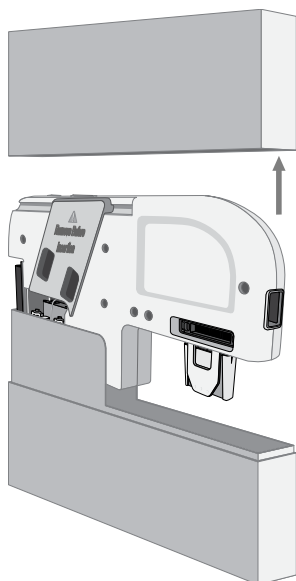
1. Touch , touch  **Eject cartridge**, then open the instrument door when prompted.
2. Hold the cartridge at the hand hold above the capillaries, then pull to remove it from the instrument.
3. Close the instrument door.

For more information, see “Store the cartridge” on page 180.

Insert the cartridge

1. If a cartridge is installed:
 - a. Touch , touch  **Eject cartridge**, then open the instrument door when prompted.
 - b. Remove the used cartridge.
2. Prepare the cartridge.
 - a. Remove the cartridge from the box.

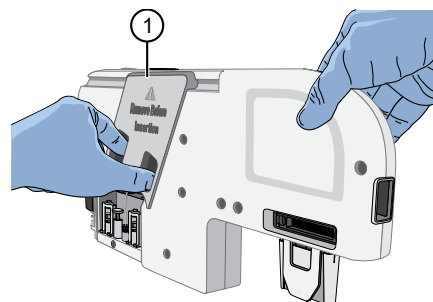
Note: Save the box for off-instrument cartridge storage.



- b. Remove the optical cover from the cartridge by grasping the finger holds, then pulling toward you.

Note: Save the optical cover for off-instrument cartridge storage.

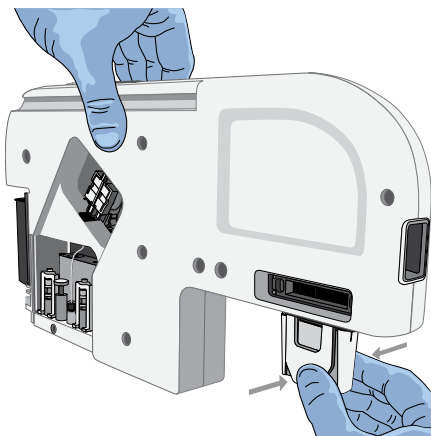
Avoid touching the capillaries that are protected by the optical cover.



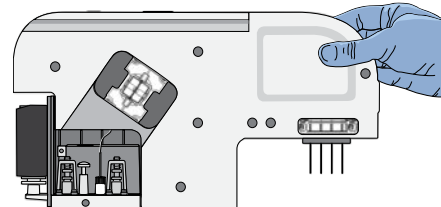
① Optical cover

- c. Pinch the clamp on the SeqStudio™ Integrated Capillary Protector, then pull down.

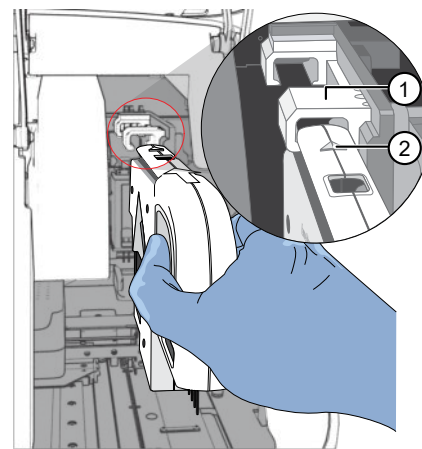
Discard the SeqStudio™ Integrated Capillary Protector. The SeqStudio™ Integrated Capillary Protector is single-use. Use a new SeqStudio™ Integrated Capillary Protector if you store the cartridge.



3. Grasp the cartridge above the capillaries.



4. Position the cartridge:
 - a. Orient the cartridge with the embossed arrow pointing toward the rear of the instrument.
 - b. Align the guides at the top of the cartridge with the insertion rails in the instrument.
5. Slide the cartridge into the instrument until it clicks into place.
6. Close the instrument door, then touch **Done** when the **Consumables Status** screen is displayed.
The cartridge is initialized.

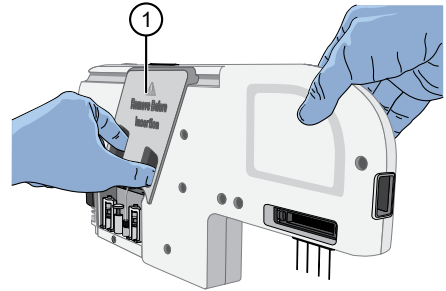


- ① Insertion rails
- ② Embossed arrow

Store the cartridge

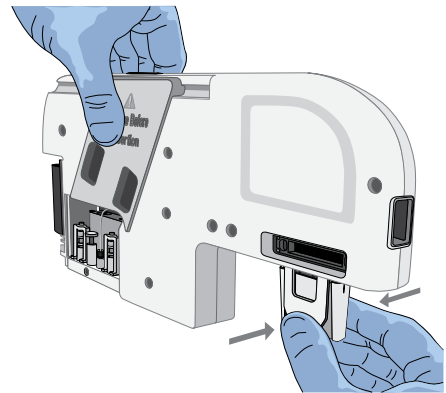
1. Touch , touch  **Eject cartridge**, then open the instrument door when prompted.
2. Hold the cartridge at the hand holds above the capillaries, then pull to remove it from the instrument.
3. Close the instrument door.

4. Carefully place the optical cover on the cartridge to avoid damage to the capillaries.



① Optical cover

5. Place a new SeqStudio™ Integrated Capillary Protector on the cartridge: pinch the clamp on the ICP, then push up on to the capillaries.
6. Place the cartridge in the white storage box.
7. Store upright at 2–8°C.



Note: Avoid cartridge exposure to ambient temperature.

IMPORTANT! Do not freeze the cartridge. It cannot be used after freezing.

SeqStudio™ Integrated Capillary Protector

The SeqStudio™ Integrated Capillary Protector (ICP) is a single-use protective cover that clamps onto the SeqStudio™ Genetic Analyzer Cartridge or the SeqStudio™ Genetic Analyzer Cartridge v2. The ICP prevents the capillary array from drying out during off-instrument storage of the cartridge, and is removed before insertion of the cartridge into the SeqStudio™ Genetic Analyzer.

Each cartridge is supplied with an ICP for shipping and one additional ICP for off-instrument cartridge storage.

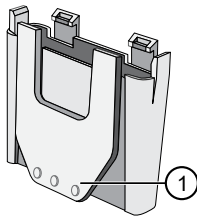


Figure 20 SeqStudio™ Integrated Capillary Protector

- ① Clamp mechanism for attaching to cartridge

IMPORTANT! Remove the Integrated Capillary Protector before installing the cartridge into the instrument. Installing the cartridge with the ICP in place can damage the capillary array.

SeqStudio™ Integrated Capillary Protector storage

Condition	Description
Shipping	Is shipped at 2–8°C. Before opening, can be stored until expiry date on label at 2–8°C. Discard the shipping ICP when you insert a new cartridge in the instrument.
Reuse	Do not reuse. Use a new ICP for off-instrument cartridge storage. IMPORTANT! If an ICP is used for off-instrument cartridge storage more than one time, the capillary array will not be kept fully hydrated.

Long-term on-instrument cartridge storage

You can leave the cartridge installed in the instrument if the instrument is powered on and a CBC is installed. Replace the CBC every 2 weeks.

IMPORTANT!

- If the instrument will be idle for longer than the shelf life of the cartridge components, remove the cartridge and cathode buffer, then power off the instrument.
- Do not power off the instrument when a cartridge is loaded.

Table 12 Cartridge storage

Cartridge	Storage
SeqStudio™ Genetic Analyzer Cartridge (Cat. No. A33671)	See Table 1 on page 18
SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. A41331)	See Table 2 on page 19

IMPORTANT! The instrument does not maintain the correct temperature conditions for the cartridge when it is powered off. Avoid cartridge exposure to ambient temperature.

Install cathode buffer

SeqStudio™ Genetic Analyzer Cathode Buffer Container storage

Condition	Description
Shipping	Is shipped at ambient temperature. Store at 2–8°C upon receipt.
On-instrument storage	After installation, can be stored for up to 2 weeks when the instrument is powered on and in Cartridge storage mode .
Off-instrument storage	Before opening, can be stored until expiry date on label at 2–8°C.
Reuse	Do not remove the CBC from the instrument for storage. Do not reuse.

Remove the SeqStudio™ Genetic Analyzer Cathode Buffer Container

Perform these steps if you are replacing the SeqStudio™ Cathode Buffer Container. It is not necessary to remove the CBC after each run.

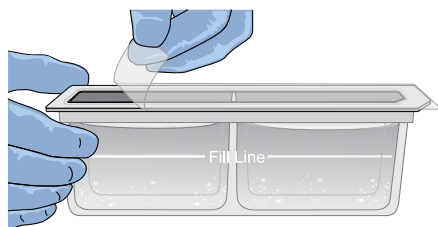
1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid.
3. Lift the CBC out of the autosampler.

Assemble the SeqStudio™ Genetic Analyzer Cathode Buffer Container (CBC)

Equilibrate the CBC to room temperature (15 minutes to overnight) before assembling.

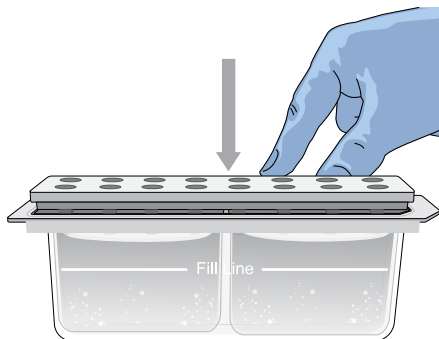
On a clean and level surface:

1. Ensure that the buffer is above the fill line.
2. Carefully peel off the seal.



3. Wipe off any buffer on top of the CBC with a lint-free tissue. Ensure that the top of the container is dry.

4. Place the reservoir septa on the CBC, then press firmly to seat the septa.



Note: The CBC is filled significantly above Fill Line to account for evaporation. Replace the CBC when the fluid level is at or below the fill line.

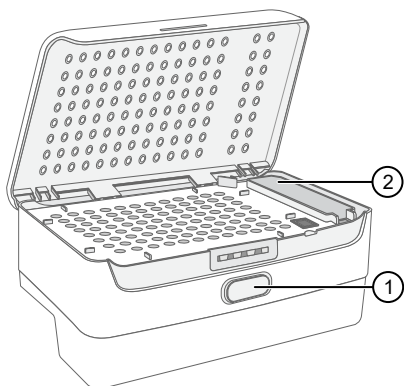
Insert the SeqStudio™ Cathode Buffer Container

Perform these steps if the SeqStudio™ Cathode Buffer Container has not been installed, if the CBC on the instrument has expired, or if the buffer level is at or below the fill line.

Install the CBC before you install the cartridge.

See Figure 2 on page 14 for the position of the cathode buffer on the instrument.

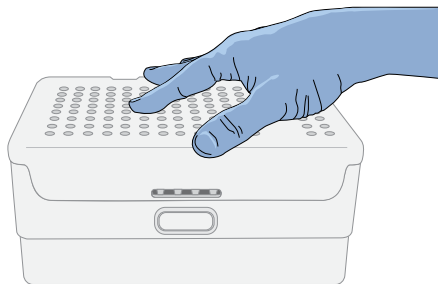
1. Touch , touch  **Eject plate**, then open the instrument door when prompted.
2. Press the release button on the autosampler to open the lid.



- ① Release button
- ② Location of CBC

3. Insert the cathode buffer in the autosampler with the notch positioned in the back right.
See Figure 8 on page 20.

4. Press the autosampler lid until it clicks shut.



5. Touch **Retract plate**, then close the instrument door.



Troubleshooting

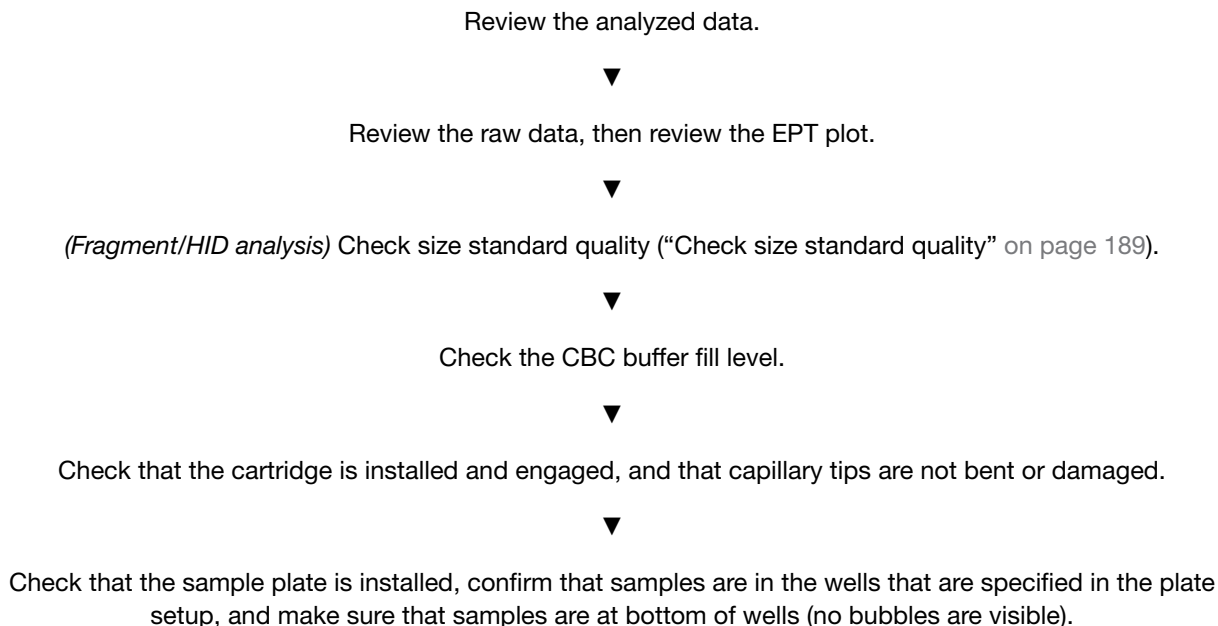
■ Troubleshooting resources	186
■ Troubleshooting workflow	187
■ Export log files for plates, install runs, injections, and instrument	187
■ View the raw data and the EPT plot	188
■ Data quality alerts	188
■ Sizecalling and basecalling quality alerts	189
■ Check size standard quality	189
■ Instrument troubleshooting	190
■ Cartridge troubleshooting	192
■ Sample and data troubleshooting	193
■ EPT data	199
■ Software troubleshooting	199
■ Service tools	202

Troubleshooting resources

Document	Pub. No.
<i>Troubleshooting Sanger sequencing data User Bulletin</i>	MAN0014435
<i>DNA Sequencing by Capillary Electrophoresis Chemistry Guide Second Edition</i>	4305080
<i>DNA Fragment Analysis by Capillary Electrophoresis User Guide</i>	4474504

Troubleshooting workflow

Follow this general workflow when you are troubleshooting:



Export log files for plates, install runs, injections, and instrument

Export log files if directed to do so by Technical Support.

In the home screen:

1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Export logs**.
2. Select one of the following options.
 - **Export logs for the last injection**—Includes all run types: analysis run, install run, and calibration run.
 - **Export logs for the last / current plate**—Includes all run types.
 - **Export logs for selected plates**—Includes options to export logs for regular runs and install check runs for the analysis run for the current user.
 - **Export recent instrument server logs**—Includes instrument export logs only.
3. Touch **Export**.
4. Select a storage location.

Note: Log files can be large. Select a location with adequate storage space.

A ZIP file containing log information is exported.

View the raw data and the EPT plot

The EPT view (ElectroPhoresis Telemetry) shows instrument data conditions (currents, temperatures, electrophoresis voltage) as a function of time.

When a run is complete, in the home screen:

1. Touch **Results**.
2. Touch **List view**.
3. Touch an injection group.
4. Touch a sample file name.
If the data triggered any quality alerts, a QC alerts screen is displayed.
Click **View data** to display the trace for the sample.
5. Touch **>** or **<** to scroll to the raw data or **EPT Plot**.

Data quality alerts

Quality alert	Description	Action
Offscale peaks. Adjust the injection parameters and/or the sample concentration.	At least 10 scans have saturated the CCD camera.	• Reduce the injection voltage or time. • Dilute the sample.
No sample was detected.	Poor signal-to-noise ratio with low signal detected.	• Verify that the sample volume follows recommendations in the user manual. • Troubleshoot upstream PCR and sequencing steps.

Sizecalling and basecalling quality alerts

Table 13 Sizecalling quality alerts

Quality alert	Description	Action
Sizing quality value is low due to poor size standard peak quality. Peak height uniformity is low or the fitting quality in sizing is poor.	Low resolution or poor quality data is present.	- Re-inject the sample. - If the problem persists, check the sample quality.
Sizecaller found broad peak(s) in the size standard peak(s).		
Sizing quality value is in the intermediate range; check size standard data quality.		
The number of size standard peaks detected is less than what is defined in the size standard.	Size standard definition includes peaks that are not present in the sample. Example: Sample peaks are detected up to 500 bp, but the size standard definition includes peak sizes that are >500 bp.	Use or create a size standard definition with the appropriate number of peaks and peak sizes.
The analysis range is too small. Correct the analysis range in analysis settings and re-analyze in secondary analysis software or re-inject sample.	Various causes.	- Analyze the data in a secondary analysis software with a corrected analysis range. - Re-inject the sample.

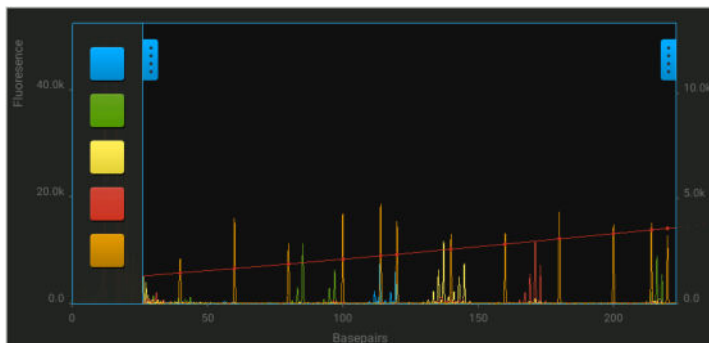
Table 14 Basecalling quality alerts

Quality alert	Description	Action
Basecalling failed due to poor quality data.	Poor quality data is present.	- Re-inject the sample. - If the problem persists, prepare fresh sample. - Troubleshoot upstream PCR and sequencing steps.

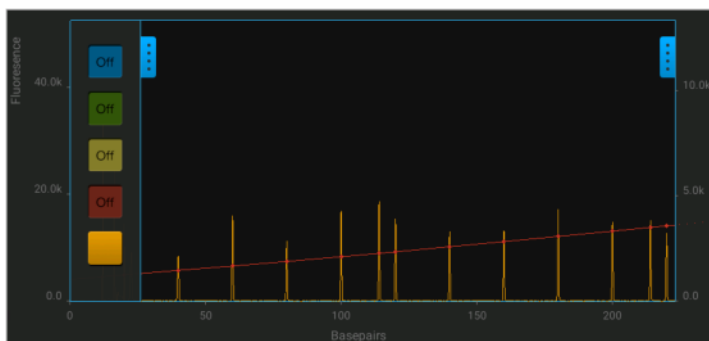
Check size standard quality

1. Touch  **Settings** ▶ **Run history**.
2. Touch a plate name, then touch **View**.
3. Touch a sample file name, then touch **View**.
If the data triggered any quality alerts, a QC alerts screen is displayed.

4. Touch **View data** to display the trace.



5. Touch  on the left border of the trace, then deselect all dyes except the size standard dye (red or orange).



6. As needed, touch  on the right border of the trace to zoom on the trace.

Instrument troubleshooting

Observation	Possible cause	Recommended action
There was a loss of power to the instrument	There was a power failure.	Restart the instrument. The Sign In screen is displayed. A run that was in progress at the time of the power failure must be restarted.
		Run a control sample to ensure that the consumables have not degraded, especially if the consumables reached room temperature.
		Replace the consumables if the results with a control sample show that the consumables have degraded.
		Inspect the cartridge syringe plunger and ensure that it is fully depressed.

Observation	Possible cause	Recommended action
The RFID tags on the cartridge or the cathode buffer are not read	The label on the cartridge or the SeqStudio™ Cathode Buffer Container is damaged, not positioned properly, or has been removed.	Ensure that the labels are present and not visibly damaged. Ensure that the consumables are installed correctly. The SeqStudio™ Cathode Buffer Container should be firmly seated in the autosampler. The cartridge will click into place when it is installed correctly.
	The RFID reader is malfunctioning.	Insert consumables that have previously had RFID tags read. Contact technical support if the RFID tags are not read.
The run stopped and Resume is displayed	A user paused the run.	Touch Resume to continue the run.
The amber light is blinking	The run was paused.	Touch Resume to continue the run.
	The instrument door is open.	Close the instrument door.
	There was an instrument error.	Follow the instructions in the error message.
		Restart the instrument.
The electrophoresis failed or Current check failed is displayed	There is insufficient cathode buffer.	Check the fill line on the SeqStudio™ Cathode Buffer Container. Replace the SeqStudio™ Cathode Buffer Container if the buffer is at or below the fill line.
	The SeqStudio™ Cathode Buffer Container has been installed on the instrument for more than two weeks or used for more than 125 injections.	Replace the SeqStudio™ Cathode Buffer Container.
	The septum on either the SeqStudio™ Cathode Buffer Container or the sample plate is not installed correctly.	Ensure that the septa are fully inserted into both the SeqStudio™ Cathode Buffer Container and the sample plate.
	Buffer or other liquid was spilled on top of the reservoir septum or on top of the autosampler.	Wipe the spill with a lint-free cloth.
	There is condensation on the SeqStudio™ Cathode Buffer Container or around the reservoir septum.	Wipe the condensation with a lint-free cloth.
		Ensure that the humidity in the lab is non-condensing.

Cartridge troubleshooting

Observation	Possible cause	Recommended action
There are crystals on the cartridge capillary array	Small amounts of leakage around the cartridge capillary array are normal.	No action is required.
There are crystals on the cartridge polymer delivery system	There is a leak in the polymer delivery system.	Run a control sample to determine if the cartridge function is affected.
		Replace the cartridge if the leak affects the cartridge function.
Poor-quality data is observed after prolonged drying of the capillary tips	The capillary tips develop blockages if they are allowed to dry out.	Use an Integrated Capillary Protector when the cartridge is off the instrument to prevent the capillary tips from drying out.
		Clear the blockage by running polymer through the capillaries. 1. Touch Settings ▶ Cartridge ▶ Cartridge maintenance ▶ Fill array. 2. Touch when the function ends. 3. Run a control sample. 4. Replace the cartridge if there is poor-quality data with the control sample.
		Follow the recommendations for cartridge storage to prevent drying of the capillary tips. See “Cartridge storage” on page 18.
Cartridge failed optical alignment	There was a sporadic cartridge engagement failure.	Remove the cartridge, then re-insert the cartridge into the instrument.
	The cartridge had optical contaminants.	Refresh the polymer delivery system (PDS). In the home screen, touch Settings ▶ Cartridge ▶ Cartridge maintenance ▶ Refresh PDS .
		Refill the capillaries. In the home screen, touch Settings ▶ Cartridge ▶ Cartridge maintenance ▶ Fill array .
	The was a communications error within the system.	Remove the cartridge, then restart the instrument.
	The cartridge was damaged.	Inspect the cartridge for damage. Replace the cartridge if there is damage, or if alignment failures continue.

Sample and data troubleshooting

See also “Data quality alerts” on page 128, “Sizecalling and basecalling quality alerts” on page 128, and *Troubleshooting Sanger sequencing data User Bulletin* (Pub. No. MAN0014435).

Observation	Possible cause	Recommended action
Offscale signal is detected is displayed Note: Samples are stable on the instrument for 16–24 hours. Determine if samples will be stable if a re-injection is recommended, then plan the re-injection accordingly.	The sample concentration is too high.	Dilute the sample, prepare a new plate, and start a new run. Quantitate the sample prior to adding reagents for capillary electrophoresis.
	The injection conditions are too strong for the sample.	Re-inject the samples with adjusted injection conditions. Injection time, voltage, or a combination of these conditions can be adjusted.
There is no signal or a low signal Note: Samples are stable on the instrument for 16–24 hours. Determine if samples will be stable if a re-injection is recommended, the plan the re-injection accordingly.	The sample volume was insufficient.	Use the recommended sample volume. See “Sample preparation guidelines” on page 38.
	The sample concentration was too low.	Re-inject the samples with adjusted injection conditions. Injection time, voltage, or a combination of these conditions can be adjusted.
	There were bubbles in the sample wells.	Centrifuge the sample plate or tubes to remove the bubbles before loading onto the instrument.
		If the samples have been run, centrifuge the plate or tubes, then set up a re-injection.
	The sequence reaction failed.	Review the sequence analysis protocol, the template quality, and the template quantity. Set up a new plate and repeat the reaction. See <i>Troubleshooting Sanger sequencing data User Bulletin</i> (Pub. No. MAN0014435).
	The Hi-Di™ Formamide used to prepare the samples was degraded.	Prepare the samples with fresh Hi-Di™ Formamide and repeat the experiment. See “Sample preparation guidelines” on page 38.
	The sample was prepared with the BigDye XTerminator™ Purification Kit but a BDX run module was not selected.	Re-inject the sample with the correct module.
	The sample was degraded.	Prepare the sample according to the protocol provided with the kits for sample preparation. See “Sample preparation guidelines” on page 38 for sample preparation guidelines. IMPORTANT! Do not resuspend samples in water.
		The sample had a high concentration of salt.
		Dilute or desalt the samples.

Observation	Possible cause	Recommended action
There is no signal or a low signal Note: Samples are stable on the instrument for 16–24 hours. Determine if samples will be stable if a re-injection is recommended, the plan the re-injection accordingly. (continued)	There was an excess of unlabeled template competing with the fragments labeled with dye during the injection.	Dilute or desalt the samples. See: • <i>DNA Sequencing by Capillary Electrophoresis Chemistry Guide Second Edition</i> (Pub. No. 4305080) • <i>DNA Fragment Analysis by Capillary Electrophoresis User Guide</i> (Pub. No. 4474504)
	A capillary array tip is blocked.	Flush the capillary array. 1. Touch Settings ▶ Cartridge ▶ Cartridge maintenance. 2. Touch Refresh PDS. 3. Touch to close the screen when the function ends.
		Replace the cartridge if the problem persists.
	The cartridge is damaged.	Inspect the cartridge for damage. Replace the cartridge if there is damage.
There was poor resolution, poor size quality, or a poor sequencing result	The samples degraded over time while on the instrument.	Analyze a maximum of 48 samples on a plate for long run modules. Additional samples will take more than 24 hours and the samples can degrade.
		Note: Thermal breakdown of samples is normal. Samples are stable for 16–24 hours on the instrument. Use Hi-Di™ Formamide to prepare the samples. Sample stability is optimal in Hi-Di™ Formamide.
	The temperature and/or humidity in the lab is too high for optimal sample stability.	Ensure that the conditions in the lab are within the operating range for the instrument (15–30°C and 20–80% relative humidity). Sample stability can be lower near the high end of the operating range of the instrument for temperature and humidity.
		Use 20 µL of sample instead of 10 µL. A larger sample volume can reduce sample breakdown under hot and humid conditions.
Poor resolution in some capillaries	There was a sporadic data quality failure.	Repeat the injection.
		Note: A sporadic data quality failure can happen occasionally.
Poor resolution in some capillaries	Poor-quality samples were used.	See “Sample preparation guidelines” on page 38.
		Use a control sample to determine if the poor resolution is due to the samples are another factor.

Observation	Possible cause	Recommended action
Poor resolution in some capillaries (continued)	A capillary was damaged or is blocked.	Try to clear the capillary, then replace the cartridge if the poor resolution is due to the cartridge and not the sample. 1. Touch Settings ▶ Cartridge ▶ Cartridge maintenance ▶ Fill array. 2. Touch when the function ends. 3. Run a control sample.
Poor resolution in all the capillaries	The cartridge has been used for more than the stated number of injections, is past the labeled expiration date, or has degraded polymer from incorrect storage.	Replace the cartridge.
	The Hi-Di™ Formamide that was used to prepare the samples was degraded.	Prepare the samples with fresh Hi-Di™ Formamide and repeat the reaction. See: • “Sample preparation guidelines” on page 38 • <i>DNA Fragment Analysis by Capillary Electrophoresis User Guide</i> (Pub. No. 4474504)
	There was too much sample injected.	Dilute the sample and re-inject it.
Spikes are present in raw and/or analyzed fluorescence data	Trace impurities passed the detector.	For fragment analysis, increasing the Minimum peak half width setting in the analysis settings (under Common settings) can reduce the identification of spikes as peaks. Note: Secondary sequencing and fragment analysis software can recognize and ignore spikes.
		Repeat the injection if necessary.
There is noise in the baseline	Fluorescent contamination has built up in the CBC.	Replace the CBC. Use a new reservoir septum when assembling the new CBC.

Observation	Possible cause	Recommended action
Sample carryover from a previous injection	Sample carryover on the SeqStudio™ Genetic Analyzer can occur in trace amounts when previous injections are off-scale. Very strong peaks, for example, primer peaks in a fragment analysis sample, are typically visible in subsequent injections.	Replace the reservoir septum on the CBC before the next injection to minimize the carryover effect.
Dye blobs are seen in the sequencing data	Impurities remained in the sample after the sample purification. The impurities cause dye blobs to appear in the sequencing data.	Improve the sample purification method. See “Sample preparation guidelines” on page 38 for guidelines.
Extra peaks are present in the sequencing traces	There was renaturation of the sample.	Heat-denature the samples prepared with fresh Hi-Di™ Formamide, then immediately place the samples on ice.
	There is low signal. With very low signal, the peaks are barely visible in the baseline noise.	Check the raw data, the raw data signal intensity, and average raw signal-to-noise ratio, then: • Increase the injection time and reinject. Or • Remake the sample. Ensure that you are using: – Enough sequencing template – Enough primer and/or a sufficient concentration of primer
	There is a heterozygous insertion-deletion (het indel) that is causing multiple peaks to appear at the same basecall position. The sequence can appear "clean" for some number of bases until the het indel is encountered.	Examine the analyzed trace. A het indel typically has single peaks at the 5' end, then part-way through the trace, two peaks appear in almost every position to the end of the trace. This pattern occurs when one copy of the gene has an insertion or deletion relative to the other copy of the gene. When aligning your sequence to a reference sequence, a series of bases may have been inserted or deleted in an allele. These indels can be encountered in any number of bases after the gene-specific priming region. To confirm that the het indel is present in both directions of your target, check the sequencing in the opposite direction.

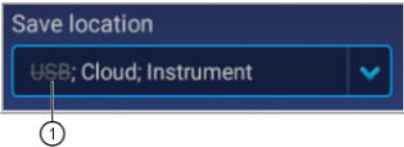
Observation	Possible cause	Recommended action
Extra peaks are present in the sequencing traces (continued)	Primer-dimer has occurred.	You can often diagnose primer-dimer by looking at the raw trace data for questionable sequences. When primer-dimer exists, the 5' sequence signal may be significantly higher for a region of bases spanning the length of the forward and reverse gene-specific primers. Primer-dimer is the annealing of the 3' end of primers during PCR. The resulting short annealed fragment may amplify more efficiently than fully extended template. Primer-dimer fragments amplified during PCR can display increased 5' signal and extra peaks when multiple PCR products are sequenced simultaneously. In some instances, the secondary or extra peaks can be read as the reverse compliment of the PCR primers in this noisy 5' region. The secondary sequence or multiple PCR product sequences appear as far as 100–200 bp into the sequence, then suddenly disappear.
	The PCR amplification primers do not have specificity and are sequencing two different regions of the genome. The analyzed trace shows extra peaks throughout the entire length of the trace.	Redesign the primers or increase the amplification temperature.
	You accidentally contaminated the DNA and are sequencing two templates at the same time. The analyzed trace shows extra peaks throughout the entire length of the trace.	Repeat the amplification and sequencing reactions with uncontaminated DNA.
	There were impure or contaminated primers.	Primer stocks may have inadvertently had other primer solution introduced. For best results, use HPLC to purify the primers.
	There was a contaminated sample well.	Use a new sample plate and buffer/wash septa whenever possible. To avoid getting sample into adjacent wells, centrifuge the plates before you remove the adhesive seal.

Observation	Possible cause	Recommended action
There are pull-up or pull-down peaks in the data	The manual dye calibration is not current or is not matched to the samples, or a high-quality sample was not run with the dye set for the first time in the absence of a manual calibration.	If the dye set is a fragment analysis dye set that is being used for the first time, run a manual dye calibration. Alternatively, if a high-quality sample is run with an uncalibrated dye set, and the automated spectral calibration successfully generates an optimized matrix, manual calibration is not absolutely required. See “Determine if manual calibration is required” on page 212.
	The incorrect dye set was selected in the Plate setup . Note: This is applicable to both sequence analysis and fragment analysis.	Correct the dye set and repeat the injection.
There are error messages about spectral issues Details: These error messages are seen more often in fragment analysis data than sequence analysis data.	A high-quality sample has not been run with the dye set.	If the dye set is a fragment analysis dye set that is being used for the first time, run a manual dye calibration. Alternatively, if a high-quality sample is run with an uncalibrated dye set, and the automated spectral calibration successfully generates an optimized matrix, manual calibration is not absolutely required. See “Determine if manual calibration is required” on page 212.
There is a short read length and uneven peak spacing in sequence data	The incorrect dye set was selected in the Plate setup .	Correct the dye set and repeat the injection.
		Select from the following: • E_BigDye™ Terminator v1.1 • Z_BigDye™ Terminator v3.1 • Z_BigDye™ Direct Reanalyze the data in the Sequencing Analysis Software, using the correct mobility file.
Fragment analysis peaks are sized differently than previously observed	Aging of the polymer in the cartridge, which can cause small (≤ 0.5 bp) changes in fragment size.	• Use a reference marker (for example, an allelic ladder) for auto bin adjustment. Or • Manually adjust the fragment bin positions to account for the size change.

EPT data

Observation	Possible cause	Recommended action
There is a fluctuating or unstable electrophoresis current, or Current check failed is displayed	There is a bubble in the polymer system.	Flush the capillary array. 1. Touch ⚙ Settings ▶ Cartridge ▶ Cartridge maintenance . 2. Touch Refresh PDS . 3. Touch ✕ to close the screen when the function ends. Replace the cartridge if the problem persists.
	The cartridge is damaged.	Inspect the cartridge for damage and replace the cartridge.

Software troubleshooting

Observation	Possible cause	Recommended action
Strikethrough text is displayed in Save location in Plate properties screen  ① Strikethrough text	The original location to which the plate was saved is no longer accessible by the instrument.	No action.
Import failed message when you select a CSV plate setup on the instrument Details: The message also indicates that a valid run module, dye set, and/or size standard is not present.	The CSV file specifies the name of a run module, dye set, and/or size standard that does not exist on the instrument. IMPORTANT! A CSV file contains only the name of the run module, dye set, and/or size standard, it does not contain the settings.	Add the run module, dye set, and/or size standard to the instrument by selecting a PSM file that contains the items, or by creating the items manually.
Import CSV fails but import PSM with same settings imports with no errors	The size standard, dye set, or run module specified in the CSV file does not exist on the instrument. The CSV file contains the <i>names only</i> of the size standard, dye set, or run module, it does not contain the settings. The PSM file contains the <i>names and settings</i> of the size standard, dye set, or run module. The size standard, dye set, or run module are automatically created when the PSM file is imported.	Create the size standard, dye set, or run module on the instrument.

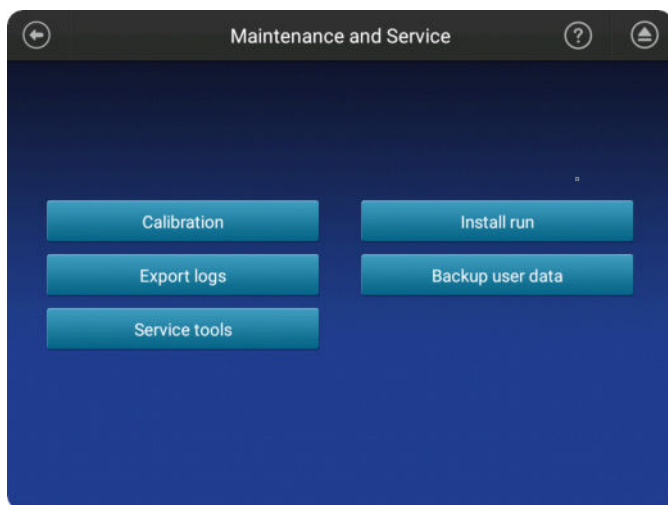
Observation	Possible cause	Recommended action
When saving a PSM file in Thermo Fisher™ Connect Platform, "You do not have edit permission to the cloud group" message is displayed	You are saving to a Thermo Fisher™ Connect Platform group and you do not have edit permissions for the group.	Save to a different location, or request edit permissions from the group administrator.

Observation	Possible cause	Recommended action
Analysis settings or run module is not available for selection by some users Details: This situation is seen on the instrument.	Analysis settings and run modules are saved per user. If the analysis settings or run modules are associated with a <i>hidden</i> plate setup, the analysis settings or run modules are not listed for selection unless the user who created the plate setup is signed in.	Change the plate setup security from Hidden to Shared (see “Hide or share a plate (Plate setup security)” on page 95).
		Create the analysis settings or run modules manually.
Your local instrument profile name is not available for sign in	Your local instrument profile is linked to your Thermo Fisher™ Connect Platform account and is replaced by your Thermo Fisher™ Connect Platform profile.	See “Local instrument profile roles and functions” on page 167.
		Sign in with your Thermo Fisher™ Connect Platform profile.
Cloud account is already linked to a local profile message	You have a local instrument profile that has previously been linked to Cloud account. You touched Get Started ▶ Connect, then typed in your Cloud account email and password.	Return to Sign in screen, then select your local profile from the Sign in list.
	Another user was signed in to the instrument. You selected  Cloud in the Set up run screen or in the Save location field in the Plate properties screen, then typed in your Cloud account email and password.	Return to Sign in screen, touch , touch Sign out, then select your local profile from the Sign in list.
The Notifications button is not displayed in Instrument Settings	You are signed in to the instrument with a local instrument profile.	Sign in to the instrument with a Thermo Fisher™ Connect Platform instrument profile.
A button is dimmed	The function is available to users with administrator role only.	None.
Sample data files from the SeqStudio™ Genetic Analyzer can be opened but not analyzed in secondary analysis software • Sequencing Analysis Software • GeneMapper™ Software • SeqScape™ Software • Variant Reporter™ Software	An updated version of the secondary analysis software is required.	Download and run the latest version of secondary analysis software. See “Secondary analysis software” on page 30. Note: The SeqStudio™ Genetic Analyzer instrument model is listed as 3200.
Files cannot be imported into Thermo Fisher™ Connect Platform applications or cannot be analyzed in Thermo Fisher™ Connect Platform sequencing apps	The sample name or file name used a special character.	Rename the sample name or file name. Note: Special characters (/ , & , @ , %) will impact file import into Thermo Fisher™ Connect Platform applications.

Observation	Possible cause	Recommended action
Unexpected error is displayed	There was a software error.	Follow the instructions in the error message.
		Export the log files to determine the potential source of the unexpected error. See “Export log files for plates, install runs, injections, and instrument” on page 187.
		Restart the system using the On/Off switch on the rear panel. See “Power on the instrument” on page 35.

Service tools

The **Service tools** function is password-protected and for service use only.





Link the instrument to your Thermo Fisher™ Connect Platform account— detailed instructions

■ Workflow: Set up the instrument for Thermo Fisher™ Connect Platform access	203
■ Network requirements	203
■ Link the instrument from a mobile device	204
■ Link the instrument using a link code	205
■ Set up email notifications from the instrument	209

Workflow: Set up the instrument for Thermo Fisher™ Connect Platform access

Register and obtain a Thermo Fisher™ Connect Platform account
(page 53)



Link the instrument to your Thermo Fisher™ Connect Platform account in any of the following ways:

- Link the instrument from a mobile device
(page 204)
- Link the instrument using a link code
(page 205)

Network requirements

The instrument is factory-configured for IPv4 TCP/IP communication and includes a fast Ethernet adapter (10/100 Mbps) with a RJ45-type connector for integrating the device into a local area network (LAN).

By default, the instrument is configured to use the Dynamic Host Configuration Protocol (DHCP) but can use a static IP address.

The instrument should be configured behind a firewall. Contact Support for information on required firewall exceptions.

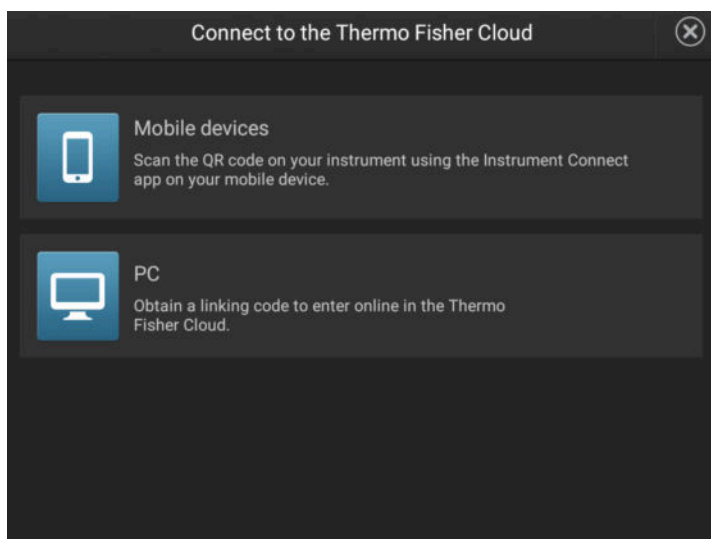
Link the instrument from a mobile device

Create a Thermo Fisher™ Connect Platform PIN and generate the QR code on the instrument

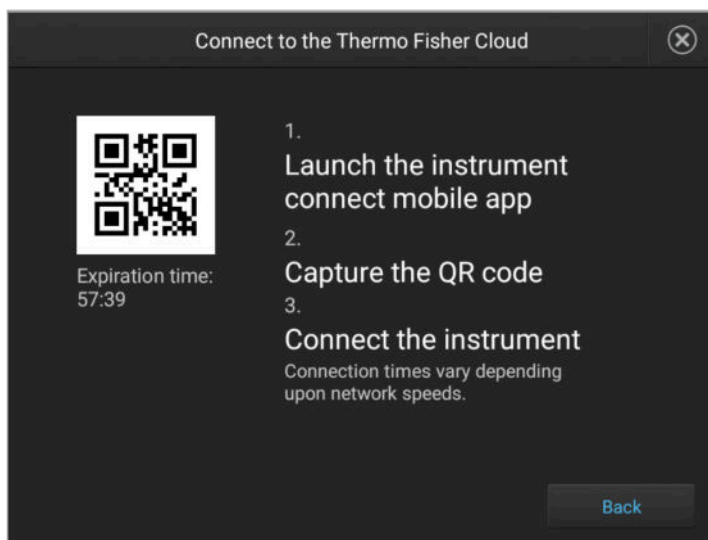
1. Sign in to thermofisher.com/connect.
2. Click  to access InstrumentConnect.
3. If you have not previously set up a PIN, click **Update PIN**, then enter a PIN that you will use to sign in to the instrument.
4. From the instrument **Sign In** screen, navigate to the **Connect to the Thermo Fisher Cloud** screen:

Do you have a local instrument profile?	Description
Yes	a. Touch Sign in , then enter your PIN. b. In the home screen, touch Setup run . c. In the Setup run screen, touch Cloud .
No	Touch Get started ► Connect .

5. Touch **Mobile devices**.



The QR code is displayed.



Register the instrument with the Instrument Connect App

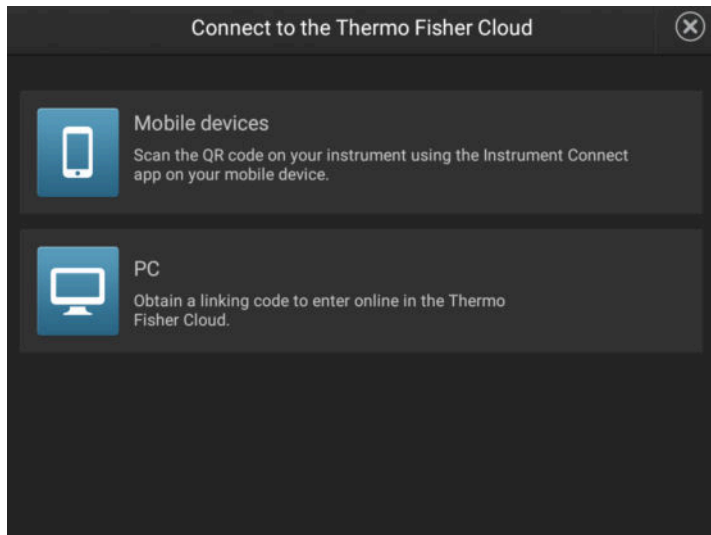
1. On your mobile device, download the InstrumentConnect from the Apple Store or from Google™ Play.
2. Launch, then sign in to the mobile app on your mobile device.
3. Register the instrument:
 - a. Touch , then touch **Register Instrument**.
 - b. Touch **QR code** on your mobile device.
 - c. With your mobile device, scan the QR code displayed in the instrument touchscreen.

Link the instrument using a link code

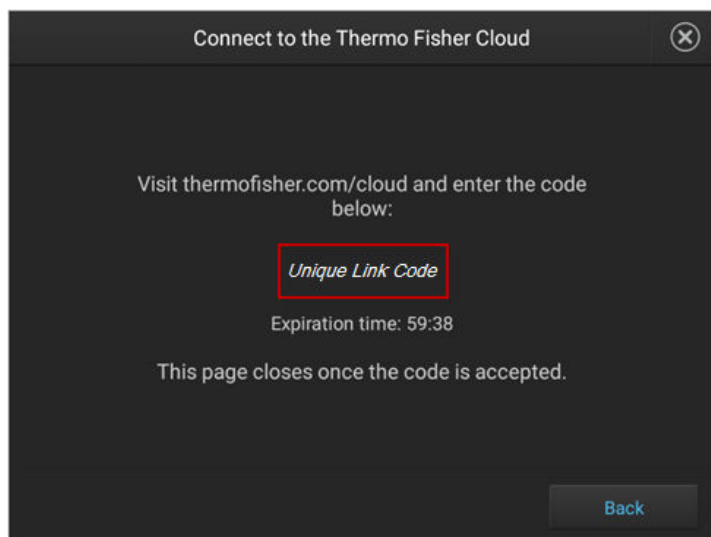
1. From the instrument **Sign In** screen, navigate to the **Connect to the Thermo Fisher Cloud** screen:

Do you have a local instrument profile?	Description
Yes	a. Touch Sign in, then enter your PIN. b. In the home screen, touch Setup run. c. In the Setup run screen, touch Cloud.
No	Touch Get started ▶ Connect .

2. Touch **PC**.

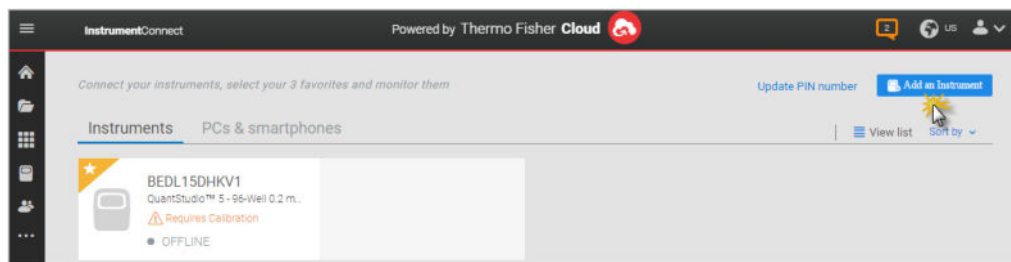
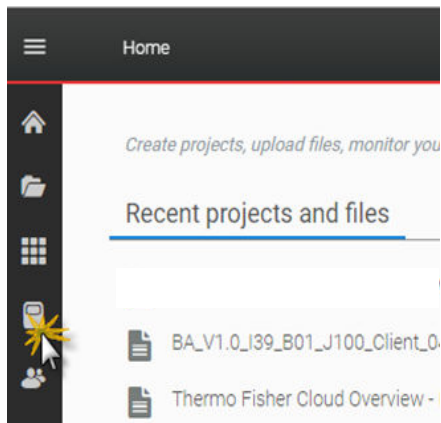


A unique link code is displayed.

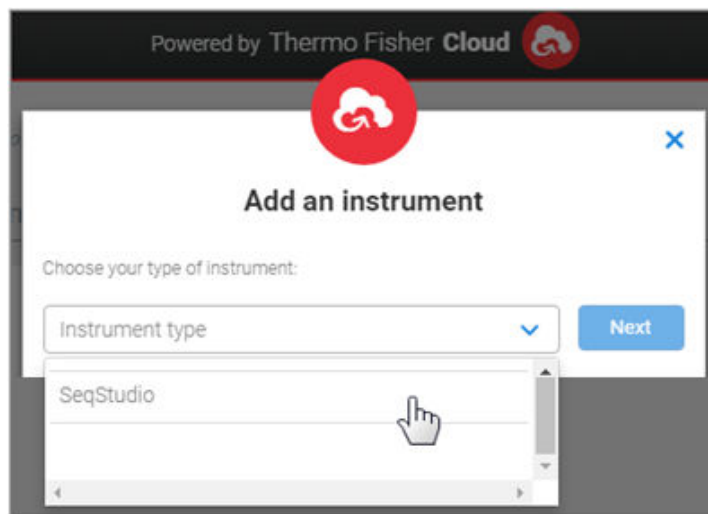


3. Sign into your Thermo Fisher™ Connect Platform account on a separate computer.
Go to thermofisher.com/connect.

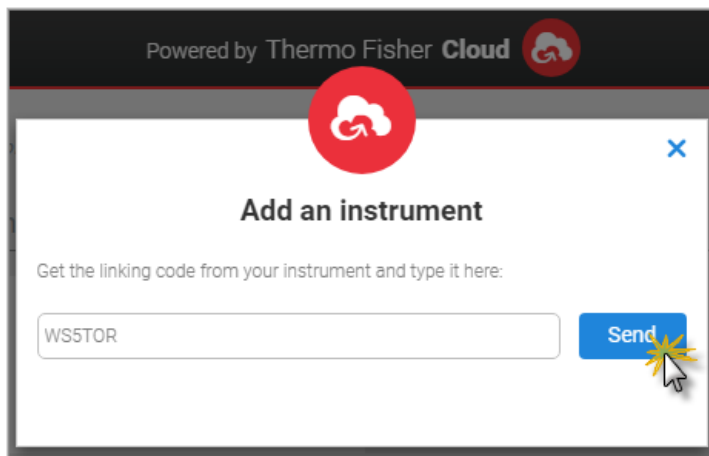
4. Click , then click **Add an Instrument**.



5. Select **SeqStudio™**, then click **Next**.

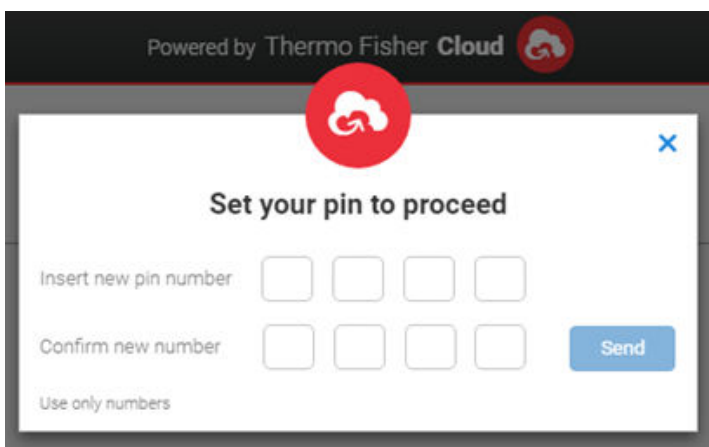


6. Enter the link code from the instrument touchscreen (from step 2), then click **Send**.



The screenshot shows a dialog box titled "Add an instrument" with a Thermo Fisher Cloud logo at the top. Below the title, it says "Get the linking code from your instrument and type it here:". There is a text input field containing the code "WS5TOR". To the right of the input field is a blue button labeled "Send". A mouse cursor is clicking the "Send" button, indicated by a yellow starburst effect.

7. If you have not previously set up a PIN, enter a PIN to use when you sign in to an instrument, then click **Send**.



The screenshot shows a dialog box titled "Set your pin to proceed" with a Thermo Fisher Cloud logo at the top. Below the title, it says "Insert new pin number" followed by four empty square input boxes. Below that, it says "Confirm new number" followed by four empty square input boxes. To the right of the confirmation boxes is a blue button labeled "Send". At the bottom left, it says "Use only numbers".

A start linking message is displayed.

A confirmation message is displayed on the instrument touchscreen when the instrument is linked and connected to your Thermo Fisher™ Connect Platform account.

The first time the instrument is linked, the software automatically:

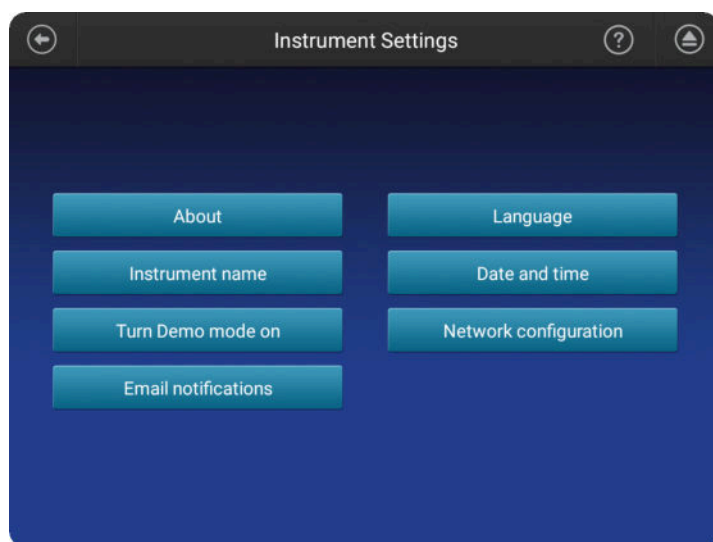
- Creates a Thermo Fisher™ Connect Platform instrument profile with the First Name and Last Name from your Thermo Fisher™ Connect Platform account.
- Registers the instrument in the InstrumentConnect software.

Set up email notifications from the instrument

When an instrument is linked to your Thermo Fisher™ Connect Platform account, email notifications are automatically sent to your Thermo Fisher™ Connect Platform account email address.

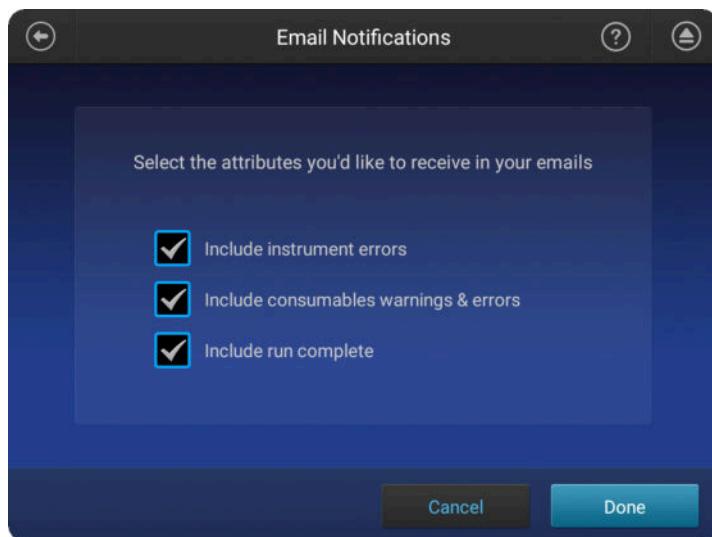
Perform this procedure to disable any of the default notifications.

1. Sign in to the instrument with your Thermo Fisher™ Connect Platform instrument profile and PIN.
2. In the home screen of the instrument, touch ⓘ **Settings** ▶ **Instrument settings** ▶ **Email notifications**.



Note: If you are signed in with a local instrument profile instead of a Thermo Fisher™ Connect Platform instrument profile, the **Email notifications** button is not displayed on the **Instrument Settings** screen.

3. In the **Email notifications** screen, select or deselect the options for which you want to receive email notifications, then touch **Done**.





Dye calibration and install standard checks

■ Calibrate dyes	211
■ Perform an install run	223

Calibrate dyes

Overview of system dye set and custom dye set calibration

Dye calibration compensates for dye emission spectral overlap.

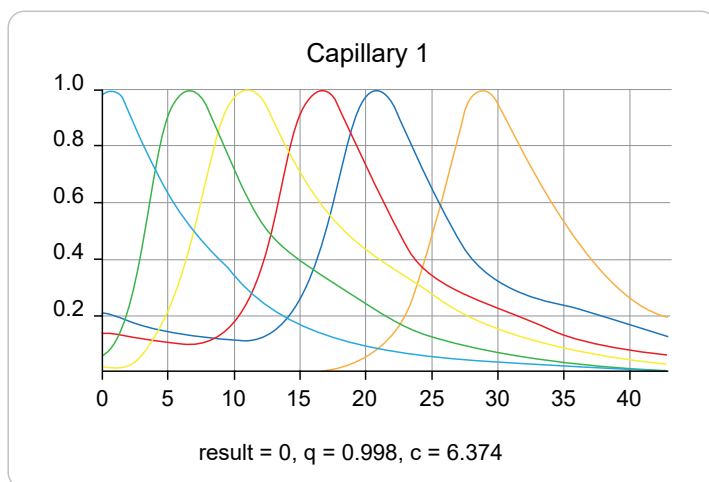


Figure 21 Dye emission spectra for the J6 (DS-36) dye set showing spectral overlap



System versus custom dyes

System dye sets are available from Thermo Fisher Scientific.

Custom dye sets are any dyes that are not available from Thermo Fisher Scientific. Custom dye sets require manual calibration.

System dye sets	
Fragment/HID analysis dye sets	Sequence analysis dye sets
• D (DS-30, DS-31) • E5 (DS-02) • F (DS-32) • G5 (DS-33) • J6 (DS-36) • J6-T (DS-37)	• E_BigDye™ Terminator v1.1 • Z_BigDye™ Terminator v3.1 • Z_BigDye™ Direct

Selecting dye sets in the instrument software

During run setup or when editing the run parameters, select the appropriate dye set.

- Select **Z_BigDye™ Terminator v3.1** for the following chemistries:
 - BigDye™ Terminator Plus Cycle Sequencing Kit
 - BigDye™ Terminator v3.1 Cycle Sequencing Kit
- Select **Z_BigDye™ Direct** for the following chemistries:
 - BigDye™ Terminator Plus Cycle Sequencing Kit with BigDye™ M13 HiRes Primers
 - BigDye™ Direct Cycle Sequencing Kit

Factory, auto, and manual calibration

Three types of calibration can occur on the instrument:

- **Factory calibration**—Default calibration provided with the instrument. It is not optimized for a specific instrument.
- **Manual calibration**—Manual procedure performed by the user that provides a baseline calibration for the instrument on which it is run. Reduces pull-up (false secondary peaks under a true peak).
- **Auto calibration**—Automatic adjustment of the baseline calibration to optimally reduce pull-up (false secondary peaks under a true peak). The instrument performs an auto calibration for system and custom dyes.

Determine if manual calibration is required

- **Sequence analysis dye sets**—Do not typically require manual calibration.
- **Fragment analysis dye sets**—Manual calibration is recommended one time before use.

Note: With high-quality sample data, it is possible for auto calibration to pass using the factory calibration.

- **HID analysis dye sets**—Manual calibration is required for each dye set one time before use.



To determine if a dye set requires manual calibration, review the calibration history for the dye set.

In the home screen:

1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Calibration** ▶ **Calibration history**.
2. Examine the entry for the dye set of interest. If the dye set does not list a date and a cartridge serial number for a manual calibration or a date for auto calibration, the dye set requires manual calibration.

Calibration History				
Dye set	Chemistry standard	Cartridge serial #	Date of manual calibration	Date of auto calibration
D (DS-30)™	Matrix		Factory	
E_BigDye™ Term...	Sequence	12345678901234...	2017/09/12 16:...	
E5 (DS-02)™	Matrix		Factory	2017/09/12 16:...
F (DS-32)™	Matrix		Factory	
G5 (DS-33)™	Matrix		Factory	
J6 (DS-36)™	Matrix		Factory	2017/09/13 13:18... Capa 1, 2
Z_BigDye™ Ter...	Sequence	12345678901234...	2017/09/05 11:...	2017/09/05 11:...

Figure 22 Fragment analysis dye sets example

- ① Manual calibration is recommended—No auto calibration or manual calibration has been performed.
- ② Manual calibration is not required—A manual calibration has been performed.
- ③ Manual calibration is not required—The factory calibration has been optimized for the instrument by auto calibration.
- ④ Manual calibration is recommended—Auto calibration has been performed, but only 2 capillaries passed.
- ⑤ Manual calibration is not required—A manual calibration and an auto calibration have been performed.



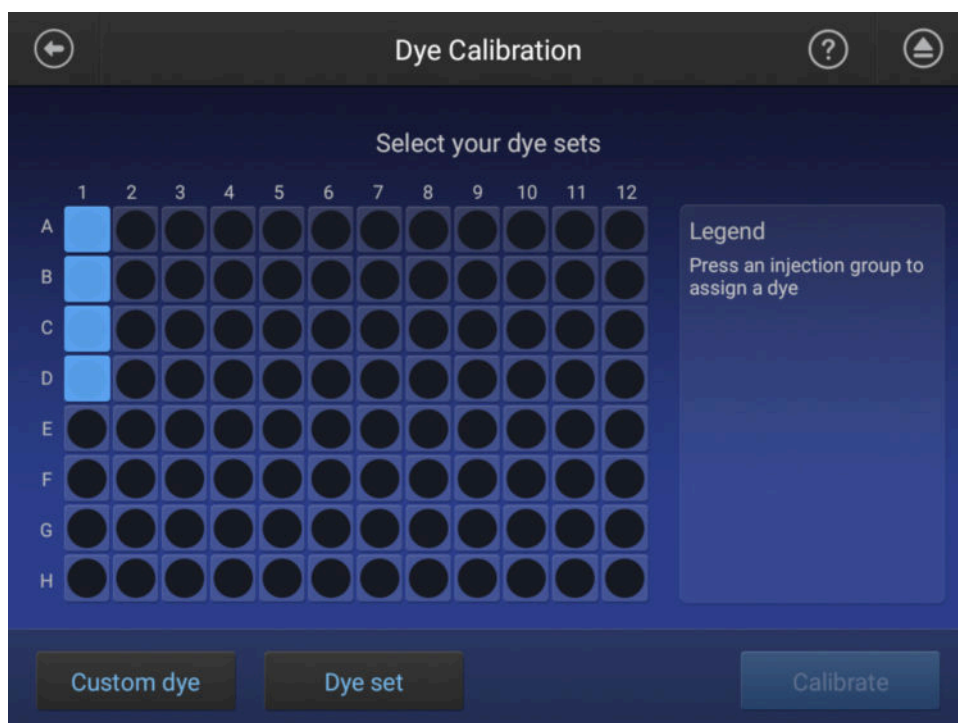
Perform a system dye calibration

A system dye calibration requires ~30 minutes to complete.

Prepare the dye set calibration standards and plate as described in the product information sheet for the dye set.

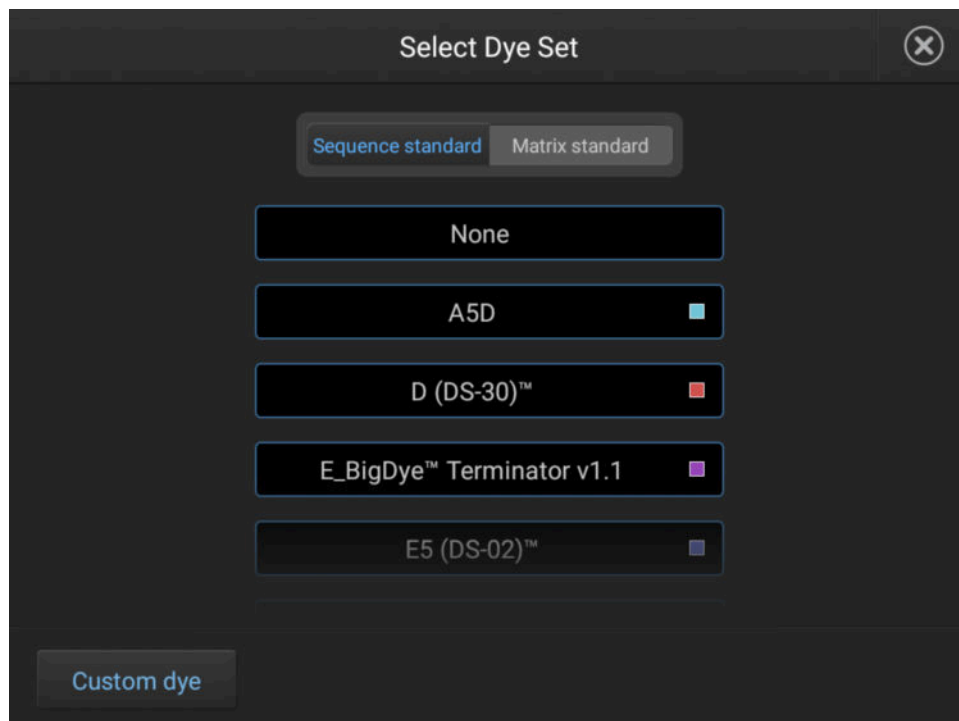
In the home screen:

1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Calibration** ▶ **Dye Calibration**.
2. Touch the injection group for the dye set in the plate, then touch **Dye set**.





3. Touch **Sequence Standard** or **Matrix Standard**, then select a system dye calibration standard provided with the instrument.



4. Touch **Calibrate**.
The calibration run starts.

IMPORTANT! If the dye calibration fails:

- The results of the calibration are not saved, and the calibration plate is not moved to **Run History**.
 - The instrument does not allow you to rerun the plate setup for a failed calibration. Close the calibration screen, then start a new calibration.
-

Spectral Quality Value

A spectral Quality Value reflects the confidence that the individual dye emission signals can be separated from the overall measured fluorescence signal. It is a measure of the consistency between the final matrix and the data from which it was computed. A Quality Value of 1.0 indicates high consistency, providing an ideal matrix with no detected pull-up/pull-down peaks.

In rare cases, a high Quality Value can be computed for a poor matrix. This can happen if the matrix standard contains artifacts, leading to the creation of one or more extra peaks. The extra peaks cause the true dye peak to be missed by the algorithm, and can lead to a higher Quality Value than would be computed with the correct peak. Therefore, it is important to visually inspect the spectral calibration profile for each capillary.



Condition number

A Condition Number indicates the amount of overlap between the dye peaks in the fluorescence emission spectra of the dyes in the dye set.

If there is no overlap in a dye set, the Condition Number is 1.0 (ideal conditions), the lowest possible value. The condition number increases with increasing peak overlap.

The ranges that the software uses to determine if a capillary passes or fails are:

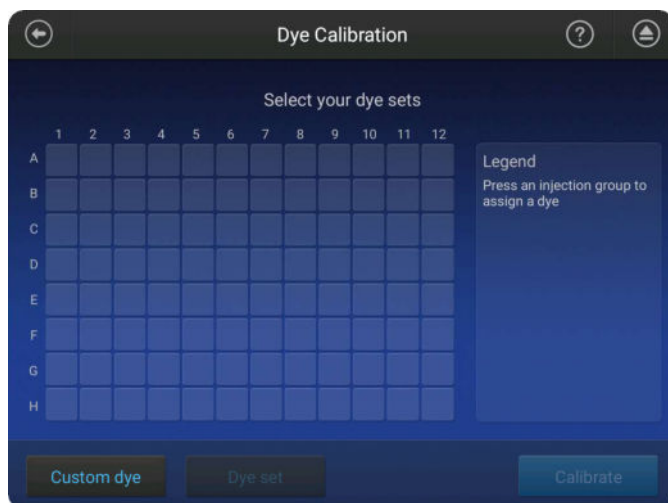
Dye Set	Quality Value Minimum	Condition Number Maximum
AnyDye ^[1]	0.8 (default)	20.0 (default)
D	0.8	8.5
E	0.95	5.5
E5	0.95	6.0
F	0.95	8.5
G5	0.95	13.5
J6	0.95	8.0
J6-T	0.95	8.0
Z	0.95	5.5

^[1] If the 8-dye option is enabled through the service tools function, **AnyDye-8** is displayed.

Perform a custom dye calibration

Add a custom dye set to the software based on a system dye

1. In the home screen, touch ⓘ **Settings** ▶ **Maintenance and Service** ▶ **Calibration** ▶ **Dye calibration**.





2. Touch **Custom Dye**.
3. Touch **Add**.
4. Touch the system dye to use as a starting point for the custom dye settings.
5. Modify the settings as needed then touch **Next**.

Dye Set Parameter

Matrix condition number upper limit: 9.0

Locate start point after scan: 500

Locate start point before scan: 5000

Limit scans To: 32500

Sensitivity: 0.15

Next

6. Enter a **Dye set name**, then touch **Done**.

Add a custom dye set to the software using the AnyDye template

1. In the home screen, touch **Settings** ► **Maintenance and Service** ► **Calibration** ► **Dye calibration**.

Dye Calibration

Select your dye sets

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

Legend
Press an injection group to assign a dye

Custom dye Dye set Calibrate

2. Touch **Custom Dye**.
3. Touch **Add**.



4. Select **AnyDye** to use as a starting point for the custom dye settings.

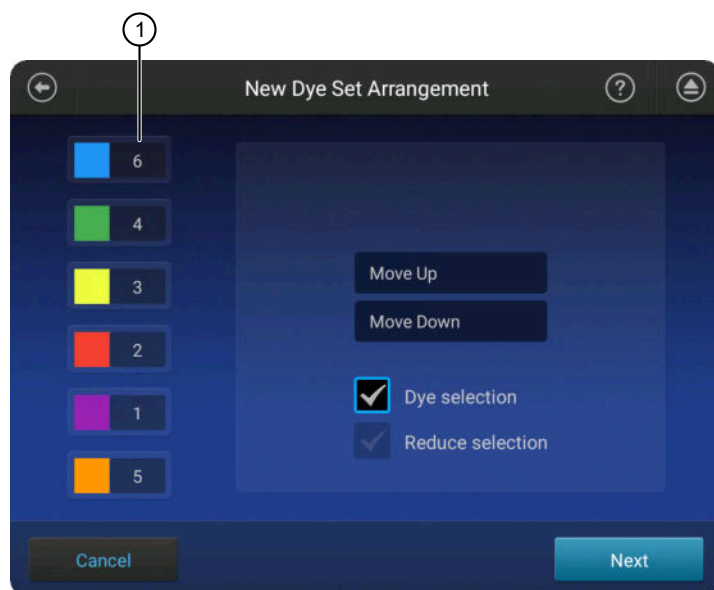
Note: If the 8-dye option is enabled through the service tools function, you can select **AnyDye-8**.

All dyes are selected by default.

5. Touch a dye color, then manage the dye set arrangement:
 - Touch **Dye selection** to deselect the dye.
 - Touch **Move up** or **Move down** to organize the dyes in the order in which they occur as peaks in the electropherogram of the custom dye set standard.
 - Touch **Reduce selection** to exclude it from the calibration.

Note: If you are calibrating with a system dye set, but will not run all system dyes in your application:

- *Enable* (select) the **Dye selection** checkbox (to indicate that the system dye is present in the matrix standard)
 - *Disable* (deselect) the **Reduce selection** checkbox (to indicate that the system dye should not be analyzed, reported, or used for auto calibration)
-



① Order of dyes in the custom dye set

6. Touch **Next**, enter dye set parameters, then touch **Next**.
7. Enter a **Dye set name**, then touch **Save**.



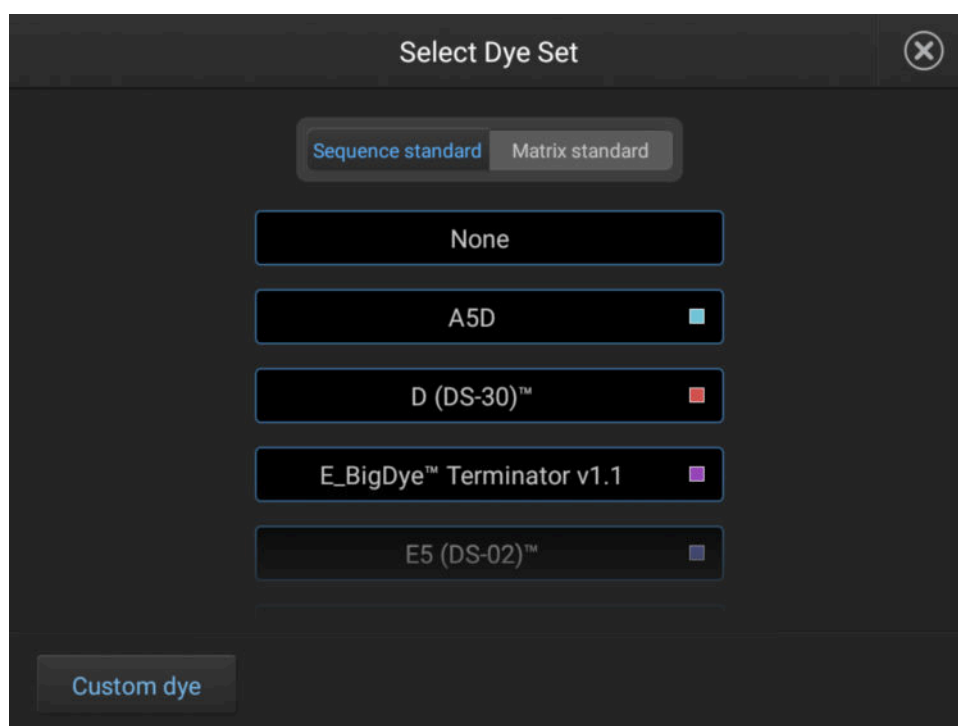
Perform a custom dye calibration

Before you begin:

- Prepare the dye set calibration standards and plate as described in the product information sheet for the dye set.
- Add a custom dye set to the software (see “Add a custom dye set to the software based on a system dye” on page 216) .

In the home screen:

1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Calibration** ▶ **Dye calibration**.
2. Touch the injection group for the dye set in the plate, then touch **Dye set**.



3. Touch a custom dye set.
4. Touch **Calibrate**.
5. When the run is complete, touch **View results**.
6. Ensure that all capillaries passed the calibration, then click **Done**.

IMPORTANT! If the dye calibration fails:

- The results of the calibration are not saved, and the calibration plate is not moved to **Run History**.
 - The instrument does not allow you to rerun the plate setup for a failed calibration. Close the calibration screen, then start a new calibration.
-



Add a custom dye to the Plate Manager or another instrument

- To add a custom dye set to the Plate Manager:
 - a. Open a plate setup that specifies the custom dye set of interest.
 - b. Export a plate setup that specifies the custom dye (see “Export or delete a plate setup (PSM file)” on page 153).
 - c. Open the exported plate setup in the Plate Manager.
- To transfer a custom dye set to another instrument:
 - a. Touch **Settings ▶ Maintenance and Service ▶ Calibration ▶ Calibration history**.
 - b. Touch **Custom dye**.
 - c. Touch **Manage**.
 - d. Touch a custom dye, then touch **Export**.
 - e. Select a location, then touch **Export**.
 - f. Import the custom dye on another instrument (⊗ **Settings ▶ Maintenance and Service ▶ Calibration ▶ Dye calibration ▶ Custom dye ▶ Manage ▶ Import**).



View the dye calibration history

In the home screen:

1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Calibration history**.

Dye set	Chemistry standard	Cartridge serial #	Date of manual calibration	Date of auto calibration
D (DS-30)™	Matrix		Factory	
E_BigDye™ Term...	Sequence	12345678901234...	2017/09/12 16:...	
E5 (DS-02)™	Matrix		Factory	2017/09/12 16:...
F (DS-32)™	Matrix		Factory	
G5 (DS-33)™	Matrix		Factory	
J6 (DS-36)™	Matrix		Factory	2017/09/13 13:18... Caps 1, 2
Z_BigDye™ Ter...	Sequence	12345678901234...	2017/09/05 11:...	2017/09/05 11:...

At the bottom of the screen are three buttons: **Filter**, **Show all**, and **Done**.

- ① Factory calibration (default calibration that is not optimized for the instrument).
 - ② Manual dye calibration performed by a user.
 - ③ Factory calibration that has been optimized by auto calibration and does not require manual calibration.
 - ④ Automatic adjustment of the baseline calibration to optimally reduce pull-up (false secondary peaks under a true peak).
 - ⑤ A manual dye calibration and an auto dye calibration have been performed.
2. Touch a dye set.
A thumbnail of each capillary calibration spectrum is displayed, with the q, and c values (see “Spectral Quality Value” on page 215 and “Condition number” on page 216).
3. Touch and to view a full-screen calibration spectrum for each capillary.
4. Touch to return to the list of dye sets on the **Calibration History** screen.
5. (Optional) Touch **Filter** to narrow the dye set list down by dye sets.
 - a. Select or deselect the dyes listed.
 - b. (Optional) Touch **Deselect All** to clear the dye set selections.
 - c. Touch **Done**.
6. Touch **OK**.



Import or export a custom dye set

In the home screen:

- Touch  **Settings** ▶ **Maintenance and Service** ▶ **Calibration** ▶ **Dye calibration** ▶ **Custom dye** ▶ **Manage**.
 1. Touch the custom dye set.
 2. Touch **Import** or **Export**.



Perform an install run

Overview of install checks

Install checks are performed by a Field Service Engineer at the time of installation.

An install check can be run at any time with the following reagents to ensure instrument performance:

Install check type	Reagent
Sequencing	- Sequencing Standard, BigDye™ Terminator v3.1, for 3500/SeqStudio™ Flex (Cat. No. 4404312) - Sequencing Standard, BigDye™ Terminator v1.1, for 3500/SeqStudio™ Flex (Cat. No. 4404314)
Fragment	DS-33 GeneScan™ Installation Standards with GeneScan™ 600 LIZ™ Size Standard v2.0 (Cat. No. 4376911)
HID Note: A manual calibration for the J6 dye is required before the install check.	GlobalFiler™ Allelic Ladder from one of the following installation kits: - SeqStudio™ HID and Sequencing Installation Kit (Cat. No. A46180) - SeqStudio™ HID Installation Kit (Cat. No. A46182)

You can include multiple injection groups, multiple applications, and/or multiple chemistries on an install run plate. For example, you can prepare an install run plate that contains replicate injections of the fragment install standard and both sequencing standards.

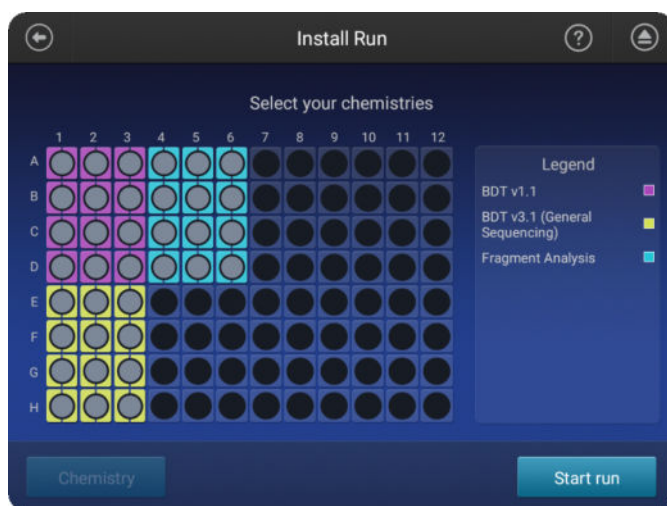


Figure 23 Example install run plate with multiple applications and replicate injections



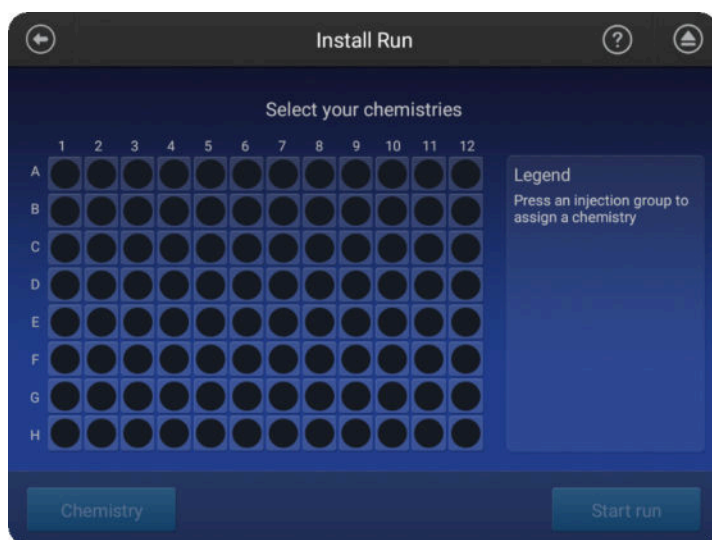
Perform an install run

An install run requires ~45 minutes to complete each injection group.

Prepare the installation standard and plate as described in the product information sheet for the installation standard.

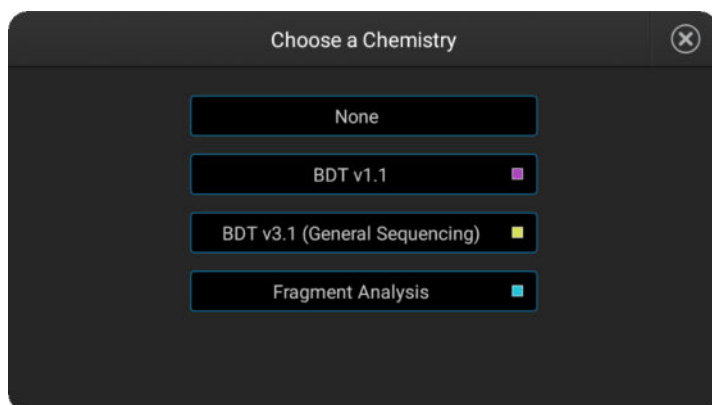
In the home screen:

1. Touch **Settings** ▶ **Maintenance and service** ▶ **Install run** ▶ **Install run**.
2. Touch an injection group on the plate to select a location for the install standards.



A plate can be set up with different injection groups for one install run.

3. Touch **Chemistry**, then select the type of install run.
Select **None** to cancel the selection.



4. Repeat the steps above for additional injection groups and/or multiple chemistries.
5. Touch **Start Run**.



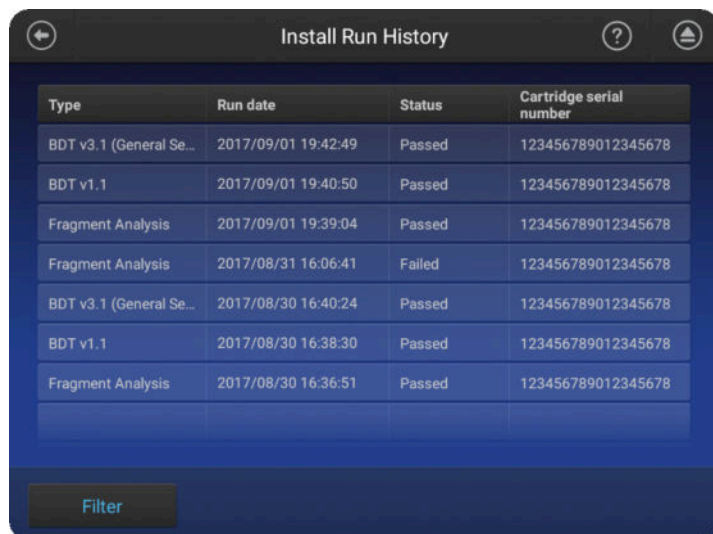
View install run results

1. Open the install run results summary:

- To view results immediately after the install run completes, touch **Results**.

Note: Using this option, you can accept a failed HID install run. For more information, see “Accept a failed HID install run” on page 228.

- To view results for a previous install run, touch  **Settings ▶ Maintenance and Service ▶ Install run ▶ Install run history**.



Type	Run date	Status	Cartridge serial number
BDT v3.1 (General Se...	2017/09/01 19:42:49	Passed	123456789012345678
BDT v1.1	2017/09/01 19:40:50	Passed	123456789012345678
Fragment Analysis	2017/09/01 19:39:04	Passed	123456789012345678
Fragment Analysis	2017/08/31 16:06:41	Failed	123456789012345678
BDT v3.1 (General Se...	2017/08/30 16:40:24	Passed	123456789012345678
BDT v1.1	2017/08/30 16:38:30	Passed	123456789012345678
Fragment Analysis	2017/08/30 16:36:51	Passed	123456789012345678

Filter



2. Touch an install run to display the results, including the pass/fail results.

Install Run

Fragment Analysis - 20170830_163651

Expected # Allele peaks: 15; Expected # Size standard peaks: 34

Cap #	# Allele peaks	# Size standard peaks	QC	Pass/Fail
1	15	34	●	Pass
2	15	34	●	Pass
3	15	34	●	Pass
4	15	34	●	Pass

Export report Detail

Install Run

BDT v3.1 (General Sequencing) - 20170830_164024

Pass when QV30 CRL >= 500

Cap #	QV30 CRL (range)	QC	Pass/Fail
1	533 (15-547)	●	Pass
2	537 (15-551)	●	Pass
3	535 (17-551)	●	Pass
4	542 (17-558)	●	Pass
Mean	536.8		
SD	2.3		

Export report Detail

Each injection group displays a QC color for each capillary:

- —All QC tests passed.
- —At least 1 warning quality alert was triggered.
- —At least 1 failing quality alert was triggered.

For information on quality alerts, see:

- “Data quality alerts” on page 128
- “Sizecalling and basecalling quality alerts” on page 128



3. Touch **Detail** to see allele-specific details for fragment/HID analysis and basecall accuracy for sequencing.

Install Run

Fragment Analysis - 20170830_163651

Expected # Allele peaks: 15; Expected # Size standard peaks: 34

	Dye	Allele	Mean size	Sizing precision	Pass/Fail
1	Green	D5S644[VIC][1]	85.13	0.04	Pass
2	Green	D5S644[VIC][2]	96.95	0.03	Pass
3	Cyan	D20S119[FAM][...]	113.68	0.00	Pass
4	Cyan	D20S119[FAM][...]	119.61	0.05	Pass
5	Yellow	D9S288[NED][1]	137.52	0.04	Pass
6	Yellow	D9S288[NED][2]	145.27	0.02	Pass

Export report Summary

Install Run

BDT v3.1 (General Sequencing) - 20170901_194249

Pass when QV30 CRL >= 500

Cap #	% QV30 CRL Basecall Accuracy	Alignment Read Length (range)	% RL Basecall Accuracy
1	100.0	553 (8-560)	99.1
2	100.0	562 (8-569)	99.1
3	99.6	555 (7-561)	99.1
4	98.7	553 (7-559)	99.1

Export report Summary

4. Touch **Export report** (see “Export the install run report” on page 229).

Fragment analysis install run pass/fail criteria

Result	Acceptance criteria
# Allele peaks	15 for all four capillaries
# Size standard peaks	34 for all four capillaries
Standard deviation of the observed allele fragment sizes for all 4 capillaries	<0.15 base pairs (bp)



HID analysis install run pass/fail criteria

Result	Acceptance criteria
# Allele peaks	343 for all four capillaries
# Size standard peaks	26 for all four capillaries
Minimum peak height	>400 RFU
Standard deviation of the observed allele fragment sizes for all 4 capillaries	<0.15 base pairs (bp)

Sequence analysis install run pass/fail criteria

Result	Acceptance criteria
QV30 CRL (contiguous read length) Note: The contiguous read length passing criteria for install checks is an uninterrupted segment of bases with an average Quality Value (QV) of 30.	BigDye™ Terminator v3.1 Capillaries with QV30 CRL ≥500 bp pass.
	BigDye™ Terminator v1.1 Capillaries with a QV30 CRL ≥500 bp pass.

The remaining results on the screen are for information only.

Accept a failed HID install run

Note: Due to the bin-offsetting feature, the GeneMapper™ *ID-X* Software may identify alleles not identified by the SeqStudio™ Data Collection Software. The bin-offsetting feature in the GeneMapper™ *ID-X* Software adjusts the reference bin locations using the observed alleles in the allelic ladder samples, while the SeqStudio™ Data Collection Software uses static bins. Troubleshoot the failed HID install run by analyzing the install standard data files in GeneMapper™ *ID-X* Software v1.6 or later using the SeqStudio™ version of the GlobalFiler™ kit panel and bin files. Accept a failed HID install run if alleles are properly called in the secondary analysis software.

1. After the HID install run completes, touch **Results**.
2. Touch **Accept**.
3. In the **Accept Install Run** screen, touch **Yes** to accept the failed install run.

The install run is accepted. The **Install Run History** screen displays the status as "Accepted Failure" in the **Status** column.



Export the install run report

1. Touch ⓘ **Settings** ▶ **Maintenance and Service** ▶ **Install run** ▶ **Install run history**.
2. Touch the install run of interest.
3. Touch **Export report**.
4. Select a destination for the report.
5. Touch **OK**.



Example fragment analysis run report



Install run report

Report name: Fragment Install Run_20170411_093309

Date: 2017/04/11

System Information

Instrument Name:	seqstudio-SVT2	Run Date:	2017/04/10
Instrument Serial Number:	132000002	Chemistry Type:	Fragment Analysis
Instrument Software Version:	SVT-1.4.21-1306-r12832	Cartridge Serial Number:	2852889004
Buffer Installation Date:	2017/03/04	Cartridge Installation Date:	2017/04/10
Buffer Expiration Date:	2017/03/18	Cartridge Expiration Date:	2017/11/16

Install Run Result

Capillary	1	2	3	4
# of Allele Peaks	15	15	15	15
# of Size Standard Peaks	34	34	34	34

Legend: ■ Passed ■ Failed

Running info

	Dye	Allele	Nominal Size	Mean	Average peak height	Peak height% > Min	Sizing precision	Sizing accuracy	Pass/Fail
1	Green	D5S644[VIC][1]	85.06	85.1	1851.75	100.0	0.02	0.04	Pass
2	Green	D5S644[VIC][2]	96.89	96.92	1184.0	100.0	0.03	0.03	Pass
3	Blue	D20S119[FAM][1]	113.7	113.67	1393.25	100.0	0.0	-0.03	Pass
4	Blue	D20S119[FAM][2]	119.6	119.56	1308.5	100.0	0.05	-0.04	Pass
5	Yellow	D9S288[NED][1]	137.5	137.38	1545.0	100.0	0.02	-0.12	Pass
6	Yellow	D9S288[NED][2]	145.24	145.09	1044.75	100.0	0.04	-0.15	Pass
7	Red	D6S289[PET][1]	171.49	171.35	1677.0	100.0	0.05	-0.14	Pass
8	Red	D6S289[PET][2]	173.41	173.25	1054.0	100.0	0.03	-0.16	Pass
9	Green	D5S424[VIC][1]	216.39	216.26	1457.5	100.0	0.04	-0.13	Pass
10	Green	D5S424[VIC][2]	218.27	218.14	955.75	100.0	0.03	-0.13	Pass
11	Blue	D9S1690[FAM][1]	236.74	236.48	2030.0	100.0	0.06	-0.26	Pass
12	Blue	D9S1690[FAM][2]	238.67	238.38	1300.5	100.0	0.04	-0.29	Pass
13	Yellow	D18S462[NED][1]	303.06	302.86	1718.25	100.0	0.06	-0.2	Pass
14	Red	D15S117[PET][1]	336.81	336.71	1331.0	100.0	0.02	-0.1	Pass
15	Red	D15S117[PET][2]	338.73	338.63	914.25	100.0	0.01	-0.1	Pass

Capillary (Raman) Uniformity 0.4

The spectral dye matrix has not been updated.



Example sequencing run report



Install run report

Report name: Sequencing Install_Run_20170424_131753

Date: 2017/04/24

System Information

Instrument Name:	seqstudio-SVT1	Run Date:	2017/04/24
Instrument Serial Number:	132000001	Chemistry Type:	BDT v3.1 (General Sequencing)
Instrument Software Version:	0.6.10	Cartridge Serial Number:	2883720037
Buffer Installation Date:	2016/12/05	Cartridge Installation Date:	2017/04/23
Buffer Expiration Date:	2016/12/19	Cartridge Expiration Date:	2017/11/17

Install Run Result

Capillary	1	2	3	4	Mean	SD
Contiguous Read Length (CRL) (range)	530 (26-555)	535 (29-563)	531 (29-559)	530 (32-561)	531.5	2.1
CRL Pass/Fail						
Comparison with Reference Sequence						
%CRL Basecall Accuracy	99.8	99.6	99.8	100.0		
%RL Basecall Accuracy	99.1	99.1	99.1	99.1		
Alignment Read Length (range)	562 (5-566)	557 (19-575)	556 (15-570)	562 (5-566)		

Legend: ■ Passed ■ Failed

Capillary (Raman) Uniformity 0.0

Signal Strength

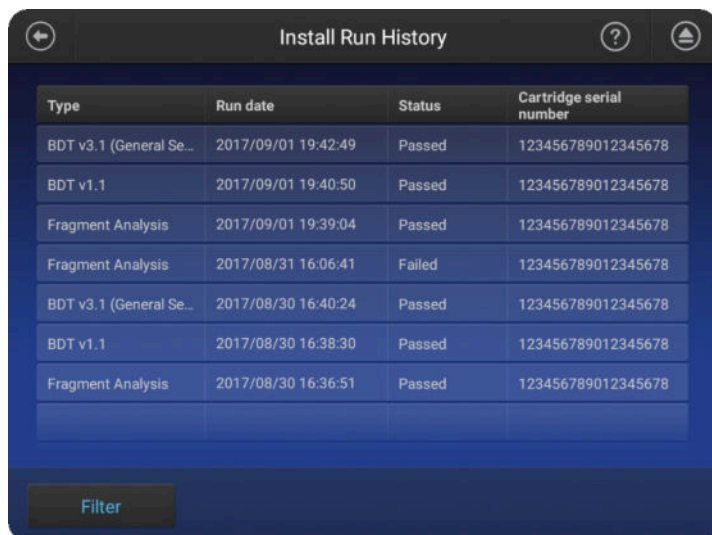
Capillary #	Sample file name	G	A	T	C
1	A2_A2_20170424_104040.ab1	792	1275	968	917
2	B2_B2_20170424_104041.ab1	1016	1625	1193	1108
3	C2_C2_20170424_104042.ab1	751	1336	1126	878
4	D2_D2_20170424_104043.ab1	635	1087	901	742



View the install run history

In the home screen:

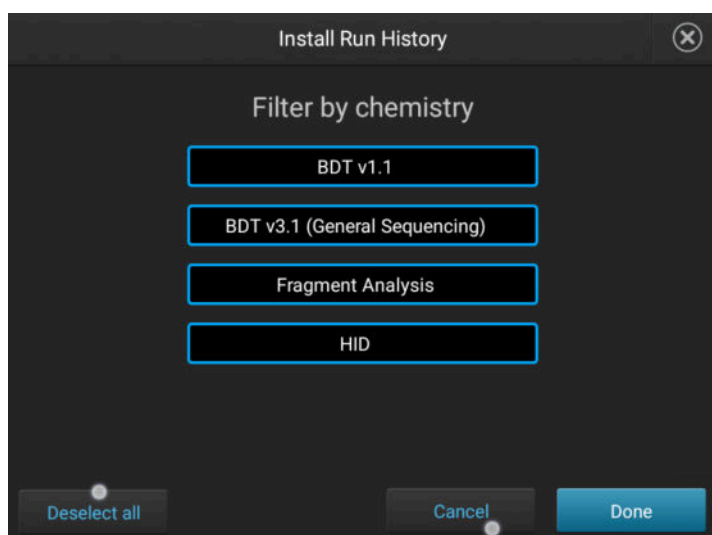
1. Touch **Settings** ▶ **Maintenance and Service** ▶ **Install run history**.



Type	Run date	Status	Cartridge serial number
BDT v3.1 (General Se...	2017/09/01 19:42:49	Passed	123456789012345678
BDT v1.1	2017/09/01 19:40:50	Passed	123456789012345678
Fragment Analysis	2017/09/01 19:39:04	Passed	123456789012345678
Fragment Analysis	2017/08/31 16:06:41	Failed	123456789012345678
BDT v3.1 (General Se...	2017/08/30 16:40:24	Passed	123456789012345678
BDT v1.1	2017/08/30 16:38:30	Passed	123456789012345678
Fragment Analysis	2017/08/30 16:36:51	Passed	123456789012345678

Filter

2. (Optional) Touch **Filter**, select the install run type, then touch **Done**.



Note: If the screen is blank when you click **Done**, no install runs are present for the chemistry you selected.



Create and run an HID plate

■ HID analysis requirements	233
■ Pull-up reduction feature	234
■ Resolution algorithm	235
■ Workflow: HID analysis	236
■ HID install run	237
■ HID run module, dye sets, size standard definitions, and kits	237
■ Manage kits (HID applications only)	238

HID analysis requirements

- **Cartridge**—Use SeqStudio™ Genetic Analyzer Cartridge v2 (Cat. No. [A41331](#)), which has been validated for HID analysis. The v1 cartridge has not been validated.
To check the Cat. No., you must remove the cartridge from the instrument. See “Store the cartridge” on page 180 for instructions.
- **Manual dye calibration and HID install check**—When the instrument is installed, the field service representative will perform a manual J6 dye calibration and an HID install check.
Before you use any other dye sets, perform a manual dye calibration to optimize the pull-up reduction function. For more information, see “Calibrate dyes” on page 211.
- **Allelic ladder**—We recommend that you inject one allelic ladder for each set of 24 samples in HID runs (1 allelic ladder per 6 injections).
Allelic ladders that are injected under the same conditions are recommended to accurately genotype samples in the secondary analysis software.

IMPORTANT! Variation in laboratory temperature can cause changes in fragment migration speed that can, in turn, cause sizing variation. We recommend the frequency of allelic ladder injections described above to account for normal variation in fragment migration speed. However, during internal HID validation studies, verify the required allelic ladder injection frequency to ensure accurate genotyping of all samples in your laboratory environment.

- **GeneMapper™ ID-X Software compatibility**—The data files that are generated by the instrument are compatible with GeneMapper™ ID-X Software v1.6 or later.

Pull-up reduction feature

The software includes two pull-up reduction features:

- Automatic pull-up reduction feature that is applied with all dye sets.
- (*HID analysis only for Applied Biosystems™ dyes*) Marker-to-marker pull-up reduction that is optimized for the markers in each kit.

You can enable/disable this feature for each injection group when you create an HID plate.

- To enable, select the appropriate kit from the **Kits** list.
- To disable, select **None** from the **Kits** list.

Note: "Kit" is displayed when **None** is selected.

To display this screen where you select a kit: Select or create a plate set up, touch the **Plate** tab, then select **Edit**. In the **Edit Plate** screen, swipe up to find the **Kit** field, then select a kit from the dropdown list.

Note: You can also select a kit from the Plate Manager.



Resolution algorithm

A resolution algorithm, unique when running an HID application, is incorporated into SeqStudio™ Data Collection Software v1.2 and later. The purpose of the algorithm is to align the resolution power traditionally generated with a 3500/3500xL Genetic Analyzer/POP-4™ Polymer configuration to the resolution power generated with a SeqStudio™ Genetic Analyzer/POP-1™ Polymer configuration.

About POP™ polymer matrices

POP™ polymer matrices separate DNA fragments of a known size range, at a desired resolution and run time. Polymers are available in different concentrations, which directly relate to the viscosity and length of the polymer molecule. Different concentrations are suited to different applications. For example:

- **Sequencing analysis applications**—Traditionally use POP-6™ and POP-7™ Polymers. POP-6™ and POP-7™ Polymers are composed of relatively longer molecules that provide greater DNA-to-polymer interaction and cause DNA fragments to move through the polymer more slowly. This design results in a longer run time and the increased resolution that is required for sequencing.
- **Fragment analysis applications**—Traditionally use POP-4™ Polymer. Because of its polymer concentration and molecule size, POP-4™ Polymer has single basepair resolving power within the size range for forensic STR amplification kits and runs faster.

About POP-1™ Polymer

We are introducing POP-1™ Polymer for the SeqStudio™ Genetic Analyzer, which can be used for all applications on the instrument.

For smaller fragments, POP-1™ Polymer resolves similarly to POP-7™ Polymer. For larger fragments, POP-1™ Polymer resolves between POP-6™ and POP-7™ Polymers. In addition to providing higher resolution for expected STR fragment peaks, peaks intrinsic to the amplification process in the $n-1/n+1$ position from the main allele may resolve as shoulders with POP-1™ Polymer. Because of the limited resolving power of POP-4™ Polymer, these shoulders were not always visible in our legacy data sets.

The resolution algorithm is applied during data collection. The algorithm maintains similar resolution for smaller fragments across the 3500/3500xL Genetic Analyzer and SeqStudio™ Genetic Analyzer, but allows for the improved resolution for larger fragments that benefit from the POP-1™ Polymer.

Workflow: HID analysis

Get started

Prepare the instrument
(page 41)

Prepare the samples
(page 38)

Create a plate setup on the instrument

Create or import a plate setup
(page 87)

Enter plate properties
(page 88)

Assign wells: run module, size standard, dye set, and kit
(page 90)

Assign wells: sample name, sample type, and custom fields
(page 92)

Start and monitor a run

On the instrument: Load the plate or the tube assembly
(page 98)

Select a plate setup and start a run
(page 99)

Monitor a run from the instrument
(page 112)

View and analyze results

View results on the instrument
(page 119)

Export results from the instrument (sample data files and QC reports)
(page 130)

Analyze data
(page 130)

(If needed) View the export status for sample data files
(page 131)



HID install run

Item	Description
Components of the install check reaction	- GeneScan™ 600 LIZ™ Size Standard v2.0 (Cat. No. 4408399) - Hi-Di™ Formamide (Cat. No. 4311320 or 4440753) - GlobalFiler™ Allelic Ladder from one of the following installation kits: - SeqStudio™ HID and Sequencing Installation Kit (Cat. No. A46180) - SeqStudio™ HID Installation Kit (Cat. No. A46182)
Volumes of install check components per reaction	- Size standard—0.4 µL - Hi-Di™ Formamide—9.6 µL - Allelic ladder—1 µL
Pass/fail criteria for HID J6 install check	343 ladder peaks 26 size standard peaks - Minimum peak height >400 RFU - Standard deviation of the observed allele fragment sizes for all 4 capillaries: <0.15 base pairs (bp)

HID run module, dye sets, size standard definitions, and kits

Run module	Resolution range	Approximate run time	Sizing precision	Compatible size standards
HIDAnalysis	60–470 bp ^[1]	39 minutes	60–470: <0.15	- GeneScan™ 500 LIZ™ Size Standard^[2] - GeneScan™ 600 LIZ™ Size Standard v2.0

^[1] Resolution Range: The range of bases over which the resolution (peak spacing interval divided by the peak width at half-max in a GS600 size standard sample sized with a third order fit) is ≥ 1 .

^[2] The GeneScan™ 500 LIZ™ Size Standard was not included in the HID validation of the instrument.

Category	Items
Dye sets	- G5 (DS-33) - J6 (DS-36) - J6-T (DS-37)
Size standards	- GS500(-250)LIZ - GS600_LIZ_(60-460) - GS600_LIZ_(80-400)

(continued)

Category	Items
Kits	- Setting of None disables the Marker-to-Marker pull-up feature ("Kit" is displayed when None is selected. - GlobalFiler™ - GlobalFiler™ Express - Huaxia™ Platinum - Identifiler™ Plus - MiniFiler™ - NGM Detect™ - NGM SElect™ - Verifiler™ Express - Verifiler™ Plus - Yfiler™ - Yfiler™ Plus

Manage kits (HID applications only)

The kit files for the kits that have been validated for the instrument are preloaded in the software. If a new or updated kit file is provided, import the kit file as described below.

1. Copy the kit file to a USB or a network drive.
2. Access the **Manage kits** screen:

From	Actions
Plate properties tab	Touch More options ▶ Manage kits .
Home screen	Touch ⓘ Settings ▶ Run settings ▶ Kits .

3. Touch **Import**, select the location of the kit file. Select the kit, then touch **Import**.
The kit file is loaded in the software.



Run modules

Run modules, read lengths, size ranges, and run times

Table 15 Sequencing run modules for standard sequencing

Run module	Contiguous read length (CRL) ^[1]	QV threshold	Approximate run time
ShortSeq	≥350	QV30	30 minutes
ShortSeq_BDX			
MediumSeq	≥500	QV30	45 minutes
MediumSeq_BDX			
LongSeq	≥800	QV20	~ 2 hours
LongSeq_BDX			

^[1] CRL was determined using the Long Read Sequencing standard. A minimum of 90% of analyzed sequences with an average QV ≥QV threshold were observed.

IMPORTANT! Use BDX run modules only if you prepare samples with BigDye XTerminator™ Purification Kit. Use non-BDX run modules for samples purified with other methods.

Table 16 Fragment analysis run modules

Run module	Resolution range	Approximate run time	Sizing precision	Compatible size standards
SNaPshot	40–120 bp	25 minutes	40–120: <0.5	GeneScan™ 120 LIZ™ Size Standard
FragAnalysis	60–460 bp ^[1]	45 minutes	60–460: <0.15	All except GeneScan™ 1200 LIZ™ Size Standard
LongFragAnalysis ^[2]	60–600 bp ^[1]	<2 hours	60–460: <0.15 461–600: <0.3 601–800: >0.45	- GeneScan™ 600 LIZ™ Size Standard v2.0 - GeneScan™ 1200 LIZ™ Size Standard

^[1] Resolution Range: The range of bases over which the resolution (peak spacing interval divided by the peak width at half-max in a GS600 or GS1200 LIZ size standard sample sized with a third order fit) is ≥1. The table shows the resolution range in ≥90% of samples.

^[2] Load a maximum of 48 samples per plate if you use a long run module.

Table 17 HID run modules

Run module	Resolution range	Approximate run time	Sizing precision	Compatible size standards
HIDAnalysis	60–470 bp ^[1]	39 minutes	60–470: <0.15	- GeneScan™ 500 LIZ™ Size Standard^[2] - GeneScan™ 600 LIZ™ Size Standard v2.0

^[1] Resolution Range: The range of bases over which the resolution (peak spacing interval divided by the peak width at half-max in a GS600 size standard sample sized with a third order fit) is ≥ 1 .

^[2] The GeneScan™ 500 LIZ™ Size Standard was not included in the HID validation of the instrument.

Note: The following size standards have not been validated for use with the instrument. A default size standard definition is not provided in the software.

- GeneScan™ 350 ROX™ Size Standard
 - GeneScan™ 400HD ROX™ Dye Size Standard
-



Parts and materials

Required materials not supplied

Unless otherwise indicated, all materials are available through [thermofisher.com](https://www.thermofisher.com). "MLS" indicates that the material is available from [fisherscientific.com](https://www.fisherscientific.com) or another major laboratory supplier.

Item	Source
Software	
(Optional) SeqStudio™ Remote Monitoring App	Available on the Thermo Fisher™ Connect Platform
(Optional) InstrumentConnect	
(Optional) SeqStudio™ Plate Manager and App	Available on the Thermo Fisher™ Connect Platform or for download at thermofisher.com
Equipment	
(Optional) Handheld Barcode Scanner	4488442
Wifi Dongle	A26774
Consumables for SeqStudio™ Genetic Analyzer	
SeqStudio™ Genetic Analyzer Cartridge	A33671
SeqStudio™ Genetic Analyzer Cartridge v2	A41331
SeqStudio™ Genetic Analyzer Cathode Buffer Container	A33401
Reservoir Septa (for SeqStudio™ Cathode Buffer Container)	A35640
SeqStudio™ Integrated Capillary Protector	A31923
Tubes, plates, and other consumables	
MicroAmp™ Optical 96-Well Reaction Plate	4316813
MicroAmp™ Optical 96-Well Reaction Plate with Barcode	4326659
MicroAmp™ Optical 8-Tube Strip, 0.2 mL	4316567
Septa for SeqStudio™ Genetic Analyzer, 96 well	A35641
Septa for SeqStudio™ Genetic Analyzer, 8 strip	A35643



(continued)

Item	Source
MicroAmp™ 96-Well Tray/Retainer Set (Adapter for 8-Tube Strip)	403081
Reagents	
Hi-Di™ Formamide	4401457
BigDye XTerminator™ Purification Kit	4376486
Sequencing kits	
BigDye™ Terminator Plus Cycle Sequencing Kit	4337461
BigDye™ Terminator Plus Cycle Sequencing Kit with BigDye™ M13 HiRes Primers	4337469
BigDye™ Terminator v3.1 Cycle Sequencing Kit	4337454
BigDye™ Terminator v1.1 Cycle Sequencing Kit	4337449
BigDye™ Direct Cycle Sequencing Kit	4458689
Fragment analysis size standards and installation standard	
GeneScan™ 600 LIZ™ Size Standard v2.0	4408399
GeneScan™ 120 LIZ™ Size Standard	4324287
GeneScan™ 1200 LIZ™ Size Standard	4379950
GeneScan™ 500 ROX™ Size Standard	401734
GeneScan™ 500 LIZ™ Size Standard ^[1]	4322682
GeneScan™ 350 ROX™ Size Standard ^[1]	401735
GeneScan™ 400HD ROX™ Dye Size Standard ^[1]	402985
DS-33 GeneScan™ Installation Standards with GeneScan™ 600 LIZ™ Size Standard v2.0	4376911
HID analysis size standards and installation standard	
GeneScan™ 500 LIZ™ Size Standard ^[2]	4322682
GeneScan™ 600 LIZ™ Size Standard v2.0	4408399
SeqStudio™ HID and Sequencing Installation Kit	A46180
SeqStudio™ HID Installation Kit	A46182

^[1] For fragment analysis: This size standard has not been validated for use with the instrument. A default size standard definition is not provided in the software.

^[2] For HID analysis: This size standard was not included in the HID validation of the instrument. A default size standard definition is provided in the software.



Instrument specifications and layout

■ Instrument dimensions	243
■ Instrument clearances	243
■ Environmental requirements	244
■ Electrical requirements	245
■ Electrical protective devices	245
■ Network requirements	246
■ Safety requirements	246

Instrument dimensions

Ensure that the installation site bench space can accommodate the dimensions and support the weight.

Configuration	Height	Length (depth)	Width	Weight
(Door Closed)	44.2 cm (17.4 in.)	64.8 cm (25.5 in.)	49.5 cm (19.5 in.)	53.5 kg (118 lbs)
(Door Open)	56.9 cm (22.4 in.)			



WARNING! PHYSICAL INJURY HAZARD. Do not attempt to lift or move the instrument without professional assistance. The crated instrument is heavy. Any incorrect lifting or moving of the crated instrument can cause serious injury.

Instrument clearances

During instrument setup and maintenance, it is necessary to access the back and sides of the instrument. If the back of the instrument faces a wall, it will be necessary to have enough space to rotate the instrument on the bench for access.

IMPORTANT! For safety, the power outlet used for powering the instrument must be accessible at all times.

Component	Top	Front	Left	Right	Back
SeqStudio™ Genetic Analyzer	30.5 cm (12.0 in)	30.5 cm (12.0 in)	10.0 cm (4.0 in)	20.0 cm (8.0 in)	10.0 cm (4.0 in)



Environmental requirements

Ensure that the installation room is maintained under correct environmental conditions.

Condition	Acceptable range
Installation site	Indoor use only
Electromagnetic interference	Do not use this device in close proximity to sources of strong electromagnetic radiation (for example, unshielded intentional RF sources). Strong electromagnetic radiation may interfere with the proper operation of the device. This equipment has been designed and tested to CISPR 11 Class A. In a domestic environment it may cause radio interference. You may need to take measures to mitigate the interference.
Altitude	Located between sea level and 2,000 m (6,500 ft.) above sea level
Humidity (instrument and computer)	• Operation: 20%–80% (noncondensing) • Transport and storage: 15%–80% (noncondensing)
Temperature (instrument and computer)	• Operation: 15°C to 30°C (60°F to 85°F) Note: The room temperature must not fluctuate more than 2°C over a 2-hour period. • Transport and storage: –20°C to 60°C (–4°F to 140°F)
Transient category	Installation categories II
Overvoltage category	Installation categories II
Vibration	Ensure that the instrument is not adjacent to strong vibration sources, such as a centrifuge, pump, or compressor. Excessive vibration will affect instrument performance.
Pollution degree	II Install the instrument in an environment designated pollution degree II (only non-conductive pollution (e.g., dust) occurs except that occasionally a temporary conductivity caused by condensation is to be expected). Typical pollution degree II environments are laboratories and office spaces.
Liquid waste collection	Dispose of the polymer, buffer, reagents and any liquid waste as hazardous waste in compliance with local and national regulations.
Other conditions	Ensure the installation site is away from any vents that could expel particulate material on the system components. Avoid placing the instrument and computer adjacent to heaters, cooling ducts, or in direct sunlight.

Electrical requirements



CAUTION! Do not unpack or plug in any components until the Field Service Engineers (FSEs) have configured the system for the proper operating voltage.



WARNING! For safety, the power outlet used for powering the instrument must be accessible at all times. See “Instrument clearances” on page 243 for information about the space needed between the wall and the instrument. In case of emergency, you must be able to immediately disconnect the main power supply to all the equipment. Allow adequate space between the wall and the equipment so that the power cords can be disconnected in case of emergency.

- Electric receptacle required: 2-prong with ground pin
- Maximum power dissipation: 380 W (approximately, not including computer and monitor)
- Mains AC line voltage tolerances must be up to ± 10 percent of nominal voltage

Rated voltage	Circuit required	Rated frequency	Rated power
100–240 $\pm 10\%$ VAC ^[1]	10 A	50–60 Hz	400 W

^[1] If the supplied power fluctuates beyond the rated voltage, a power line regulator may be required. High or low voltages can adversely affect the electronic components of the instrument.

Electrical protective devices

We recommend several protective devices to protect the system in environments with large voltage and power fluctuations.

Device	Description
Power line regulator	We recommend the use of a 1.5-kVA power line regulator in areas where the supplied power fluctuates in excess of $\pm 10\%$ of the normal voltage. Power fluctuations can adversely affect the function of the instrument and computer. Note: A power line regulator monitors the input current and adjusts the power supplied to the instrument or computer. It does not protect against a power surge or failure.

(continued)

Device	Description
Uninterruptible power supply (UPS)	We recommend the use of a 1.5-kVA uninterruptible power supply (UPS), especially in areas prone to power failure. Power failures and other events that abruptly terminate the function of the instrument and computer can corrupt data and possibly damage the system.  WARNING! PHYSICAL INJURY HAZARD. Do not attempt to lift the UPS unit without assistance of at least two people. Improper lifting can cause painful and permanent back injury. Refer to the UPS manufacturer user guide for more information. IMPORTANT! UPSs provide power for a limited time. They are meant to delay the effects of a power outage, not to serve as replacement power sources. In the event of a power loss, power off the instrument and computer unless you expect to regain power within the battery life of the UPS.
Surge protector	We recommend the use of a 10-kVA surge protector (line conditioner) in areas with frequent electrical storms or near devices that are electrically noisy, such as refrigerators, air conditioners, or centrifuges. Short-duration, high-voltage power fluctuations can abruptly terminate the function of, and thereby damage the components of, the computer and the instrument. Note: A dedicated line and ground between the instrument, computer, and the building's main electrical service can also prevent problems caused by power fluctuations.

Network requirements

The instrument is factory-configured for IPv4 TCP/IP communication and includes a fast Ethernet adapter (10/100 Mbps) with a RJ45-type connector for integrating the device into a local area network (LAN). If the instrument will be connected to a LAN, an active, tested network jack must be in place before the scheduled installation date. Also, a representative from your information technologies department must complete and return the *SeqStudio™ Genetic Analyzer IT Checklist* (Pub. No. MAN0016055) before installation, and be available during the installation to help connect the instrument to your network.

A wireless adapter (also referred to as a Wifi Dongle) is provided with the instrument. The wireless connection conforms to 802.11 ac/a/b/g/n wireless standards.

Safety requirements

Safety practices

A safety representative from your facility must ensure that:

- Personnel establish and follow all applicable safety practices and policies to protect laboratory personnel from potential hazards.
- All applicable safety devices and equipment are available at all times.



Required safety equipment

Your laboratory has specific safety practices and policies designed to protect laboratory personnel from potential hazards that are present. Follow all applicable safety-related procedures at all times.

The following safety equipment and protection from hazards must be available at the installation site:

- Protection from any sources of hazardous chemicals, radiation (for example, lasers, radioisotopes, radioactive wastes, and contaminated equipment), and potentially infectious biological material that may be present in the area where the service representative will work.
- Appropriate fire extinguisher:
 - You are responsible for providing an appropriate fire extinguisher for use on or near the equipment.
 - The types and sizes of fire extinguishers shall be suitable for use on electrical and chemical fires as specified in current codes, regulations, and/or standards, and with approval of the Fire Marshall or other authority having jurisdiction.
 - The installation of appropriate fire extinguishers shall be in addition to other fire-protection systems and not as a substitute or alternative to them.
- Eyewash
- Safety shower
- Eye and hand protection
- Adequate ventilation, including vent line/fume hood, if applicable
- Biohazard waste container, if applicable
- First-aid equipment
- Spill cleanup equipment
- Applicable Safety Data Sheets (SDSs)



Safety



WARNING! GENERAL SAFETY. Using this product in a manner not specified in the user documentation may result in personal injury or damage to the instrument or device. Ensure that anyone using this product has received instructions in general safety practices for laboratories and the safety information provided in this document.

- Before using an instrument or device, read and understand the safety information provided in the user documentation provided by the manufacturer of the instrument or device.
- Before handling chemicals, read and understand all applicable Safety Data Sheets (SDSs) and use appropriate personal protective equipment (gloves, gowns, eye protection, and so on). To obtain SDSs, visit thermofisher.com/support.

Symbols on this instrument

Symbols may be found on the instrument to warn against potential hazards or convey important safety information. In this document, the hazard symbol is used along with one of the following user attention words:

- **CAUTION!**—Indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices.
- **WARNING!**—Indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.
- **DANGER!**—Indicates an imminently hazardous situation that, if not avoided, will result in death or serious injury.

Symbol	English	Français
	Caution, risk of danger Consult the manual for further safety information.	Attention, risque de danger Consulter le manuel pour d'autres renseignements de sécurité.
	Caution, risk of electrical shock	Attention, risque de choc électrique
	Caution, piercing hazard	Attention, danger de perforation

(continued)

Symbol	English	Français
	Caution, hot surface	Attention, surface chaude
	Potential biohazard	Danger biologique potentiel
	On	On (marche)
	Off	Off (arrêt)
	On/Off	On/Off (marche/arrêt)
	Protective conductor terminal (main ground)	Borne de conducteur de protection (mise à la terre principale)
	Terminal that can receive or supply alternating current or voltage	Borne pouvant recevoir ou envoyer une tension ou un courant de type alternatif
	Do not dispose of this product in unsorted municipal waste  CAUTION! To minimize negative environmental impact from disposal of electronic waste, do not dispose of electronic waste in unsorted municipal waste. Follow local municipal waste ordinances for proper disposal provision and contact customer service for information about responsible disposal options.	Ne pas éliminer ce produit avec les déchets usuels non soumis au tri sélectif.  MISE EN GARDE ! Pour minimiser les conséquences négatives sur l'environnement à la suite de l'élimination de déchets électroniques, ne pas éliminer ce déchet électronique avec les déchets usuels non soumis au tri sélectif. Se conformer aux ordonnances locales sur les déchets municipaux pour les dispositions d'élimination et communiquer avec le service à la clientèle pour des renseignements sur les options d'élimination responsable.

Conformity symbols

Conformity mark	Description
	Indicates conformity with safety requirements for Canada and U.S.A.



(continued)

Conformity mark	Description
	Indicates conformity with European Union requirements.
	Indicates conformity with United Kingdom requirements.
	Indicates conformity with Australian standards for electromagnetic compatibility.

Safety alerts on this instrument

Additional text may be used with one of the symbols described above when more specific information is needed to avoid exposure to a hazard. See the following table for safety alerts found on the instrument.

English	Français
 CAUTION! Hazardous chemicals. Read the Safety Data Sheets (SDSs) before handling.	 MISE EN GARDE ! Produits chimiques dangereux. Lire les fiches signalétiques (FS) avant de manipuler les produits.
 CAUTION! Hazardous waste. Refer to SDS(s) and local regulations for handling and disposal.	 MISE EN GARDE ! Déchets dangereux. Lire les fiches signalétiques (FS) et la réglementation locale associées à la manipulation et à l'élimination des déchets.
 DANGER! Class 3B (III) visible and/or invisible laser radiation present when open and interlocks defeated. Avoid exposure to beam.	 DANGER ! Rayonnement laser ou DEL visible ou invisible de classe 3B (III) présent en position ouverte et avec les dispositifs de sécurité non enclenchés. Éviter toute exposition au faisceau.

Location of safety labels

The instrument label with safety symbols is located on the rear panel of the instrument.

IMPORTANT! If any of the labels are not present on the instrument, contact Technical Support for replacements.



Figure 24 Instrument label for the SeqStudio™ Genetic Analyzer

Safety information for instruments not manufactured by Thermo Fisher Scientific

Some of the accessories provided as part of the instrument system are not designed or built by Thermo Fisher Scientific. Consult the manufacturer's documentation for the information needed for the safe use of these products.

Instrument safety

General



CAUTION! Do not remove instrument protective covers. If you remove the protective instrument panels or disable interlock devices, you may be exposed to serious hazards including, but not limited to, severe electrical shock, laser exposure, crushing, or chemical exposure.

Physical injury



CAUTION! Moving Parts. Moving parts can crush, pinch and cut. Keep hands clear of moving parts while operating the instrument. Disconnect power before servicing.



Electrical safety



WARNING! Ensure appropriate electrical supply. For safe operation of the instrument:

- Plug the system into a properly grounded receptacle with adequate current capacity.
- Ensure the electrical supply is of suitable voltage.
- Never operate the instrument with the ground disconnected. Grounding continuity is required for safe operation of the instrument.



WARNING! Power Supply Line Cords. Use properly configured and approved line cords for the power supply in your facility.



WARNING! Disconnecting Power. To fully disconnect power either detach or unplug the power cord, positioning the instrument such that the power cord is accessible.

Cleaning and decontamination



CAUTION! Cleaning and Decontamination. Use only the cleaning and decontamination methods specified in the manufacturer's user documentation. It is the responsibility of the operator (or other responsible person) to ensure the following requirements are met:

- No decontamination or cleaning agents are used that could cause a HAZARD as a result of a reaction with parts of the equipment or with material contained in the equipment.
- The instrument is properly decontaminated a) if hazardous material is spilled onto or into the equipment, and/or b) prior to having the instrument serviced at your facility or sending the instrument for repair, maintenance, trade-in, disposal, or termination of a loan (decontamination forms may be requested from customer service).
- Before using any cleaning or decontamination methods (except those recommended by the manufacturer), users should confirm with the manufacturer that the proposed method will not damage the equipment.

Instrument component and accessory disposal

To minimize negative environmental impact from disposal of electronic waste, do not dispose of electronic waste in unsorted municipal waste. Follow local municipal waste ordinances for proper disposal provision and contact customer service for information about responsible disposal options.

Laser



WARNING! LASER HAZARD. Under normal operating conditions, the SeqStudio™ Genetic Analyzer is categorized as a Class 1 laser product. However, removing the protective covers and defeating the interlock(s) may result in exposure to the internal Class 3B laser. Lasers can burn the retina, causing permanent blind spots. To ensure safe laser operation:

- Never look directly into the laser beam.
- Do not remove safety labels, instrument protective panels, or defeat safety interlocks.
- The system must be installed and maintained by a Thermo Fisher Scientific Technical Representative.
- Remove jewelry and other items that can reflect a laser beam into your eyes or those of others
- Wear proper eye protection and post a laser warning sign at the entrance to the laboratory if the laser protection is defeated for servicing
- DO NOT operate the laser when it cannot be cooled by its cooling fan; an overheated laser can cause severe burns on contact.



Safety and electromagnetic compatibility (EMC) standards

The instrument design and manufacture complies with the following standards and requirements for safety and electromagnetic compatibility.

Safety compliance

Reference	Description
EU Directive 2014/35/EU	European Union “Low Voltage directive”
EN 61010-1 UL 61010-1 CSA C22.2 No. 61010-1	<i>Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements</i>
EN 61010-2-010	<i>Safety requirements for electrical equipment for measurement, control and laboratory use – Part 2-010: Particular requirements for laboratory equipment for the heating of materials</i>
EN 61010-2-081	Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 2-010: Particular requirements for automatic and semi-automatic laboratory equipment for analysis and other purposes
EN 60825-1	Safety of lasers products – Part 1: Equipment classification and requirements

EMC

Reference	Description
EU Directive 2014/30/EU	European Union “EMC Directive”
EN 61326-1	Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements
AS/NZS CISPR 22	Limits and Methods of Measurement of Electromagnetic Disturbance Characteristics of Industrial, Scientific, and Medical (ISM) Radiofrequency Equipment
ICES-003, Issue 5	Industrial, Scientific and Medical (ISM) Radio Frequency Generators
FCC Part 15 Subpart B (47 CFR)	<i>U.S. Standard Radio Frequency Devices</i>



(continued)

Reference	Description
RFID	FCC Notice (for U.S. Customers): This device complies with Part 15 of the FCC Rules: Operation is subject to the following conditions: 1. This device may not cause harmful interference, and 2. This device must accept any interference received, including interference that may cause undesired operation Changes and Modifications not expressly approved by Thermo Fisher Scientific can void your authority to operate this equipment under Federal Communications Commissions rules. Canada (English): This device complies with Industry Canada licence-exempt RSS standard(s). Operation is subject to the following two conditions: (1) this device may not cause interference, and (2) this device must accept any interference, including interference that may cause undesired operation of the device. Canada (Français): Le présent appareil est conforme aux CNR d'Industrie Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes : (1) l'appareil ne doit pas produire de brouillage, et (2) l'utilisateur de l'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

Environmental design

Reference	Description
Directive 2012/19/EU	European Union “WEEE Directive” —Waste electrical and electronic equipment
Directive 2011/65/EU	European Union “RoHS Directive” —Restriction of hazardous substances in electrical and electronic equipment
SJ/T 11364-2014	“China RoHS” Standard —Marking for the Restricted Use of Hazardous Substances in Electronic and Electrical Products

Radio compliance

Reference	Description
Directive 2014/53/EU (as of June 12, 2017)	European Union “RE Directive” — Radio equipment



Chemical safety



WARNING! GENERAL CHEMICAL HANDLING. To minimize hazards, ensure laboratory personnel read and practice the general safety guidelines for chemical usage, storage, and waste provided below. Consult the relevant SDS for specific precautions and instructions:

- Read and understand the Safety Data Sheets (SDSs) provided by the chemical manufacturer before you store, handle, or work with any chemicals or hazardous materials. To obtain SDSs, see the “Documentation and Support” section in this document.
- Minimize contact with chemicals. Wear appropriate personal protective equipment when handling chemicals (for example, safety glasses, gloves, or protective clothing).
- Minimize the inhalation of chemicals. Do not leave chemical containers open. Use only with adequate ventilation (for example, fume hood).
- Check regularly for chemical leaks or spills. If a leak or spill occurs, follow the manufacturer's cleanup procedures as recommended in the SDS.
- Handle chemical wastes in a fume hood.
- Ensure use of primary and secondary waste containers. (A primary waste container holds the immediate waste. A secondary container contains spills or leaks from the primary container. Both containers must be compatible with the waste material and meet federal, state, and local requirements for container storage.)
- After emptying a waste container, seal it with the cap provided.
- Characterize (by analysis if necessary) the waste generated by the particular applications, reagents, and substrates used in your laboratory.
- Ensure that the waste is stored, transferred, transported, and disposed of according to all local, state/provincial, and/or national regulations.
- **IMPORTANT!** Radioactive or biohazardous materials may require special handling, and disposal limitations may apply.

Biological hazard safety



WARNING! BIOHAZARD. Biological samples such as tissues, body fluids, infectious agents, and blood of humans and other animals have the potential to transmit infectious diseases. Conduct all work in properly equipped facilities with the appropriate safety equipment (for example, physical containment devices). Safety equipment can also include items for personal protection, such as gloves, coats, gowns, shoe covers, boots, respirators, face shields, safety glasses, or goggles. Individuals should be trained according to applicable regulatory and company/ institution requirements before working with potentially biohazardous materials. Follow all applicable local, state/provincial, and/or national regulations. The following references provide general guidelines when handling biological samples in laboratory environment.

- U.S. Department of Health and Human Services, *Biosafety in Microbiological and Biomedical Laboratories (BMBL)*, 6th Edition, HHS Publication No. (CDC) 300859, Revised June 2020
[cdc.gov/labs/bmbi](https://www.cdc.gov/labs/bmbi)
- Laboratory biosafety manual, fourth edition. Geneva: World Health Organization; 2020 (Laboratory biosafety manual, fourth edition and associated monographs)
[who.int/publications/i/item/9789240011311](https://www.who.int/publications/i/item/9789240011311)



Documentation and support

Related documentation

Document	Pub. No.
<i>SeqStudio™ Genetic Analyzer Instrument and Software Getting Started Guide</i>	MAN0018654
<i>SeqStudio™ Genetic Analyzer for HID User Bulletin—New Software Features and Verification/Validation Studies</i>	MAN1001221
<i>SeqStudio™ Genetic Analyzer Site Preparation Guide</i>	MAN0016143
<i>SAE Administrator Console v2.2 User Guide for Capillary Electrophoresis Products</i>	MAN1000261
<i>SAE Administrator Console v2.1 User Guide for Capillary Electrophoresis Products</i>	MAN0025849
<i>DNA Fragment Analysis by Capillary Electrophoresis User Guide</i>	4474504
<i>DNA Sequencing by Capillary Electrophoresis Chemistry Guide Second Edition</i>	4305080
<i>Troubleshooting Sanger sequencing data User Bulletin</i>	MAN0014435

Customer and technical support

Visit thermofisher.com/support for the latest service and support information.

- Worldwide contact telephone numbers
- Product support information
 - Product FAQs
 - Software, patches, and updates
 - Training for many applications and instruments
- Order and web support
- Product documentation
 - User guides, manuals, and protocols
 - Certificates of Analysis
 - Safety Data Sheets (SDSs; also known as MSDSs)

Note: For SDSs for reagents and chemicals from other manufacturers, contact the manufacturer.



Limited product warranty

Life Technologies Corporation and its affiliates warrant their products as set forth in the Life Technologies' General Terms and Conditions of Sale at www.thermofisher.com/us/en/home/global/terms-and-conditions.html. If you have questions, contact Life Technologies at www.thermofisher.com/support.

Numerics

3200 data, SeqStudio Genetic Analyzer 30, 130

A

AB1 file
 export automatically 86
 export manually 130, 171
amber light blinking 15
analysis software 30
analyze data, SeqStudio Genetic Analyzer (3200)
 30, 130
audit reason, specify 64
auto calibration 212
auto export
 failed 131
 sample data files (AB1 and FSA) 86
 set network location 149
 set Thermo Fisher Connect Platform location
 147
autospectral 212

B

base calling settings 143, 161
biohazard safety 256

C

calibration. *See* dye calibration
cancel injection 114
cartridge
 history 177
 instruments on which a cartridge has been in-
 stalled 177
 load 43
 maximum days allowed on instrument 42
 remove 178
 reuse 18, 175
 storage 18, 175
 storage, on-instrument long-term 182
cartridge storage mode 177
cathode buffer
 assemble 183

 check fill line 43
 load 43
 maximum days allowed on instrument 42
CBC. *See* cathode buffer
Cloud. *See* Thermo Fisher Connect Platform
computer, connect directly to instrument (hardware)
 27
condition number 215
Connect cloud-based platform
 browser requirements 25
 operating system requirements 25
connect instrument to the Connect cloud-based
 platform. *See* link
consumables status
 check 42
 updates 21
CSV file
 import 87
 save plate setup 137
custom dye 212, 216
 See also dye calibration, custom
custom field 82, 96

D

date and time 164
delete, plate 153, 170
Demo mode 164
disposal of instrument components and accessories
 252
documentation, related 186, 257
dye calibration
 auto calibration 212
 custom dye 212
 custom dye, add 216, 217
 custom, add to Plate Manager or another instru-
 ment 220
 custom, import or export 222
 custom, perform 219
 factory 212
 history 221
 manual 212
 manual calibration, determine if required 212
 overview 211
 perform 214

system dye 212

E

E-signature records E-signature records

dye calibration 67

install run 67

EPT plot, remote monitor 108

export

failed 131

plate 153, 170

Export Status screen 131

F

file name convention 81, 94, 162

fill level 173

firmware version 163

fragment analysis, run module, size ranges and run times 156, 239

front panel 15

FSA file

export automatically 86

export manually 130, 171

H

HID analysis, results 121

HID kit. *See* kit

I

import plate setup, CSV or PSM file 87, 154

injection

cancel 114

options 97, 112, 129

order, change 97, 112, 129

order, default 39

re-inject 97, 112, 129

injection group, definition 39

install check. *See* install run

install run

history 232

log file, export 187

overview 223

perform 224

results 225, 228

sequence and fragment in same run 223

instrument

connect to SAE server 60

connect to wired or wireless network (hardware) 27

disconnect from Connect cloud-based platform 51

lock 101, 152

log file 187

remove user 51

view results 119

instrument conditions during run 126

instrument name 163

instrument profile

delete 168

PIN, change 55, 169

Instrument-cartridge history, view or export 177

IP address 144, 163

K

kit (HID analysis only), marker-to-marker pull-up reduction 71, 85

kit, HID, import 238

L

limited product warranty 258

load

cartridge 179

CBC, sample plate, and cartridge order 43

plate 98

tubes 98

lock instrument 101, 152

log file

export 187

install run 187

M

manual dye calibration 212

marker-to-marker pull-up reduction (HID analysis only), set for injection group 71, 85

mobile device

monitor a run 109

view results 109

My plates. *See* hidden

N

network

auto export location 149

connect instrument to 165

connect to 146

connection (hardware) 27

open plate setup default location 148

network configuration and security 27

P

password security 28

pause plate 114
 PCR amplicons from diploid genes, sequencing settings 143, 161
 PDF, plate setup 137
 plasmid templates, sequencing settings 143, 161
 plate
 load 43
 loading guidelines 39
 pause 114
 Plate Manager
 add custom dye calibration 136
 desktop, access 73
 operating system requirements 25
 plate setup
 create on instrument 87
 CSV file, import 87
 delete 138
 guest 75, 85
 hidden (my plates) 81, 95
 hidden (my plates), definition 85
 hidden (my plates), set in Plate Manager 75
 import from CSV or PSM file 154
 import on instrument 87
 open 87
 PSM file, import 87
 security 81, 95
 security, definition 85
 security, set in Plate Manager 75
 set default open location on network 148
 set default open location on Thermo Fisher Connect Platform 146
 shared (public) 81, 95
 shared (public), definition 85
 shared (public), set in Plate Manager 75
 template 136, 137
 plate type supported 38
 polymer delivery system (PDS), refresh 178
 private plate setup. *See* hidden
 probability of error 125
 profile. *See* instrument profile
 PSM file
 See also plate setup
 create or open in Plate Manager 74
 import 87
 See also plate setup
 public plate setup 81, 95
 See also shared
 pull-up reduction (HID analysis only) 71, 85

Q

Q and Condition 216
 QC report, export 129
 quality flag 121

quality value
 See also QV
 probability of error 125
 spectral calibration 215
 See also QV
 QV, probability of error 125
 QV ranges 125
 QV30 CRL 228

R

raw data 188
 raw trace, remote monitor 107
 re-inject, remote monitor 106
 re-injection 97, 112, 129
 related documentation 186, 257
 remote monitor
 analyzed trace 108
 EPT plot 108
 pause or cancel run 106
 raw trace 107
 re-inject 106
 view results 105, 118
 remove plate associated with inactive user 171
 replicate injection 80, 93
 results
 export automatically 86
 export manually 130, 171
 view, instrument 119
 view, mobile device 109
 view, remote monitor 105, 118
 results report, export 129
 RFID read and write 21
 run, pause or cancel 114
 run history, delete 171
 run module
 read length and run time 156, 239
 settings 155
 size range and run time 156, 239

S

SAE
 disable on instrument 62
 enable on instrument 61
 overview 59
 server, specify 61
 version compatibility with instrument software versions 60
 safety, biohazard 256
 safety practices 246
 safety, required equipment 247
 sample
 plates and tubes supported 38

- purification 38
 - stability on instrument 39
 - volume requirements 38
- sample plate. *See* plate
- secondary analysis software 30
- sequence quality 125
- sequencing
 - results 124
 - run module, read lengths and run times 156, 239
- sequencing settings
 - description 143, 161
 - PCR amplicons from diploid genes 143, 161
 - plasmid templates 143, 161
- serial number 163
- set in injection group 71, 85
- sign
 - dye calibration 67
 - install run 67
- sign in 36
- sign out 37
- size calling settings 141, 159
- Size Quality 121
- size standard quality 189
- sleep mode 177
- smart phone. *See* mobile device
- software version, compatibility between SAE and instrument software 60
- spectral calibration
 - condition number 215

- quality value 215
- SQ 121
- support, customer and technical 257
- symbols, safety 248
- system dye 212

T

- template, plate setup 136, 137
- terms and conditions 258
- Thermo Fisher Cloud. *See* Connect cloud-based platform
- Thermo Fisher Connect Platform
 - analysis software 30
 - auto export default location 147
 - disconnect instrument 51
 - open plate setup default location 146
- troubleshooting, log file, export 187
- tube type supported 38

U

- update software 166

W

- warranty 258
- wired or wireless instrument connection 165
- wired or wireless network, connect instrument (hardware) 27

